

simple machines are levers, inclined planes, wheel-and-axle, gears and pulley systems.

MACHINE CODE (MACHINE LANGUAGE; OBJECT CODE): A collection of binary digits or bits comprising a long series of zeros and ones that the computer's CPU (central processing unit) reads and understands. All programs must be converted into machine language before they can be run. Because machine language is difficult to read by humans, programmers usually use assembly language, which is a human-readable notation for the machine language in which the instructions represented by patterns of zeros and ones are replaced with alphanumeric symbols called mnemonics.

MACHINE CYCLE TIME: 1. The basic unit of time at the Medium Scale Integration (MSI) complexity level. It is equal to one clock period. A register transfer operation can be completed during a single clock period. 2. The time for a sequential machine to complete a single state transition.

MACHINE-READABLE: Of or relating to data that is stored on a disk or tape that can be understood directly by a computer without the need for retyping. Typed or printed text that can be scanned for optical character recognition or bar codes are also machine-readable.

MACHINE TOOL: An automatic or semi-automatic power-operated or manually operated machine used for cutting and shaping objects. The use of precise machine tools in mass production assembly methods ensure that all duplicate parts produced are virtually identical. The basic functions performed by a machine tool are turning, facing, milling, drilling, boring, broaching, threading and tapping. Example of a power-operated machine, which can perform several functions using machine tools is the **lathe machine**.

MACINTOSH: 1. Often abbreviated as **Mac**, it is a range of personal computers introduced in 1984 by Apple Computers, which were the first widely used computers with graphical user interface (GUI). Each Macintosh computer runs a version of the Mac OS, Apple's desktop operating system. Since 2001, all Macs have run Mac OS X, a redesigned version of the original Mac OS that was built from the NeXTSTEP operating system. 2. Also known as Mackintosh and abbreviated as Mac or Mack, it is a lightweight waterproofed raincoat, made with rubberised cotton fabric. It is named after its Scottish inventor Charles Macintosh (1766-1843).

MACKEREL: A common name for a number of different species of pelagic fish, mostly from the family Scombridae. Some of the types include the Chub mackerel, *Scomber japonicus*; Blue mackerel, *Scomber australasicus*; the Atlantic mackerel, *Scomber scombrus*; and the Pacific mackerel, *Pneumatophorus diego*, with a swim bladder. Mackerel is found in both temperate and tropical seas, mostly living along the coast or offshore in the oceanic environment. It is characterised by a long slender body with vertical silver and blue stripes on their backs and deeply forked tails, pointed head and large mouth. Some stocks migrate in large schools along the coast to suitable spawning grounds on land, where they spawn in fairly shallow waters. After spawning they return to the sea. Mackerel is rich in essential oils, vitamins and minerals. Omega-3 and Omega-6 fatty acids occur in high quantities in this fish. It is one of the highly recommended oily fish for a healthy diet.

MACLAURIN, COLIN (1698 - 1746): Scottish mathematician, who published and extended Newton's work in calculus, geometry and gravitation. He graduated from the University of Glasgow, aged only fourteen, with a thesis on the power of gravity. In 1717 he became professor of mathematics at Marischal College in Aberdeen. In 1725 Maclaurin was appointed deputy professor at the University of Edinburgh.

MACLAURIN'S FORMULA (MACLAURIN'S THEOREM): The theorem of mathematical analysis stating that if a real-valued function is infinitely differentiable in an open neighbourhood of the origin, then $f(x)$ can be approximated locally as the sum of $f(0)$ and an initial segment of the series of terms of the form $f_n(x) = \frac{1}{n!} f^{(n)}(0)x^n$, where $f^{(n)}(x)$ is the n^{th} derivative of $f(x)$.

MACLAURIN'S SERIES: A series of expansion in which all terms are non-negative integer powers of the variable, named after the Scottish mathematician Colin Maclaurin. The **Taylor series** is a representation of a function as an infinite sum of terms that are calculated from the values of the function's derivatives at a single point, introduced by the English mathematician Brook Taylor in 1715. The Maclaurin series is

$$\text{given by: } f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n.$$

MACRE: The inner layer of the shell of a mollusc. It is an iridescent substance, composed mainly of calcium carbonate.

MACRO: 1. A prefix meaning large. 2. In computer science, it is a small program or script, that automates common tasks. These scripts are usually run within programs and can often be created by the user. For example, a user might record a macro for Microsoft Word that inserts his entire address when he presses a custom key combination. A Microsoft Excel user might record a macro to format the data in the selected column of a spreadsheet.

MACROCLADOUS: Having long branches.

MACROCYCLIC: A term describing a chemical compound whose molecules have a large ring structure containing a ring of seven or more atoms, for example, streptomycin and rapamycin.

MACROEVOLUTION: Evolution such as the emergence of entire or large groups of organisms, for example, flowering plants or mammals on a relatively large scale, including the origins of novel organism designs, evolutionary trends, adaptive radiations and extinctions.

MACROFAUNA: Large animals, which can be seen by the naked eye without the use of a microscope. Small soil-dwelling invertebrates such as annelids and nematodes, may be separated into an intermediate category called the **meso-fauna**.

MACROFIBRILS: Fibrils within the cell wall that are large enough in some cases to be visible under light microscope. They are formed from microfibrils lying parallel to each other.

MACROLECITHAL: Of an egg, having a large yolk in the cytoplasm of ovums, as found in birds, reptiles, fish, sharks, many insects, crustaceans and monotremes. Macrolecithal eggs are described as telolecithal.

MACROMETEOROLOGY: The study of the largest-scale aspects of the atmosphere, for example, general global circulation.

MACROMOLECULAR CRYSTAL: A crystalline solid in which the atoms are all linked together by covalent bonds. Examples of substances with macromolecular crystals are carbon as it occurs in diamond, boron nitride and silicon carbide. The crystal is a large molecule, which accounts for the hardness and high melting point of such materials.

MACROMOLECULAR STRUCTURE (GIANT COVALENT STRUCTURE; GIANT MOLECULAR STRUCTURE): A solid structure of covalent substances such as diamond, graphite and silicon dioxide (silicon(IV) oxide), made up of a huge number of similar atoms joined by covalent bonds in one dimension (long chains, such as polymers); two dimensions found in flat sheets as in graphite, where each layer is a giant molecule; or in three dimensions as found in diamond and silica. Giant molecules do not conduct electricity, as they have no electrons as solids and no free ions when molten. They are some of the hardest structures known because of the strong bonds between their atoms. These bonds also give them high melting and boiling points. Graphite, an allotrope of carbon, is an anomaly here as free electrons between its layers enable it to conduct electricity.

MACROMOLECULE: 1. A very large, usually organic polymer such as starch, haemoglobin and polythene made up of many smaller structural units called monomers linked together. Many natural and synthetic polymers and many nucleic acids and proteins are macromolecules. 2. A crystal such as a diamond, in which the atoms are identical and held by covalent bonds of equal strength. 3. A giant molecule of living matter formed by the joining of smaller molecules,

usually by condensation synthesis. Polysaccharides, proteins and nucleic acids (DNA and RNA) are examples of macromolecules.

Macromolecular is a term used to describe the structure of chemical compounds that consists of macromolecules, for example, proteins, haemoglobin and polypeptide.

MACROMONOMER: Polymer or oligomer whose molecules each have one end-group that acts as a monomer molecule, so that each polymer or oligomer molecule contributes only a single monomer unit to a chain of the product polymer.

MACROMONOMER MOLECULE: A macromolecule that has one end-group which enables it to act as a monomer molecule, contributing only a single monomeric unit to a chain of the final macromolecule.

MACROMONOMERIC UNIT (MACROMONOMER UNIT): The largest constitutional unit contributed by a single macromonomer molecule to the structure of a macromolecule.

MACRONUTRIENT (ESSENTIAL NUTRIENT; MAJOR NUTRIENT; ESSENTIAL ELEMENT): An inorganic nutrient other than carbon, hydrogen and oxygen, consisting of nitrogen (N), potassium (K), calcium (Ca), phosphorus (P), magnesium (Mg) and sulfur (S) present in the air, soil or food sources like carbohydrates, proteins, lipids, mineral salts, vitamins and water that are required in large amounts by organisms (plants and animals) for normal growth and development. They are used in formation of structural components of organisms or their complex forms carbohydrates, lipids and proteins are used as sources of energy.

MACROPHAGE: A large phagocytic cell, which is a type of white blood cell that is present in lymph nodes, connective tissues, lungs, lining of liver sinusoids, skin and nervous tissues, forming an important part of the body's immune system. It attacks foreign substances and helps the rest of the immune system identify harmful toxins, bacteria and viruses. The sinusoids of the spleen are lined with macrophages that remove worn-out red cells and platelets from the blood and destroy them. Macrophages also recognise and bind to microbial surfaces that have been coated with complement proteins. Signals from helper T cells, play a crucial role in stimulating macrophages to destroy their ingested bacteria by increasing the expression of toxic effector molecules including inducible oxide synthase, hydrogen peroxide (H_2O_2), superoxide anion (O_2^-) and nitric oxide (NO), which are all toxic effector molecules produced by macrophages to kill engulfed microorganisms.

MACROPHAGOUS: A method of feeding in heterotrophic organisms in which food is ingested in the form of relatively large pieces.

MACROPHYLL (MEGAPHYLL): The relatively large and expanded type of leaves. This is usually typical of seed plants and ferns. They are thought to have evolved from early leafless plants such as *Rhynia*. On the other hand, a **Microphyll** is the relatively small type of leaves.

MACROPINOCYTOSIS: A mechanism of endocytosis by which dendritic cells engulf relatively large volumes of fluid from their surroundings to take in antigens.

MACROPORE: Large soil pores, usually between aggregates from which water drains readily by gravity and helps to aerate the soil. Macropores are generally more than 0.08 mm in diameter.

MACROSCLEREID: A sclereid cell that is rod-like and formed from the embryonic epidermis of certain seed coats. Macrosclereids form the outer layer of the seed coat in many plants.

MACROSCOPIC: Designating a size scale very much larger than that of atoms and molecules and observable by the naked eye. For example, the macroscopic view of an object such as an orange is just as the naked eye sees it. Macroscopic objects and systems are described by classical physics although quantum mechanics can have macroscopic consequences.

MACROSCOPIC CROSS-SECTION: The cross-section per unit volume of a given material for a specified process. For a pure nuclide, it is the product of the microscopic cross-section and the number of target nuclei per unit volume. For a mixture of nuclides, it is the sum of such products.

MACROSCOPIC FILM: A film which has lateral dimensions above 10 μm .

MACROSCOPIC KINETICS: A term applied to a kinetic study in which the interest is in the behaviour of bulk systems, for example, in changes in concentrations of reactants and products.

MACROSPORANGIUM (MEGASPORANGIUM): A structure in which megaspores are formed, which corresponds to the ovule seed plants.

MACROSPORE (MEGASPORE): The larger of the two types of spores that gives rise to a female gametophyte present in heterosporous plants, which include seed plants (gymnosperms and flowering plants), water ferns (Salviniales) and spike mosses (Selaginellaceae). It contains food reserves for the gametophyte. In angiosperms, the cell that gives rise to the nuclei of the embryo sac is termed a megaspore, though it is usually smaller than a pollen grain (microspore).

MACROSPOROPHYLL (MEGASPOROPHYLL): A leaf-like structure that bears the megasporangia. In angiosperms and gymnosperms it is represented by the carpel and ovuliferous scale respectively.

MACROSTYLOUS: A flower that possesses a long style and short stamens.

MACULA: 1. A patch of sensory hair cells in the utricle and saccule of the inner ear that provide information about the position of the body in relation to gravity. The hair of the cell is embedded in an otolith, a gelatinous cap containing particles of calcium carbonate. 2. Also called **macula lutea**, it is an oval-shaped highly pigmented yellow spot near the centre of the retina of the human eye with increased visual activity. They occur in some animals that lack foveae and often surround the fovea of animals that possess them. Because the macula is yellow in colour it absorbs excess blue and ultraviolet light that enters the eye and acts as a natural sunblock or sunglasses for this area of the retina.

MAD COW DISEASE, MCD (BOVINE SPONGIFORM ENCEPHALOPATHY, BSE): A neuro-degenerative disease that affects cattle's nervous system making it behave abnormally. It is caused by an abnormal form of a cellular protein called prion that is a microorganism, many times smaller than virus. It causes a spongy degeneration in the brain and spinal cord as well as redness of the eyes. There is a human type of the disease, called Creutzfeldt-Jakob disease, believed to be caused by infection of prion.

MADELUNG CONSTANT: A constant, first calculated by Erwin Madelung, a German physicist in 1918, used in determining the cohesion of ionic crystals by approximating the ions by point charges.

The electrostatic interaction per ion pair, U is given by $U(r) = \frac{\alpha e^2}{r}$,

where α is the Madelung constant and $\frac{e^2}{r}$ is the Coulomb interaction

between the ions, with r being the lattice constant. The value of α depends on the type of lattice. For the sodium (NaCl) chloride lattice, Madelung constant, α_M has a value of about 1.75. A more realistic calculation of cohesion is obtained if short-range repulsion

with an inverse power law are included, i.e. $U(r) = \frac{\alpha e^2}{r} - \frac{C}{r^n}$, where C and n are constants. The value of α can be used in calculations to determine C and n .

MADELUNG'S RULE (THE MADELUNG ENERGY ORDERING RULE): An approximate rule about atomic orbital stating that the order of their being filled is the order of increasing $n + l$ and that when two or more orbitals have the same value of $n + l$ then the order of filling is increasing n . This rule was first put forward by Erwin Madelung (1881-1972) in 1936. There are some exceptions to it but it does have some theoretical justification.

MADS BOX GENE: Any of a super family of genes that encode transcription factors having crucial roles in the development of plants, animals and fungi. They are characterised by a sequence of about 56

amino acids that forms part of a DNA-binding domain of the protein. MADS is the acronym formed from the first letter of the genes: MCM1 (from yeast), agamous and deficiens (from flowering plants) and serum response factor (SRF) from mammals. Some act as homeotic genes analogous to the Hox genes of animals, with mutations causing the transformation of one flower part into another.

MAEDI-VISNA (GRAFF–REINERT DISEASE; LA OVINE PROGRESSIVE PNEUMONIA; LYMPHOID INTERSTITIAL PNEUMONIA): A chronic virus disease of sheep caused by a lenti virus, which causes pneumonia and breathing difficulties.

MAFIC: Relating to igneous rock composed chiefly of dark-coloured ferromagnetic minerals or such minerals. Mafic minerals include amphibole, olivine and pyroxene.

MAGELLANIC CLOUDS: Two dwarf galaxies, Large Magellanic Cloud (LMC) and Small Magellanic Cloud (SMC) situated in the constellation Dorado and constellation Tucana respectively in the far southern sky, close to the Milky Way. They are visible to the unaided eye from the Southern Hemisphere only and named after, Ferdinand Magellan (1480-1521), Portuguese-born Spanish explorer and navigator.

MAGENTA (FUCHSINE; FUCHSIN): A bright or dark purplish red dyestuff, with the chemical formula $C_{20}H_{19}N_3HCl$ and melting point 200 °C (473 K). It is used in colouring textile and leather.

MAGGOT: The worm-shaped larva of various members of the fly family order Diptera, found in decaying matter. Maggots are used as baits by anglers.

MAGHEMITE (GAMMA Fe_2O_3):

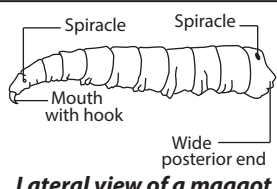
A form of iron oxide, consisting of ferric oxide with the crystalline structure and strongly magnetic nature of magnetite but the composition of hematite, which can act as a powerful catalyst. On Earth, it is usually found only around the edges of regions of hydrothermal or magnetic activity, where the temperature ranges from 300 to 400°C. The abundance of water on Earth has converted much of the maghemite into a noncatalytic form, but on Mars it has survived virtually unaltered. Synthetic maghemite is being studied as part of an electrically conductive and magnetisable film in nanocomposite structures.

MAGIC CUBE: By analogy with a magic square, a magic cube is defined as an $n \times n \times n$ array of integers in which the n^2 rows, n^2 columns, n^2 pillars and four space diagonals each sums to a single number $M(n)$ known as the cube's magic constant. If a magic cube of order n is made up of the consecutive numbers 1, 2, ..., n , it is known as a normal magic cube and the magic constant is given by $M(n) = 1/2(n^3 + 1)$.

MAGIC NUMBERS: Numbers of neutrons or protons that occur at particle numbers in atomic nuclei to produce very stable nuclei, which are required to fill major quantum shells. They include 2, 8, 20, 28, 50, 82 and 126

MAGIC SQUARE: A square array of different numbers in which the rows, columns and diagonals add up to the same total. A simple example is obtained by employing the numbers 1 – 9 with a total of 15, as shown in the diagram.

MAGIC-ANGLE SPINNING: A technique used in solid-state nuclear magnetic resonance (NMR) for making the line widths smaller. Both the

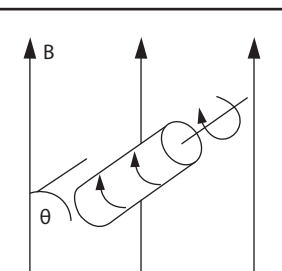


Lateral view of a maggot

4	9	2
3	5	7
8	1	6

A 3 x 3 Magic Square

dipole-dipole interaction and the chemical shift anisotropy have the angular dependence $1 - 3\cos^2\theta$, where θ is the magic angle between the principal axis of the molecule and the applied magnetic field that satisfies $1 - 3\cos^2\theta = 0$ and is given by $\theta = 54.7450$. In magic angle spinning the material is spun very rapidly at the magic angle to the applied magnetic field so that the dipole-dipole interactions and chemical shift anisotropies average to zero. It is necessary for the frequency of spinning to be at least as large as the width h of the spectrum. This technique has been extensively used, with the spinning between 4 and 5KHz.



Magic-Angle Spinning

MAGLEV (MAGNETIC LEVITATION): A transportation system in which trains glide above a track, supported by magnetic repulsion and propelled by a linear motor.

MAGMA: Molten rock material usually consisting of silicate liquid beneath the Earth's surface from which igneous rocks are formed.

Lava is magma that has reached the surface and solidified from volcanic eruptions.

MAGNADUR: A trade name for a ceramic material used to make permanent magnets. It consists of sintered iron oxide and barium oxide.

MAGNALIUM: Trade name for an aluminium alloy with 1.5 to 2% magnesium and small amounts of copper, nickel and tin. Magnalium is corrosion resistant and have lower density than pure aluminium. It is workable and easier to weld than pure aluminium. Magnalium alloys are used to make aircraft bodies and automobile parts.

MAGNESITE: A white, colourless or grey mineral of magnesium carbonate, with the chemical formula $MgCO_3$, mined as source of magnesium carbonate. It crystallises in the trigonal system. It is formed as a replacement mineral of magnesium rich rocks when carbon dioxide is available. Magnesite can be used as a slag in steelmaking furnaces, in conjunction with lime, in order to protect the magnesium oxide lining. It can also be used as a catalyst and filler in the production of synthetic rubber and in the preparation of magnesium chemicals and fertilisers.

MAGNESIUM: A light silvery white, very ductile and malleable metallic element, a member of Group 2 (IIA), alkaline earth metals with the symbol Mg, atomic number 12, atomic weight 24.305, melting point about 648.8 °C (922 K) and boiling point about 1090 °C (1363 K). It is the eighth most abundant element in the earth's crust but does not occur by itself in nature. It has a hexagonal close-packed crystalline structure. It occurs in abundance in the minerals brucite, magnesite, dolomite and carnalite; and also in silicate form in asbestos, meerschaum, serpentine, talc and seawater, which is the main source. It is extracted by electrolysis of molten magnesium chloride. It is chemically active and burns fiercely with a brilliant white flame. It reacts with the halogens, sulfur and nitrogen. The element is used in a number of light alloys such as magnalium, used to make aircraft bodies and automobile parts. Magnesium forms many compounds. The oxide, hydroxide, chloride, carbonate and sulfate are commercially important and used in ceramics, cosmetics, fertilisers, insulation, leather tanning and textile processing and also Epsom salts (magnesium sulfate heptahydrate, $MgSO_4 \cdot 7H_2O$), milk of magnesia (magnesium hydroxide, $Mg(OH)_2$) and citrate of magnesia are used in medicine. Magnesium is an essential element for living organisms. It functions as a cofactor for various enzymes and is a constituent of chlorophyll in plants. In animals, it is involved in the transmission of nerve impulses.

MAGNESIUM CARBONATE: A white compound with the molecular formula $MgCO_3$ that occurs in nature as a mineral. The anhydrous

materials (trigonal) are found in the mineral magnesite. There is also a trihydrate, $\text{MgCO}_3 \cdot 3\text{H}_2\text{O}$ (rhombic), which occurs naturally as nesquehonite and a pentahydrate, $\text{MgCO}_3 \cdot 5\text{H}_2\text{O}$ (monoclinic), which occurs as lansfordite. Magnesium carbonate also occurs in the mixed salt dolomite ($\text{CaCO}_3 \cdot \text{MgCO}_3$) and as basic magnesium carbonate in the two minerals artinite ($\text{MgCO}_3 \cdot \text{Mg}(\text{OH})_2 \cdot 3\text{H}_2\text{O}$) and hydromagnesite ($3\text{MgCO}_3 \cdot \text{Mg}(\text{OH})_2 \cdot 3\text{H}_2\text{O}$). The anhydrous salt can be formed by stream of carbon dioxide: $\text{MgO}_{(s)} + \text{CO}_{2(g)} \rightarrow \text{MgCO}_{3(s)}$. It is a commercial antacid and the anhydrous form is used as a drying agent. Magnesium carbonate is used in making magnesium oxide and is a drying agent, for example, in table salt; and also as a refractory for furnace lining and various grades of magnesium oxide. It is also used as a laxative (the basic carbonate is used) and is a component of certain inks and glasses.

MAGNESIUM CHLORIDE: A white solid compound, with the chemical formula MgCl_2 . Its molar mass is 5.211 g/mol (anhydrous), 203.31 g/mol (hexahydrate), density 2.32 g/cm³ (anhydrous), 1.569 g/cm³ (hexahydrate), melting point 714 °C (987 K) on rapid heating: slow heating leads to decomposition from 300 °C (573 K), boiling point 1,412 °C (1,685 K). The anhydrous salt (hexagonal) can be prepared by the direct combination of dry chlorine with magnesium: $\text{Mg}_{(s)} + \text{Cl}_{2(g)} \rightarrow \text{MgCl}_{2(s)}$. The compound also occurs naturally as a constituent of carnalite ($\text{KCl} \cdot \text{MgCl}_2$), a deliquescent compound that commonly forms the hexahydrate, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (monoclinic). When heated, this hydrolyses to give magnesium oxide and hydrogen chloride gas. The hydrated magnesium chloride can be extracted from brine or seawater. Magnesium chloride is used for a variety of applications such as the production of magnesium, the manufacture of textiles, paper, fireproofing agents, cements and refrigeration brine and for dust and erosion control.

MAGNESIUM HYDROGEN CARBONATE (MAGNESIUM BICARBONATE): A compound with the chemical formula $\text{Mg}(\text{HCO}_3)_2$ that is stable only in solution and one of the compounds responsible for temporary hardness in water. It is formed by the action of carbon dioxide on a suspension of magnesium carbonate in water. The process can be reversed by heating. It can also be produced by treating a suspension of magnesium hydroxide with pressurised carbon dioxide: $\text{Mg}(\text{OH})_2 + 2\text{CO}_2 \rightarrow \text{Mg}(\text{HCO}_3)_2$.

MAGNESIUM HYDROXIDE: A white crystalline solid compound with the chemical formula $\text{Mg}(\text{OH})_2$ (trigonal), relative density 2.36 and decomposes at 350 °C (623 K). It is often called **milk of magnesia** when suspended in water. Magnesium hydroxide occurs naturally as the mineral brucite and can be prepared by reacting magnesium sulfate or chloride with sodium hydroxide solution. A common component of antacids (milk of magnesia) and laxatives, it is also used in refining sugar and in the processing of uranium.

MAGNESIUM OXIDE (MAGNESIA): A white solid mineral with the chemical formula MgO , density 3.58 g/cm³, melting point 2852 °C (3125 K) and boiling point 3600 °C (3873 K), which occurs naturally as the mineral periclase. It is prepared commercially by thermally decomposing the mineral magnesite: $\text{MgCO}_{3(s)} \rightarrow \text{MgO}_{(s)} + \text{CO}_{2(g)}$. It has a wide range of uses, including as moisture absorbent for preserving books in libraries; for making cement in dry process plants; as reflective coatings on optical instruments and aircraft windscreens and in semiconductors; as a refractory lining in metal and glass furnaces; and to treat acidity of the stomach.

MAGNESIUM PEROXIDE: An odourless, white or slightly yellow, amorphous, fine powder with the chemical formula MgO_2 . It decomposes at 100 °C (373 K) to release oxygen. It is similar to calcium peroxide (CaO_2), which also releases oxygen on reaction with water. It is prepared by reacting hydrogen peroxide with magnesium oxide solution. It contains not less than 22% and not more than 28% of hydrogen peroxide. Magnesium peroxide releases oxygen slowly in contact with atmospheric moisture and used to generate

contaminated soils and lakes. It is used as a slow oxygen release source in agricultural for bioremediation of contaminated soil and can improve the soil quality for plant growth and metabolism and in aquaculture. It is used as a disinfectant in cosmetics, detergents, toothpaste and pharmaceuticals; and used as bleach for cotton and silk, disinfecting, deodorising and as a mild laxative.

MAGNESIUM SULFATE (MAGNESIUM TETRAOXOSULFATE(VI): A white soluble compound with the chemical formula MgSO_4 , which exists as the anhydrous compound (rhombic) and in hydrated crystalline forms. The monohydrate $\text{MgSO}_4 \cdot \text{H}_2\text{O}$ (monoclinic) occurs naturally as the mineral kieserite. The commonest hydrate is the heptahydrate, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (rhombic), which is called **Epson salt** and occurs naturally as the mineral epsomite. This is a white powder with a bitter saline taste, which loses $6\text{H}_2\text{O}$ at 150 °C (423 K) and $7\text{H}_2\text{O}$ at 200 °C (473 K). It is used in sizing and fireproofing cotton and silk, in tanning leather and in the manufacture of fertilisers, explosives and matches. In medicine, it is used as a laxative. It is also used in veterinary medicine for treatment of local inflammations and infected wounds.

MAGNET: An object that produces a magnetic field and attracts iron and able to repel other magnets. This attraction property occurs naturally or by passing an electric current through it, called **excited magnets**. It points roughly north and south when freely suspended. Any magnet that retains appreciable magnetisation indefinitely is called a **permanent magnet**, but can be reversed by heating, hitting it or exposing it to extraneous magnetic field. It is often made in the form of a bar or a horse shoe. Magnets are essential for the generation of electric power and are used in motors, generators, information storage, recording and numerous specialised applications, for example, seals of refrigerator doors.

MAGNETAR: A neutron star with an intense magnetic field, of the order of 10^{11} tesla, which is about a thousand trillion times stronger than that of Earth and over 1000 times greater than that of a pulsar.

MAGNETIC ANISOTROPY: A phenomenon in which magnetic properties of a substance are different in different directions relative to a given frame of reference in the substance.

MAGNETIC BOTTLE: A non-uniform magnetic field used as a vessel for plasma in a thermonuclear experimental device.

MAGNETIC BUBBLE MEMORY: A form of computer memory in which a small magnetised region of a substance is used to store information. It holds its content without power. It consists of materials such as magnetic garnets that are easily magnetised in one direction but hard to magnetise in the perpendicular direction.

MAGNETIC CARD: A card that contains a magnetisable strip or surface on which data can be recorded. A magnetic card may contain information about an individual such as available credit on a credit card or personal information about an individual, which can be used for identification and security purposes. An example is a bank card for use in an automated teller machine.

MAGNETIC CIRCUIT: A closed circuit containing a group of magnetic flux lines. There may be air gaps or other materials in the path. The flux is usually generated by permanent magnets or electromagnets and confined to the path by magnetic cores consisting of ferromagnetic materials such as iron. Magnetic circuits are employed to efficiently channel magnetic fields in many devices such as electric motors, generators, transformers, relays, lifting electromagnets, galvanometers and magnetic recording heads.

MAGNETIC CIRCULAR DICHROISM (MCD): Magnetic circular dichroism is observed when a sample differentially absorbs left- and right circularly polarised light in a magnetic field parallel to the light beam.

MAGNETIC CONSTANT (ABSOLUTE PERMEABILITY): The permeability of free space (μ_0) with the value of $4\pi \times 10^{-7}$ henry per metre.

MAGNETIC CORE: 1. Computer storage device made of ferromagnetic coil wound with wires. 2. Ferrous material such as iron oxide or ferrite particles placed in a coil or transformer containing a magnetic flux. This provides better path than air for magnetic flux, thereby increasing the inductance of the coil and increasing the coupling between the windings of a transformer.

MAGNETIC CORE MEMORY (CORE MEMORY): In computer science, it is a form of random-access memory consisting of an array of magnetic cores. It uses tiny magnetic toroids (rings), the cores, through which wires are threaded to write and read information. Each core represents one bit of information. It is now replaced by semiconductor memories.

MAGNETIC DECLINATION (MAGNETIC VARIATION): The angle between magnetic north and true north at a particular point on the Earth's surface, measured in degrees East (+) or West (−) according to whether magnetic north lies east or west of true north. The magnetic variation indicated on a map or a chart is not a fixed quantity but is subject to change. Magnetic variation is indicated by dashed lines on maps and charts. Most maps and charts also indicate the rate of annual change of variation.

MAGNETIC DEFLECTION: The deflection of an ion beam as a result of the motion of the ions in a magnetic field (magnetic sector). Generally, the direction of motion of the ions is at right angles to the direction of the magnetic field and the motion is uniform.

MAGNETIC DIP (ANGLE OF DIP; DIP ANGLE; MAGNETIC INCLINATION): The angle made with the horizontal by the Earth's magnetic field lines at any specific location or by a compass needle when the compass is held in a vertical orientation. It varies at different points on the Earth's surface, for example, it is 0° at the magnetic equator and 90° at each of the magnetic poles. If the value is positive at the point of measurement, the magnetic field of the Earth is pointing downward, into the Earth; and if negative, it is pointing upward.

MAGNETIC DIPOLE: An object, such as a permanent magnet, current loop or particle with angular momentum, which experiences a torque in a magnetic field and gives rise to a magnetic field, as if it consisted of two magnetic poles of opposite sign separated by a small distance.

MAGNETIC DISK: A memory device, such as a floppy disk, a hard disk or a removable cartridge, consisting of a round, flat piece of flexible plastic that is covered with a magnetic coating on which digital information is stored in the form of microscopically small, magnetised needles. Compact discs (CD-ROMs) and new microfloppy designs with more storage capacity are now replacing the earlier designs.

MAGNETIC DOMAIN: A region of a ferromagnetic material within which atomic or molecular magnetic moments are aligned parallel. When heated above a temperature called the Curie temperature, a piece of ferromagnetic material undergoes a phase transition and the uniform magnetisation within a domain spontaneously disappears. Each atom has its own direction of magnetic moment, independent from its neighbouring atoms. Magnetic domain structure is responsible for the magnetic behaviour of ferromagnetic materials such as iron, nickel, cobalt and their alloys and ferrites. The regions separating magnetic domains are called **domain walls**, where the magnetisation rotates coherently from the direction in one domain to that in the next domain.

MAGNETIC DRUM: Computer storage device used in the 1950s and 1960s consisting of a rotating drum coated with magnetic material such as iron oxide with tracks around its circumference. Each track had its own read/write head.

MAGNETIC EFFECT OF CURRENT: The power of electric current, used to create a magnetic field in its immediate surroundings.

Maxwell's right hand grip rule is used to determine the direction of the magnetic field in relation to the current. As shown in the diagram, the magnetic field is like concentric circles having their centre at the axis of the conductor. The direction of these circular magnetic lines is dependent upon the direction of the current. If a straight current-carrying conductor is held in the right hand, so that the fingers wrap around the conductor and the thumb is stretched, the thumb points in the direction of the current and the fingers point in the direction of the magnetic field. For a solenoid, if you wrap your right hand fingers in a circle, to resemble the coils in a solenoid, with the thumb stretched, the fingers will point in the direction of the current and the thumb points towards the end of the solenoid, which is the north-seeking pole.

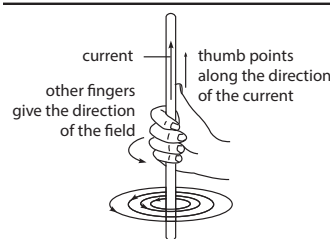


Fig 1: Right hand grip rule for a straight current-carrying conductor

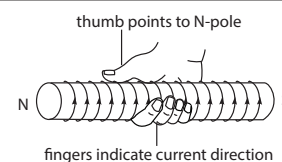


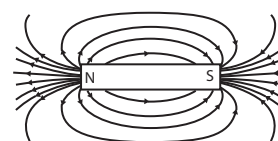
Fig 2: Right hand grip rule for a solenoid carrying current

MAGNETIC ELEMENTS: Elements such as iron, nickel, cobalt, chromium and manganese, which are capable of becoming magnetic.

MAGNETIC EQUIVALENCE: Nuclei having the same resonance frequency in nuclear magnetic resonance spectroscopy and also identical spin-spin interactions with the nuclei of a neighbouring group are magnetically equivalent. The spin-spin interaction between magnetically equivalent nuclei does not appear and thus has no effect on the multiplicity of the respective NMR signals. Magnetically equivalent nuclei are necessarily also chemically equivalent, but the reverse is not necessarily true.

MAGNETIC FIELD: A vector field of force produced by a permanent magnet or around a current carrying conductor detected by the force it exerts on other magnetic materials and moving electric charges. This is a vector field because the magnetic field at any given point is specified by direction and a magnitude or strength. The Earth has its own magnetic field, which is important in navigation using the magnetic compass, with one or more magnetic needles pointing in the general direction of the magnetic North Pole under the influence of the magnetic field of the Earth. Rotating magnetic fields also form the basis of electrical motors and generators.

MAGNETIC FIELD LINES: Imaginary lines in space, used for visually representing the strength and direction of magnetic fields. At any point in space, the local field line points in the direction of the magnetic force, which an isolated magnetic pole at that point would experience. In plasma, magnetic field lines also guide the motion of ions and electrons and direct the flow of some electric currents.



Magnetic field lines near a bar magnet

MAGNETIC FIELD SCAN: The method of producing a momentum (mass) spectrum, by varying the strength of the magnetic field.

MAGNETIC FIELD STRENGTH (MAGNETIC INTENSITY): The magnetic fields generated by currents. It is one of two ways that the intensity of a magnetic field can be expressed, which are magnetic field strength H , measured in amperes per metre (A/m) and magnetic flux density B , measured in Newton-metres per ampere (Nm/A), also called **tesla** (T). When the generated fields pass through magnetic materials which themselves contribute internal magnetic fields, ambiguities can arise about what part of the field comes from the external currents and what comes from the material itself. It has been a common practice to define another magnetic field quantity,

usually called the **magnetic field strength** designated by H . It can be defined by the relationship $H = B_0 / \mu_0 = B_0 / \mu_0 - M$ and has the value of unambiguously designating the driving magnetic influence from external currents in a material, independent of the material's magnetic response. The relationship for B can be written in the equivalent form $B = \mu_0(H + M)$, where $\mu = \mu_m = K_m \mu_0$, μ_0 being the magnetic permeability of space and K_m the relative permeability of the material. If the material does not respond to the external magnetic field by producing any magnetisation, then $K_m = 1$.

MAGNETIC FLUX: Measurement of the total amount of magnetic field in a given region. This is dependent on the strength and the extent of a magnetic field. The unit of magnetic flux density is the weber, with the symbol Φ . It is expressed as: Magnetic flux (Φ) = magnetic field strength (B) multiplied by the component $A \sin \theta$ of the area A that is at right angle to the magnetic field: $\Phi = B A \sin \theta$, where θ is the angle between the area and the magnetic field direction, B is the magnetic field, A is the cross-sectional area and Φ is the flux density. **Magnetic flux density** with the symbol B and also known as the **magnetic induction** is the magnetic flux per unit area of a magnetic field at right angles to the magnetic force and a vector quantity. It is one of two quantities that specifies both the strength and direction of the magnetic field; the other one is magnetic field strength (H). B and H are related as follows: $B = \mu H$, where μ is the permeability.

MAGNETIC FORCE: The attractive or repulsive force exerted on a magnetic pole or a moving electric charge in a magnetic field. This is responsible for the action of electric motors and the attraction between magnets and iron.

MAGNETIC FORCE MICROSCOPY (MFM): A variety of atomic force microscope, where a sharp magnetised tip scans a magnetic sample. The tip-sample magnetic interactions are detected and used to reconstruct the magnetic structure of the sample surface. Many kinds of magnetic interactions are measured by MFM, including magnetic dipole-dipole interaction. MFM scanning often uses non-contact (N-C) atomic force microscopy (AFM) mode.

MAGNETIC INDUCTION: The production of magnetic properties in a ferromagnetic material such as iron or steel, when it is brought close to a magnet, produced by the force of the field radiating from the poles of the magnet.

MAGNETIC LENS: A device that uses a magnetic field to focus a beam of charged particles. The field is produced between two annular pole pieces around the beam driven by an electromagnet. Its imaging properties were first studied by Hans Busch in 1926.

MAGNETIC MATERIAL: A substance such as iron, nickel and cobalt that is strongly attracted by magnets or in which magnetism can be induced.

MAGNETIC MERIDIAN: An imaginary line in the vertical plane passing through the magnetic axis of a freely suspended magnet, influenced by the Earth's magnetic field and also passing through the magnetic north and south poles.

MAGNETIC MIRROR (MAGNETIC BOTTLE): A device used to confine plasma in thermonuclear experimental devices or controlled-fusion experiments.

MAGNETIC MOMENT (ELECTROMAGNETIC MOMENT; MAGNETIC AREA MOMENT; MAGNETIC DIPOLE MOMENT): A vector quantity having a magnitude and direction (south to north pole of a magnet), which is the ratio between the maximum torque exerted on a magnet, a current carrying coil or a moving charge as it moves through a magnetic field and the strength of that field with the symbol m . In other words the magnetic moment of a magnet is a measure of its tendency to align with a magnetic field. It is a function of the strength and orientation of the magnetic field created by the object; and therefore a measure of the strength of a magnet or current-carrying coil.

MAGNETIC MONOPOLE: A hypothetical magnetic entity consisting of an isolated elementary North or South Pole.

MAGNETIC NORTH: The direction in which the north end of a compass needle or other freely suspended magnet points in response to the Earth's magnetic field. The **magnetic north pole** is the northern pole of the Earth's magnetic field, where the Earth's magnetic field points vertically downwards. Its position is not fixed and changes constantly due to changes in the Earth's magnetic field.

MAGNETIC PERMEABILITY (PERMEABILITY): The measure of the ability of a material to support the formation of a magnetic field within itself, classified by the value of their magnetic permeability. Its symbol is μ . In empty space the magnetic permeability is 1, because there is no matter to modify the field. **Relative permeability**, also known as **dielectric constant** is the ratio of the permeability of a substance to the permeability of a vacuum at the same magnetic field strength. Its symbol is μ_r . It is expressed as $\mu_r = \mu / \mu_0$, where μ is the permeability of the medium and μ_0 is the permeability of free space or vacuum. Diamagnetic materials have constant relative permeabilities of slightly less than 1. Paramagnetic materials have constant relative permeabilities of slightly more than 1. The relative permeability of ferromagnetic materials increases as the magnetising field increases, reaches a maximum and then decreases. Pure iron and some alloys have relative permeabilities of 100,000 or more. The reciprocal of magnetic permeability is **magnetic reluctivity**.

MAGNETIC POLES: 1. The regions of a magnet from which the magnetic forces appear to originate or a place where the lines of force of magnetic field enter or leave a magnet or a region of a magnet in which its magnetic properties are strongest. A magnetised bar has a pole at each end; if it is freely suspended in the Earth's magnetic field it will rotate so that one end points approximately towards the Earth's geographical North Pole. This end is called the north-seeking end or the north pole of the magnet. The other end is called the south seeking end or south pole. 2. The **Earth's magnetic poles** are the two nearly opposite ends of the planet, north and south, where the Earth's magnetic intensity is the greatest. They are not the same as the **geographic poles**, which are the ends of the axis that the Earth revolves around. The magnetic poles do not have a fixed position and change constantly due to changes in the Earth's magnetic field. The north magnetic pole is now about 78°N and 104°W in the Queen Elizabeth Islands of northern Canada. The south magnetic pole is now about 66°S and 139°E on the Adélie Coast of Antarctica.

MAGNETIC QUANTUM NUMBER: One of the four quantum numbers used to describe the orbitals in which electrons can be found in an atom. It is symbolised m_l and describes the orientation in space of the orbitals. The magnetic quantum number, m_l , can take integer values between $-l \dots, 0, \dots, +l$. If

$l = 0, m_l = 0$; $l = 1, m_l = -1, 0, +1$; $l = 2, m_l = -2, -1, 0, +1, +2$. The other quantum numbers are: (i) Principal quantum number (n), which describes the energy level of an electron in an orbital; n takes values from 1, 2, 3 $\dots, \infty$. (ii) Angular momentum quantum number (l) takes values from 0, 1, 2, $\dots, n - 1$. It describes the shapes of the orbitals. (iii) Spin quantum number (s) takes values of $+\frac{1}{2}$ and $-\frac{1}{2}$; s describes the direction of spin of an electron (clockwise or anticlockwise).

MAGNETIC RELUCTANCE (RELUCTANCE): A measure of magnetic resistance in a closed magnetic circuit. It is the ratio of the magnetomotive force or magnetic potential difference to the total magnetic flux in a magnetic circuit. Its unit is henry and the symbol is R .

MAGNETIC RESONANCE IMAGING (MRI): A non-invasive diagnostic technique that uses radio waves instead of x-rays or other radiation to obtain images of invisible parts of a structure or the body's soft tissue. The body or part to be imaged is placed in a magnetic resonance (MR) scanner, which is a large, hollow cylinder with a strong magnetic field. This causes the alignment of certain tissue nuclei, notably hydrogen nuclei in water and lipids. Pulses of high frequency radio waves are generated and images of internal tissues are created by a computer according to how the radio waves are absorbed and transmitted. MRI can generate computerised images of any part of the body such as the heart, arteries and veins from any angle and direction in a relatively short period of time. It is used in the diagnosis

of brain tumours and disorders, spinal disorders, multiple sclerosis and cardiovascular disease.

MAGNETIC SOUTH: The direction in which the south-seeking end of a compass needle or other freely suspended magnet points in response to the Earth's magnetic field. The **magnetic south pole** is the southern pole of the Earth's magnetic field, where the Earth's magnetic field points vertically upwards. Its position is not fixed and changes constantly due to changes in the Earth's magnetic field.

MAGNETIC STORM: A sudden and temporary disturbance affecting the Earth's magnetic field. It is caused by solar shock wave that strikes the Earth's magnetic field; causing anomalies in radio transmissions, magnetic compasses, artificial satellites, electronic communications and affecting electric power transmission.

MAGNETIC STRATIGRAPHY (PALAEOMAGNETIC STRATIGRAPHY): The use of the change in the direction of the remanent magnetisation of rocks, caused by reversals in the polarity of the Earth's magnetic field magnetism retained in rocks at the time of their formation to date them.

MAGNETIC STRIP (MAGNETIC STRIPE): A thin strip of plastic card in which iron-based magnetic particles are attached, used for storing data. Magnetic strips are read or decoded by magnetic stripe readers such as ATMs and card swipe machines by detecting magnetic variations on the strips. It is commonly used to read information on items such as credit cards, bank cards, telephone cards and railway tickets.

MAGNETIC SUSCEPTIBILITY: The degree of magnetisation of a substance when it is subjected to a magnetic field. All ferromagnetic materials exhibit paramagnetic behaviour above their ferromagnetic Curie points. The **Curie point** also known as **Curie temperature** is the temperature marking the transition between ferromagnetism and paramagnetism or between the ferroelectric phase and paraelectric phase. The general behaviour of the susceptibility of ferromagnetic materials at temperatures well above the ferromagnetic Curie temperature follows the Curie-Weiss law. The paramagnetic Curie temperature is usually slightly greater than the temperature of transition.

MAGNETIC TAPE: A narrow strip of plastic coated with a ferromagnetic material-iron oxide powder, chromium dioxide or particles of pure iron, used for recording data in tape recorders, video tape recorders and computers.

MAGNETIC TRANSITION: A transition between disordered and ordered arrays of atomic magnetic moments. Where the ordered phase has a net spontaneous magnetisation, the magnetic-ordering temperature is called a **Curie temperature**. Where the net spontaneous magnetisation of the ordered phase remains zero, the ordering temperature is called a **Néel temperature**. The temperature at which the two ferromagnetic subarrays of a ferrimagnet just cancel each other is called the **compensation point**. Materials exhibit ferromagnetic behaviour when unpaired electron spins are aligned in parallel, antiferromagnetic behaviour when the alignment is antiparallel and ferrimagnetic behaviour if the alignment of the spins is antiparallel with unequal numbers in the two orientations or if the spins are canted. Therefore, ferromagnetic, ferrimagnetic and weak ferromagnetic transitions involve a net magnetic moment change, whereas antiferromagnetic transitions have zero net magnetic moment change.

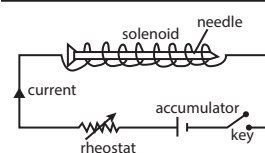
MAGNETIC TRAP: A device consisting of a spherical magnetic field set up by two magnets. It is used to trap electrically neutral particles that have magnetic moments. Only atoms with magnetic energies that decrease as the strength of the magnetic field surrounding them decrease get trapped in magnetic traps. Atoms with magnetic energies that increase as the strength of the field decreases are not trapped. Magnetic traps have been used in research on low temperatures and the formation of Bose-Einstein condensates.

MAGNETIC VARIATION (MAGNETIC DECLINATION): The angle between magnetic north and true north at a particular point on the Earth's surface, measured in degrees East (+) or West (–) according to whether magnetic north lies east or west of true north. The magnetic variation indicated on a map or a chart is not a fixed quantity but is subject to change. Magnetic variation is indicated by dashed lines on maps and charts. Most maps and charts also indicate the rate of annual change of variation.

MAGNETIC-INK CHARACTER RECOGNITION (MICR): A technique that enables special characters printed in magnetic ink to be read and put into a computer. MICR is used to verify the legitimacy or originality of paper documents, especially cheques, as magnetic characters are difficult to forge. It is used mainly in banks.

MAGNETISATION (MAGNETIZATION): The process of making a material or an object magnetic or to become magnetic. The origin of the magnetic moments responsible for magnetisation can be either microscopic electric currents resulting from the motion of electrons in atoms or the spin of the electrons or the nuclei. Net magnetisation results from the response of a material to an external magnetic field, together with any unbalanced magnetic dipole moments that may be inherent in the material itself; for example, in ferromagnets. Magnetisation also describes how a material responds to an applied magnetic field as well as the way the material changes the magnetic field and can be used to calculate the forces that result from those interactions.

MAGNETISM: The phenomenon of physical attraction for magnetic materials, inherent in magnets or induced by a moving electric charge or current or the force exerted by a magnetic field. Whenever an electric current flows a magnetic field is produced. As the orbital motion and the spin of atomic electrons are equivalent to tiny current loops, individual atoms create magnetic fields around them, when their orbital electrons have a net magnetic moment as a result of their angular momentum. The magnetic moment of an atom is the vector sum of the magnetic moment of the orbital motions and the spins of all the electrons in the atom. The macroscopic magnetic properties of a substance arise from the magnetic moments of its component atoms and molecules. Different materials have different characteristics in an applied magnetic field. The four main types of magnetic behaviour are: (i) **diamagnetism** in which the magnetisation is in the opposite direction to that of the applied field, i.e. the susceptibility is negative. (ii) **Paramagnetism**, in which the atoms or molecules of the substance have net orbital or spin magnetic moments that are capable of being aligned in the direction of the applied field. They, therefore have a positive but small susceptibility and a relative permeability slightly in excess of 1. Paramagnetism occurs in all atoms and molecules with unpaired electrons, for example, free atoms, free radicals and compounds of transition metals containing ions with unfilled electrons shells as a result of the magnetic moments associated with the spins of the conducting electrons. (iii) In **ferromagnetic** substances, within a certain temperature range, there are net atomic magnetic moments, which line up in such a way that magnetisation persists after the removal of the applied field. Below a certain temperature, called the Curie point (or Curie temperature) an increasing magnetic field applied to a ferromagnetic substance will cause increasing magnetisation to a high value, called the **saturation magnetisation**. This is because a ferromagnetic substance consists of small (1 – 0.1 mm across) magnetised regions called **domains**. (iv) **Antiferromagnetism** is a character exhibited by some metals, alloys and transition-element. This occurs below a certain temperature, called the **Néel temperature**, when an ordered array of atomic magnetic moments spontaneously forms in which



Using electricity to magnetise a nail

alternate moments have opposite directions. There is, therefore, no net resultant magnetic moment in the absence of an applied field. In manganese fluoride, for example, this antiparallel arrangement occurs below a Néel temperature of -201°C (72 K). Below this temperature the spontaneous ordering opposes the normal tendency of the magnetic moments to align with the applied field. Above the Néel temperature the substance is paramagnetic.

MAGNETITE (FERROUS-FERRIC OXIDE; HAEMATITE): A black ferrimagnetic mineral form of iron oxide crystallising in the cubic system with the chemical formula Fe_3O_4 or $\text{FeO}\cdot\text{Fe}_2\text{O}_3$ and Mohs scale hardness of 5.5-6.5. It is one of the major ores of iron. It is strongly magnetic and the naturally magnetised ones known as lodestone were once used as compasses. Magnetite is widely distributed and occurs as an accessory mineral in almost all igneous and metamorphic rocks. The largest deposits of the mineral are in Sweden.

MAGNETISABILITY: A tensor quantity relating the induced magnetic moment to the applied magnetic flux density.

MAGNETISATION TRANSFER: NMR method for determining kinetics of chemical exchange by perturbing the magnetisation of nuclei in a particular site or sites and following the rate at which magnetic equilibrium is restored. The most common perturbations are saturation and inversion and the corresponding techniques are often called 'saturation transfer' and 'selective inversion-recovery'.

MAGNETO: An alternating current generator used as a high-tension source in the ignition systems of some engines in which there are no batteries. Examples are found in some tractors, marine, industrial and aircraft engines.

MAGNETO HYDRODYNAMICS (MHD; MAGNETOFLUIDDYNAMICS; HYDROMAGNETICS): The study of the interactions between conducting fluids such as plasmas, liquid metals and salt water and a magnetic field. MHD was initiated by Hannes Alfvén, for which he received the Nobel Prize in Physics in 1970. One of the important applications includes the magneto hydrodynamic (MHD) power generator. MHD generator operates on the same basic principles as the present rotating generator except that the conductor used is a gas, while compression and expansion of the hot gas serve as the prime mover. In MHD, magnetic fields can induce currents in a moving conductive fluid, which create forces on the fluid and also change the magnetic field itself. In the open-cycle MHD generator a fossil fuel, burnt in oxygen or preheated compressed air, is seeded with an element of low ionisation potential such as potassium temperature (usually over 2227°C , 2500K) producing sufficient free electrons, for example, $\text{K} - \text{K} \rightarrow \text{K}^+ + \text{e}$ to provide adequate electrical conductivity. MHD phenomena occur naturally in the Earth's interior, constituting the dynamo that produces the Earth's magnetic field, in the magnetosphere that surrounds the Earth and in the Sun and throughout astrophysics. In the laboratory, magneto hydrodynamics is important in the magnetic confinement of plasmas in experiments on controlled thermonuclear fusion. Magneto hydrodynamic principles are also used in plasma accelerators for ion thrusters for spacecraft propulsion, for light-ion-beam powered inertial confinement and for magneto hydrodynamic power generation.

MAGNETO-OPTICAL EFFECTS: Effects resulting from the influence of a magnetic field upon matter that is in the process of emitting or absorbing light.

MAGNETOBREMSTRAHLUNG: The German word for magnetic braking radiation. It is a class of radiation mechanisms in which light charged particles, usually electrons, are more easily accelerated than the more massive protons and heavier ions, so electrons and possibly positrons produce virtually all of the radiation observed. The light charged particles follow a spiral path in a magnetic field and thereby radiate energy in the form of electromagnetic waves. If the velocity of the particle is much less than the speed of light, the emission is known as **cyclotron radiation**. If the particle's speed is an appreciable fraction of the speed of light the emission is known as

gyrosynchrotron radiation. For speeds very close to that of light it is called **synchrotron radiation**.

MAGNETOCALORIC EFFECT: A reversible change of temperature resulting from a change in the magnetisation of a ferromagnetic or paramagnetic substance.

MAGNETOCHEMISTRY: The branch of physical chemistry concerned with measuring and investigating the magnetic properties of compounds. It is used particularly for studying transition-metal complexes, many of which are paramagnetic because they have unpaired electrons. Measurement of the magnetic susceptibility allows the magnetic moments of the metal atoms to be calculated, giving information about the bonding in the complex.

MAGNETOGRAM: A chart that shows the strength, polarity and distribution of magnetic fields across the Sun's disk. It is produced by a magnetograph, which measures Zeeman splitting by taking two narrowband images in a spectral line that is sensitive to the magnetic field. Grey areas in a magnetogram indicate that there is no magnetic field along the line of sight, while black and white areas indicate regions where there are magnetic fields.

MAGNETOGRAPH: An instrument attached to a solar telescope for the purpose of producing magnetograms. The first solar magnetograph was developed by Horace Babcock, an United States astrophysicist in 1953.

MAGNETOGRIC RATIO: The ratio of the magnetic moment to the angular momentum.

MAGNETOMECHANICAL RATIO: With the symbol γ , it is the ratio of the angular momentum (μ) of a rotating charged particle to its magnetic moment L expressed as $\gamma = \frac{\mu}{L}$. It is the inverse of the

gyromagnetic ratio.

MAGNETOMETER: An instrument used for measuring the magnitude and sometimes the direction of a magnetic field. Magnetometers may be classified as either scalar or vector instruments. A **scalar magnetometer** measures the strength of the total magnetic field. A **vector magnetometer** measures one or more vector components of the magnetic field.

MAGNETOMOTIVE FORCE (MMF): Formerly called the **magnetic potential**, it is the force required to drive the magnetic flux in the magnetic circuit, and therefore, the analogy of electromotive force (emf) in electricity. Mathematically, it is the circular integral of $H\cos\theta ds$, where $H\cos\theta$ is the component of the magnetic field strength in the direction of a path of length ds . The line integral is taken around any closed path in the field. Its unit is ampere-turn (AT), which is given by the electrical current (I) flowing through a coil multiplied by the number of turns (N): IN .

MAGNETON (BOHR MAGNETON): A unit for measuring magnetic moments of nuclear, atomic or molecular magnets, which results from the electron's spin. The Bohr magneton, with the symbol μ_B , named after Danish physicist Niels Bohr has the value of the classical magnetic moments of an electron, given by

$$\mu_B = \frac{eh}{4\pi m_e} = 9.274 \times 10^{-24} \text{ Am}^2, \text{ where } e \text{ and } m_e \text{ are the charge and}$$

mass of the electron and h is the Planck constant. The nuclear magneton, μ_N is obtained by replacing the mass of the electron by the mass of the proton and is, therefore, given by

$$\mu_B = \frac{\mu_N m_p}{m_e} = 5.05 \times 10^{-27} \text{ Am}^2.$$

MAGNETORECEPTOR: A device or organ that detects magnetic fields, especially one used by animals such as insects, fishes, amphibians, reptiles, birds and mammals to sense the Earth's magnetic field when navigating over long distances during migration. In the brains of some species there are small particles of the magnetic material, magnetite, which is postulated to transduce the magnetic field into nervous impulse. Certain fishes, particularly sharks and stingrays use their electroreceptors as magnetoreceptors.

MAGNETORESISTANCE: An increase in the resistance of a metal when an external magnetic field is applied to it, altering the paths

of the electrons. It was first discovered by Lord Kelvin (1824-1907), British mathematician and physicist in 1856. At normal temperatures the change in resistance resulting from the magnetic field is small, but at very low temperatures the increase is considerable.

MAGNETOSOME: Small iron-rich magnetic particles (which serve as compass), present in cells of magnetotactic bacteria that enable them to orient themselves towards magnetic field to search for favourable habitats, usually microaerophilic environments (environments containing lower levels of oxygen than are present in the atmosphere).

MAGNETOSPHERE: A comet-shaped cavity or bubble surrounding the Earth and other magnetic planets in which the charged particles of the solar wind are controlled by the planet's magnetic field rather than the Sun's magnetic field. This cavity is formed because the Earth's magnetic field represents an obstacle to the solar wind, which is a supersonic flow of plasma blowing away from the Sun.

MAGNETOSTATICS: The study of magnetic fields produced by steady currents.

MAGNETOSTRICTION: The change in length of a ferromagnetic substance when it is magnetised, depending on the direction and extent of magnetisation. It results from changes in the boundaries of the domains. It is used in transducers for the reception and transmission of high-frequency sound vibrations.

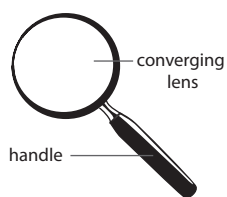
MAGNETOTACTIC BACTERIA (MTB): Bacteria that have small iron-rich magnetic particles in their cells which serve as internal compasses and help to orient themselves with respect to a magnetic field. MTB orient and migrate along the geomagnetic field towards favourable habitats, a behaviour known as **magnetotaxis**.

MAGNETRON: A vacuum tube oscillator that generates high-power electromagnetic signals in the microwave frequency range (10^9 - 10^{11} hertz). It is generated by a heated cathode moving under the combined force of an electric field and a magnetic field. The magnetron is extensively used as a generator for radar installations and can produce microsecond pulses of up to 10 MW.

MAGNIFICATION: A measure of the extent to which an optical system enlarges an image or how much larger an image is than the object. This refers only to enlarging in appearance, not in physical size. Its formula is the image height divided by the object height:

$$\text{Magnification} = \frac{\text{image height (size)}}{\text{object height (size)}} = \frac{\text{image distance (size)}}{\text{object distance (size)}}$$

MAGNIFYING GLASS: A handheld converging lens, used to produce a magnified effect and virtual image of a small object. It is the simplest optical instrument. The object must be between the pole and focal point of the lens or the distance between the lens and the object must be shorter than the focal length of the lens for this to occur. This lens is the basis of the simple microscope. It can be used to examine small or finely detailed things such as fingerprints and fine print.



Magnifying glass

MAGNIFYING POWER: The ratio of the tangent of the angle subtended at the eye by an image formed by an optical system to the tangent of the angle subtended at the eye by the corresponding object at a distance for convenient viewing.

MAGNITUDE: 1. A numerical value that describes the amount of something, usually expressed in terms of a multiple of standard units or the item measured in this way. 2. A non-negative scalar associated with a quantity, such as its absolute value or length. For example, x and $-x$ have the same magnitude but opposite polarity. 3. A measure of the energy of an earthquake, specified on the Richter scale. 4. Also called apparent magnitude, it is a numerical measure of the brightness of an astronomical object such as a star as seen by the ordinary eye from Earth, on a scale in which a lower number represents greater

brightness. A decrease of one magnitude represents an increase in apparent brightness by a factor of 2.512.

MAGNITUDE ESTIMATION: 1. Also known as **direct scaling**, it is a method of evaluating stimuli above threshold that involves directly assigning a number to represent the intensity of the sensation according to the apparent magnitudes of the stimuli. For example, in an experiment measuring perceived brightness of light, the experimenter presents a series of lights. The subject then assigns a number to each stimulus, depending on how bright he thinks each light is. If the first light was moderately bright, the subject could assign a value of 5. If the next one was perceived to be twice as bright, it could be given a value of 10. If the next was just a flicker, it could be a 1, and so on. 2. **Magnitude estimate** is a rough estimate of whether a number is in the tens, hundreds, thousands, or other powers of 10.

MAGNOLIIDAE (RANALES): A subclass of the dicotyledons containing plants in which the flower parts are numerous and often inserted spirally and the gynoecium is apocarpous.

MAGNOLIOPHYTA: In certain classifications, a division containing the flowering plants. It is divided into the two classes Magnoliopsida (dicotyledons) and Liliopsida (monocotyledons).

MAGNON (QUANTISED SPIN WAVE): A quasi-particle which is introduced to describe small departures from complete ordering of electronic spins in ferro-, ferri-, antiferro- and helimagnetic arrays. As a quasiparticle, a magnon carries a fixed amount of energy and lattice momentum. It also possesses a spin of $\hbar$, where $\hbar$ is Planck's constant divided by 2π .

MAGNOX (MAGNESIUM NON-OXIDISING): The condensed form for **Magnesium non-oxidising**. It is an alloy composed mainly of magnesium with small amounts of aluminium and other metals used to enclose uranium fuel elements in a nuclear reactor. It has the advantage of low neutron capture cross section. Its two major disadvantages are that it limits the thermal efficiency of the plant; and it also reacts with water preventing long-term storage of spent fuel.

MAGNUS EFFECT: The effect on a spinning cylinder or sphere moving through a fluid, in which force acts perpendicular to the direction of motion and to the direction of spin.

MAHOGANY: Any of several tropical trees with more than 500 species in about 45 genera; mainly of the genus *Swietenia*, family Meliaceae, especially *Swietenia Mahagoni* also known as West Indian or Cuban mahogany, *Swietenia Macrophylla* also known as Honduran or big-leaf mahogany and *Swietenia Humilis*. These three genera are known collectively as American mahogany and are considered the true or genuine mahogany trees. Some of the other important genera of Mahogany, also in the family Meliaceae include *Khaya senegalensis*, with the English common names of African mahogany, dry zone mahogany, Senegal mahogany, etc; the West African genus, *Khaya ivorensis* with the common names African Mahogany or Lagos Mahogany; *Dysoxylum spectabile*, the New Zealand mahogany; and *Toona sinensis*, the Chinese mahogany. *Shorea*, the Philippine mahogany belongs to the family Dipterocarpaceae. Most genera of Mahogany have buttresses swollen at the base and grow from between 15 to 50 m on maturity, with the dominant trees forming emergent layer above the general canopy. Some of the genera have straight, unbranched boles of up to 30 m high. In some genera, matured trees have boles that are cloaked in thick, deep furrows. Some have large leaves, which alternate and are pinnately compound. The leaflets are arranged in 3-4 pairs on either side of the central midrib. The flowers are generally small, white or pale yellow, fragrant and monoecious. The fruits are capsules and woody, growing in panicles at the ends of branchlets. When ripe, they split open to release many winged seeds. Mahogany wood is hard, with the colour of matured trees varying from golden to deep red brown. They are mostly close-grained and resistant to termites and do not decay. The wood is commercially important and used in

making cabinets, high-class furniture, boats, pianos and other musical instruments. Traditionally, various parts of Mahogany such as the bark and the seeds have been used medicinally to treat disorders and ailments such as fever, diarrhoea, high cholesterol, to improve sexual vitality in men, and similar such conditions.

MAIL (E-MAIL): In computing, messages transmitted and received by computers through a network. An e-mail system allows computer users on a network to send text, graphics, and sometimes sounds and animated images to other users.

MAIL BOMB (E-MAIL BOMB): In computing, sending large amounts of e-mail to a specific person or system to fill the recipient's hard disk with large useless files, causing irritation and sometimes computer failure. It also fills the allotted space on an e-mail server for the users e-mail, which can result in crashing the e-mail server.

MAILBOX: In computing, it is a folder in which e-mail is stored for a user.

MAIL GATEWAY: A machine that connects two or more electronic mail systems (often, mail systems on different networks) and transfers messages between them.

MAIL SERVER (E-MAIL SERVER): In computing, it is a computer or software in a network that stores incoming e-mail and distributes it to users and also sends outgoing mail.

MAIL TRANSFER AGENT, MTA (MAIL CLIENTS; MAIL USER AGENTS): A software program that transfers messages between computers by receiving incoming emails and delivering the messages to individual users.

MAIN CONNECTIVE: In a logical statement, the connective that has the widest scope. For example, in the expression "not every one is either rich or poor", the main connective is 'not' and in "either everyone is rich or everyone is poor", the main connective is 'or'.

MAIN DIAGONAL (LEADING DIAGONAL; PRIMARY DIAGONAL; PRINCIPAL DIAGONAL): The elements of a matrix running from the top left corner and proceeding down to right edge of the bottom. The main diagonal of a square matrix is the diagonal which runs from the top left corner to the bottom right corner. The main diagonal of a rectangular matrix is the diagonal which runs from the top left corner and steps down and right, until the right edge or the bottom edge is reached. For example, in the matrix below, the main diagonal has 1

entries: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

MAIN SEQUENCE STAR: A star that spends most of its active live in a stable state fusing hydrogen in its core to helium. It has a stable balance of outward pressure from core nuclear fusion and gravitational forces pushing inward. This stable state is known as the star's main sequence.

MAIN-SMITH-STONER RULE: An empirical rule in the theory of atomic structure stating that for a principal quantum number n the number of electronic quantum states that can have the orbital quantum number l is $2(2l + 1)$. This rule described the subshells of atoms. Main-Smith-Stoner rule is a consequence of the Pauli Exclusion Principle. The rule was one of the key developments that led to the enunciation of the Pauli Exclusion Principle in 1925.

MAINBOARD (MOTHERBOARD; LOGIC BOARD): The main circuit board of a computer to which the CPU, ROM, memory RAM expansion slots, PCI slots and USB ports are attached. It also includes controllers for devices like the hard drive, DVD drive, keyboard and mouse. Basically, the motherboard is what makes everything in the computer work together. Each motherboard has a collection of chips and controllers known as the chipset.

MAINFRAME COMPUTER (MAINFRAME): A large and expensive computer that can support hundreds or thousands of users simultaneously. Mainframe computers are just below supercomputers. The difference between a mainframe computer and a supercomputer is that a mainframe is designed to execute many programs simultaneously at a high speed, whereas a supercomputer is designed to execute a single or few processes at an extremely fast rate. Mainframes are typically used by large businesses for commercial data processing and for scientific purposes and other large scale operations.

MAINTENANCE RATION: The minimum quantity of feed needed to keep a farm animal healthy, in the same condition and at the same weight but not productive for an indefinite period.

MAIZE (CORN): A tall annual cereal plant, *Zea mays* that originated from Central America, but now cultivated in most tropical countries as well as some parts of temperate regions as a summer crop. It varies greatly in height but is more commonly 2-3 m high. It has a single main culm with several internodes, often with prop roots from the lower internodes. The apex of the culm ends in the tassel, an inflorescence of male flowers. The sword-shaped leaf blades are broad and alternate, usually drooping. Its upper surface is hairy, with a hairless lower surface. The leave colour is usually green, but may be variegated white, yellow or purple-red. Inflorescence is monoecious, born in separate parts of the plant. The pistillate (female) inflorescence called **ears** arise from axillary bud apices; while the staminate (male) inflorescence called **tassels** arise from the apex of the culm. The ovary develops a long style or silk, which looks like tufts of hair and extends from the cob receiving the pollen from the tassel. The ears consist of a robust rachis

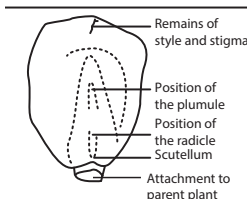


Fig. I: External structure of a maize fruit

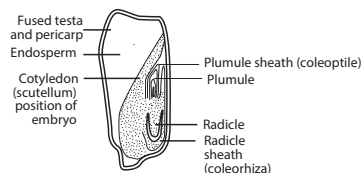


Fig. II: Longitudinal section of a maize fruit

(cob), with 4-30 rows of almost sessile spikelets bearing the kernels (caryopses). The kernels are of various colours such as white, yellow, blackish, purple, green and red; and produce a range of commercial types including dent, flint, flour, waxy, pop and sweet corn. Corn is used in many forms as a staple food in many regions of the world as vegetable; used as livestock feed; and also produces cooking oil known as corn oil. It is also used to produce alcohol such as Bourbon whiskey and also used to brew beer. It is used in many industrialised food products; and as a feedstock for the production of ethanol fuel.

MAJOR ARC: An arc of a circle, which is equal to more than half the length of its circumference. It bounds the major sector or segment of a circle. It is also the longer of the two arcs produced by a secant of a circle.

MAJOR AXIS: The longer of two axes of symmetry of a geometric figure. It is a line that runs through the centre and both foci, its ends being at the widest points of the shape.

MAJOR GENE: A gene whose effects are readily identifiable in the phenotype. The existence of different alleles of such genes results in instances of qualitative variation in the phenotypes. All Mendelian genes are examples of major genes.

MAJOR HISTOCOMPATIBILITY COMPLEX (MHC): A large gene cluster found in most vertebrates. It encodes various components

of the immune system, including the histocompatibility protein component of the complete system. In humans, the MHC is located on chromosome 6 and includes the HLA (Human Leucocyte Antigen system) system. Other vertebrate species have similar MHC regions. Certain MHC genes can have many variant alleles.

MAJOR PREMISE: The sentence in syllogism that asserts a general relationship between classes of objects.

MAJOR SECTOR: A sector of a circle greater in area than a semicircle.

MAJOR SEGMENT: The larger region of a circle bounded by a chord and an arc.

MAJOR TERM: In logic, the predicate of the conclusion of a syllogism.

MAJORITY CARRIER: The conduction band electrons in an n-type material and the valence band holes in a p-type material. They are produced by pentavalent impurities in n-type material and trivalent impurities in p-type material. When an electron meets with a hole, they recombine and these free carriers effectively vanish. The energy released can be either thermal, heating up the semiconductor, called **thermal recombination**, which is a source of waste heat in semiconductors or released as photons called **optical recombination**, used in LEDs and semiconductor lasers.

MAJORISE: To dominate with respect to some measurement scale, especially with respect to a partial ordering; or to subject to majorisation (partial order over vectors of real numbers).

MAKSUTOV TELESCOPE: A type of catadioptric telescope that combines a spherical mirror with a weakly negative meniscus lens. It was developed by the Russian optical specialist, Dmitri Maksutov (1896-1964). It uses a spheroidal primary mirror, a deeply curved spheroidal meniscus corrector plate and a small spheroidal secondary mirror fixed to or simply silvered onto, the back of the corrector plate, to give a triple-folded optical path. The result is a very compact telescope of long focal length that gives a wide field of view and images of excellent quality.

MALACHITE: A bright-green secondary mineral form of copper carbonate-hydroxide with the chemical formula $\text{Cu}_2\text{CO}_3(\text{OH})_2$. A source of copper, it is formed by the alteration of copper sulfide minerals and found in association of similar minerals such as azurite. It crystallises in the monoclinic system but usually occurs as aggregates of fibres or in massive form. It is used as an ornamental stone and as a gemstone.

MALACOPHYLLOUS: Having soft or fleshy leaves in which water is stored. This is common in plants that grow in dry regions. Examples include cactus, creosote bush and chain fruit.

MALADAPTATION: Any process that results in changes in natural or human systems that inadvertently increase vulnerability to climatic stimuli; an adaptation that does not succeed in reducing vulnerability but increases.

MALAR BONE (CHEEKBONE; ZYGOMATIC BONE): A pair of bones of the human skull situated on the side of the head or the cheek. It forms the prominence of the cheek, part of the lateral wall and floor of the orbit and parts of the temporal and infratemporal fossae.

MALARIA: A tropical relapsing infectious parasitic disease caused by the parasitic protozoan *Plasmodium*, transmitted by the bite of the

MAJOR SEGMENT

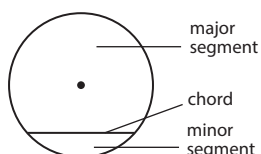


Fig 1:
A circle showing the major segment, minor segment and chord

MINOR ARC

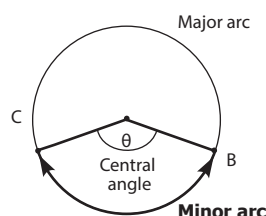


Fig 2: A circle showing the minor arc

female anopheles mosquito. It requires two hosts, the blood sucking female anopheles mosquito and a human to complete its life cycle. Symptoms of malaria in humans are fever and anaemia due to the invasion and destruction of the red blood cells during an asexual phase of the life cycle, painful joints, loss of appetite, dizziness and chills. Frequent attacks cause enlargement of the spleen.

MALARIA PARASITE: The organism that causes malaria. It is a protozoan called the *Plasmodium*.

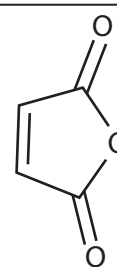
MALATHION: An organophosphate insecticide which binds irreversibly to cholinesterase. It is used to kill aphids mainly on flowers grown in glass houses, but can also be used to kill other insects such as fruit flies.

MALE: Denoting the sex of an organism which produces small mobile gametes, called spermatozoa that during sexual reproduction fuse with female gametes (ova) in the process of fertilisation. Male gametes are usually smaller than the female gametes and are motile in many organisms. The male gamete of angiosperms is carried by the pollen tube into the ovary.

MALE REPRODUCTIVE SYSTEM: A series of organs used during sexual intercourse and procreation by males. It consists of a pair of testes, the penis and a network of secretory ducts, which include epididymis, ductus deferens or vas deferens, ejaculatory ducts, seminal vesicles, the prostate gland and the bulbourethral glands. Although all men are born with all of their sexual organs, they do not begin to function for reproduction until puberty.

MALE STERILITY: 1. Generally, it is the condition in which either male organs or gametes are absent, deficient in number or nonfunctional. In flowers, it is described as carpellate or pistillate. 2. In botany, the term refers to a condition in which pollen production is prevented by mutation of one or more genes governing its formation. Male sterility has been employed by plant breeders as a method of ensuring cross pollination and hence F1 hybrid production. Male-sterile cultivars can be maintained by crossing ss (female parent) with SS (male parent) giving 50% ss and 50% SS offspring.

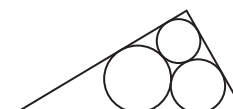
MALEIC ANHYDRIDE (FURAN-2,5-DIONE): A colourless solid with an acrid odour having the molecular formula $\text{C}_4\text{H}_2\text{O}_3$, density 1.48 g/cm^3 , melting point 52.8°C (326 K) and boiling point 202°C (475 K). It is the anhydride of cis-butenedioic acid (maleic acid) and a cyclic compound with a ring containing four carbon atoms and one oxygen atom. It is made by the catalytic oxidation of benzene or its derivatives at high temperature. It is used mainly in the manufacture of alkyd, polyester resins and copolymers.



Chemical structure of Maleic Anhydride

MALEIC HYDRAZIDE: A growth inhibitor often used as a herbicide or to inhibit sprouting. In some species it promotes flowering by inhibiting vegetative growth.

MALFATTI CIRCLES: Three circles constructed inside a given triangle such that each circle is tangent to the other two and to two sides of the triangle.



Malfatti Circles

MALIC ACID

(2-HYDROXYBUTANEDIOIC ACID): Organic crystalline acid with the molecular formula $\text{C}_4\text{H}_6\text{O}_5$, density 1.609 g cm^{-3} and melting point 130°C (403 K). It occurs in living organisms as salts and esters known as malates. The malate anion is an intermediate metabolite in the Krebs cycle and also in certain plants in photosynthesis and contributes to the pleasantly sour taste of fruits. Malic acid can be extracted from unripe

apples, grapes and other fruits. It is used as a flavouring agent and in the aging of wine.

MALIGNANT: 1. A mutant cell or group of cells that proliferate at a faster rate than normal cells and has the capacity to spread to other sites in the body. 2. Describing a disease or condition that is liable to cause death or serious disablement unless effectively treated.

MALLEABILITY: The property of a metal to be pressed, beaten, hammered, forged or rolled into new shapes without cracking or breaking. Pure gold is the most malleable; some others are silver, copper, aluminium, lead, tin, zinc and iron. Applying heat usually increases malleability. Drink cans are made from aluminium because it is sufficiently malleable and do not break during pressing.

MALLEUS (HAMMER): A hammer-shaped bone or ossicle of the middle ear attached to the tympanic membrane that transmits sound waves from the eardrum to the incus of the mammalian middle ear.

MALLOPHAGA: An order of permanently wingless insects comprising the bird lice or the biting lice. Bird lice are minute with flattened ovoid bodies, five-segmented antennae, reduced eyes and biting mouthparts. They are ectoparasites of birds feeding on fragment particles of dead skins and feather and sometimes blood, but rarely transmit any disease to their host. The eggs hatch to form nymphs resembling the adults.

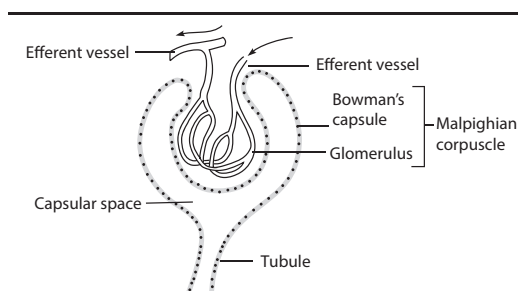
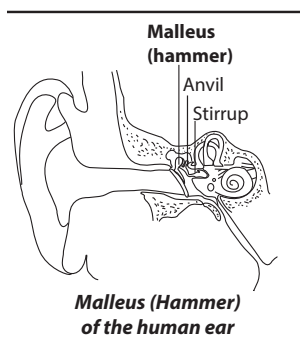
MALNUTRITION: Inadequate nutrition caused by not taking a balance diet or lack of one or more nutritional components such as protein, carbohydrate, fat or vitamins necessary for health and well-being. For example, beriberi, a diet-deficiency disease caused by a lack of vitamin B₁ is especially prevalent in those parts of the world where the diet consists mainly of polished rice. Kwashiokor, another diet-deficiency disease develops when the diet lacks proteins and hence essential amino acids. There are three stages in the process of malnutrition: first, the carbohydrate stores are depleted. Secondly, the fat reserves are metabolised. Finally, protein levels are drastically reduced. In third world countries, diseases associated with protein and carbohydrate deficiency cause many deaths among children.

MALODOROUS: Having an unpleasant odour or bad smell.

MALONYL CoA: The coenzyme A derivative of the dicarboxylic acid malonic acid. It is synthesised from acetyl CoA and bicarbonate in a reaction catalysed by the enzyme acetyl CoA carboxylase. Malonyl CoA provides the carbons for the synthesis of fatty acids - the malonyl moiety reacts with acyl carrier protein (ACP) to form malonyl ACP, which in turn combines with the growing fatty acid molecules, with elimination of carbon dioxide, to add two more carbons to the chain.

MALONYL-ACYL-CARRIER-PROTEIN (MALONYL ACP): The three-carbon dicarboxylic acid malonic acid bound to acyl carrier protein (ACP). Malonyl ACP, with malonyl CoA is central to fatty acid synthesis. It is formed from malonyl CoA and ACP and then reacts with enzyme-bound acetate to form acetoacetyl ACP, with concomitant release of carbon dioxide. Acetoacetyl ACP is then reduced to butyryl ACP before the addition of another molecule of malonyl ACP to form a six-carbon chain. Malonyl ACP thus provides the two-carbon building blocks with which long-chain fatty acids are progressively synthesised.

MALPIGHIAN BODY: 1. Also known as **malpighian corpuscle** or **renal corpuscle**, it is the expanded end of the nephron enclosed in a capsule called the Bowman's capsule attached to the end of each of the tiny urine-secreting tubule nephrons of the kidney called the glomerulus. It is responsible for blood filtration. 2. A compact aggregation of lymphoid tissue surrounding an arteriole in the spleen.



Malpighian Body
(composed of glomerulus and Bowman's capsule)

MALPIGHIAN HAIR: 1. Shaped like a pick-axe or T. 2. Short T-shaped hair which tapers to both free ends, with the base stalk attached near the middle of the upper part.

MALPIGHIAN LAYER (BASAL CELL LAYER; STRATUM GERMINATION): The innermost layer of the outermost part of the epidermis of the mammalian skin separated from the underlying dermis by a fibrous base membrane, where mitosis or active cell division occurs. As the cells produced by this division age and mature, they migrate upwards through the layers of the epidermis to replace the cells being continuously worn away at the surface. It was named after Marcello Malpighi (1628-1694), Italian physician and anatomist.

MALPIGHIAN TUBULE: A unique excretory organ made up of long blind-ending tubes or organs lying in the haemocoel of insects and other terrestrial arthropods. It is attached to the insect's alimentary canal, at the point where the midgut and hindgut are joined. It empties into the digestive tract and removes nitrogenous wastes from the blood in the form of uric acid and functions in osmoregulation. Although primarily involved in excretion and osmoregulation, Malpighian tubules have been modified in some insects to serve accessory functions such as in the secretion of the sticky substance that surrounds eggs of certain leaf beetles, or in the secretion of silk by certain immature beetles. It was named after Marcello Malpighi (1628-1694), Italian physician and anatomist. Some reabsorption can occur in the tubule but most reabsorption occurs in the hindgut.

MALT: The product of the hydrolysis of starch by β -amylase that occurs during the germination of barley in brewing, adding flavour and nutrients. It is made by steeping grain, usually barley, in water and allowing partial germination to occur. The enzymes produced within the barley seed during germination break down starch into malt sugar or maltose, which is then fermented by yeast to yield alcohol and carbon dioxide.

MALT CULMS: Dried rootlets and shoots of partly germinated malting barley. The culm is a by-product of the malting process and used as livestock feed.

MALTASE: A membrane bound enzyme found in the pancreas, intestine, liver, kidney and blood serum of animals and also in bacteria, fungi and in mono-cotyledonous and dicotyledonous plants that breaks down or hydrolyses the disaccharide maltose into glucose.

MALTED MEAL: Brown wheat flour mixed with flour made from barley.

MALTESE-CROSS TUBE: An evacuated tube that is used to demonstrate some of the properties of cathode rays or beam of electrons. The cathode rays or beam of electrons emitted by the tube's electron gun are directed towards a fluorescent screen in front of which hangs a metal Maltese cross. The electrons that hit the cross are stopped by the metal, but those that hit a fluorescent screen at the side of the tube give up their kinetic energy and cause its phosphor coating to fluoresce. This demonstrates that cathode rays travel in straight lines and cannot pass through metal.

MALTHUSIAN PARAMETER: The rate of increase that a population would attain if its age-specific birth and death rates were to continue indefinitely. The exponential equation $N(t) = N_0 e^{\beta t}$ represents change in population, where β is the population growth, N_0 is the initial population size ($t = 0$) and t is the elapsed time.

MALTING: A process of preparing grain for the production of beer and alcohol.

MALTOSE (MALT SUGAR): A disaccharide sugar composed of two units of glucose, with the chemical formula $C_{12}H_{22}O_{11}$. It occurs in high concentration in germinating seeds. Maltose is produced by hydrolysis of starch in the presence of diastase, an enzyme present in malt. It is used in the brewing of beer.

MALUS' LAW: A law, discovered by the French scientist Etienne Louis Malus (1775-1812) in 1809 concerning the polarisation of light. It states that if linearly polarised light passes through a polariser, the intensity of the light transmitted is given by $I_1 = I_0 \cos^2(q)$, where q is the angle between the polarisation direction of the light and the transmission axis of the polariser.

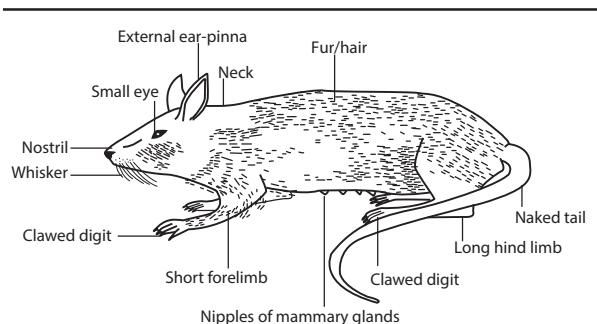
MALVACEOUS: Belonging to the Malvaceae, the mallow family of plants, which includes mallow, cotton, okra, althaea and abutilon.

MALVERTISING: A type of advertising done by malicious users to spread malware through legitimate online advertising or by hijacking websites to display advertisements that look like regular ones, but contain codes that attack the visitor's computer system with malicious software that can damage data, steal personal information or control a user's computer.

MALWARE (MALICIOUS SOFTWARE): A software program that can damage or is harmful to a computer system. Examples of malware include viruses, worms, trojan horses and spyware.

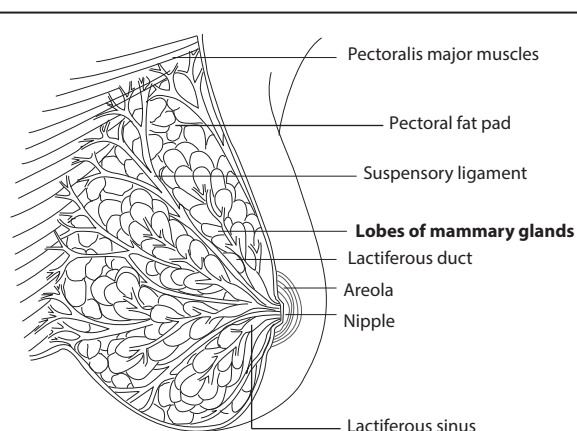
MAMMAL: A vertebrate animal that has hair on its body, sweat glands and mammary glands and gives birth to live young (viviparity). It has different types of teeth (heterodont) and possesses a diaphragm that separates the chest cavity from the abdomen. Mammals form the class Mammalia, the highest class of Vertebrata with some 4250 species. There are three sub-classes of the Mammalia: (i) The Monotremata includes the genera *Echidna* and *Ornithorhynchus*, the female lays large eggs resembling those of a bird or lizard and the young, which are hatched like those of birds, are nourished by a watery secretion from the imperfectly developed mammary glands; (ii) Marsupialia (Metatheria), for example, kangaroo in which the young are tiny when born and move into a pouch on the mother's abdomen, where they attach themselves to nipples and grow slowly; (iii) Placentalia (Eutheria) includes all the higher orders, including man, in which the foetus is attached to the uterus by a placenta.

MAMMALIA: The vertebrate class of mammals belonging to the phylum Chordata and kingdom Animalia, used in taxonomy. Mammalians are characterised by body hair, heterodont dentition and mammary glands that produce milk to nourish the young. Mammalia is divided into three subclasses: (i) **Placentalia** (Eutheria), which embraces all the higher orders, including man, having foetus attached to the uterus by a placenta. (ii) **Marsupialia** (Metatheria), with no placenta is formed and the young, are born at an early state of development, are carried for a time attached to the teats and usually protected by a marsupial pouch. Examples are opossum, kangaroo and wombat. (iii) **Monotremata**, which includes the genera echidna and ornithorhynchus, with the female laying large eggs resembling those of a bird or lizard and the young, which are hatched like those of birds, are nourished by a watery secretion from the imperfectly developed mammary glands.



Lateral view of a rodent, an example of a mammal

MAMMARY GLANDS (MAMILLARY DUCTS; MAMMARY DUCTS; MILK DUCTS; LACTIFEROUS DUCTS; GALACTOPHOUS CANALS; GALACTOPHOUS DUCTS): Milk producing glands possessed by mammals that secrete milk for the nourishment of the young. It is active only after the production of the young. Their number varies from 2 to 20 and the position is usually located on the chest or abdomen varies according to species. In most mammals, except the duckbill or platypus, the gland projects with nipples or teats. The mammary gland contains thousands of milk-producing units called alveoli, each of which consists of a unicellular layer of epithelial cells arranged in a spheroid structure. Each alveolus is connected to a duct through which milk flows. The mammary system is the milk production system in females.



Left breast of a human showing the mammary glands

MAMMIFORM: Having the shape of the mammalian breast.

MAMMILLA: A projection like the nipple.

MAMMOTH: A large extinct prehistoric relative of modern elephants belonging to the genus *Mammuthus* that had long curved tusks and was covered with abundant hair. It existed mainly in the Northern Hemisphere over Eurasia and North America in the Pleistocene epoch and died out more than 10,000 years ago.

MANAGEMENT INFORMATION SYSTEM (MIS): Focus on developing business applications that provide management with reports that would give managers the information needed for decision making purposes.

MANCUDE: An acronym for **maximum non-cumulative double**. It describes an organic compound that contains the maximum possible number of noncumulative double bonds, as in the annulenes.

MANCUDE-RING SYSTEMS (MANCUNIDE-RING SYSTEMS): Rings having (formally) the maximum number of noncumulative double bonds, for example, benzene, indene, indole and 4H-1,3-dioxine.

MANDATORY ENTRY FIELD: A field in which a user must enter at least one character.

MANDATORY-FILL FIELD: A field a user must fill in completely or leave blank.

MANDELBROT SET: A fractal that produces complex self-similar patterns, named after its discoverer, the Polish-born French mathematician Benoit Mandelbrot (1924-). In mathematical terms, it is the set of value of c that makes the series $Z_{n+1} = (Z_n)^2 + c$ converge, where c and z are complex numbers and z begins at the origin (0,0). For example, letting $c = 1$ gives the sequence 0, 1, 2, 5, 26, ..., which tends to infinity. As this sequence is unbounded, 1 is not an element of the Mandelbrot set. On the other hand, $c = i$ (where i

is defined as $i^2 = -1$ gives the sequence $0, i, (-1+i), (-1+i), -i, \dots$, which is bounded and so i belongs to the Mandelbrot set.

MANDELIN TEST: A presumptive test for amphetamines and alkaloids. The Mandelin reagent is a 1% solution of ammonium vanadate (NH_4VO_3) in concentrated sulfuric acid. Different substances give different colours. Mescaline, for example, produces an orange colour, heroin a brown colour and amphetamine a blue-green colour.

MANDIBLE: 1. Either of a pair of mouthparts in insects and crustaceans such as lobsters, crabs, shrimp, crayfish, grasshoppers and myriapods used for biting and cutting food. 2. The lower jaw of vertebrates, which holds the lower teeth in place. 3. Either of the two parts of a bird's beak.

MANET (MOBILE AD HOC NETWORK): A type of ad hoc network that can change locations and configure itself on the fly. MANETS use wireless connections such as Wi-Fi connection or another medium, such as a cellular or satellite transmission to connect to various networks.

MANGANATE(VI): A salt containing the ion MnO_4^{2-} . Manganate(VI) ions are dark green. They are produced by the reduction of manganate(VII) ions in basic solution.

MANGANESE: 1. A grey brittle metallic transition element with the chemical symbol Mn, melting point 1245°C (1518 K), boiling point 2061°C (2334 K), specific gravity 7.4, atomic number 25 and atomic weight 54.9380 g/mole. It was first isolated in 1774 by the Swedish chemist Johan Gottlieb Gahn. The main sources are pyrolusite (MnO_2) and rhodochrosite (MnCO_3). Manganese ranks about 12th in abundance among elements in the Earth's crust. The metal can be extracted by reduction of the oxide using magnesium (Kroll process) or aluminium (Goldschmidt process). Often the ore is mixed with iron ore and reduced in an electric furnace to produce ferromanganese for use in alloy steels. The element is fairly electropositive. It combines with oxygen, nitrogen and other non-metals when heated (but not with hydrogen). The most common oxidation states of manganese are +2, +3, +4, +6 and +7. But other oxidation states such as -1, -2, -3 and even +1 have been observed. Manganese is used in iron and steel alloys and much of the rest in nonferrous aluminium and magnesium alloys to improve their corrosion resistance and mechanical properties. Manganese compounds with various valences are used in fertilisers, textile printing and as reagents and raw materials. Potassium permanganate is used for disinfecting, deodorising and bleaching and as a reagent in analysis. 2. A grey brittle metallic element that is a trace element for living organisms. It functions as a cofactor for several enzymes.

MANGANESE ORE: Any mineral such as pyrolusite (MnO_2) and rhodochrosite (MnCO_3) from which manganese is produced.

MANGANESE(II) OXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for **manganese monoxide**, with the chemical formula MnO . It forms green crystals and it is odourless. It is the simplest ionic manganese oxide. It is insoluble in water but dissolves in acids, forming manganese salts. It can be prepared by the reduction of any higher oxide with hydrogen. It is prepared commercially by the reduction of MnO_2 with hydrogen, CO or methane. Although it is classified as a strong base, dissolved MnO is non-corrosive and is harmless to the skin.

MANGANESE(IV) OXIDE (MANGANESE DIOXIDE): A black or brown solid, with the chemical formula MnO_2 made by heating manganese(II) nitrate. It occurs naturally as pyrolusite. It is a strong oxidising agent used as a depolarising agent in voltaic cells and also acts as a depolariser in dry batteries such as the alkaline battery and the zinc-carbon battery by oxidising the hydrogen gas produced to water. It is also used as a pigment and as a precursor to other manganese compounds, such as potassium permanganate (KMnO_4).

MANGANIN: A copper alloy containing 86% copper, 12% of manganese and 2% of nickel, that exhibits high electrical resistance as well as being relatively insensitive to temperature changes. It was first developed by Edward Weston in 1892. Manganin foil and wire is used in the manufacture of resistors and also used in gauges for studies of high-pressure shock waves such as those generated from the detonation of explosives.

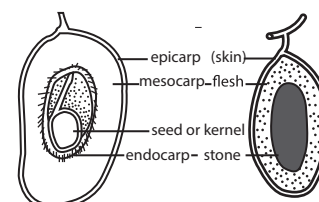
MANGANOUS COMPOUNDS: Compounds of manganese such as manganese monoxide (manganese(II) oxide), MnO in its +2 oxidation state.

MANGE: Infectious skin disease of animals and sometimes humans characterised by hair loss, scabs and itching that is caused by mites.

MANGEL FLY: A fly whose yellow-white legless larvae cause blistering of the larvae of mangels and sugar beet. This holds back plant growth and causes death in severe conditions.

MANGER: A trough in a stable, from which cattle and horses feed.

MANGO: Evergreen tree (*Mangifera indica*) cultivated for its edible fruit, also called mango. It is native to India but now found in all tropical countries. Best valued for its fruits, it also makes an excellent shade tree. The fruit is a large drupe with a smooth tough skin, thick whitish-yellow aromatic juicy flesh and a single large hard seed. The fruit is eaten raw or processed.



Longitudinal section of a mango

MANGROVE: Specialised tropical trees forming shrubby thickets in estuaries. The red mangrove, belongs to the family Rhizophoraceae and is classified as *Rhizophora mangle*. It has stilt roots which grow down from the branches into the mud giving increasing support. The white mangrove, *Laguncularia racemosa*, on the other hand, grows further from the open water and lacks stilt roots but possesses specialised roots for breathing, called pneumatophores which grow vertically upwards through the mud from the horizontal root system and project into the atmosphere. They have lenticels through which oxygen from the air passes into the roots buried in the mud. The black mangrove, *Avicennia germinans*, belongs to the family Verbenaceae and grows in tropical and subtropical regions of the Americas, on both the Atlantic and Pacific coasts and on the Atlantic coast of tropical Africa, where it thrives on the sandy and muddy shores that seawater reaches.

MANGROVE SWAMP: A region of vegetation found along tropical coasts in which mangrove trees predominate. The waterlogged area is highly saline to which mangrove trees are adapted. Mangroves play a vital role in protecting coastal regions from effects of tropical storms, erosion and high tides.

MANIA: A psychological and biological disorder characterised by extreme behaviours such as irritability, excessive happiness, over-reaction to the point of physical exhaustion, difficulty in concentrating, reduced sleep, overly optimistic, involvement in dangerous activities, increased sexual desire, etc. It may be a symptom of bipolar disorder or of other psychiatric conditions or may be caused by substance abuse.

MANIC DEPRESSION (BIPOLAR DISORDER): A mental disorder characterised by recurring periods of high (mania) or low (depression) moods. In high moods, the symptoms are similar to that of mania and include behaviours such as irritability, excessive happiness, over-reaction to the point of physical exhaustion, difficulty in concentrating, reduced sleep, overly optimistic, involvement in dangerous activities, increased sexual desire, etc. In low moods the symptoms include

depressed behaviours such as sadness, loss of energy, loss of interest or pleasure in most activities, lacking self-confidence and feeling hopeless, trouble concentrating, crying, irritability, insomnia, thoughts of death and attempting suicide, etc. The true cause of the disease is not known, but factors such as genetic and defective neurotransmitter can be contributors.

MANICATE: Covered with hairs or growth that is thickened and interwoven together to form a mass that is easily removed.

MANIFOLD: 1. In mathematics, it is a collection of objects of a set. For example, an affine manifold is an affine subset of a vector space. 2. A topological space that resembles Euclidean space near each point. Each point of an n -dimensional manifold has a neighbourhood that is homeomorphic to the Euclidean space of dimension n .

MANIPULATED VARIABLE: The independent variable that is changed for a particular effect in an experiment.

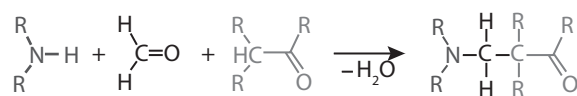
MANIPULATIVES: In mathematics education, objects, such as blocks, counters, beads, etc., that can be used within instruction to represent mathematical concepts by manipulating them.

MANIPULATOR: A hand-operated or controlled device for remotely handling radioactive materials.

MANN-WHITNEY TEST: A statistical test of the difference between the medians of the distributions of data collected in two experimental conditions applied to unmatched groups of subjects by comparing the distributions of the ranks of the scores.

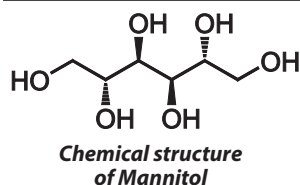
MANNAN: A polysaccharide in which the major monosaccharide subunit is mannose. Galactose and glucose are also often present, forming galactomannans, glucomannans and galactoglucomannans. Mannans frequently occur in hemicellulose. They are also found as reserve polysaccharides in some higher plants, for example, the ivory palm (*Phytelephas macrocarpa*), in which the extremely hard endosperm, known as vegetable ivory, is composed of mannans.

MANNICH REACTION (MANNICH CONDENSATION REACTION): An organic reaction named after the German chemist, Carl Mannich (1877-1947) in which a primary or secondary amine reacts with methanal (formaldehyde) and a carbonyl compound to produce an amino-carbonyl compound. It takes place in two stages. First, the amine reacts with methanal to form a Schiff base: $R_2NH + H_2CO \rightarrow R_2N^+ = CH_2$. This then reacts with the carbonyl compound: $R_2N^+ = CH_2 + R_1R_2CHCOR_3 \rightarrow R_2N-CH_2-C(R_1R_2)COR_3 + H_2O$.



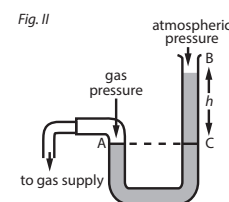
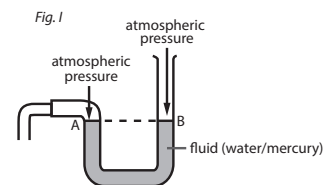
Mechanism of Mannich Reaction

MANNITOL (MANNITE; MANNA SUGAR): A soluble hexahydric alcohol with the molecular formula $[C_6H_8(OH)_6]$ that occurs in many plants and derived mainly from mannose or fructose. It is the main soluble sugar in fungi and an important carbohydrate reserve in brown algae. It is used as a sweetener in certain foodstuff such as chewing gums, as an osmotic diuretic agent to relieve fluid retention, to treat acute glaucoma in veterinary medicine, etc.



MANNOSE: A monosaccharide with the molecular formula $C_6H_{12}O_6$ that occurs naturally in small amounts in legumes, manna and some gums, fungi and bacteria. It stores food energy.

MANNONE-BINDING LECTIN (MBL; MANNAN-BINDING PROTEIN, MBP): An important factor in innate immunity in which acute-phase protein is secreted into blood plasma by the liver in response to infection. It binds to bacterial cells, enhancing their susceptibility to ingestion by phagocytic cells. The MBL molecule has up to six clusters of carbohydrate recognition domains, which bind to appropriately spaced mannose or fucose residues on bacterial surfaces.

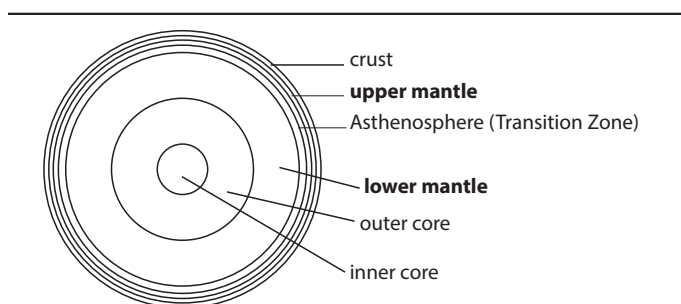


The manometer

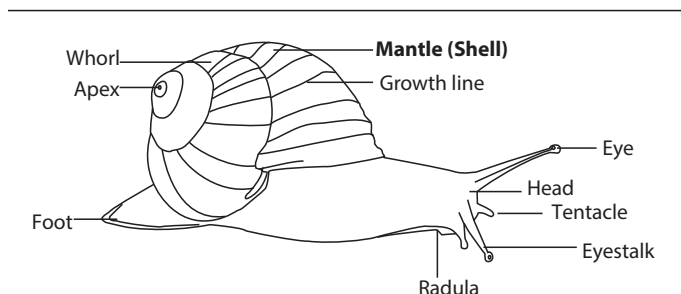
MANOMETER: The most common fluids used in manometers are water or mercury. The simplest type is the U-tube manometer, which consists of a glass tube bent into the shape of a U. If a pressure to be measured is fed to one side of the U-tube and the other is open to the atmosphere, the difference in the level of the liquid in the two limbs gives a measure of the unknown pressure. In Fig. I in the diagram, each surface of the liquid is acted on equally at atmospheric pressure and the levels are the same. If one side is connected to a gas supply as in Fig. II, the gas exerts a pressure on the surface A and level B rises until the pressure of gas is equal to the sum of the atmospheric pressure due to liquid column BC. The pressure of the liquid column BC, therefore, equals the amount by which the gas pressure exceeds the atmospheric pressure. The total pressure (P_T) is expressed as pressure due to liquid column (ρ) + atmospheric pressure (P_A): $P_T = P_A + \rho gh$, where h is the difference between the two surface in the u-tube, ρ is the density of the liquid and g is the acceleration due to gravity.

MANTISSA: The decimal or fractional part of a logarithm. For example, in the expression $\log 3300 = 3.5185...$, 0.5185 is the mantissa.

MANTLE: 1. One of the three main layers of the Earth crust that lies between the innermost layer, the core and the thin outermost layer, the crust. The mantle is divided into several layers, which include the following: the **lithosphere**, which is the thin outermost shell of the upper mantle, similar to the crust, though cooler and more rigid; the **asthenosphere**, which is composed of molten rock moving constantly in convection current, with a density of up to 3.3 kg/m³ and a thickness of some 2800 km; the **upper mantle**, which is a layer that is stronger and more solid than the asthenosphere. All layers below the crust down to a depth of about 670 km are known as the upper mantle. The **lower mantle** is the layer between the upper mantle and the core. It is denser and hotter than the upper mantle. 2. The fold of skin covering the dorsal surface of the body of molluscs. It extends into the lateral flap that protects the gills in the mantle cavity. The outer surface of the mantle secretes the shell. 3. Single or paired set of feathers located on the mid-back of a bird. 4. Chemically prepared, incombustible network hood for a gas jet, kerosene wick, etc., that, when the jet or wick is lighted, becomes incandescent and gives off a brilliant light.



Earth crust showing the layers



Mantle of a snail

MANTLE KEEL: A relatively cold slab of mantle material attached to the underside of a continental craton that protrudes down into the mantle like the keel of a boat. The presence of mantle keels is a major reason for some features such as tectonic stability, elevation above the ocean floor and the occurrence of diamonds that are associated with continents. The kimberlite eruptions that transport diamonds to the surface also carry samples of lithospheric mantle rocks called xenoliths. Mantle keels also contain about 5% of the high-pressure form of basalt known as eclogite and hosts nearly all of the world's gem diamonds.

MANURE: Waste matter from animals including excreta, which may include other natural materials such as straw or some artificial material. It is often rich in nitrogen and phosphate content and applied to land to make it fertile. Its composition varies greatly depending upon the animals that produce it. **Green manure** is any fast-growing green vegetation such as mustard, grasses, legumes or rape which is ploughed back into the soil to rot for its beneficial effects to the soil and crops. They add nitrogen to the soil, increase the general fertility level, reduce erosion, improve the physical condition of the soil and reduce nutrient loss from leaching.

MANURE CYCLE: The process by which waste materials from plants, animals and humans are returned to the soil to restore nutrients.

MANY: In botanical descriptions, this term usually means more than ten.

MANY-ONE: In mathematics, the many-one of a mapping or function, is one that is capable of associating the same element of the range of a function with more than one member of the domain or holding between more than one first argument and the same second argument of a relation.

MANY-BODY PROBLEM: The problem that it is very difficult to obtain exact solutions to systems involving interactions between more than two bodies, using either classical mechanics or quantum mechanics. To understand the physics of many-body systems it is necessary to make use of approximation techniques or model systems. For some problems, such as the three-body problem in classical mechanics, it is possible to obtain qualitative information about the system. If there are a great many bodies interacting, such

as the molecules in a gas, the problem can be analysed using the techniques of statistical mechanics.

MANY-VALUED LOGIC: 1. A logical system in which the truth values that a statement may have are not restricted to simple truth and falsehood. If the designated values are interpreted as species of truth and the antidesignated values as species of falsehood, a truth-value gap may still remain. 2. The study of such systems of many-valued logic.

MANY-WORLDS INTERPRETATION: An interpretation of quantum mechanics in which it is supposed that any measurement leads to all the possible outcomes of the measurement coexisting in different versions of the universe. One common way of thinking about the many-worlds view is to assume that every quantum event results in a split. For example, the spin of a particle may have two states; 'up' and 'down'. In the standard Copenhagen interpretation of quantum mechanics the state is indeterminate until a measurement is made, at which point the particle becomes either 'up' or 'down'. In the many-worlds interpretation, the measurement creates two universes, one in which the spin is 'up' and the other in which the spin is 'down'. The idea is that the universe is constantly multiplying into an enormous number of parallel worlds, which all exist together for a multiverse.

MAP: 1. A visual representation on a plane surface prepared to show various quantitative and qualitative facts, including boundaries, physical features, patterns and distribution. It also represents such comparative data as industrial power, population density and birth and death rates. **Map legend** or **map key** is a diagram that explains the symbols, markings, and colours on a map. A **map scale** is the ratio of a distance on a map, globe, or drawing to an actual distance on the ground. For example, on a 1:100000 scale map, 1cm on the map equals 1 km on the ground. 2. A diagrammatic drawing of a route or area made to show the location of a place or how to get there. 3. In mathematics, a transformation that takes the points of one space into the points of the same or another space. 4. A relation such that each element of a given set, which is the domain of the function is associated with an element of another set, which is the range of the function. 5. The image of a given element or set under a mapping. 6. A representation of the stars or the surface of a planet, usually in the form of a diagrammatic drawing.

MAP KINASE (MITOGEN-ACTIVATED PROTEIN KINASE; MAPK): An intracellular signal molecule that forms part of a key signal transduction pathway in eukaryotic cells. It enters the nucleus to activate transcription factors that control gene expression. This signal pathway is involved in diverse cellular activities including response to extracellular stimuli such as mitogens, osmotic stress, heat shock and proinflammatory cytokines. It also regulates various cellular activities, such as gene expression, mitosis, differentiation, proliferation and cell survival/apoptosis.

MAP PROJECTION: Systematic methods of depicting the spherical surface of the Earth on a flat piece of paper or plane surfaces to transform the spherical representation of parallels, meridians and geographic features of the Earth's surface to a nonspherical surface, usually a plane. The circles of latitude and longitude are represented by a network of lines. Directions, distances, areas, shapes and recreation depend on the purpose of the map and the part of the world represented. The three main groups of projections are cylindrical, conic (conical) and azimuthal (zenithal) projections.

MAP UNIT (M.U.; CENTIMORGAN): A unit for measuring distance between genes on a chromosome according to frequency of recombination due to crossing over. They are used in constructing linkage map; and measure relative genetic distance between loci and not absolute physical distance. They are additive over short distances, but are not reliable over distantly linked genes because of the possibility of multiple crossovers.

MAPLE: Hard wood trees and shrubs of the genus *Acer* largely of the northern temperate regions. The wood is used for furniture. The sap of *Acer saccharum* is used for making sugar and syrup.

MAPPING: 1. Transformation of mathematical information from one form to another. A relation in which each element in the first set (domain) has an image in the second set (codomain). 2. The act or process of making maps. 3. In computer science, a process of converting data that is transmitted in one format by the sender into the data format that can be accepted by the receiver.

MAQUIS: A stunted form of woodland (scrub woodland) found in semiarid regions that have been deforested for agricultural purposes or by fire. The poor soil supports a mass of tangled bushes and shrubs growing to about three metres, with scattered twisted dwarfed trees. The species include rockroses (*Cistus*), broom (*Cytisus scoparius*), gorse (*Ulex*), etc., together with herbs, such as thymes (*Thymus*) and heathers (*Erica*).

MARANGONI EFFECT: The effect that a disturbance of the liquid-liquid interface (due to interfacial tension) has on mass transfer in a liquid-liquid extraction system.

MARBLE: A metamorphic limestone with crystalline texture resulting from heat and pressure that contains calcium carbonate and magnesium carbonate, recrystallised calcite or dolomite. Pure marbles are white but the presence of impurities such as silica or clay minerals result in variations of colour. It is easy to cut, carve and polish and used. It is used as a decorative wall covering in buildings and as a flooring material. The pure white marble from Carare in Italy is especially prized by sculptors.

MARCESCENT: Retention of organs, without being shed. Organs which undergo marcescent include the sepals and petals in some flowers or leaves. It is usually noticed in the winter, when certain deciduous tree species do not lose their dead leaves. When photosynthesis slows in fall, the base of the leaf stem in deciduous plants produce an enzyme that forms a separation layer known as **abscission zone**. This enables the leaf stem to separate from its branch and fall to the ground. Marcescent leaves, however, do not form an abscission layer and only drop in heavy winds or when new leaf buds push them off in the spring. One advantage of marcescence is that, it prevents herbivores from feeding on leaves since dead and dry leaves make the twigs less nutritious. Examples are seen in Oak (*Quercus*), Beech (*Fagus*) and Hornbeam (*Carpinus*).

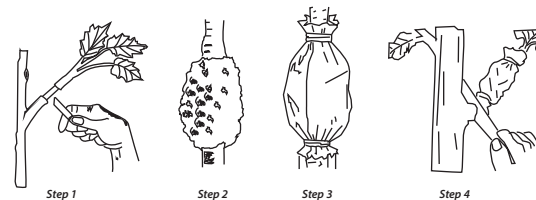
MARCHANTIA: A genus of liverworts in the family, Marchantiaceae which are found in moist clay or silty soils. It consists of simple plants without true roots, stems, leaves and vascular systems (xylem and phloem). Male and female plants have cup-like reproductive structures called gemma cups, which are used for asexual reproduction. The male structures are cup-shaped with scalloped edges; the female structures have nine fingerlike projections. Sperm produced by male plants are splashed by raindrops onto the female plants. Sporophytes develop from fertilised eggs in the female structures. Vegetative reproduction occurs in both male and female plants by means of rounded, fringed gemmae (asexual buds) or by pieces of the plant body that may break off and grow. Examples are *Marchantia polymorpha*, known as common liverwort, a dense, fleshy mat that grows in wet places and looks like green seaweeds or leafy mosses and *Marchantia berteroana* that contains the flavones isoscutellarein and hypolaetin 8-glucuronides.

MARCHIGIANA: A breed of cattle with a short white/grey coat, native to Italy. It is used for cross-breeding to improve beef-calf quality in dairy cattle. Either polled or horned, it matures fast.

MARCONI, GUGLIELMO (1874-1937): Irish-Italian electrical engineer and one of the founding fathers of radio communication.

In 1894, he began experimenting with Hertzian waves, making the first radio transmissions. He moved to London in 1896 and improved the range and reliability of his equipment. In 1901, he transmitted Morse signals across the Atlantic Ocean, establishing radio telegraphy and more importantly the use of radio waves as a communications medium. In 1909, he and Karl Braun were awarded the Nobel Prize for Physics.

MARCOTTING (AIR LAYERING): Plant propagation by girdling a plant stem, wrapping with sphagnum moss and protecting it with plastic to encourage root formation.

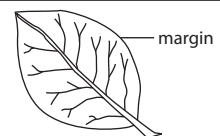


Air layering (marcoting)

MARE: 1. Mature female horse or other female animal of the horse family. 2. Any of the several large, dark plains on the moon and Mars.

MAREK'S DISEASE: A highly contagious viral neoplastic disease affecting poultry, caused by an alpha herpesvirus known as Marek's disease virus (MDV) or gallid herpesvirus 2 (GaHV-2). It is characterised by the presence of T cell lymphoma, paralysis of wings and legs as well as infiltration of nerves and organs by lymphocytes. Viruses related to MDV appear to be benign and can be used as vaccine strains to prevent Marek's disease.

MARGIN: The border or edge. In botany, it refers to the edge or border of, for example, a leaf. Marginal means pertaining to the border or edge.



Margin of a leaf

MARGIN OF ERROR: The number of acceptable errors in an experiment. In statistics, it is the range of values below and above the sample statistic in a confidence interval. The confidence interval is a way to show what the uncertainty is with a certain statistic (i.e., from a poll or survey).

MARGIN SQUEEZE: A margin squeeze occurs when the cost of farm inputs such as fertiliser, fuel, and labour increases and the price of farm outputs falls.

MARGINAL DISTRIBUTION: The probability distribution function of a component of a random vector. For example, if $X = (x_1, x_2)$ is a two-dimensional continuous random vector with probability distribution function $P_X = (x_1, x_2)$, then the marginal probability distribution function of x_1 is $P(x_1) = \int_R P_X(x_1, x_2) dx_2$.

MARGINAL EFFECT: In botany, it is a phenomenon exhibited by the plants at the edge of a stand of vegetation whereby they grow more vigorously than those in the centre.

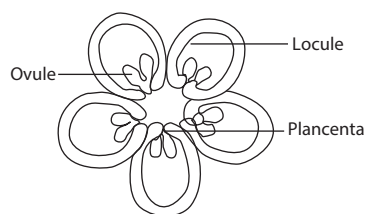
MARGINAL EXPECTATION: In statistics, the expectation of a given component of a random vector. This is related to the conditional expectation by $E[E(X_1 | X_2)] = E(X_1)$, where $X = E(X_i)$, is the random vector.

MARGINAL LAND: Land that is of little agricultural value because it is almost too unproductive to be cultivated profitably because of poor soil or other undesirable characteristics. This land is often located at the edge of deserts or other desolate areas.

MARGINAL MERISTEM: A region of meristematic tissue located along the edges of a leaf primordium. Repeated divisions of initials in this area give rise to the mesophyll and epidermal tissues of the blade.

MARGINAL PLACENTATION (VENTRAL PLACENTATION):

Arrangement of ovules along the ventral suture of the carpel. There is only one elongated placenta on one side of the ovary and ovules are attached to the fusion line of the carpels' margin.



*Marginal placentation
of apocarpous gynoecium*

**M A R G I N A L
P R O B A B I L I T Y:**

In a multivariate distribution, the probability of one variable or function of several of the variables, taking a specific value or falling in a specific interval, irrespective of the values of the other variables.

MARGINAL PROBABILITY MEASURE: In statistics, a measure constructed from an outer measure on a product space, by restriction to one of the two factors: if μ^* is an outer measure on $X \times Y$, the marginal probability measure is a measure α^* that satisfies $\alpha^*(A) = \mu^*(A \times Y)$.

MARGINAL RELATIVE FREQUENCIES: Total frequencies for rows or columns in a two-way table.

MARGINATE: In botany, it is having a distinct margin, as seen, for example, in the pods of the Leguminosae.

MARGIN: In computer science, it refers to the spacings that keep texts or elements away from the edges of the page. Margin defines where a line of text begins and ends. Most programs allow the widths of top (header), bottom (footer), left, and right margins to be specified.

MARGO: In a pollen grain, the zone of thickened sexine around a colpus.

MARIANAS TRENCH: The deepest trench in the world that lies in the western Pacific Ocean about 200 kilometres east of the Mariana Islands. It is about 2,550 km long and 69 km wide with a depth of about 11 km.

MARIJUANA (HEMP; INDIAN HEMP; *Cannabis*): Common name for annual herbal flowering plants of the genus *Cannabis* of the Hemp family having erect stems. The three main subspecies are *Cannabis sativa*, which is widely used for recreation; *Cannabis indica*, which has a poor fibre quality and used to make drugs for recreation and medicine and *Canabis ruderalis*. *C. sativa* plants grow taller and thinner than *C. indica* plants, with *C. sativa* growing to about 6 metres tall. *C. indica* grows mainly in cool highland regions, while *C. sativa* grows in warmer, lowland equatorial areas of the world. *C. ruderalis* is generally smaller than the other two subspecies, as the mature plant reaches heights of around 90 centimetres. This subspecies, found in Southeast Russia, Poland, the Czech Republic, some parts of China and the Caucasus mountains is autoflowering and unlike *C. sativa* and *C. indica*, does not require certain light patterns in order to flower. *C. sativa*'s leaves are long and slender, often with pronounced serrations, giving the leaves a jagged, almost spiky appearance with bright, lime green to blackish-green colours. *C. indica* leaves are much wider and the largest leaves usually have fewer leaflets than the largest *sativa* leaves. *C. indica* leaves are commonly deep olive-green. *Cannabis* leaves are palmately compound with toothed leaf margin. The male and female flowers are borne on separate plants of the same species and are pollinated by wind. The fruit is an ovate, one-seeded achenium; the seeds are roundish-ovate, slightly flattened. The major differences between the *C. sativa* and *C. indica* are in appearance and the amount of delta-9-tetrahydrocannabinol (THC), the active ingredient of marijuana with *C. sativa* having high THC levels and low or no cannabidiol (CBD) levels. *C. indica* has moderate THC levels and higher levels of CBD. *C. sativa* produces more of a euphoric high, lifting the consumer's mood and therapeutically relieving stress. *C. indica* strains relax muscle and work as general analgesics and is often used to treat insomnia. *C.*

ruderalis is lower in THC than most hybrids, with a comparatively high proportion of CBD. Oil, called hash oil, which is extremely high in THC is extracted from the cannabis plant. The effect of long term use of cannabis can affect one's memory and cognition.

MARINE: Living in, pertaining to or obtained from the sea.

MARINE BIOME: A major community of plants or flora and cold-blooded vertebrate animals having similar life forms or morphological features existing in waters containing 3.5% salt on average. It includes the oceans and covers more than 70% of the Earth's surface. It is divided into the **littoral zone**, found in shallow water; the **pelagic zone** in the open ocean; and the benthic zone on the ocean floor.

MARINE FLORA (MARINE PLANTS): Plants such as algae, for example, *Sargassum* and kelp that are adapted to live in the sea. Marine flora form mostly the base of all food chains in the sea and provide important habitats for life, sometimes acting as hiding and foraging places for larval forms of larger fish and invertebrates. The non algae plants that survive in the sea are often found in shallow waters, such as the seagrasses, for example, eelgrass (*Zostera*) and turtle grass (*Thalassia*). In the open ocean, marine plants are components of the plankton, known as phytoplankton.

MARIOTTE LAW (BOYLE'S LAW; BOYLE-MARIOTTE LAW): A gas law that refers to the relationship between pressure and volume of a perfect gas. It states that at a constant temperature, the volume (V) of a given mass of gas is inversely proportional to the pressure (P) acting on the gas: $V \propto 1/P$. Thus, increasing the pressure of a gas causes a proportional decrease in its volume. Conversely, decreasing the pressure causes an increase in the volume of the gas, for example, when a diver rises to the surface, the gases within the lungs expand.

MARK SENSING: In computing, it is a technique to detect pencil marks made in predetermined positions on specially prepared boxes or paper forms that can be read by a mark sense reader and input to a computer. Because pencil marks contain graphite, which conduct electricity. When a mark sense reader scans the form by passing small metal brushes over the paper surface, the brush touches a pencil mark, which contains graphite that conducts electricity and a circuit is completed, which is converted into digital code that can be detected by a computer. This technique has been replaced by the more efficient optical mark reading (OMR), which detects the marks photoelectrically.

MARKER: 1. **DNA Marker** is a fragment of known size used as reference for analytical purposes. 2. **Genetic Marker** is a gene with known phenotype and mapped position. 3. In chromatography, **marker** refers to a reference substance co-chromatographed with the sample to assist in identifying the components.

MARKET IMPACTS: Impacts that are linked to market transactions and directly affect Gross Domestic Product (GDP), for example, changes in the supply and price of agricultural goods.

MARKET POTENTIAL: An estimate of the maximum sales of a product or service, representing the upper limits of the market for a product. In climate change, it is the portion of the economic potential for greenhouse gas emissions reductions or energy-efficiency improvements that could be achieved under forecast market conditions, assuming no new policies and measures.

MARKOV CHAIN: A stochastic process $\{X_t\}$, $t = 0, 1, 2, \dots$, where X_t takes values in the finite set $S = \{1, 2, 3, \dots, N\}$ and is such that $P(X_n = i_n | X_0 = i_0, \dots, X_{n-1}) = P(X_n = i_n | X_{n-1} = i_{n-1})$. The equation given implies that to make predictions about the future behaviour of the system it suffices to consider only its present state and not its past. The probability $P(X_n = i_n | X_{n-1} = i_{n-1})$ is known as a one step transition probability. The more general transition probability, $P(X_n = i | X_m = j)$ satisfies the famous Chapman Kolmogorov equation by $P(X_n = i | X_m = j) = \sum_k P(X_n = i | X_{m+1} = k) P(X_{m+1} = k | X_m = j)$. The homogeneous Markov chain is one for which the transition

probabilities depend only on the difference $n-m$ rather than on n or m . In particular, the one-step transition probability, $P(X_i = i | X_{i-1} = j)$ can be written simply as P_{ij} . The $N \times N$ matrix, P , with ij th element P_{ij} is a stochastic matrix, i.e., $0 \leq P_{ij} \leq 1$, $1 \leq i, j \leq N$. $\sum_j P_{ij} = 1$, $1 \leq i \leq N$. P is known as transition matrix. The ij element of P^n gives the n -step transition probability if the limit $n \rightarrow \infty$ is π_{js} constitutes the stationary distribution of the chain.

MARKOV PROCESS: In statistics, it is a random process, in which the rate of change of a time-dependent quantity $da(t)/dt$ depends on the instantaneous value of the quantity $a(t)$, where t is the time, but not on its previous history. If a random process can be assumed to be a Markov process, an analysis of the process is greatly simplified enabling useful equations in nonequilibrium statistical mechanics and disordered solids to be derived. It was named after the Russian mathematician Andrey Markov (1856-1922).

MARKOVNIKOFF'S RULE (MARKOVNIKOFF'S RULE): Formulated by the Russian chemist, Vladimir Vasilevich Markovnikov in 1870, it states that in the electrophilic addition of a hydrogen halide across a double bond in an unsymmetrical alkene, the hydrogen will add to the carbon atom with the greatest number of hydrogen atoms. When an acid HA adds to an alkene, a mixture of products can be formed if the alkene is not symmetrical. For instance, the reaction between $C_2H_5CH=CH_2$ and HCl can give (i) $C_2H_5CH_2CH_2Cl$ and (ii) $C_2H_5CHClCH_3$. In general, a mixture of product occurs in which one predominates over the other. In this reaction, product (ii) would be the major product. This reaction proceeds through the formation of a carbocation and so the most stable carbocation intermediate would lead to major product formation.

MARKSMAN: In statistics, an estimate that has smaller variance than any other estimate of the same parameter. It is formally defined as an estimate $M(X)$ associated with another estimate $S(X)$, such that $M(X) = E[S(X) | T(X)]$, where $T(X)$ is any complete sufficient statistic.

MARKUP: 1. A markup language is a computer language such as HTML and XML that uses embeds processing instructions called tags to define elements within a document. For example, with HTML (Hyper Text Markup Language) that is used to create web pages on the Internet the `<b></b>` tag (markup) is used to bold text on a page. Markup languages also allow inclusion of additional descriptive information called metadata about the encapsulated data to facilitate exchange of complex data types between different systems. **Markup files** contain standard words, rather than typical programming syntax and are readable. 2. With a word processor and some other text editors, markups are comments and tracked changes that help keep track of changes made to documents being modified by multiple people. You can enable or disable what markup is shown in Microsoft Word by clicking the **Review tab** and selecting the "Show Markup" option to view the markup you want to view.

MARKUSH STRUCTURE: A description of a compound class by generic notations, primarily used in patent applications. It is named after Eugene Markush (1888-1968), an American manufacturer of dyes and pharmaceuticals. It can be described for compounds with substituents at several positions and often many thousands of possible compounds are defined in this way. An important part of chemical database searching is the ability to find possible Markush structures to rule out priority in the patent application. Chemical drawing programs can present such structure. For example, a bond to the centre of a ring indicates substitution at any position on the ring.

MARL: Fine-grained sediment that consists of a mixture of calcium carbonate ($CaCO_3$) mixed with silt and clay with any other constituents in varying proportions. The dominant carbonate mineral in most marls is calcite, but other carbonate minerals such as aragonite (another form of calcium carbonate), dolomite [$Ca, Mg(CO_3)_2$] and siderite ($FeCO_3$) may be present.

MARQUIS TEST: A widely used presumptive test that gives a variety of colour changes with a range of compounds. It is used to identify opiate alkaloids as well as other compounds such as amphetamines and methamphetamine. It is a mixture of methanol (formaldehyde)

solution in water with concentrated sulfuric acid. Different compounds produce different colour reactions. Mescaline gives an orange colouration. With morphine, a violet colour is produced. Amphetamines give an orange-red colour and methamphetamine gives an orange colour. The two can be distinguished by the Simon test. The mechanism involves attack of the aldehyde on a substituted aromatic ring to form a carbocation. Further reaction forms a coloured dimer of the original molecule.

MARRAM GRASS (MARRAM): A perennial grass of the genus *Ammophila*, especially *A. breviligulata*, planted on sand dunes to prevent them from being spread by the wind.

MARRIAGE THEOREM (HALL'S MARRIAGE THEOREM): The theorem that a perfect matching exists for a bipartite graph of which all the vertices have the same non-zero degree. It gives a necessary and sufficient condition for finding a matching that covers at least one side of the graph. The theorem also deals with a collection of finite sets, giving a necessary and sufficient condition for selecting a distinct element from each set.

MARROW: A soft tissue that fills the spongy interiors of animal bones. Yellow marrows consist mostly of cells containing fat. It produces some types of white blood cells. Red marrow, found mainly in the ribs, the vertebrae, the pelvic bones and the skull is the principal organ that forms blood cells in mammals, including humans. Erythrocytes or red blood cells, platelets and all but one kind of leukocyte or white blood cell are manufactured in human red marrow, which releases about 10 million to 15 million new erythrocytes every second, while an equivalent number are destroyed by the spleen. In children, the bones contain only red marrow but as the skeleton matures, fat-storing yellow marrow displaces red marrow in the shafts of the long bones of the limbs.

MARS: The fourth planet from the Sun with a mean distance of 228 million km. It has two small moons, Phobos and Deimos. It has an equatorial diameter of 3,396 km (about half that of Earth), with average daytime surface temperature of $-20^\circ C$ (253 K). Its day is 24 hours 37 minutes and its year about 687 Earth days. Its mass is about one-tenth of Earth's and its surface gravity about one-third as strong.

MARSH: Low-lying wetland characterised by grassy vegetation and often forming a transition zone between water and land.

MARSH GAS: Combustible gas, consisting mostly of methane. It is produced in swamps or marshes by the action of bacteria on decaying vegetation.

MARSH SPOT: A disease that affects pea caused by manganese deficiency.

MARSH'S TEST: A chemical test for arsenic; named after its inventor, the English chemist James Marsh in which hydrochloric acid and zinc are added to the sample, arsine being produced by the nascent hydrogen generated. It is very sensitive and can be used to detect minute amounts of arsenic in foods or in stomach contents. Gas from the sample is led through a heated glass tube and, if arsine is present, it decomposes to give a brown deposit of arsenic metal. The arsenic is distinguished from antimony, which gives a similar result by the fact that antimony does not dissolve in sodium chlorate(I) hypochlorite).

MARSUPIAL: A mammal such as kangaroo, koala and opossum of the order Marsupialia, in which the female has a pouch on the anterior abdomen with its mammary gland, where she carries her young.

MARTENSITE: A solid solution of carbon in alpha-iron formed when steel is cooled too rapidly for pearlite to form from austenite, named after the German metallurgist Adolf Martens (1850-1914). It is responsible for the hardness of quenched steel.

MARTENSITIC TRANSITION: A diffusionless transition (first studied in the steel alloy, Martensite), at constant composition, generated by coordinated atomic, ionic or molecular displacements over distances smaller than interatomic distances in the parent phase. The cooperative rearrangements of the crystal structure generally take place progressively by the movement of a two-dimensional interface through the solid. Examples are the face-centred-cubic to

body-centred-tetragonal transition of iron containing some carbon; and the transition of tetragonal zirconium dioxide (ZrO_2) to monoclinic ZrO_2 .

MARTINGALE: 1. In statistics, a sequence of random variables $\{x_n\}$ in which the conditional expectation of each x_{n+1} , given $x_1, x_2, \dots, x_n$, is x_n . 2. A betting system in which one doubles the stakes after each loss and halves them after each win.

MASCAGNITE: A rare mineral form of ammonium sulfate, with the chemical formula $(\text{NH}_4)_2\text{SO}_4$. It is soft, not higher than 2.5 on the Mohs Scale and water soluble. It is named after Italian anatomist, Paolo Mascagni (1752-1815), who first described it. It occurs in fumaroles and associated with coal seam fires.

MASCON: A gravitational anomaly on the surface of the Moon resulting from a concentration of mass below the lunar surface.

MASER: Acronym for microwave/molecular amplification by simulated emission of radiation, a device that works on the same principle as the laser by producing coherent electromagnetic waves through amplification of stimulated emission. It is also known as **optical maser**. Some common types of masers are the ammonia maser, rubidium maser and ruby maser. Masers uses include high precision frequency references, electronic amplifiers in radio telescopes, etc.

MASH: A mixture of feeding meals combined to provide all the elements for a balanced diet required for normal growth and development. An example is a layer starter mash.

MASHAM: A crossbreed type of sheep which results from a Wensleydale or Teeswater ram mated with a hill ewe of the Swaledale type. It has a black marking on the face and leg.

MASHLUM: A mixture of oat and barley sown to provide grain for feeding livestock.

MASKED (PERSONATE): In botany, it refers to tubular corolla possessing two lips, with the throat closed by a prominent projection known as palate. It is common in the snapdragon.

MASQUERADE: In ecology, it is the resemblance of an organism to some inanimate object in its environment so that it remains effectively hidden from predators. For example, some larvae look like bird droppings. Certain butterflies are leaf-like. Many insects have evolved particular shapes and colours so they resemble leaves, twigs or other features of their natural surroundings.

MASS: A measure of the quantity or amount of matter in an object or a body as measured by its inertia. The unit of mass is the kilogram or gram; a measure of a body's inertia or its resistance to acceleration. According to Newton's laws of motion, if two unequal masses, m_1 and m_2 are allowed to collide, in the absence of any other forces both will experience the same force of collision. If the two bodies acquire accelerations a_1 and a_2 as a result of the collisions then $m_1a_1 = m_2a_2$. This equation can be compared if one of the masses is regarded as a standard of mass, measured in terms of this standard. The body used for this standard is a 1-kg cylinder of platinum-iridium alloy, called the international standard of mass.

MASS ACTION EXPRESSION: For a reversible reaction, $aA + bB \rightleftharpoons cC + dD$ the product of the concentrations of the products (species on the right), each raised to the power that corresponds to its coefficient in the balanced chemical equation, divided by the product of the concentrations of reactants (species on the left), each raised to the power that corresponds to its coefficient in the balanced chemical equation. At equilibrium the mass action expression is equal to K ; at other times it is Q .

MASS ACTION LAW (GULDBERG AND WAAGE LAW): It states that the rate of a chemical reaction for a uniform system at constant temperature is proportional to the product of the **active masses** of the reactants. The active mass of a reactant is taken to be its molar concentration. For example, for a reaction $xA + yB \rightarrow \text{products}$, the rate is given by $R = k[A]^x[B]^y$, where k is the rate constant.

MASS ANALYSIS: A process by which a mixture of ionic or neutral species is identified according to the mass-to-charge ratios (ions)

or their aggregate atomic masses (neutrals). The analysis may be qualitative and/or quantitative.

MASS BALANCE: Summation of the masses of a given element in its various compounds before and after reaction (changes) in the atmosphere. It provides a test of the completeness of the accounting of the various reaction paths for this element, which can be obtained through the compounds which have been analysed.

MASS CONCENTRATION: The mass of solute per unit volume of solvent by dividing the mass of the component to the volume of solution. It can also be defined as the mass of a substance dissolved in 1 dm³ of solution. Its symbol is ρ . It is expressed as $\rho_i = m_i/V$, where m_i is the mass of the constituent i in the mixture or solution and V is the volume of the mixture or solution. It has units of kg/dm³ or g/dm³ or g/cm³.

MASS DEFECT (MASS DECREMENT): The difference between the rest mass of an atomic nucleus and the sum of the masses of its individual components in the unbound state or the difference between the rest mass of radioactive nucleus decay and the total rest mass of the decay products. Except for the very lightest nuclei, the mass defect amounts to about 1 percent of the total nuclear mass. When mass defect is expressed in energy units, it is called the **binding energy**.

MASS DEFICIENCY: The amount of matter that would be converted into energy if an atom were formed from constituent particles.

MASS DENSITY GRADIENT: Change in mass density in a small distance divided by that distance.

MASS DISTRIBUTION RATIO: The fraction of a component in the stationary phase divided by the fraction in the mobile phase.

MASS ENERGY EQUIVALENCE: The energy that would be needed to pull apart all the neutrons and the protons of the nucleus of the atom. This equation relates the mass of an object to the energy change, according to the theory of relativity that mass and energy are equivalent and interchangeable according to the equation $E = \Delta mc^2$, where E is energy, m is mass and c is the speed of light in a vacuum. If the object gains energy its mass rises and vice versa.

MASS EXCESS: Mass of an atom minus the product of mass number and atomic mass constant.

MASS EXTINCTION: The extinction of a large number of species of Earth within a relatively short interval of the geological time scale, opening ecological niches for other species to fill. The fossil record provides evidence for several mass extinction, comprising at least ten such events since the start of Phanerozoic eon about 570 million years ago. The most devastating extinction occurred at the end of the Permian period (about 250 million years ago), when over 80% of marine invertebrate genera and 8 of 27 insect orders disappeared; and at the end of Cretaceous (65 million years ago), when the dinosaurs and many other plants and animals disappeared and up to 75% of all marine genera were lost.

MASS FLOW (PRESSURE FLOW; MASS TRANSFER; BULK FLOW): The movement of substances such as sugar at equal rates or as a single body. Examples of such movements include blood circulation and the transport of water in xylem vessels and sugar in phloem tissue of plants. This movement relies upon the cohesion of water molecules to each other and adhesion to the vessel's wall by hydrogen bonding. For example, at source, sugars are actively secreted from the phloem companion cells into the sieve elements, causing water to flow by osmosis. The pressure of water in the tubes causes sugar to move to a sink where the reverse process occurs. Sugars diffuse or are actively transported from the sieve elements into the companion cells and to surrounding tissues establishing a concentration gradient from source to sink. If an air bubble occurs the flow will be stopped as the column is broken and the pressure difference in the vessel cannot be transmitted. This is called an **air lock**.

MASS FLOW HYPOTHESIS: In botany, it refers to the theory that translocation of sugars in the phloem is brought about by a continuous flow of water and dissolved sugars between sources and sinks. At the source, osmotic pressure is high due to the continuous formation of

sugars and at the sink osmotic pressure is low as the sugars are used up. Thus water from the xylem enters the phloem at the source and leaves it at the sink, returning to the source via the xylem. This tends to drive the contents of the phloem towards the sink.

MASS FLOW RATE: Mass of a material crossing a surface divided by the time.

MASS FRACTION: Mass of a constituent divided by the total mass of all constituents in the mixture.

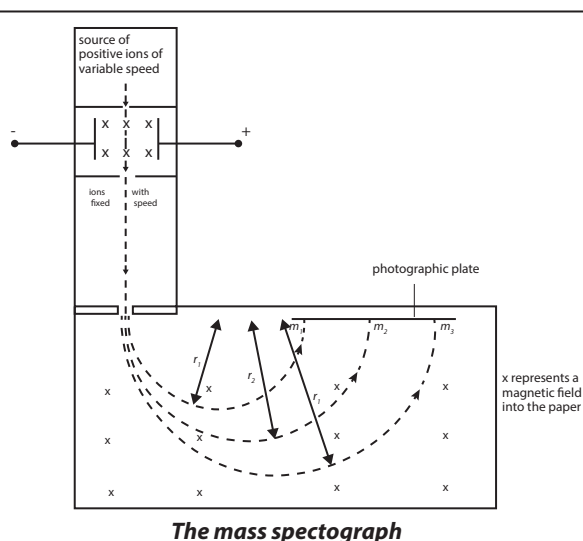
MASS MOMENT OF INERTIA (ANGULAR MASS; MOMENT OF INERTIA; ROTATIONAL INERTIA): A measure of an object's resistance to changes to its rotation, which is the sum of all the products formed by multiplying the magnitude of each element of mass by the square of its distance from the axes. Its symbol is I . The moment of inertia of an object about a given axis describes how difficult it is to change its angular motion about that axis. If a body consists of particles of mass m , a perpendicular distance r from an axis of rotation, the moment of inertia I of the body about that axis of rotation is given by $I = \sum mr^2$. The basic equation is $T = I\alpha$, where T is the torque causing angular acceleration α about the specified axis. The moment of inertia of a body is an important quantity because it is the analogue for rotational motion of mass for linear motion.

MASS NUMBER (NUCLEON NUMBER; ATOMIC MASS NUMBER): The total number of protons and neutrons (nucleons) in the nucleus of an atom, written either after the element name or as a superscript to the left of an element, for example, carbon-12 or ^{12}C . Its symbol is A . The difference between the mass number (A) and the atomic number (Z) gives the number of neutrons (N) in a given nucleus: $N = A - Z$ and so $A = N + Z$.

MASS PEAK: A record of the ion current, at a specific mass to charge ratio, received by the collector.

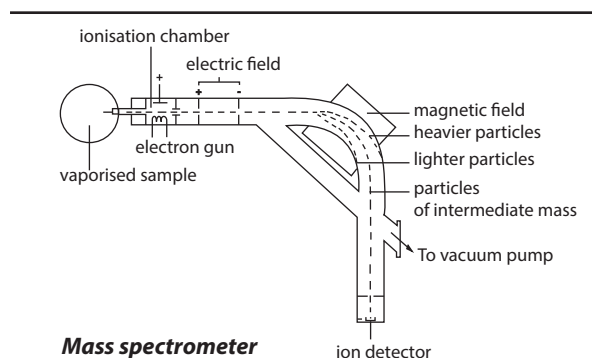
MASS PRODUCTION: The production of large numbers of identical objects, usually by use of mechanisation. The root of mass production is the assembly line, essentially a conveyor belt which transports the product so that each worker or robot may perform a single function on it such as adding a component.

MASS SPECTROGRAPH: A system of vacuum in which positive rays of charged atoms are passed through electric and magnetic fields to separate electrically charged particles according to their mass-to-charge ratios. The amount of deflection of the particles, which are registered on a photographic plate depends on their mass and velocity. The greater the mass or velocity of the ion, the less it is deflected. An instrument called mass spectrometer is often used to measure the masses of isotopes of elements with precision.



MASS SPECTROMETER: An apparatus used to identify atoms and isotopes of elements and to measure their masses with precision by converting molecules into ions and then separating the ions

according to their mass-to-charge ratio. It consists of three modules: the ion source which ionizes and splits the molecule into ions; the mass analyser, which sorts the ions by their masses by applying electromagnetic field; and the detector, which measures the value of an indicator quantity and thus provides data for calculating the abundance of each ion present. Other types of mass spectrometer exist. In a quadrupole mass spectrometer the ions pass along a region surrounded by four parallel rods. Variable voltages applied to the rods produce an oscillating electric field. Varying the frequency of oscillation allows different ions to pass through to a detector. In a time-of-flight mass spectrometer the ions are accelerated by an electric field and then enter a drift tube through which they pass to a detector. Different types of ion are distinguished by their time of flight in the drift tube.



MASS SPECTROMETER FOCUSING SYSTEM (DEFLECTION SYSTEM): An assembly permitting the separation of ions according to their mass-to-charge ratio.

MASS SPECTROMETRIC DETECTOR: A mass spectrometer used as a detector to give qualitative and quantitative data on the various eluted substances. The mass spectrum of the eluted compounds provides evidence beyond the elution time as to the chemical nature of the species.

MASS SPECTROMETRY (MS): An analytical technique used to identify atoms and isotopes to determine the chemical composition of a sample or molecule. It is also used for elucidating the chemical structures of chemical compounds such as identification of metabolites, drugs and other molecules isolated by biological samples. The technique consists of ionising chemical compounds to generate charged molecules or molecular fragments followed by separation and measurement of their mass-to-charge ratios.

MASS SPECTROSCOPY: An analytical technique used to determine relative atomic masses, the relative abundance of isotopes and for identification of metabolites, drugs and other molecules isolated by biological samples. It produces spectra of the masses of the atoms or molecules, comprising a sample of the material. The spectra are used to determine the elemental or isotopic signature of a sample, the masses of particles and molecules. It works by ionising chemical compounds to generate charged molecules and measuring their mass-to-charge ratio.

MASS SPECTRUM: Indication of the distribution in mass or in mass-to-charge ratio of ionised atoms or molecules produced by a mass spectrograph. For example, a beam of ions passed through a mass spectrograph in which they are deflected according to their charge-to-mass ratios will have a range of masses called a mass spectrum.

MASS STRUCTURE: A structure that is made by combining similar materials into a certain shape or design. Mountains are an example of a natural mass structure, and a pyramid is an example of a man-made mass structure.

MASS TRANSFER: Spontaneous (irreversible) process of transfer of mass across non-homogeneous fields. The driving force can be the difference in concentration (in liquids) or partial pressure (in gases) of

the component. In fluids, mass transfer may be enhanced by turbulent flow. In biological systems, mass transfer through membranes may result from normal diffusion, facilitated diffusion or active transport.

MASS WASTING (MASS MOVEMENT; SLOPE MOVEMENT):

Any of the processes of movement of large masses of rocks, soil and other earth materials downward due to the force of gravity. This usually occurs when gravitational force overcomes the shear strength (resistive forces), which holds the material in place. The three main types of mass wasting are **falls**, like rock fall and rock avalanche; **slides (land slides)**, such as rock slide and slump (sliding of coherent rock material along a curved surface); **flows**, like rock avalanche, debris flow, earth flow, and creep or soil creep (gradual downhill movement, under the force of gravity of soil and loose rock material on a slope). Many factors contribute to, or cause mass wasting. Examples are earthquake, very steep slope, scarce slope vegetation, presence of water in slope material and biological activity such as movement of animals. Earthquakes, for example, can break off sections of mountains or hills, causing them to slide down the slope. Plants or vegetation tend to resist mass wasting as it stabilises soil particles on the surface and anchors soil under the surface through its root system.

MASS-DISTRIBUTION FUNCTION (WEIGHT-DISTRIBUTION FUNCTION):

A distribution function in which the relative amount of a portion of a substance with a specific value or a weight distribution range of values, of the random variable(s) is expressed in terms of mass fraction.

MASS-ENERGY EQUATION: The equation $E = mc^2$, where E is the energy in joules, m is the mass in kilograms and c , the speed of light in a vacuum that equals 300,000 km per second. It was formulated by German-born US physicist, Albert Einstein (1879-1955) in his special theory of relativity to denote the equivalence between mass and energy. The conversion of mass into energy in accordance with this equation occurs in ordinary chemical reactions, but a much larger percentage occurs in nuclear reactions. This forms the basis of atomic energy.

MASS-ENERGY RELATIONSHIP: The famous equation, discovered by Einstein: $E = \Delta mc^2$, where E is energy, m is the equivalent amount of mass and c is the velocity of light. This relation shows that, effectively, mass and energy are two sides of the same coin. It is also the basis for understanding the principle of nuclear fusion.

MASS-FLOW SENSITIVE DETECTOR: A device whose response is proportional to the amount of sample component reaching the detector in unit time.

MASSETER: A thick, quadrate muscle in the cheek which arises from the lower border and deep surface of the zygomatic arch and extends to the mandible. It is important for mastication (chewing) and acts by closing the jaws.

MASSIC: Attribute to a physical quantity obtained by division by mass. It is synonymous with specific.

MASSIVE TRANSITION: A transition in which a crystal structure is changed into a new structure during cooling or heating, predominantly by interface-controlled reactions and the usual nucleation and growth characteristics associated with thermal diffusion.

MASSULA (POLLINIUM): A mass of pollen grains that originate from a single anther, but sticks together during pollination and is transported as a single unit. This is a characteristic of orchid plants (Orchidaceae) and many species of milkweeds (Asclepiadoideae).

MAST: 1. The small hard fruits that fall from a beech tree and used as food by pigs and other animals. 2. A tall broadcasting antenna. 3. Vertical spar that supports sails, riggings or flags on a ship. 4. Any vertical support, especially of a flag.

MAST CELL: A wide cell with densely granular cytoplasm, capable of amoeboid movement that is found in connective and fatty tissues. Examples are found around blood vessels and in the skin. They release substances such as leukotrienes, histamine and heparin during allergic

reactions and inflammation response, which causes a local increase in blood flow attracting white blood cells to the area.

MASTECTOMY: Surgical removal of all or part of a breast in order to treat or prevent breast cancer.

MASTER BOOT RECORD, MBR (MASTER BOOT BLOCK; MASTER PARTITION BOOT SECTOR; SECTOR 0): A small section of about 512 bytes in the first sector of a computer's hard drive mass storage media or other storage device that is performed during boot-up to partition the hard drive, to find the operating system and load it into memory, etc.

MASTER BOOT RECORD VIRUS (MBR VIRUS): A computer virus that modifies or infects the Master Boot Record program causing the computer to lose CD-ROM support, cause Microsoft Windows to run in compatibility mode or damages the hard disk drive. It infects a computer only if the virus is used to boot-up and does not infect it if the virus is introduced after the boot-up process or when the computer is running a Macintosh Operating System (Mac OS).

MASTER CRYPTOGRAPHY KEY: In computing, a key used to encipher operational keys that are to be used at a node.

MASTER GLAND (PITUITARY GLAND; HYPOPHYSIS): A pea-sized endocrine gland attached to the hypothalamus at the base of the brain by a thin stalk. The pituitary gland secretes hormones regulating homeostasis, including tropic hormones that stimulate other endocrine glands and also influences protein synthesis and growth of the bone structure, sexual maturing, oxidative metabolism, development of gametes and milk secretion. It also enhances the secretion of thyroxine and adrenal cortical hormones.

MASTIC: 1. A resinous exudate obtained from an evergreen shrub native to the Mediterranean region. Mastic was used by the ancient Egyptians for embalming and is now used in varnish, as a theatrical fixative and for temporary teeth. 2. The bushy evergreen Mediterranean tree which yields mastic and has aromatic leaves and fruit, closely related to the pistachio.

MASTICATION: The first step of digestion, which involves chewing or grinding food by the teeth to break it into smaller particles to allow more efficient break down by enzymes, providing a greater surface area for digestion and enabling the formation of a bolus, which is small enough to pass through the oesophagus.

MASTITIS: Inflammation of the breast or mammary glands of cattle, goat and other milk producing animals. Mastitis most commonly occurs in nursing mothers between the first and third weeks usually after the birth of the first child. It may also occur in women who are not breast-feeding. In breastfeeding women, it is caused by the bacteria commonly found on the skin, which enter the nipple through small cracks. The bacteria then multiply in the fatty tissue of the breast surrounding the milk ducts, causing swelling, warmth or pain.

MASTOID BONE: The lower part of the temporal bone of the skull. Jutting from the lower part of the mastoid bone is a bony projection called the **mastoid process**.

MASTOID PROCESS: A conical outgrowth projecting from the temporal bone of the skull containing air cavities that communicate with the cavity of the middle ear. In humans, its size is larger in males than in females. It is a route through which infection may spread from the middle ear.

MATCHED CELLS: A pair of cells with closely similar optical properties. One cell is the sample cell while the other, the reference (or blank) cell contains the solvent or a reference solution. In double beam spectrometers, radiation is passed either simultaneously or alternately through the cells. In single beam instruments the cells are moved sequentially into the radiation beam.

MATCHED-PAIRS DESIGN: In statistics, an experimental design concerned with measuring the differences in the values of the dependent variables for pairs of subjects that have been matched to eliminate individual differences and that are respectively subjected to the control condition and the experimental condition.

MATCHES: Small, thin sticks of wood chemically coated at the tip with a mixture of fire-producing substances that produce a flame when struck against a rough surface. The reaction occurs between phosphorus sulfide and potassium chlorate.

MATCHING: In mathematics, a subset of the edges of a bipartite graph consisting of edges joining each of the vertices in one of the sets of vertices defining the bipartite structure such that no two edges share the same vertex. Vertices matched by an edge are called mates. A matching is perfect if it is complete in the sense that all vertices are mated.

MATERIAL CONDITIONAL: A logical connective or a binary operator that is often symbolised by a forward arrow, $\rightarrow$. It is used to form conditional statements of the form ' $P \rightarrow Q$ ' (termed a conditional statement) that is read as 'if P then Q ' or ' P only if Q '.

MATERIAL DERIVATIVE: The derivative that computes the time rate of change of any quantity such as temperature or velocity for a portion of a material moving with a velocity, v . If the material is a fluid, then the movement is the flow field.

MATERIAL DESCRIPTION (LAGRANGIAN DESCRIPTION): The description of physical phenomena associated with the deformation of a body in terms of fields that are defined over the reference rather than the current configuration.

MATERIAL IMPLICATION (IMPLICATION): 1. In logic, the binary truth-functional sentential connective that assigns the value false to a compound sentence only when its first argument, the antecedent, is true and the second, the consequent, is false and the value true otherwise, without consideration of relevance etc. The closest approximation in natural language is 'if ... then ...'. 2. A compound sentence formed with this connective, written ' $P \rightarrow Q$ ' or ' $P \supset Q$ ', where P is the antecedent and Q the consequent. 3. The relationship that holds between an ordered pair of sentences when there are no circumstances in which the first is true and the second false. 4. **Paradoxes of material implication** is defined as a number of inferential patterns that follow immediately from the definition: a falsehood materially implies any statement whatever and anything whatever materially implies a truth.

MATERIALS SCIENCE: An applied science concerned with the relationship between the structure and properties of materials. Chemists who work in the field study how different combinations of molecules and materials result in different properties. They use this knowledge to synthesise new materials with special properties.

MATERNAL EFFECT GENE: Gene expressed in maternal follicle cells whose products diffuse into the egg cell to influence its early development. Gradients of the products are established in the egg cytoplasm following fertilisation and subsequent cell division of the zygote. These gradients influence zygotic gene expression and cause regional differentiation of the embryo. For example, in many types of embryo, maternal effect genes are responsible for determining polarity.

MATH (MATHEMATICS) JOURNAL: A book in which a student records his/her mathematics work and thinking. It can be used to record the solutions to mathematics problems, along with the strategy and thought processes used to arrive at the solution. Journal pages give models for conceptual understanding, problems to solve, and directions for individual and small-group activities.

MATHEMATICAL: Of, relating to, pertaining to or characterised by mathematics or its methods and principles, especially its precision and exactness.

MATHEMATICAL EXPECTATION (EXPECTED VALUE): In statistics, the sum or integral of all possible values of a random variable or any given function of it, multiplied by the respective probabilities of the values of the variable. Its symbol is $E(X)$. $E(X)$ is the mean of the distribution; $E(X - c) = E(X) - c$, where c is a constant.

MATHEMATICAL INDUCTION (FINITE INDUCTION): A two-part method of proving a theorem involving an integral parameter that all integers have a certain property. First the theorem is verified for the smallest admissible value of the integer. Then it is proven that if

the theorem is true for any value of the integer, it is true for the next greater value. The final proof contains the two parts.

MATHEMATICAL LOGIC: Formal logic, especially that branch concerned with the foundations of mathematics. Its fields include fields of set theory, model theory, recursion theory and proof theory.

MATHEMATICAL MODELS: Physical objects used to represent mathematical abstractions; or, more frequently, mathematical constructions such as formulae, functions and graphs, used to express physical phenomena. Such models occur throughout applied mathematics and physics, their greatest value being heuristic (i.e., the model may suggest the existence of unsuspected properties in the phenomenon).

MATHEMATICAL PLATONISM (PLATONISM): The philosophical theory that mathematical objects such as numbers and sets exist and are independent of us and our language, thought and practices and therefore that mathematical truth does not consist in, but is the aim of, the construction of proofs.

MATHEMATICAL PROBABILITY (CLASSICAL PROBABILITY): The probability of an event consisting of n out of m possible equally likely occurrences, defined to be n/m . 2. The study of such probabilities.

MATHEMATICAL PROBLEM (PROBLEM): In mathematics, a question or statement that requires proof or a method of solution, for example by mathematical operation or geometric construction.

MATHEMATICAL PROGRAMMING: The theory and application of the optimisation of functions, often subject to constraints given in terms of functions. The most basic problem studies optimisation of a real-valued function over a prescribed set. This involves establishing the existence of optimal points, characterising such points and finding algorithms for their computation.

MATHEMATICS: The science of spatial and numerical relationships or the study of the relationships among numbers, shapes and quantities or the science of structure order and relation that has evolved from counting, measuring and describing the shapes of objects. It uses signs, symbols and proofs and includes algebra, analysis, arithmetic, combinatorics, Euclidean and non-Euclidean geometries, game theory, number theory, numerical analysis, optimisation, probability, set theory, statistics, topology and trigonometry.

MATHEMATIZE: To interpret a real-world situation by organising, sorting, and coordinating information to develop a mathematical representation of a given problem.

MATHIEU'S DIFFERENTIAL EQUATION: The differential equation originating in the study of vibrations in the two-dimensional wave equation. It is of the form $\frac{d^2 y}{dt^2} + (a + 116q \cos^2 t)y = 0$. For a

appropriately related to q , periodic solutions with period 2π exist; of these, the even and odd solutions are Mathieu functions.

MATING: Part of the process of sexual reproduction in some animals that brings the male and female sex cells (gametes) together. Mating may not be physical, where a male and female copulate. For example, in some aquatic animals, the union of sperm and egg takes place outside the body of the female, a process called **external fertilisation** by releasing great quantities of sperm and eggs into the water, which ensures fertilisation even if most of the gametes are dispersed by currents. Frog stimulates the female to lay eggs by the male climbing on the back of the female, which the male sheds with its sperm.

MATING RITUAL: In most animal species, the close approach of one individual to another is an aggressive action. Yet, for reproductive purposes such a male-female approach must be made. Thus complicated behaviour patterns have evolved from a combination of submissive gestures and ritualised displacement activities, to appease the sexual partner. In many cases these rituals also involve the development of morphological display characters. Mating rituals may also serve as an isolating mechanism to ensure that mating is with another member of the same species.

MATING STRAIN: A group of organisms within a species that are characterised by not being able to interbreed with each other. However, they are able to breed with members of other physiologically different but morphologically identical groups. Different mating strains are usually denoted by the symbols + and - since male and female forms cannot be distinguished as such. The differences are genetically determined.

MATING TYPE: Distinct strains that are compatible in sexual reproduction. This occurs in microorganisms, fungi and algae that undergo sexual reproduction via isogamy. Mating types look alike morphologically (in form and structure) and therefore difficult to distinguish. For this reason different mating types of the same species are designated by numbers, letters or + and - instead of "male" and "female". They are able to give rise to zygote capable of developing into new individuals.

MATRIC (MATRICAL; MATRICIAL): Of or pertaining to a matrix. A matrix polynomial is constructed by replacing each power of the indeterminate in a polynomial by the same power of a given matrix. The Cayley-Hamilton theorem ensures that a matrix is always a zero of the matrix polynomial arising from the characteristic polynomial of the given matrix.

MATRIC POTENTIAL: A component of water potential due to the adhesion of water molecules to non-dissolvant structures of the system (the matrix, such as plasma membranes or soil particles). It is always negative and is significant only outside living cells in relatively dry systems, for example, soils, where much of the water is tightly bound to soil particles.

MATRIX: 1. A rectangular array of numbers or algebraic variables. Matrices are a means of condensing information about mathematical systems and can be used for solving certain mathematical operations such as simultaneous linear equations and transformations. An $a \times b$ matrix has a rows and b columns, for example, a 2×2 matrix is

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ or $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The entry denoted a_{ij} lies at the intersection of

the i^{th} row and the j^{th} column. A matrix with the same number of rows and columns is a square matrix. A square matrix with all entries zero except those at the intersection of the pairs of rows and columns with the same number is a **diagonal matrix**; an $n \times n$ diagonal matrix with all non-zero entries equal to 1 is the **identity matrix of order n** . 2. A component of tissues such as bone and cartilage in which the cells of the tissue are embedded. 3. The fine grained material of rock in which the larger coarser grained material is embedded. 4. The non-living component of connective tissue, consisting of a web of fibres embedded in homogeneous ground substance that may be liquid, jellylike or solid. An example is as found in fluid-filled matrix of the mitochondrion.

MATRIX ABSORPTION LASER DESORPTION IONISATION (MALDI): A technique for producing ions for mass spectroscopy, used especially in the analysis of large biological species such as proteins, peptides and sugars and large organic molecules such as polymers, dendrimers and other macromolecules, which tend to be fragile and fragmented, when ionised by other ionisation methods. The ionisation is triggered by a laser beam, usually a nitrogen laser. The sample is absorbed on an inert matrix from which ions are desorbed by a laser.

MATRIX BAND: A cylindrical metal or plastic band with a special clamp or holder (the matrix retainer), filled with softened impression compound and seated over a tooth so that the compound flows into the prepared cavity and an impression of the tooth can be obtained. It is also used for placement and contouring of certain restorative materials.

MATRIX CALCULUS: A yellowish-white to light tan urinary calculus containing calcium salts, with the consistency of putty. It is composed chiefly of an organic matrix consisting of a mucoprotein and a sulfated mucopolysaccharide and usually associated with chronic infection.

MATRIX EFFECT: 1. In analytical chemistry, it refers to the combined effect of all components of the sample other than the analyte on the measurement of the quantity. If a specific component can be identified as causing an effect, it is referred to as **interference**. 2. In surface analysis, it refers to the effects which cause changes in Auger-electron, photoelectron, secondary ion yield or scattered ion intensity, the energy or shape of the signal of an element in any environment as compared to these quantities in a pure element. Chemical matrix effects refer to changes in the chemical composition of the solid which affect the signals as described above. Physical matrix effects refer to topographical and/or crystalline properties which affect the signal as described above.

MATRIX ELEMENT: 1. In mathematics, it is one of the set of numbers which form a matrix. 2. The scalar product of a member of a complete orthogonal set of vectors, representing states, with a vector which results from applying a specified operator to another member of this set.

MATRIX MECHANICS: A formulation of quantum mechanics using matrices to represent states and operators. Matrix mechanics was the first formulation of quantum mechanics to be stated by Werner Heisenberg (1901-1976) in 1925 and was developed by him, German physicists Max Born (1882-1970) and the Pascual Jordan (1902-80).

MATRIX OF COEFFICIENTS: The $m \times n$ matrix A in the set of linear equations $Ax = b$, where x is a vector of n variables and b is a vector of m constants. This is equivalent to a system of m simultaneous equations in n unknowns.

MATROID: In mathematics, a collection of subsets of a set such that any subset of a member of the collection belongs to the collection and such that if the sets $\{a_1, \dots, a_k\}$ and $\{b_1, \dots, b_{k+1}\}$ are in the collection, so is the set $\{a_1, \dots, a_k, b_i\}$ for some $i \leq k+1$. Given any graph, the set of all sets of edges containing no cycle is a matroid.

MATTER: Any substance made up of atoms of which all physical objects are made and which has mass and volume and can be detected and measured. In other words, matter is anything that has both mass and volume and occupies space. Solid, liquid, gas and plasma are the four states of matter. In chemical reactions, matter is conserved, so that no matter is lost or destroyed and the sum of the mass of the reactants is always equal the sum of the end products.

MATTHIAS, JAKOB SCHLEIDEN (1804-1881): German botanist (1804-1881), who together with the German physiologist Theodor Schwann (1804-1881) formulated the cell theory.

MATTOCK: One of the oldest tools of agriculture, it is a heavy tool with a long handle and a metal head, one end of which is sharp and the other blunt, used for breaking soil, cutting roots, etc.

MATURE SOIL: A well-developed soil that has passed through the major natural processes of soil formation and has a soil profile with clearly defined horizons.

MATURITY: 1. The stage in life cycle that is reached when a developing organism has taken on the appearance of the adult form and is capable of reproduction. 2. The stage reached in the formation of gametes following meiotic division of precursor cells and their development into fully functional gametes. 3. The final differentiation processes in biological systems, such as the final ripening of a seed or the attainment of full functional capacity by a cell, a tissue or an organ. **Maturation** is the process of reaching maturity. To **mature** is to attain maturity or having reached maturity.

MATUTINAL: Pertaining to or occurring in the morning. It describes an organism that is primarily active only in the early morning. Numerous genera and species of bees exhibit this behaviour. A flower such as the morning glory also exhibits this behaviour.

MAUVE (MAUVEINE): A purple dye obtained from aniline. It was the first synthetic dye, made by William Perkin in 1856.

MAW: 1. A stomach, especially the last of a ruminant's four stomachs. 2. The mouth, stomach, jaws or gullet of a voracious animal, especially a carnivore.

MAX-FLOW MIN-CUT THEOREM: The important duality theorem, which states that in a flow network, the maximum amount of flow passing from the source to the sink is equal to the minimum capacity that, when removed in a specific way from the network, causes the situation that no flow can pass from the source to the sink.

MAXICELL: A bacterial cell that has been irradiated with ultraviolet light to intentionally damage the cell's own DNA so that plasmid genes, which are relatively spared, can be preferentially expressed.

MAXILLA: 1. One of a pair of mouthparts in insects, crustaceans, centipedes and millipedes. They lie behind the mandibles. The lateral movements assist in feeding. 2. One of the pair of large tooth bearing bones in the upper jaw of vertebrates. In mammals, they carry all the upper teeth except the incisor.

MAXILLARY ARTERY: The larger terminal branch of the external carotid artery in the parotid gland, which supplies structure of the face. It arises behind the neck of the mandible and runs forward medial to it, to reach the lower border of the lateral pterygoid muscle.

MAXILLARY NERVE: The second of the three divisions of the trigeminal nerve, that supplies sensory fibres to the skin areas of the middle part of the face, the upper jaw and its teeth and the mucous membranes of the palate, nasal cavities and nasopharynx.

MAXILLARY SINUS: The largest of the paranasal sinuses (air sinuses) that communicates with the middle meatus of the nose.

MAXILLIPED: One of the three pairs of head appendages in crustaceans that contribute to the mouthparts located behind the mandibles and used to move food to the mouth.

MAXILLULA: A paired appendage that contributes to the mouthparts in certain crustaceans. Lying between the mandibles and the maxillae, they may bear hairs or spines designed to move food particles deeper into the mouth.

MAXIMAL: 1. A maximal element of an ordering or lattice, is an element with no element greater than it. A maximal element need not be the unique greatest element unless the ordering is total; for example, the set of proper subsets of a given set ordered by inclusion has no greatest element, but every set formed by removing a single member from the given set is maximal. 2. A maximal element of an orthogonal or orthonormal sequence is such that whenever the inner products of any given element with the sequence elements are zero, so is the given element. Unless the inner product space is complete, not every maximal orthonormal sequence need be a Schauder basis, but clearly every basis is maximal.

MAXIMAL DOMAIN (DOMAIN): The set of all real numbers for which a function is defined. In other words, it is a set from which all the first elements of the ordered pair that constitute the function or relation are drawn. It is composed of the set of x -values that give rise to real y -values.

MAXIMAL IDEAL: A proper ideal that is maximal with respect to inclusion.

MAXIMIN: 1. In mathematics, the highest of a set of minimum values. 2. Being or relating to a strategy or value that maximises the minimum value of a function; for example, the maximin of a function $\{f_n\}$ is the function f_i .

MAXIMISE (MAXIMIZE): 1. To increase, for example, a value or profit to the greatest degree or amount possible. 2. In mathematics, of a function, to find, the maximum value.

MAXIMISED LIKELIHOOD: The probability of randomly drawing a given sample from a population, maximised over the possible values of the population parameters.

MAXIMUM: 1. The largest element of a set S , usually denoted $\max S$; for example, negative numbers have no maximum, but non-positive numbers have maximum 0, although both have 0 as supremum. 2. The highest value of a function, usually denoted $\max f$. It is a **global maximum** if it is the highest value of a set or function over the entire domain in comparison with all other values. It is the value of the function, f at a point, a that is greater than (or equal to) any other point in the entire domain of comparison with all other values. It is a **local maximum**, if it is a value greater than any other in a

neighbourhood of comparison. In the case of a real differentiable function on an open set, it is identified by having zero derivative and negative second derivative since the tangent to the curve changes from rising to falling at this point.

MAXIMUM ALLOWABLE CONCENTRATION: The maximum concentration of a pollutant which is considered harmless to healthy adults during their working hours, assuming they breathe uncontaminated air at all other times.

MAXIMUM AND MINIMUM: Points at which the slope of a curve representing a function changes from positive (minimum) to negative (maximum) or from negative to positive. A tangent to the curve at a maximum or minimum has zero gradient.

MAXIMUM AND MINIMUM THERMOMETER (SIX'S THERMOMETER): A self-registering thermometer used to measure and record maximum and minimum temperatures over a period of time. It was named after British born scientist and retired business man, James Six (1731 1793), who invented it in 1782. It consists of a U-shaped glass tube whose bottom half is filled with mercury and graduated right-hand and left-hand tubes for recording the minimum and maximum temperatures registered. The left-hand tube is filled with alcohol, which expands (when the temperature rises) or contracts (when the temperature drops). The right-hand tube is partially filled with alcohol and a portion is filled with gas. When the temperature increases, the alcohol in the left side expands and forces the mercury and the alcohol column on the right side into the gas chamber. When the temperature decreases, the alcohol in the left side contracts and the gas expands, forcing the liquid back into the left side and pushing the index along with it. In all cases, the maximum and minimum temperatures recorded remain at their positions until they are reset.

MAXIMUM CONDITION: In mathematics, the condition on a module that every non-empty set of submodules has a maximal member.

MAXIMUM EMISSION CONCENTRATION: Standards for maximum concentration of air pollutant emission from stationary or mobile sources.

MAXIMUM LATENT PERIOD: The length of time after which no causal relationship can reasonably be established. This length of time is the period from the last exposure to the point of time at which an exposed person has demonstrated the initial signs or symptoms.

MAXIMUM LIKELIHOOD ESTIMATION: A statistical method for parameter estimation using a pre-specified model and a set of empirical data. The parameter values are chosen to maximise the probability of observing the data, given the hypothesis. In phylogeny, the method is used to find the tree topology with maximum likelihood, given the data and the model of sequence evolution and to estimate branch lengths and other evolutionary parameters. The values of the parameters obtained from using the maximum likelihood estimation is called **maximum likelihood estimator**.

MAXIMUM MODULUS THEOREM: The theorem of complex analysis that if a function is analytic and attains its maximum modulus in some open region, it is constant. As a consequence, if the region D is bounded and the modulus of the function f is continuous on the closure of D , then $|f(z)|$ achieves its maximum on the boundary of D . If the function vanishes nowhere in the domain, the minimum modulus is also achieved on the boundary.

MAXIMUM PERMISSIBLE DOSE: The recommended maximum amount of ionising radiation that one may receive in a specified period without significant harm to the body.

MAXIMUM PERMISSIBLE LEVEL (MPL): Level, usually a combination of time and concentration, beyond which any exposure of humans to a chemical or physical agent in their immediate environment is unsafe.

MAXIMUM STORAGE LIFE: A period during which there is no change in concentration in excess of the value of the uncertainty of the concentration.

MAXIMUM SUSTAINABLE YIELD (MSY): The maximum yield or catch that can be taken from a species' stock over an indefinite period without harming growth. Fundamental to the notion of sustainable harvest, the concept of MSY aims to maintain the population size at

the point of maximum growth rate by harvesting the individuals that would normally be added to the population, allowing the population to continue to be productive indefinitely.

MAXIMUM TOLERABLE CONCENTRATION (MTC): Highest concentration of a substance in an environmental medium that does not cause death of test organisms or species. It is denoted by LC_{50} .

MAXIMUM TOLERABLE EXPOSURE LEVEL (MTEL): Maximum amount or concentration of a substance to which an organism can be exposed without leading to an adverse effect after prolonged exposure time.

MAXIMUM TOLERATED DOSE (MTD): High dose used in chronic toxicity testing that is expected on the basis of an adequate subchronic study to produce limited toxicity when administered for the duration of the test period. It should not induce: (i) overt toxicity, for example appreciable death of cells or organ dysfunction; or (ii) toxic manifestations that are predicted materially to reduce the life span of the animals except as the result of neoplastic development; or (iii) 10% or greater retardation of body weight gain as compared with control animals. In some studies, toxicity that could interfere with a carcinogenic effect is specifically excluded from consideration.

MAXIMUM TRANSMISSION UNIT (MTU): A networking term that defines the largest packet size that can be sent over a network connection. The MTU is typically limited by the type of connection, but may sometimes be adjusted in a computer's network settings. Because a higher MTU allows for more data to be transferred at once, connections with high MTUs typically have more bandwidth than connections with lower MTUs.

MAXIMUM VALUE THEOREM: The theorem that for a real-valued continuous function f on a closed bounded interval or any compact set, there is at least one element x in the domain of f for which $f(x)$ achieves its maximum value. For example, on the closed interval $[0, 1]$, the function $y = x^3 - x^2 + 1$ achieves its maximum at $x = 1$, but on the open interval $]0, 1[$ it does not, as for all $x < 1$ there is an $x < x' < 1$ for which $f(x') > f(x)$.

MAXIMUM WATER HOLDING CAPACITY: The average moisture content of a disturbed sample of soil, 1 cm high that is at equilibrium with a water table at its lower surface.

MAXWELL: A unit of magnetic flux in the centimetre–gram–second (cgs) system. It is equal to the magnetic flux, which produces an electromotive force of 1 abvolt in a circuit of one turn linking the flux, as the flux is reduced to zero in 1 second at a uniform rate. One Maxwell is equal to 10^{-8} Weber. The International System of Units (SI) is the currently used system of units, and the weber (Wb) has replaced the maxwell.

MAXWELL-BOLTZMANN DISTRIBUTION: A law explaining the distribution of speeds among the molecules of a gas, where the particles do not constantly interact with each other but move freely between short collisions. It was named after James Clerk Maxwell and Ludwig Boltzmann. In a system made up of N molecules that are independent of each other except that they exchange energy on collision, it is clearly impossible to say what velocity any particular molecule will have. However, statistical statements regarding certain functions of the molecules were worked out by James Clerk Maxwell

and Ludwig Boltzmann. One form of their law states that $n = \exp\left(\frac{-E}{RT}\right)$,

where n is the number of molecules with energy in excess of E , T is the thermodynamic temperature and R is the gas constant and N the total number of molecules present.

MAXWELL, JAMES CLERK (1831-79): Scottish physicist, born in Edinburgh, who published his first scientific paper at the age of 14, entered the University of Edinburgh at 16 and graduated from Cambridge University, where he supervised the building of Cavendish Laboratory. He held academic posts in Universities of Aberdeen, London and Cambridge. He is distinguished for his contributions to the electromagnetic theory of light and the kinetic theory of gases. His best-known work was a mathematical analysis of electromagnetic radiation, published in 1864.

MAXWELL'S DEMON: An imaginary creature that is able to open and shut a partition dividing two volumes of a gas in a container, when the two volumes are initially at the same temperature. The partition operated by the demon is only opened to allow fast molecules through. Such a process would make the volume of gas containing the fast molecules hotter than it was at the start. The volume of gas remaining would accordingly become cooler. This process would be a violation of the second law of thermodynamics and therefore cannot occur. Maxwell's demon was invented by James Clerk Maxwell in a letter written in 1867 to show that the second law of thermodynamics has its origins in statistical mechanics, although the name was suggested by the Scottish scientist, Lord Kelvin (1824-1907).

MAXWELL'S DISPLACEMENT CURRENT (DISPLACEMENT CURRENT): The rate of change of the electric displacement vector, which must be added to the current density to extend Ampère's law to the case of time-varying fields (metre-kilogram-second units).

MAXWELL'S DISTRIBUTION (MAXWELLIAN DISTRIBUTION): A function giving the number of molecules of a gas in thermal equilibrium whose velocities lie within a given, infinitesimal range of values, assuming that the molecules obey classical mechanics and do not interact.

MAXWELL'S EQUATIONS: A set of four partial differential equations, named after the Scottish physicist and mathematician James Clerk Maxwell. The equations relate the electric field and magnetic field vectors E and B to their sources, electric charge, current and changing fields. These four equations are known as Gauss's law, Gauss's law for magnetism, Faraday's law of induction and Ampère's law with Maxwell's correction. The Maxwell's equations are given by

$$\nabla \cdot E = 4\pi\rho, \quad \nabla \times E = -\frac{1}{c}\frac{\partial B}{\partial t}, \quad \nabla \cdot B = 0, \quad \nabla \times B = \frac{4\pi}{c}J + \frac{1}{c}\frac{\partial E}{\partial t} \quad \text{where}$$

∇ is the divergence, $\nabla \times$ is the curl, π is a constant, E is the electric field, B is the magnetic field, ρ is the charge density, c is the speed of light and J is the vector current density.

MAXWELL'S RELATIONS: 1. According to Maxwell's electromagnetic theory, that relation wherein the dielectric constant of a substance equals the square of its index of refraction. 2. One of four equations for a system in thermal equilibrium, each of which equates two partial derivatives, involving the pressure, volume, temperature and entropy of the system.

MAXWELL'S SCREW RULE: A rule formulated by Scottish physicist, James Maxwell that predicts the direction of the magnetic field produced around a wire carrying electric current. It states that, if a right hand screw is turned so that it moves in the same direction as the current, its direction of rotation will give the direction of the magnetic field.

MAXWELL'S THERMODYNAMIC EQUATIONS: Equations in the thermodynamics for a given mass of a homogeneous system, relating the entropy (S), pressure (p), volume (V) and the thermodynamic temperature (T). The four equations are; $\left(\frac{dT}{dV}\right)_s = -\left(\frac{dp}{dS}\right)_V$; $\left(\frac{dT}{dp}\right)_s = -\left(\frac{dV}{dS}\right)_p$; $\left(\frac{dV}{dT}\right)_s = -\left(\frac{dS}{dp}\right)_T$; and

$$\left(\frac{dS}{dV}\right)_s = -\left(\frac{dp}{dT}\right)_V.$$

MAYER'S F-FUNCTION: A quantity that occurs in the calculation of virial coefficients. It is defined by $f = \exp\left(\frac{-V_2}{kT}\right) - 1$, where V_2 is the two body interaction potential energy, k the Boltzmann constant and T is the thermodynamic temperature. It is related to the second virial coefficient B by: $B = \left(\frac{-N_A}{V}\right) \int dr_1 dr_2 f$, where N_A is the Avogadro number and V is the volume of the system. This equation is simplified to: $B = -2\pi N_A \int_0^\infty f r^2 dr$ in the case of closed-shell atoms and octahedral and tetrahedral molecules. When particles are so far apart from the

interaction $V \rightarrow 0$, then $f \rightarrow 0$. When the particles are so close together the interaction $V \rightarrow \infty$ then $f \rightarrow -1$. This enables strong repulsive interactions between particles to be analysed in terms of f but not of V .

MAYER'S TEST: A general presumptive test for cocaine, morphine, heroin and other alkaloids. Mayer's reagent is a solution of potassium mercury iodide in water. A positive result is indicated by a cream precipitate.

MBONE (MULTICAST BACKBONE): In computing, it a virtual network on the Internet that supports the IP multicast protocol. It is composed of many networks and delivers packets of multi-media data to multiple users at the same time, enabling video and audio communication as well as video conferencing.

MCLEOD GAUGE: A vacuum pressure gauge to measure very low pressures, down to 10^{-6} mbar, in which a relatively large volume of a low-pressure gas is compressed to a small volume in a glass apparatus. It was invented in 1874 by Herbert G. McLeod (1841–1923). The volume is reduced to an extent that causes the pressure to rise sufficiently to support a column of fluid high enough to read.

McMILLAN-MAYER THEORY: A theory of solutions of non-electrolytes that show that there is one to-one correspondence between the equations describing a non-ideal gas and those describing dilute solutions of non electrolytes. In particular, they showed that there is a correspondence between the pressure of the gas and the osmotic pressure of the solution. This enables an expansion for solutions to be written, which is analogous to the virial expansion of non-ideal gases with analogues of the virial coefficients. These coefficients can be calculated with the analogue of potential being the potential of mean force of N solute molecules in the pure solvent. The McMillan Mayer theory can also be extended to distribution functions.

MEADOW: Field of grass and wild plants sometimes grown for fodder.

MEAL WORM: The larva of various beetles of the genus *Tenebrio* that infects and pollutes grain product.

MEALY: Powdery; grainy texture.

MEALY BUG: A scale insect that is covered with a powdery white secretion and feeds on plants, often causing significant damage to both greenhouse plants and crops in the field. It is notorious for the transmission of viruses.

MEAN: 1. Also known as **average**, it is the central value in a set of data, especially the mode or median. To find the mean of a set of values, add them and divide the sum by the number of values in the set. The arithmetic mean or average of $x_1, x_2, x_3, \dots, x_n$ is given

by mean $(\bar{x}) = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$. If $x_1, x_2, x_3, \dots, x_n$ occur with

frequencies $f_1, f_2, f_3, \dots, f_n$ then the arithmetic mean $(\bar{x}) = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$.

2. A parameter or statistic, for the population mean and for a sample mean. The population mean, denoted $\mu(X)$ is calculated by summing or integrating the product of each possible value of a variable with its probability, the sum or integral being taken over the whole range of the variable. The sample mean, denoted $\bar{X}$ is calculated by summing the observed values in the sample and dividing by their number. 3. The integral of a continuous function, f , over an interval $]a, b[$ divided by the length of the interval, $\frac{1}{b-a} \int_a^b f(x) dx$; since the integral is the

area between the graph of f and the axis between a and b , the mean of the function is the height of the rectangle on the same base and with the same area and so, in an obvious sense, is the average value of f on this interval. 4. The second and third terms in a proportion; for example, the terms b and c in the proportion $a : b = c : d$. 5. Any function of two or more positive variables of which the value always lies between the minimum and the maximum values of the arguments.

MEAN AND EXTREME PROPORTION: The relationship between four numbers or quantities, $a : b : c : d$, when $ad = bc$. It is written as $a : b : c : d$.

MEAN ABSOLUTE DEVIATION (AVERAGE DEVIATION): It is the arithmetic mean of the absolute deviations. Its symbol is μ . Statistically,

it is defined as $\mu = \frac{\sum_{i=1}^N |X_i - \bar{X}|}{N} = \frac{|X_1 - \bar{X}| + |X_2 - \bar{X}| + \dots + |X_N - \bar{X}|}{N}$, where

$X_1, X_2, \dots, X_N$ are the sample points and $\bar{X}$ is the sample mean based on $X_1, X_2, \dots, X_N$.

MEAN CATALYTIC ACTIVITY RATE: The catalytic activity of a specified component in or moved from a system divided by the time during which the component was changed or moved. This kind of quantity should not be confused with rate of reaction.

MEAN CURVATURE (MEAN NORMAL CURVATURE): The mean curvature at a point of a surface is the sum of the principal curvatures.

MEAN DEVIATION: 1. Also known as **mean deviation from the mean** or **average deviation**, it is the mean of the absolute values of the differences between individual values and the mean, used as a measure of dispersion. This is given by the formula:

M.D. = $\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$, where $\bar{x}$ is the mean. 2. The difference between an observed value of a variable and its mean.

MEAN ERGODIC THEOREM (WEAK ERGODIC THEOREM): A weaker form of the Birkhoff ergodic theorem, in which one only obtains functions convergent in mean square.

MEAN FREE PATH: The average distance travelled between collisions by the molecules in a gas, the electrons in a metallic crystal, the neutrons in a moderator or the average distance that a gas molecule travels before it collides with another molecule or with the containing vessel, etc. According to the kinetic theory the mean free path between elastic collision of gas molecule of diameter d (assuming they are rigid spheres) is $\frac{1}{\sqrt{2}n\pi d^2}$, where n is the number of molecules per unit volume in the gas. The mean free path is inversely proportional to the pressure. Its symbol is λ .

MEAN FREE TIME: The average time that elapses between the collisions of the molecules in a gas, the electrons in a moderator, etc. It is used in the kinetic theory of gases to compute transport coefficients such as the viscosity.

MEAN LIFETIME: Average time for which the unstable nucleus of a radioisotope exists before decaying, which is the sum of the lifetimes of all the individual unstable nuclei in a sample, divided by the total number of unstable nuclei present. It is the reciprocal of the decay constant. For a given isotope, the mean life is always 1.443 times its half-life. For example, lead-209 decays to bismuth-209 with a half-life of 3.25 hours and a mean life of 4.69 hours.

MEAN LINEAR RANGE: In a given material, for specified charged particles of a specified energy, the average displacement of the particles before they stop.

MEAN MASS RANGE: The mean linear range multiplied by the mass density of the material.

MEAN MASS RATE: The mass of a specified component changed in or moved to or from a system divided by the time during which the component was changed or moved.

MEAN MOTION: The constant angular velocity that an object, such as a planet, would have if it were moving in a circular orbit (instead of an elliptical one) of radius equal to its mean distance from the Sun equal to its actual revolution period. This is a hypothetical concept used in celestial mechanics.

MEAN MOTION RESONANCE: The dynamical situation where the ratio of the orbital periods of two orbiting objects can be expressed

as the ratio of two small integers. For example, plutinos are Kuiper Belt objects that are in a 3:2 resonance with Neptune, meaning that they orbit twice in the time Neptune completes three orbits. Mean motion resonances may lead to major changes in the orbit of one or both of the bodies or may enhance orbital stability, depending on the precise nature of the resonance.

MEAN NORMAL CURVATURE (MEAN CURVATURE): The mean curvature at a point of a surface is the sum of the principal curvatures.

MEAN RESIDENCE TIME: The mean time during which the molecules remain on the surface of the adsorbent, i.e. the mean time interval between impact and desorption.

MEAN SEA LEVEL: The average height of the surface of the sea for all stages of tide. It is used as a standard in reckoning land elevation. It also plays an important role in marine navigation as a chart datum and aviation, where standard sea level pressure is used as the measurement datum of altitude at flight levels.

MEAN SOLAR DAY (CIVIL DAY; MEAN DAY; SOLAR DAY): The period of time between two successive transits of the mean sun, which is the standard for the 24-hour day measured from midnight to midnight, symbol d . The length of a mean solar day is 24 h 3 min 56.555 s.

MEAN SQUARE: Of or pertaining to the sum or integral of the squares of the absolute values of a given series or integral. The mean square norm or Hilbert norm of a function on an interval $]a, b[$ is given by

$$\int_a^b f(t)^2 dt.$$

MEAN SQUARE SPEED ($\overline{C^2}$): Average of the squares of a set of given speeds.

MEAN SUBSTANCE RATE: The amount of substance of a specified component changed in or moved to or from a system divided by the time during which the component was changed or moved. This kind of quantity usually should be preferred to mean mass rate or mean volume rate.

MEAN VALUE THEOREM: If the derivative of a function f is defined everywhere between two points, $(x, f(x))$ and $(y, f(y))$, then the mean value theorem states that there will be at least one value of x between x and y such that the value of the derivative is equal to the slope of the line between $(x, f(x))$ and $(y, f(y))$. This means that there is at least one point in the interval where the tangent line to the curve is parallel to the secant line that passes through the curve at the two endpoints of the interval.

MEAN VOLUME RATE: The volume of a specified component changed in or moved to or from a system divided by the time during which the component was changed or moved.

MEAN-SPHERICAL APPROXIMATION MODEL (MSA): An approximation used in the theory of liquids that relates pair distribution functions to correlation functions. The mean-spherical approximation enables exact solutions to be found for several potential describing the interactions between molecules in a liquid.

MEANDER: A winding curve or bend in a river, caused by erosion and deposition of silt. It is caused when moving water in a stream erodes and undercuts the outer banks and widens its valley and the inner part of the river has less energy and deposits silt.

MEASLES (RUBEOLA): A highly contagious viral disease, especially of children, caused by a filterable virus and spread by droplet spray from the nose, mouth and throat of individuals in the infective stage. It is characterised by fever and small red spots that cover the whole body. One attack of measles confers lifelong immunity. However, it renders the patient susceptible to other more serious infections such as bronchial pneumonia and encephalitis.

MEASURABILITY-PRESERVING TRANSFORMATION: A one-to-one transformation between measure spaces such that both the given map and its inverse are measurable.

MEASURABLE: 1. A set is measurable if it belongs to the sigma-algebra in a given measurable space. 2. Of a set A with respect to a given outer measure μ^* , having the property that for every subset E of the space $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E)$. 3. Of a function or transformation between measure algebras, such that the inverse image of a measurable set in the range space is measurable in the domain space. Thus a real-valued function is Borel measurable if the inverse image of every open (or Borel) set is Borel measurable and a real function is Lebesgue measurable if the inverse image open (or Borel) set is Lebesgue measurable.

MEASURABLE COVER: In mathematics, for a set, it is a collection of measurable sets whose union contains the given set.

MEASURABLE KERNEL: In mathematics, a set K contained in a given set E , such that every measurable subset of $E \setminus K$ has measure zero.

MEASURABLE QUANTITY: An attribute of a substance which may be distinguished qualitatively and determined quantitatively. In the context of analytical chemistry, the attribute may refer to a physical quantity such as X- or γ -ray energy or it may refer to a measure of amount such as mass or concentration.

MEASURABLE SPACE: In mathematics, a set together with a sigma-algebra on the set.

MEASURAND: Particular quantity subject to measurement.

MEASURE: 1. Dimensions, quantity or capacity as ascertained by comparison with a standard. 2. A reference standard or sample used for the quantitative comparison of properties.

MEASURE OF CENTRAL TENDENCY: A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data, for example, mean (average), median and the mode. The mean is the most commonly used measure of central tendency.

MEASURE OF DISPERSION: A measure of dispersion allows the description of a set of data concerning a particular variable, giving an indication of the variability of the values inside the data set. The measure of dispersion completes the description given by the measure of central tendency of a distribution. If different distributions are observed, for some of them, all the data are grouped in a more or less short distance from the central value; for others the distance is much greater. Measure of dispersion can be classified into two groups: (i) Measures defined by the distance between two representative values of the distribution: Range, also called the interval of variation and Interquartile Range. (ii) Measures calculated depending on deviation of each datum from a central value: Geometric deviation, Median deviation, Mean absolute deviation and Standard deviation. Among measures of dispersion, the most important and most used is the standard deviation.

MEASURE PRESERVING TRANSFORMATION: A one-to-one transformation between measure spaces that is measurability preserving and preserves the measure of sets. More generally, a measurable transformation such that the measure of every measurable set and of its inverse image agree.

MEASURE RING: A Boolean sigma ring endowed with a measure.

MEASURE SPACE: A measurable space endowed with a non-negative measure. The collection of all subsets and the counting measure of any non-empty set define a measure space. The measure is often desired to be positive except on the zero element, which may be achieved by identifying sets with null symmetric difference; the sets of finite measure in the resulting ring form a metric space in which the metric is given by the measure of the symmetric difference of two sets.

MEASURE THEORY: The field of mathematics that investigates the conditions under which integration can be carried out. It focuses mainly on the various ways in which the size or measure, of a set can be estimated.

MEASURED EXCITATION SPECTRUM: The spectrum observed by measuring the variation of the luminescence flux from an analyte as a function of the excitation wavelength.

MEASURED VALUE: The observed value of weight, volume, meter-reading or other quantity, found in the analysis of a material.

MEASUREMENT: A description of a property of a system by means of a set of specified rules that maps the property onto a scale of specified values, by direct or mathematical comparison with specified reference(s). The demand for rules makes measurement a scientific concept in contrast to the mere colloquial sense of description. However, in the present definition, measurement has a wider meaning than given in elementary physics. Even a very incomplete description of, for instance, a patient (at a stated time) has to be given by a set of measurements, that are easier to manage and grasp.

MEASUREMENT ERROR: The difference between the true value of a quantity and the value obtained by measurement.

MEASUREMENT RESOLUTION: The minimum value above which the difference of two values of air quality characteristic can be distinguished with 95% probability.

MEASUREMENT RESULT: The outcome of an analytical measurement (application of the chemical measurement process) or value attributed to a measurand. This may be the result of direct observation, but more commonly it is given as a statistical estimate derived from a set of observations. The distribution of such estimates (estimator distribution) characterises the chemical measurement process, in contrast to a particular estimate, which constitutes an experimental result.

MEASUREMENT SOLUTION: The solution that is presented to the measuring device. It comprises the test solution or an aliquot thereof, after it has undergone any treatment required prior to the presentation to the measuring device.

MEASUREMENT THRESHOLD: The minimum concentration of a substance which produces a quantifiable signal with a given uncertainty.

MEASURING CYLINDER (GRADUATED CYLINDER): A graduated cylindrical glass or transparent plastic container used to measure the volume of a sample of liquid.

MEATUS: A tube shaped cavity within a bone or a small canal or passage in the body. Examples include the external auditory meatus of the outer ear in mammals, which connects the exterior opening to the eardrum; the urinary meatus, which is the opening of the urethra, situated on the glans penis in males and in the vulva in females; and the internal auditory meatus, a canal in the temporal bone of the skull.

MEBIBYTE: A unit of data storage written in binary number as 2^{20} that is equal to 1,048,576 bytes (1024×1024). Its symbol is MiB. In 1998 the International Electrotechnical Commission (IEC), approved new binary prefixes to accurately reflect the difference between decimal and binary numbers. For example, the prefixes kilo, mega and giga were changed to kibi, mebi and gibi, respectively. The unit, megabyte with the symbol MB, which is 1000000 (10^6) bytes, is still commonly used.

MECHANICAL: 1. Something moved, worked or produced by machinery. 2. Involving or located in or on a machine or engine, for example, mechanical failure. 3. Relating to, involving or typical of the science of mechanics, for example, mechanical energy. 4. Relating to, involving or done by physical forces such as wind and rain, for example, mechanical erosion

MECHANICAL ADVANTAGE (MA): For any machine such as a lever, the ratio between the force moving the load compared to the effort applied. It is defined mathematically as $MA = \text{Load/Effort}$ or $\text{Output Force/Input Force}$.

MECHANICAL BONDING: Bonding that involves a mechanical constraint preventing two parts of a molecule separating, rather than a chemical linkage based on transfer or sharing of electrons. It is found in rotaxanes, catenanes and molecular knots.

MECHANICAL ENERGY: Energy produced or used by a machine, where a force causes part of the machine to move or the sum of a system's kinetic energy (KE) and gravitational potential energy (GPE). Mechanical energy (ME) can be either kinetic energy (energy of motion) or potential energy (stored energy of position). It is constant in a system that experiences no dissipative forces such as friction or air resistance. For example, a swinging pendulum that experiences only gravitation has greatest KE and least PE at the lowest point on the path of its swing, where its speed is greatest and its height least. It has least KE and greatest PE at the extremities of its swing, where its speed is zero and its height is greatest. Mechanical energy (ME) is expressed as: $ME = KE + PE = \frac{1}{2}mv^2 + mgh$, where ME is the

Mechanical Energy in Joules, m is the mass of an object in kilograms, v is the velocity of the object in metres per second (m/s), g is the gravitational acceleration of Earth (9.8m/s^2) and h is the height from the ground in metres.

MECHANICAL ENTRAPMENT: 1. The process of random incorporation of comparatively small quantities of other phases, for example, water, dust and particles in the bulk of a precipitate during its formation. 2. The deliberate capture of small quantities of such phases by the deliberate addition of solids to a liquid phase.

MECHANICAL EQUIVALENT OF HEAT: The ratio of a unit of mechanical energy to the equivalent unit of thermal energy, when a system of units is used. Its symbol is J.

MECHANICAL MOUSE: A mouse that uses a metal or rubber ball underneath it to send signals to mechanical sensors inside the unit when the ball is rolled on a pad or desktop in any direction along two planes. The mechanical sensors convert the motion of the ball in any direction into cursor or pointer movements on the screen.

MECHANICAL PUZZLE: A puzzle, involving several objects or a single object composed of one or more movable parts, whose solution requires moving from an initial state to a predefined final state.

MECHANICAL TISSUE (STRENGTHENING TISSUE, SUPPORTING TISSUE): In botany, it refers to any tissue consisting of cells with thickened cell walls, such as collenchyma and sclerenchyma.

MECHANICAL WAVE: Wave that is not capable of transmitting its energy through a vacuum and requires a medium in order to be transmitted. It is a disturbance in matter that transfers energy through the matter. It is therefore, transmitted only when matter is disturbed. For example, to create a mechanical wave in a pool of calm water, you need to disturb the water, for example, by dropping an object like stone in it.

MECHANICAL WEATHERING (PHYSICAL WEATHERING): The form of weathering responsible for the mechanical breakdown of rocks, which involves no chemical change. Processes include exfoliation, frost shattering, granular disintegration, salt weathering, spalling and thermal weathering.

MECHANICALLY RECOVERED MEAT (MECHANICALLY SEPARATED MEAT, MSM): The scraps of meat which remain on an animal's carcass produced by forcing meat with attached edible meat, under high pressure through a sieve or similar device to separate the bone from the edible meat tissue. It is ground and used as cheap fillers for pies, sausages, etc.

MECHANICS: A branch of physics concerned with forces and their effects on whole objects or a branch of physics dealing with the motions of bodies and the forces causing these motions and also with the forces acting on bodies in equilibrium. In other words, it is the application of mathematical methods to the study of the equilibrium and motion of bodies in a particular frame of reference, including statics, dynamics and kinematics.

MECHANISATION (MECHANIZATION): To change a process so that it is performed by machinery instead of animal or human labour.

MECHANISM: 1. The means of transmitting and modifying motion in a machine or an assembly of mechanical parts composed of parts, which move only in certain ways in relation to each other. Motion is transmitted by chain drive or belt drive, by direct contact as in a cam or gear or by a pin-connected linkage. 2. Also known as **reaction mechanism**, it is the way in which a particular chemical reaction occurs, described in terms of the steps involved. For example, the hydrolysis of an alkyl chloride proceeds by the S_N1 mechanism. **S_N1 reaction** is a substitution reaction in which " S_N " stands for nucleophilic substitution and "1" the unimolecular rate-determining step or the reaction of an electron pair donor, which is the nucleophile (Nu) with an electron pair acceptor (the electrophile). **S_N2 reaction** means that two molecules are involved in the actual transition state.

MECHANISM-BASED INHIBITION (SUICIDE INHIBITION): Irreversible inhibition of an enzyme due to its catalysis of the reaction of an artificial substrate.

MECHANO-CHEMICAL REACTION: Chemical reaction that is induced by the direct absorption of mechanical energy. Shearing, stretching and grinding are typical methods for the mechano-chemical generation of reactive sites, usually macroradicals, in polymer chains that undergo mechano-chemical reactions.

MECHANORECEPTOR: A receptor that responds to such mechanical stimuli as touch, sound and pressure. The human hairless skin or glabrous skin is rich in four main types of mechanoreceptors, which are Pacinian corpuscles, Meissner's corpuscles, Merkel's discs and Ruffini corpuscles. There are also mechanoreceptors in the hairy skin and the hair cells in the cochlea are the most sensitive mechanoreceptors, transducing air pressure waves into sound.

MECKE'S TEST: A presumptive test for amphetamines, methamphetamines and heroin. Mecke's reagent consists of 1 gram of selenious acid in 100ml of concentrated sulfuric acid. Different substances give different results, ecstasy, for example, gives a light blue colour, turning to turquoise and then dark blue. Heroin gives a yellow colour changing to green. LSD gives an olive-green colour, changing to black. Mescaline gives a brownish-orange colour.

MECONIUM: The semisolid dark green first faecal material passed by a newborn mammal.

MECOPROP (METHYLCHLOROPHENOXYPROPIONIC ACID, MCP): A commonly used herbicide mostly for controlling weeds in cereal and grass.

MEDIA: 1. Material that provides plants with nourishment and support through their root system. 2. The middle, muscular layer of the wall of a blood or lymph vessel. 3. A primary vein in an insect's wing. 4. In communications, the term refers to the various means of communication, for example, television, radio and the newspaper are different types of media. Computer media can be hard drives, removable drives (such as Zip disks), CD-ROM or CD-R discs, DVDs, flash memory, USB drives and floppy disks.

MEDIA ACCESS PORT: A hardware-addressable component (such as a communication adapter) of a node that gives the node access to a transmission medium and enables data to pass into and out of the node.

MEDIAL: In anatomy, it means toward the midline or nearer the midline plane. The opposite of medial is lateral.

MEDIAL MORaine: A type of moraine formed from two lateral moraines when two glaciers merge.

MEDIAL TRIANGLE: The triangle whose vertices are the midpoints of the sides of a given triangle.

MEDIAN: 1. The middle number of an ordered group of numbers or variables. It is the item or score which divides a distribution into two equal halves. If there is no middle number (because there is an even number of terms) the median is the mean (average) of the two middle numbers. For example, the median of a set of numbers arranged in order of magnitude 2, 3, 7, 11, 12, is 7; and that of 3,

4, 7, 9, 11, 13 is 8 (the average of 7 and 9). For a grouped data the

median is given by, $\text{Median} = L_1 + \left(\frac{\frac{1}{2} \sum f - \sum fa}{f_m} \right) c$, where L is the

lower boundary of median class, $\sum f$ is the total frequency, $\sum fa$ is the sum of frequencies below median class, f_m is the frequency of median class and c the width of median class. It can be estimated using the cumulative frequency curve. 2. A straight line in a triangle that joins the vertex to the mid-point of the sides. The triangle, therefore has three medians.

MEDIAN EFFECTIVE DOSE (ED_{50}): Statistically derived dose of a chemical or physical agent (radiation) expected to produce a certain effect in 50% of test organisms in a given population or to produce a half-maximal effect in a biological system under a defined set of conditions.

MEDIAN EYE (PINEAL EYE; PARIETAL ORGAN; THIRD-EYE): An eye-like structure, with a lens and retina found on the top of the head of some lizards, pelagic sharks and in many fossil vertebrate, where it is visible as a light-sensitive spot on top of their heads. It corresponds to the pineal gland of other vertebrates and is thought to act as a photoreceptor, detecting changes in light intensity and modifying the physiological and behaviour of the animal.

MEDIAN LEAF: A leaf on top of stem, as occurs in heterophyllous species of Selaginella.

MEDIAN LETHAL CONCENTRATION (LC_{50}): Statistically derived concentration of a substance in an environmental medium expected to kill 50% of organisms in a given population under a defined set of conditions.

MEDIAN LETHAL DOSE (LD_{50}): Statistically derived dose of a chemical or physical agent (radiation) expected to kill 50% of organisms in a given population under a defined set of conditions.

MEDIAN LETHAL TIME (TL_{50}): Statistically derived average time interval during which 50% of a given population may be expected to die following acute administration of a chemical or physical agent (radiation) at a given concentration under a defined set of conditions.

MEDIAN NARCOTIC CONCENTRATION (NC_{50}): Statistically derived concentration of a substance in an environmental medium expected to cause narcotic conditions in 50% of a given population under a defined set of conditions.

MEDIAN NARCOTIC DOSE (ND_{50}): Statistically derived dose of a substance expected to cause narcosis in 50% of test animals under a defined set of conditions.

MEDIAN NERVE: One of the main nerves of the arm, which runs down the arm's full length into the hand. It is a branch of the brachial plexus. The median nerve controls the muscles of the forearm and hand, which carry out bending movements of the wrist, finger and thumb. It also conveys sensations from the thumb, index finger, middle finger, part of the ring finger and the region of the palm at the base of these digits.

MEDIAN SAGITTAL PLANE: A vertical plane passing through the centre of the body (including the navel and spine), dividing it into equal right and left halves. Planes situated to one or the other side of the median sagittal plane and parallel to it are known as **sagittal planes** or **paramedians**. A structure situated nearer to the median plane of the body than another is said to be **medial**. Similarly, a structure that lies farther away from the median plane than another is said to be **lateral** to the other.

MEDIAN TRIANGLE (MEDIAL TRIANGLE): A triangle formed by joining the midpoints of the sides of the original triangle.

MEDIASTINUM: A membrane in the middle of the thorax of mammals surrounded by loose connective tissue that separates the lungs. It contains the heart, the great vessels of the heart, oesophagus, trachea, phrenic nerve, cardiac nerve, thoracic duct, thymus and lymph nodes of the central chest.

MEDIATED GENERALISATION: The act or process by which an organism responds to a stimulus not physically similar to the conditioned stimulus and not previously encountered in conditioning.

MEDIATOR: 1. In mathematics, the plane through the midpoint of a line segment and perpendicular to that segment or any axis of symmetry. 2. In biology, a substance or structure that mediates a specific response in a bodily tissue. 3. In physics, a subatomic particle that effects or conveys a force between subatomic particles.

MEDICAL: Referring to the art of medicine, of curing disease or thorough physical examination or relating to, involving or used in medicine or treatment given by doctors.

MEDICAL JURISPRUDENCE (FORENSIC MEDICINE; FORENSIC PATHOLOGY): The branch of medicine involving the application of medical knowledge to establish facts such as the cause and time of death in civil or criminal legal cases. The primary tool used is autopsy.

MEDICAMENT: A substance used in or on the body to cure illness.

MEDICINE: 1. The science of preventing, diagnosing, alleviating or curing diseases, both physical and mental. 2. A drug or remedy used for treating illness.

MEDITERRANEAN CLIMATE: A type of climate typical of the regions along the Mediterranean Sea but also found in some parts of the world such as parts of California, Chile, South Africa and outh Australia. It is characterised by warm, wet winters and calm, hot, dry summers. Plants that thrive in this climate must be able to survive long dry summers. Its vegetation include evergreen trees such as pines, cypresses and oaks; deciduous trees such as sycamores and oaks; fruit trees such as olives, figs, citrus, walnuts and grapes; shrubs; grasses; and herbs.

MEDIUM: 1. Any substance or space through which a form of radiation travels. 2. A substance such as air or water or in an experimental situation such as staining medium, culture medium, etc, in which specimens of animals and plants are preserved or mounted. It also refers to a substance or environment in which an organism naturally lives or grows. 3. A solvent mixed with a pigment or paint to make it thinner. 4. A material on which data is stored or printed, for example, paper, tape or disk.

MEDIUM FREQUENCY: The band of frequencies of electromagnetic radiation in the range of 300 kHz-3MHz and having a wavelength in the range of 100-1000 metres. It is used mainly for nondirectional beacons (NDB).

MEDIUM-TEXTURED SOIL: In soils, it is the intermediate between fine-textured and coarse-textured soils, which includes very fine sandy loam, loam, silt loam, and silt.

MEDIUM-SCALE INTEGRATION, MSI: The technology that enables hundreds of electronic components to be etched into a single integrated circuit that is just a few square millimetres.

MEDLINE (MEDICAL LITERATURE ANALYSIS AND RETRIEVAL SYSTEM ONLINE): A bibliographic database of life sciences and biomedical information administered by the US National Library of Medicine (NLM) that contains over 15 million references to journal articles in life sciences and medicine.

MEDULLA: The innermost part of an organ or body structure. The adrenal medulla is the central region of an adrenal gland and the medulla of bone is the bone marrow.

MEDULLA OBLONGATA (MYELENCEPHALON): The lowest part of the vertebrate brain stem. It is continuous with the spinal cord, forms the major pathway for nerve impulses entering and leaving the skull and contains the cardiac, respiratory, vomiting and vasomotor centres and also controls involuntary vital functions such as heart beat and breathing.

MEDULLARY RAY (RAY): Any of the thin vertical plate of parenchyma cells or connective tissue that radiates between the pith and bark in the stems of some higher woody plants. Primary medullary rays occur in young plants and in those not showing secondary

thickening and run from pith to the cortex between primary vascular bundles. Secondary medullary rays are produced by the cambium and terminate in xylem and phloem tissues. Medullary rays store and transport food materials.

MEDULLARY SHEATH (MYELIN SHEATH): The layer of fatty acid material that surrounds and electrically insulates the axons of nerve cells of most vertebrates and some invertebrate neurones, formed from the cell membrane of the Schwann cell in the peripheral nervous system and from oligodendroglia cells. It allows for a more rapid transmission of nerve impulses (at speeds of up to 120 ms^{-1}). If damaged, the nerve impulse is impaired and can cause multiple sclerosis.

MEDULLATED NERVE FIBRE: A nerve fibre that is characterised by a myelin sheath, which insulates the axon.

MEDULLATED PROTOSTELE: A protostele in which the central core of xylem consists mainly of tracheids interspersed with numerous parenchyma cells. The medullated protostele shows how the siphonostele may have evolved from a protostele, with the pith originating in the xylem itself. Medullated protosteles are exhibited in some primitive ferns such as *Gleichenia*.

MEDUSA: The free swimming stage in the life cycle of the coelenterate or Cnidaria, the form usually taken during the sexual stage of these animals. This form alternates with a bottom-dwelling polyp stage in which reproduction is asexual. Medusae are umbrella-shaped, with tentacles round the edge with mouth in the centre underneath. They swim by pulsation of the body and they reproduce sexually. In the Scyphozoa, which includes all the common jelly fish, the medusa is the dominant form and the polyp is reduced or absent.

MEERSCHAUM: Very fine-grained claylike mineral composed of hydrated magnesium silicate, used for making smoking pipes and ornamental sculptures.

MEET: 1. In mathematics, the binary lattice operator whose value is the infimum of a pair of a lattice. Given a pair of elements x and y of the lattice, their meet, expressed as $x \wedge y$, is the element j such that $j \leq x$, $j \leq y$ and there is no $lc > j$ that has the same relations to x and y . 2. To intersect, as in two sets or line.

MEGA: 1. A prefix used in the metric system to denote one million times. For example, 10^6 volts = 1 megavolt (MV). 2. A prefix denoting a large size. Examples are meganucleus, megadose, mega sporangium, etc.

MEGABASE: A unit of length used to measure the size of polynucleotides (DNA or RNA) or segments of such molecules, which is equal to one million nucleotides or approximately 1 cm. Its symbol is Mb. $1\text{Mb} = 10^6$ bases or base pairs.

MEGABYTE: The measurement of a computer memory size. One megabyte is equal to 1 048 576 bytes. A byte is computer memory space needed to store one character, which is normally 8 bits.

MEGACATASTROPHE: An event of cataclysmic proportions that causes death and destruction, regionally or globally and may be large enough to threaten the existence of human civilisation, all or much of terrestrial life, the Earth itself or, in some extreme scenarios, the entire universe. Natural megacatastrophes have happened in the past and will inevitably occur again in the future. Added to these is now the potential for artificial or human-caused megacatastrophes.

MEGAHERTZ (MHZ): One million hertz (cycles per second).

MEGAKARYOCYTES: A bone marrow cell, 10 to 15 times larger than a typical red blood cell, averaging $50\text{--}100 \mu\text{m}$ in diameter. It is responsible for the production of blood clotting particles called thrombocytes or platelets, normally accounting for 1 out of 10,000 bone marrow cells but can increase 10 times in certain diseases.

MEGAPASCAL (MPa): A unit of pressure equivalent to 10 atmospheres of pressure.

MEGAPHONE: A device used to amplify the sound of a voice. It amplifies the sound because of its shape (a hollow cone of metal or plastic) and increases the effective sound power by concentrating

the sound in a single direction. Electric megaphones combine a microphone, amplifier and loudspeaker.

MEGAPHYLL (MACROPHYLL): A type of foliage leaf in ferns and the leaves of gymnosperms and angiosperms that have branched or parallel vascular bundles running through the lamina and connected by a network of smaller veins. The megaphylls of ferns are large pinnate leaves called **fronds**.

MEGAPIXEL: A measure of resolution of digital cameras, which is equal to one million pixels. The higher the megapixel number, the more detail the camera can capture. A camera's megapixel number is calculated by multiplying the number of vertical pixels by the number of horizontal pixels captured by the camera's sensor or CCD.

MEGASPORANGIOPHORE: A stalklike structure that bears megasporangia. It is also the ovule-bearing structures (megasporephylls) making up the female cones of cycads.

MEGASPORANGIUM (MACROSPORANGIUM): A structure in which megaspores are formed. In the seed plants it corresponds to the ovule.

MEGASPORE (MACROSPORE): The larger of the two types of haploid spores formed after meiosis in heterosporous species and usually designated female. It is immobile and contains food reserves for the gametophyte. In angiosperms, the cell that gives rise to the nuclei of the embryo sac is termed a megaspore, though it is usually smaller than a pollen grain (microspore).

MEGASPORE MOTHER CELL (MEGASPOROCYTE): A diploid cell in plants in the genera Selaginella, Marsileaceae, Azolla, Salvinia and gymnosperms that divides by meiosis to give rise to haploid megaspore. In flowering plants the megaspore mother cell is situated in the ovule.

MEGASPOROPHYLL (MACROSPOROPHYLL): A leaf-like structure that bears the megasporangia. In angiosperms and gymnosperms it is represented by the carpel and ovuliferous scale respectively.

MEGATON: A unit of energy used to measure the explosive power of nuclear explosion or warhead, equivalent to a million tonnes of TNT or 4.184 petajoules (10^{15} joules). A nuclear weapon with an explosive power equivalent to one million tons of TNT is called a **megaton weapon**.

MEIOCYTE: A cell that divides by meiosis to produce haploid spores. The microspore mother cells (pollen mother cells) in anthers and the megaspore mother cells in nucellar tissue are the meiocytes of flowering plants.

MEIOFAUNA (MEIOBENTHIC FAUNA): Animals such as small polychaetes, bivalves and nematodes just visible to the naked eye, which inhabit the bottom of a river, lake or sea. They have dimensions in the range of 0.1 to 1mm, intermediate between microfauna and macrofauna.

MEIOSIS: A part of sexual reproduction that occurs only in reproductive cells in which cells divide to form sperm cells in a male and egg cells in a female. It is a type of cell division that reduces the number of chromosomes in the parent cell by half and produces four gamete cells.

MEIOTIC DRIVE: A process occurring during meiosis that results in the two kinds of gametes produced by a heterozygote not being equally common. It is more often seen in the formation of megaspores than microspores. An example is seen in maize plants that are heterozygous for a chromosome mutation, in which one of the homologues of chromosome 10 has an abnormal terminal knob. The knob acts as a centromere and causes the abnormal chromosomes to move to the poles of the spindle earlier in both the first and second divisions of meiosis.

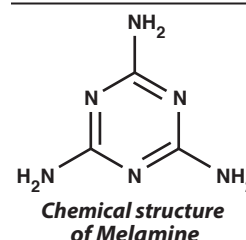
MEISSNER EFFECT: The falling off of the magnetic flux from the interior of a superconducting metal, when it is cooled in a magnetic field to a temperature below the critical temperature.

MEISSNER'S CORPUSCLES (TACTILE CORPUSCLE): Tiny egg-shaped touch receptors that respond to light pressure. They are found in the papillary region of the dermis of the skin, in the palms of hands, fingertips and soles of the feet as well as in the nipples, tip of the penis and other hairless sensitive areas.

MEITNER, LISE (1878-1968): Austrian-born Swedish physicist, who first identified nuclear fission. She went to Berlin to study radioactivity with German physical chemist, Otto Hahn. In 1917 they discovered protactinium. After World War I, Meitner and Hahn returned to Berlin, where in 1935 they bombarded uranium with neutrons. In 1938 she left Germany, with other Jewish scientists and went to the Noble Institute in Stockholm. In 1939 she and her nephew, British physicist Otto Frisch (1904-79) explained Hahn's results in terms of nuclear fission. She shared the 1966 Enrico Fermi Award with Hahn and Strassmann. Element 109, meitnerium, is named in her honour.

MEITNERIUM (UNNILENNIUM): A radioactive transactinide element with the atomic number 109, symbol Mt. It was first made in 1982 by Peter Armbruster and a team in Darmstadt, Germany, by bombarding bismuth-209 nuclei with iron-58 nuclei. Only a few atoms have ever been detected.

MELAMINE (1,3,5-TRIAZINE-2,4,6-TRIAMINE; 2,4,6-TRIAMINO-S-TRIAZINE; CYANUOTRIAMIDE; CYANUOTRIAMINE; CYANURAMIDE): A colourless, crystalline compound produced from urea and containing 66% nitrogen by mass with the chemical formula $C_3H_6N_6$, density 1574 kg/m³, melting point 345 °C, 618 K (decomposes) and sublimates at boiling point. Melamine is a cyclic



compound having a six-membered ring of alternating C and N atoms, with three NH₂ groups. It is used in the manufacture of synthetic resins and in leather tanning. It can be copolymerised with methanol to give thermosetting melamine resins, which are used particularly for laminated coatings. 2. A plastic made from copolymerising a white crystalline solid with formaldehyde.

MELANIN: Any of a group of water-insoluble polymers, derived from the amino acid tyrosine, that causes pigmentation of the eyes, skin, hair in vertebrates and hyphae and spores of fungi. Melanins are produced by specialised cells called melanophores or melanocytes. Their dispersion is controlled by melanocyte-stimulating hormone and melatonin. Certain invertebrates, fungi and microorganisms also produce melanin pigments. The ink of octopus is a notable example. Hereditary albinism is caused by the absence of the enzyme tyrosinase, which is necessary for melanin production. In humans and fungi it helps protect the skin against ultraviolet radiation from sunlight.

MELANISM (MELANOSIS): Dark colouration of the skin, feathers, eyes or hair caused by over production of the pigment melanin, often caused by genetic mutation or as a reaction to the environment such as exposure to abnormal temperature changes during gestation which transiently alter gene transcription or translation. Industrially polluted areas have several species of melanic moths.

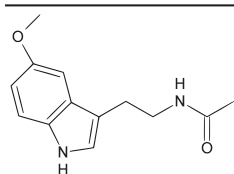
MELANOCYTE-STIMULATING HORMONE (MSH): Any of several related peptide hormones that affect the skin and colouration and also act as anti-inflammatory cytokines and as neuropeptides in the brain. They were first isolated by the Yale University professor, Aaron B. Lerner. MSHs are produced from the precursor pro-opiomelanocortin (POMC) and secreted by the anterior or the intermediate lobe of the pituitary gland. MSH stimulates dispersal of melanin in the chromatophores of amphibian skin, causing the skin to darken. In humans it stimulates production and dispersal of melanin in the pigment cells of the skin. MSH signals to the brain have effects on appetite and sexual arousal.

MELANOMA: A form of skin cancer that begins in melanocytes, the cells that make the brown pigment melanin. Melanoma usually starts in a mole.

MELANOPHYLLOUS: Possessing dark leaves.

MELANOXYLON: A dark wood or black timber, for example, *Acacia melanoxylon*.

MELATONIN (N-ACETYL-5-METHOXYTRYPTAMINE): A naturally occurring compound found in animals, plants and microbes. It is a hormone derived from serotonin and secreted by the pineal gland and retina of vertebrates. Melatonin secretion by the pineal is linked to the dark-light cycle of the organism's environment, being greatest at night and lowest by day. Production is regulated by signals from the suprachiasmatic nucleus of the hypothalamus, which itself receives input from the retina. The hormone is involved in regulating certain diurnal and seasonal changes in the body such as the reproductive cycles in seasonally breeding animals. Melatonin also controls pigmentation changes, triggers aggregation of the pigment melanin into melanophores in the skin, causing the skin to turn pale. It is used as a drug to treat sleep disorders and symptoms of jet lag.



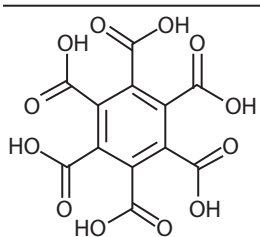
Chemical structure of Melatonin

MELITTOPHILY: Pollinated by bees.

MELLIN TRANSFORM (MELLIN INVASION FORMULA): Named after Robert Mellin (1854-1933), it is the integral transform defined

by $g(s) = \int_0^{\infty} x^{s-1} f(x) dx$, and that is invertible under reasonable conditions. The Mellin transform of the negative exponential is the gamma function.

MELLITIC ACID (BENZENEHEXACARBOXYLIC ACID; GRAPHITIC ACID; BENZENE-1,2,3,4,5,6-HEXACARBOXYLIC ACID): A colourless crystalline compound with the chemical formula $C_6(COOH)_6$ and melting point 287 °C (560 K). It was discovered in 1799 by M. H. Klaproth in the mineral mellite (honeystone), which is the aluminium salt of the acid. Its molecules consist of a benzene ring in which all six hydrogen atoms have been substituted by carboxyl (-COOH) groups. It is made by oxidising charcoal with concentrated nitric acid. It decomposes on heating to form pyroellitic anhydride. It is used in making epoxy resins and dyes.



Chemical structure of Mellitic Acid

MELT: A change of state from a solid to a liquid, associated with an intake of energy. A solid melts to form a liquid when the energy of the particles is sufficient to break up the bonds holding them in a lattice.

MELTING POINT: The specific temperature at which a substance changes its state from pure solid into a liquid. A pure substance under standard conditions of pressure has a single reproducible melting point. If heat is gradually and uniformly supplied to a solid the consequent rise in temperature stops at the melting point until the fusion process is complete. At the melting temperature, latent heat of fusion does not permit a temperature rise or fall to be visibly registered.

MELTWATER: Water produced by the melting of snow and ice, particularly in glaciated areas including glacial ice and ice shelves over oceans.

MEMBER (ELEMENT): In mathematics, any individual object belonging to a set or logical class. The member of a set is denoted by $\in$.

MEMBRANE: A thin sheet-like flexible tissue composed mostly of lipid and protein that forms the external boundary of cells and of major structures within cells of a plant or animal.

MEMBRANE BONE: Bone such as face bones, some cranial bones and part of the clavicle formed directly in membranous connective tissue rather than by replacing cartilage. Small areas of membrane become jelly-like and attract calcium salts. Bone-forming cells break down these areas, forming a bone lattice, which eventually fill in.

MEMBRANE EMF: The potential difference between two saturated potassium chloride (KCl) bridges inserted into two solutions separated by a membrane. The solutions need not be in equilibrium with one another and need not contain any colloidal material.

MEMBRANE POTENTIAL: The charge difference between the cytoplasm and extracellular fluid in all cells, due to the differential distribution of ions. Membrane potential affects the activity of excitable cells and the transmembrane movement of all charged substances.

MEMBRANE TRANSPORT: The facilitated transport of solutes such as ions and small molecules through biological membranes. This is regulated by selective membrane permeability, which allows biological membranes to separate substances of distinct chemical nature.

MEMBRANE-ATTACK COMPLEX (MAC): A large cylindrical multiprotein complex formed by the complement system which kills invading microorganisms by embedding in their plasma membrane, creating a pore causing the cell to burst. It is one of the mechanisms of innate immunity that follows activation of the complement cascade. Certain complement components (C5b, C6 and C7) bind to the lipid bilayer of the target cell, insert into it and then induce polymerisation of the C9 component to form a cylindrical channel about 10 nm in diameter. This disrupts membrane integrity and electrochemical properties and kills the cell. The MAC is significant for defence against only certain bacterial infections such as *Neisseria* species responsible for gonorrhoea, but is important in certain immune system diseases.

MEMBRANOUS LABYRINTH: The soft tubular sensory structures for the senses of equilibrium and hearing filled with fluid (the endolymph). It is found in the inner ear of vertebrates, housed within the bony labyrinth.

MEME: 1. A term coined by Richard Dawkins in his book "The Selfish Gene" in 1976. It is a self-replicating unit of cultural inheritance such as habits, skills, songs, stories or any kind of behaviour analogous to a gene and passed from person to person by imitation, which can change overtime. Unlike genes, memes get stored in our memories and passed on to someone else. Such a concept can provide useful insight into the nature of cultural development. 2. **Internet meme** is a concept that is propagated through the Internet, which may take the form of a hyperlink, video, picture, website, hashtag or just a word or phrase. Examples of behavioural Internet memes include using Facebook and checking email several times a day.

MEMORY: 1. Part of a system, used to store data and programmes in a computer either on a temporary or permanent basis. Most digital computer systems have two types of memory, the main memory, which may be a high-speed RAM; and one or more auxiliary storage units, which include hard disks, floppy disks, magnetic tape drives, ROM and optical storage media such as videodiscs and compact discs. 2. The ability of the brain to store and retrieve information. Various models of memory exist, with different types of information and levels of processing. Memory is essential to the process of learning and recognition of individuals and objects.

MEMORY CARD: 1. A removable module that contains non-volatile memory chips. The term typically refers to flash memory cards used for storage in digital cameras and other portable devices. The primary memory cards are SD Cards, Memory Sticks and CompactFlash. 2. A module in a laptop computer that contains volatile RAM chips and is user replaceable.

MEMORY CARD READER: An input device that reads flash memory cards. It can be a standalone device that connects to a computer via USB or it may be integrated into a computer, printer or multifunction device. Most multifunction printer/scanner/copiers now have built-in card readers.

MEMORY CELL: A clone of long-lived lymphocytes, formed during the primary immune response that remains in a lymph node until activated by exposure to the same antigen that triggered its formation. Activated memory cells mount the secondary immune response.

MEMORY CHIP: A chip that holds programs and data either temporarily or permanently. RAM chips are the computer's temporary workspace, while flash memory chips are used like disk drives (permanent until erased).

MEMORY DUMP (BARF; CRASH DUMP; DISASTER DUMP): The data that is written to computer's memory and displayed when a computer crashes. It helps in the diagnosis and identification of the problem that led to application or system failure, in order to resolve it.

MEMORY EFFECT: In instruments used for atmospheric trace component analysis, the dependence of an instrument reading on one or several previous sample(s).

MEMORY LEAK: A leak in a program's dynamic store allocation logic that causes it to fail to reclaim memory in the heap after it has finished using it, eventually causing the program to fail due to lack of memory.

MEMORY LOCATION: A small area in a computer's memory or a cell, where bits of data, usually a byte (or 8-bits) are held or stored. It is identified by its starting address and size.

MEMORY MODULE: A narrow printed circuit board that holds memory chips. It is often used as a general term to describe SIMM, DIMM and SO-DIMM memory.

MEMORY SLOT (MEMORY SOCKET; RAM SLOT): A connector on the motherboard of a computer, typically of between three or four slots that allow extra computer memory (RAM) to be inserted into the computer.

MEMORY STICK: A type of flash memory developed by Sony. It is used to store data for digital cameras, camcorders and other kinds of electronics. High-speed versions of Memory Stick media support data transfer rates up to 80Mbps or 10 MB/sec, which is fast enough to record high-quality digital video.

MEMORY T CELL: A long-lived T cell that bears receptors for an antigen during its encounter with a prior infection or vaccination. Memory T cells are important to the immunity of an organism against pathogens. The receptors they possess make for a rapid immune action against these foreign bodies. Upon recognition of pathogen previously encountered these cells clone themselves immediately after re-exposure.

MEMORY-SCRAPING MALWARE: A type of malware that attacks the RAM inside point-of-sale (POS) devices at places such as retail points, restaurants and hotels. It is used by hackers to find personal data, which can be used, for example, to withdraw money from a victim's accounts.

MENARCHE: The first menstrual cycle in human females, experienced usually between 12 and 15 years. It is the monthly shedding of the uterine wall; considered as the onset of puberty and signals the possibility of fertility and pregnancy. Certain factors, for example, female biology, genetic and environmental factors, especially nutritional factors influence the onset of menarche.

MENDEL, GREGOR JOHANN (1822 – 1884): Austrian monk, botanist and founder of genetics. In 1865, Mendel presented a paper before the Brunn Society for the Study of Natural Science explaining how variations are transmitted in the form of unit determiners, a term he used for what is now called genes. Mendel was accepted in 1843 as a novice at the Augustinian monastery of St. Thomas, in Brno

ordained in 1847, followed by studies at the University of Vienna and returned to the monastery in 1854 for the rest of his life. It was at the monastery that he performed experiments on genetics of garden pea, *Pisum sativum*, that were to make him posthumously famous. The garden pea was a good plant for his research. He knew that the garden peas were available in a number of true-breeding races with distinctive characters and sought to determine how these characters were transmitted in crosses of plants with contrasting characters in stem length (tall and dwarf), flower position (axial and terminal), flower colour (red and white), form of the seed (smooth and wrinkled) and seed colour (green and yellow). Mendel did something that was uncommon for a biologist of his time. He counted the number of the different types of the products of his numerous crossings. In so doing he laid the groundwork for the continued collection of quantitative data that he could later mathematically analyse. Undoubtedly, his report in 1865 was read by many biologists, but it was also ignored by many until 1900 when the value of his work was appreciated and he was only given the credit he deserved long after his death. The scientists generally credited with independently discovering Mendel's paper were Dutch botanist, Hugo De Vries (1848-1935), the German Karl Correns (1864-1933) and the Austrian Erich Tschermak (1871- 1962).

MENDEL'S LAW: Law of heredity proposed by Gregor Mendel to explain the transmission of characteristics from one generation to the next. These are: (i) the Law of Segregation, which states that the members of a pair of homologous chromosomes segregate during meiosis and are distributed to different gametes. (ii) The Law of Independent Assortment states that pairs of factors segregate independently of each other when germ cells are formed.

MENDELEEV, DMITRI IVANOVICH (1834-1907): Russian chemist, noted for compiling the first valid periodic table of the elements in 1869. He studied in England and Germany for his post-graduate degrees and became professor of chemistry at St. Petersburg in 1866.

MENDELEEV'S LAW: Properties of elements listed in order of increasing relative atomic masses repeating at regular intervals of 2, 8, 18 and 32. For example, the 2nd element (helium) is similar in its chemical behaviour to the 10th (neon), as well as to the 18th (argon), etc.

MENDELEVium: A synthetic, radioactive element with the symbol Md, atomic number 101, mass number of most stable isotope 258 and melting point 827 °C (1100 K). It belongs to the transuranic elements of the actinoids in Group 3 of the Periodic Table. Only 15 isotopes of mendelevium are known with Md-258 being the most stable with a half-life of 55 days.

MENDELIAN INHERITANCE: The manner in which genes and traits are passed from parents to children. Examples of Mendelian inheritance include autosomal dominant, autosomal recessive and sex-linked genes. It was named after Gregor Mendel (1822 – 1884), Austrian monk, botanist and founder of genetics.

MENDELISM (MENDELIAN GENETICS; MENDELIAN INHERITANCE): The theory of hereditary formulated by Gregor Mendel that forms the basis of classical genetics. He suggested that individual characteristics were determined by inherited factors. According to him, individual characteristics are passed on from one generation to the other through genes.

MENDIUS REACTION: A reaction in which an organic nitrile is reduced by nascent hydrogen, for example, from sodium in ethanol to a primary amine: $RCN + 2H_2 \rightarrow RCH_2NH_2$.

MENINGES: The three membranes that surround the brain and spinal cord of vertebrates giving them protection. They are the dura mater, a tough, fibrous, double-layered structure that is adherent to the skull; the arachnoid membrane, a delicate serous layer loosely investing the brain; and the pia mater, a vascular layer which closely follows each

convolution of the brain. The pia and arachnoid are separated by the subarachnoid space, which contains cerebrospinal fluid.

MENINGITIS: Inflammation of the meninges caused by bacteria such as meningococcus, pneumococcus, hemophilus or viruses. Bacterial meningitis has a sudden onset with headache, vomiting, fever, neck stiffness and avoidance of light. Viral meningitis is a milder illness. Tuberculous meningitis is an insidious chronic type which responds slowly to anti tuberculous drugs.

MENISCOID: 1. One side concave, the other side convex. 2. A cartilage disk that acts as a cushion between ends of bones that meet in a joint.

MENISCUS: The curve at the surface of a liquid at the point of contact with its container or the rise or fall of a liquid surface edge, caused by adhesion. A meniscus results from the forces of attraction (adhesive forces) between the liquid and the walls of the container. A meniscus may be either convex or concave depending on the liquid. A **convex meniscus** occurs when the molecules have a stronger attraction to each other (cohesion) than to the container (adhesion). Example may be seen between mercury and glass in barometers. A **concave meniscus** occurs when the molecules of the liquid attract those of the container, for example, when water is put in a glass.

MENOPAUSE: The time at the end of the reproductive period of human females, characterised by irregular and the eventual cessation of menstruation. It happens between the ages of 45 and 55. The effects of the gonadotrophic hormones, follicle stimulating hormones and luteinising hormones in the ovaries decrease so the follicles do not develop properly. There is a change in the balance of the hormones oestrogen and progesterone, secreted by the ovaries, which may be associated with certain physical symptoms such as weight gain and hot flushes and mood changes.

MENSTRUAL CYCLE: A series of physiological changes that can occur in fertile females. It commences with the first day of menstruation when the endometrium is shed through a process of repair and hypertrophy till the next period. Necessary for reproduction, it is governed by the anterior pituitary gland and the ovarian hormones, oestrogens and progesterone. The lining of the uterus becomes progressively thicker with more blood vessels in preparation for the implantation of a fertilised egg cell. Ovulation occurs in the middle of the cycle. If fertilisation does not occur, the uterine lining breaks down and is discharged from the body. In women the fertile period is 11-15 days after the end of the last menstruation. Menstrual cycles are counted from the first day of menstrual bleeding. The average menstrual cycle is 28 days, which may vary from woman to woman and cycle to cycle.

MENSTRUATION: The periodic (occurring monthly) shedding of the uterine lining (endometrium) resulting in the discharge of blood and endometrium and unfertilised egg from the uterus. Starting from the age of puberty, it occurs on a regular basis in reproductive-age females of certain mammal species such as humans and lasts until menopause. It ceases during pregnancy, but resumes some time after delivery.

MENSURATION: Geometric measurement in science involving the calculation of geometric quantities such as length, area and volume from dimensions and angles already known.

MENTAL ARITHMETIC (MENTAL MATHEMATICS): A method to calculate an estimate quickly in the mind rather than on paper, using arithmetical facts such as addition, subtraction, multiplication, or division that a student has committed to memory.

MENTAL ILLNESS (PSYCHIATRIC DISEASE): Disorders characterised by abnormal function of the higher centres of the brain responsible for thought, perception, mood and behaviour. Although the exact cause of most mental illnesses is not known, it is becoming clear through research that many of these conditions are caused by a combination of biological, psychological and environmental factors.

For example, some mental illnesses have been linked to abnormal functioning of nerve cell circuits or pathways that connect particular brain regions.

MENTHOL (3-p-MENTHANOL; HEXAHYDROTHYMOL; MENTHOMENTHOL; PEPPERMINT CAMPHOR): A white crystalline terpene alcohol with the chemical formula $C_{10}H_{19}OH$. It has a minty taste and found in certain essential oils such as peppermint and used as a flavouring.

MENU BAR: A horizontal strip that contains lists of available menus for a certain program. In Windows programs, the menu bar resides at the top of each open window, while on the Mac, the menu bar is always fixed on the top of the screen. A window bar contains options such as File, Edit, View, Favorites, Tools and Help. Usually, each option in a menu bar is associated with a pull-down menu.

MEPHITIC: Having a strong, offensive odour.

MERACIDIUM: The small free-living larval stage of parasitic flatworms in the class Trematoda, which hatches from eggs excreted in the faeces of its vertebrate host. Its leaf-like body is covered with cilia enabling the larva to swim towards a secondary host, usually a mollusc in which it continues its development.

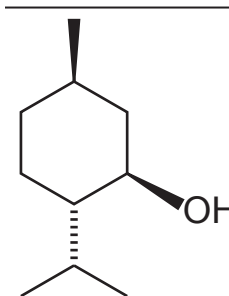
MERCALLI SCALE: A type of scale devised by Italian seismologist Giuseppe Mercalli (1850-1914) to measure the intensity of an earthquake in terms of its impact on a particular area and its inhabitants and buildings based on observed phenomena. The scale has intensity values from I (detectable only with instruments) to XII (causing almost total destruction).

MERCATOR'S PROJECTION: Cylindrical projection of the Earth's surface onto a flat sheet surface in which the meridians and latitudes are shown as straight lines that cross at right angles. Mercator projection distorts the size and shape of large objects, as the scale increases from the Equator to the poles, where it becomes infinite.

MERCURIC COMPOUNDS: Compounds of mercury in its +2 oxidation state, for example, mercuric chloride is mercury(II) chloride, $HgCl_2$.

MERCUROUS COMPOUNDS: Compounds of mercury in its +1 oxidation state. Example is mercury(I) chloride, also known as mercurous chloride (Hg_2Cl_2).

MERCURY: 1. Also known as **quicksilver**, it is a heavy silvery liquid metallic element belonging to the zinc group with the chemical symbol Hg, atomic number 80, atomic weight 200.59, melting point $-38.842^\circ C$ (312 K), boiling point $356.58^\circ C$ (629.73 K) and specific gravity 13.55 at $20^\circ C$ (293 K). It is the only common metal existing as a liquid at ordinary temperatures. When spilled, it breaks up into tiny beads which often become lodged in cracks. The main ore is the sulfide cinnabar (HgS), which can be decomposed to the elements. The element is less reactive than zinc and cadmium and will not displace hydrogen from acids. It is also unusual in forming mercury(I) compounds containing the Hg^+ ion, as well as mercury(II) compounds containing Hg^{2+} ions. It also forms a number of complexes and organomercury compounds. Mercury is used in thermometers, barometers, in electric switches, in mercury-vapour lamps, in certain electric batteries as an electrode in the production of chlorine and sodium hydroxide, etc. 2. The innermost planet of the Solar System and the second smallest after Pluto, named after a Roman god. It is the closest planet to the Sun and is slightly smaller in diameter than the moons Ganymede and Titan but more than twice as massive. Mercury's orbit is highly eccentric; at perihelion it is 46 million km from the Sun but at aphelion it is 70 million km. Mercury speeds around the Sun every 88 days, travelling through space at nearly 50 km per second, faster than any other planet. The length of one Mercury day



Chemical structure of Menthol

(sidereal rotation) is equal to 58.646 Earth days. Mercury has no moons and atmosphere and possesses an exosphere containing 42 percent oxygen, 29 percent sodium, 22 percent hydrogen, 6 percent helium, 0.5 percent potassium, with possible trace amounts of argon, carbon dioxide, water, nitrogen, xenon, krypton and neon. It has been visited by two spacecrafts, Mariner 10 and MESSENGER.

MERCURY CELL (MERCURY BATTERY; MERCURIC OXIDE BATTERY): A primary or non-rechargeable voltaic cell consisting of a zinc anode and a cathode of mercury(II) oxide (HgO) mixed with graphite. The electrolyte is potassium hydroxide (KOH) saturated with zinc oxide, the overall reaction is $\text{Zn} + \text{HgO} \rightarrow \text{ZnO} + \text{Hg}$. The e.m.f is 1.35 volts and the cell can deliver about 0.3 ampere-hour per cm^3 . It is made in button types for watches, cardiac pacemakers, hearing aids and calculators and in larger forms for other applications.

MERCURY FLOW SYSTEM: For mercury released directly as atomic vapour, different direct transfer systems are used. In an open dynamic system the liberated analyte is transported by a carrier gas through the sampling or excitation source and swept away. In an open static system the equilibrated gaseous phase is forced into the absorption cell by displacement, for example, by water. During measurement the gaseous phase is thus static. In a closed system the analyte and carrier gas are circulated through the absorption cell and the generator vessel until equilibrium between the liquid and gaseous phases is established.

MERCURY-VAPOUR LAMP: A type of gas discharge tube in which a glow discharge takes place when electricity is passed through mercury vapour. They are relatively efficient; offering a very long lifetime, as well as intense lighting for several special purpose applications.

MERCURY-XENON LAMP (MERCURY-XENON ARC): Non-coherent intense source of ultraviolet, visible and near-infrared radiation produced by an electrical discharge in a mixture of mercury vapour and xenon under high pressure.

MERCURY(I) CHLORIDE (MERCUROUS CHLORIDE; CALOMEL): A white, crystalline, poisonous and insoluble compound, very slightly soluble in water with the chemical formula Hg_2Cl_2 , density 7.150 g/cm^3 , melting point 525°C (798 K) and sublimes at its boiling point of 383°C (656 K). It forms by the reaction of elemental mercury and mercuric chloride: $\text{Hg} + \text{HgCl}_2 \rightarrow \text{Hg}_2\text{Cl}_2$. It can be prepared via metathesis reaction involving aqueous mercury(I) nitrate using various chloride sources including sodium chloride (NaCl) or hydrogen chloride (HCl). It is used as an insecticide, as a purgative and in mercury cell.

MERCURY(II) CHLORIDE: A white salt with the chemical formula HgCl_2 , density 5.43 g/cm^3 , melting point 276°C (549 K) and boiling point 304°C (577 K). It is made by reacting mercury with chlorine and used in making other mercury compounds and also used as an antiseptic and a disinfectant, in insecticides, preservatives and batteries and in metallurgy and photography.

MERCURY(II) FULMINATE: A grey crystalline solid, with the chemical formula $\text{Hg}(\text{CNO})_2$ made by dissolving mercury in nitric acid and adding ethanol to the solution. Highly sensitive to friction and shock, it can be handled safely only under cold water. It is used as a detonator for cartridges.

MERCURY(II) OXIDE (MERCURIC OXIDE): A red or yellow compound, with the chemical formula HgO . It is slightly soluble in water, which reduces to metallic mercury on heating. The red form is made by heating mercury in oxygen at 350°C (623 K) or by pyrolysis of $\text{Hg}(\text{NO}_3)_2$. The yellow form, which differs from the red in particle size, is precipitated when sodium hydroxide solution is added to a solution of mercury(II) nitrate. Both forms decompose to the elements at high temperature. The black precipitate formed when sodium hydroxide is added to mercury(I) nitrate solution is sometimes referred to as mercury(I) oxide (Hg_2O) but is probably a mixture of HgO and free mercury. It is sometimes used in the production of mercury as it decomposes quite easily. It is also used as a material for cathodes for mercury batteries

MERCURY(II)SULFIDE (MERCURY SULFIDE; MERCURIC SULFIDE; VERMILION): A red or black compound composed of mercury and sulfur with the chemical formula HgS . It occurs naturally as cinnabar and used as a pigment and source of mercury. It can be obtained as a black precipitate by bubbling hydrogen sulfide through a solution of mercury (II) nitrate. The red form is obtained by sublimation.

MEREOLGY: The abstract study of the logical properties of the relation of whole and part.

MERICARP: Any of the one-seeded portions that result when a compound fruit divides at maturity. Mericarps may be dehiscent or indehiscent.

MERIDIAN: 1. An imaginary great circle or line of longitude on the Earth's surface, perpendicular to the Equator that passes through the north and south magnetic poles. It marks a system of location and timekeeping whose zero, called Greenwich Meridian or Longitude 0° passes through Greenwich, England and other cities such as Tema in Ghana. 2. A great circle of the celestial sphere that passes through the zenith and the celestial poles. 3. In mathematics, also known as **meridian curve** or **meridian section**, it is a plane section of a surface of revolution, such as a paraboloid, that contains the axis of revolution.

MERISTEM: A localised tissue in all plants made up of actively dividing cells responsible for growth in plants or actively dividing cells that give rise to cells that differentiate into new tissues of the plant. The principal meristems are **apical meristems** found at shoot and root tips which are responsible for the primary growth of the plant; and the **lateral meristem** which gives rise to secondary growth.

MERISTEM CULTURE (SHOOT-TIP CULTURE): A tissue culture containing meristematic dome tissue without adjacent leaf primordia or stem tissue. The term may also imply the culture of meristemoid regions of plants or meristematic growth in culture. Meristem culture may be used to propagate infertile plants, nursery stock or F1 hybrids that do not breed true. As few virus diseases affect the plant apex, the technique can be used to produce virus-free stock.

MERISTEMATIC TISSUE: Plant tissue made up of actively dividing cells. The cells of meristematic tissues have thin and elastic primary cell wall made up of cellulose and are compactly arranged without inter-cellular spaces between them. Each cell contains a dense cytoplasm and a prominent nucleus. Normally the meristematic cells are oval, polygonal or rectangular in shape. It consists of cells which mainly divide in one plane and cause the organ to increase in diameter and growth; and intercalary meristem, which is located in between permanent tissues. It is usually present at the base of node, inter node and on leaf base. They are responsible for growth in length of the plant, adding growth in the girth of stem. Depending on the region of occurrence of meristematic tissues they are classified as **apical meristem**, which is present at the growing tips of stems and roots and increases the length of the stem and root; and **lateral meristem** occurring beneath the bark of the tree in the form of cork cambium and in vascular bundles of dicots in the form of vascular cambium.

MERISTODERM: The outer meristematic layer of the thallus of certain brown algae. The cells produced by the meristoderm increase the width of the thallus.

MERKEL'S DISC: One of the four main types of mechanoreceptors in the epidermis of hairless skin or glabrous skin that responds to sustained pressure. Merkel's discs are found in large numbers in the fingertips where they are important in discriminating textures and shapes. Each consists of a sensory Merkel cell and associated sensory nerve ending which conveys impulses to a sensory nerve.

MERMAPHRODITIC: Inflorescence with hermaphroditic flowers inside or above and pistillate outside or below.

MERO: A prefix meaning part, partial or fragment, used in formation of compound words, for example, merocyanines are compounds related to cyanines but have a nitrogen atom at only one end of the chromogenic system instead of at both ends.

MEROBLASTIC CLEAVAGE (INCOMPLETE CLEAVAGE; PARTIAL CLEAVAGE): A type of cleavage in which there is incomplete division of yolk-rich egg such as occurring in birds and reptiles. In meroblastic cleavage, only a portion of the egg divides. This is because of the large amount of yolk in the egg cell that the cleavage furrows do not penetrate the yolk.

MEROMORPHIC: Of a complex function in a domain, analytic except for poles.

MEROMORPHISM: 1. A complex function all of whose singularities are poles; it is analytic elsewhere in the given domain. 2. A monomorphism from a group into itself.

MEROZYGOTE: A bacterial cell that contains more than the haploid number of chromosomes but less than the full diploid number. Merozygotes arise when the genetic material from one bacterial cell is only partially transferred into another cell during conjugation, transduction or transformation.

MERRY-GO-ROUND REACTOR (TURNABLE REACTOR): An apparatus in which several samples are rotated around a radiation source in order to expose each to equal amounts of radiation.

MERSENNE NUMBER (MERSENNE PRIME): A number of the form $2^p - 1$ where p is a prime number or a positive integer that is one less than a power of two: $M^p = 2^p - 1$. The numbers 3, 7, 31 and 127 are Mersenne numbers.

MERTEN'S THEOREM: The theorem that the product of the values of two complex series, one of which is absolutely convergent, is the value of the series of which the coefficients are the Cauchy product

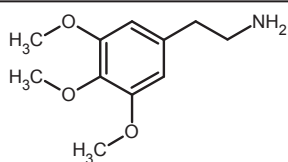
of the terms of the given series $\left(\sum_n a_n \right) \left(\sum_n b_n \right) = \sum_n \left\{ \sum_{j+k=n} a_j b_k \right\}$. If

both series are absolutely convergent, then so is the Cauchy product.

MESA: A flat-topped hill, less extensive than a plateau, having steep sides found in arid and semiarid parts of the southwestern U.S. and Mexico.

MESARCH: A term used to describe xylem maturation in which the older cells (protoxylem) are in the centre of the xylem strand because maturation has progressed both centrifugally and centripetally.

MESCALINE (3,4,5-TRIMETHOXYPHENETHYLAMINE): A powerful naturally occurring hallucinogenic compound with the chemical formula $C_{11}H_{17}NO_3$. It is obtained from peyote cactus (*Lophophora williamsii*), the San Pedro cactus (*Echinopsis pachanoi*) and the Peruvian Torch cactus (*Echinopsis peruviana*). Mescaline is a class A drug in the UK. It can be detected by the Mecke test.



Chemical structure of
Mescaline

MESENCHYME: A type of unspecialised tissue consisting of loosely packed, unspecialised cells set in a gelatinous ground substance. They develop into the tissues of the lymphatic and circulatory systems, as well as connective tissues throughout the body, such as bone and cartilage. A sarcoma is a malignant cancer of mesenchymal cells.

MESENTERIC ARTERY: The artery that transports blood to the intestine. It consists of two main arterial branches arising from the abdominal aorta. The inferior artery supplies the descending colon and the rectum. The superior supplies the small intestine, the caecum and the ascending and transverse colon.

MESENTERY: A thin sheet of tissue, bounded on each side by peritoneum that supports the gut and other organisms in the body cavities of animals. Dorsal mesentery is well developed in vertebrates and it anchors the stomach and intestines. It contains the blood vessels and nerves supplying the alimentary canals. The reproductive organs and their ducts are also supported by mesenteries.

MESH NETWORK TOPOLOGY: A type of physical topology that employs either of two schemes, called full mesh and partial mesh.

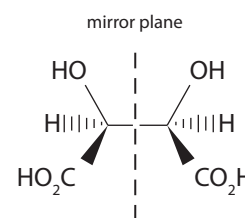
In the full mesh topology, each workstation is connected directly to each of the others. In the partial mesh topology, some workstations are connected to all the others and some are connected only to those other nodes with which they exchange the most data.

MESH-FINENESS/NORM: Of a partition of an interval, the supremum of the set of distances between successive members of the partition.

MESIC: Growing in conditions of medium water supply or a well-balanced supply of moisture.

MESO: 1. A prefix to indicate the presence of a symmetry element of the second kind in a chemical species which is usually one of a set of diastereoisomers that also includes a chiral member. 2. A prefix meaning middle or intermediate.

MESO COMPOUND: A compound with two chirality centres, where there is the same set of four groups at each chirality centre, with the combination of the four groups arranged such that the centres are mirror images of each other. Such a compound has plane of symmetry and is achiral (it is superimposable on its mirror image) and optically inactive. An example of the achiral structure of a meso compound is shown in the diagram.



Plane of Symmetry of
Meso Compound

MESOCARP: The middle layer of a fruit wall (pericarp) which may be succulent as in peach, hard as in almond or fibrous as in coconut.

MESOCIMATE: The climate over a specific locality such as a hillside or valley, extending not more than a few kilometres in radius which may not be representative of the general climate of the district.

MESOCOTYL: The elongated part of the axis between the cotyledon and coleoptile. This is seen in grass embryo.

MESODERM: The middle layer of tissue of the three cell layers in an embryo, from which connective tissue, muscle, blood, dermis and bone develop or the layer of cells in the gastrula that lies between the ectoderm and endoderm. It develops into the muscles, circulatory systems and sex organs and in vertebrates also into the excretory system and skeleton.

MESOFAUNA (MEIOFAUNA): Invertebrates residing in the soil or in a leaf litter layer on the soil surface which are 200 to 1000 microns (between 0.1 mm and 2 mm) in length. This group includes nematodes, oligochaete worms, smaller insect larvae, and certain arthropods that feed upon microorganisms, other soil animals, decaying plant or animal material, living plants, or fungi.

MESOGEN: A component of a molecule that induces a mesomorphic or liquid crystalline phase.

MESOGENIC GROUP: A part of a molecule or macromolecule endowed with sufficient anisotropy in both attractive and repulsive forces to contribute strongly to mesophase or, in particular, liquid-crystal mesophase formation in low-molar-mass and polymeric substances.

MESOGENIC MONOMER: A monomer which can impart the properties of liquid crystals to the polymers formed by its polymerisation.

MESOGENIC PITCH: A pitch with a complex mixture of numerous essentially aromatic hydrocarbons. It does not contain anisotropic particles detectable by optical microscopy. Mesogenic pitch is low in quinoline-insoluble fractions and capable of transforming into mesophase pitch during continuous heat treatment above by the formation of optically detectable carbonaceous mesophase.

MESOGLEA: The translucent gelatinous non-cellular layer separating the endoderm and ectoderm in the body wall of coelenterates such as jellyfish, comb jellies and certain primitive sea creatures in the phylum Cnidaria. It acts as the creatures' structural support in water, as they

lack bones or cartilage, endoskeletons or exoskeletons. It may be thin as in hydra or tough and fibrous as in the larger jellyfish and sea anemones. It often contains cells that have migrated from the two body layers but these do not form tissues and organs.

MESOIONIC COMPOUNDS: Dipolar five- (possibly six-) membered heterocyclic compounds in which both the negative and the positive charge are delocalised, for which a totally covalent structure cannot be written and which cannot be represented satisfactorily by any one polar structure. The formal positive charge is associated with the ring atoms and the formal negative charge is associated with ring atoms or an exocyclic nitrogen or chalcogen atom. Mesoionic compounds are a subclass of betaines.

MESOLECITHAL: Of an egg, having an intermediate amount of yolk, concentrated at one part of the cytoplasm. This is characteristic of lampreys and amphibians such as frogs and salamanders. Mesolecithal eggs are described as telolecithal.

MESOLYTIC CLEAVAGE: Cleavage of a bond in a radical ion whereby a radical and an ion are formed. The term reflects the mechanistic duality of the process, which can be viewed as homolytic or heterolytic depending on how the electrons are attributed to the fragments.

MESOMERIC EFFECT: The effect (on reaction rates, ionisation equilibria, etc.) attributed to a substituent due to overlap of its p- or π -orbitals with the p- or π -orbitals of the rest of the molecular entity. Delocalisation is thereby introduced or extended and electronic charge may flow to or from the substituent. The effect is symbolised by M. Strictly understood, the mesomeric effect operates in the ground electronic state of the molecule. When the molecule undergoes electronic excitation or its energy is increased on the way to the transition state of a chemical reaction, the mesomeric effect may be enhanced by the electromeric effect, but this term is not much used and the mesomeric and electromeric effects tend to be subsumed in the term resonance effect of a substituent.

MESOMERISM (RESONANCE): A phenomenon in which a chemical compound can adopt several contributing structures by the alternation of bonds while the atoms in the molecules remain in the same relationship to each other, so that the bonding cannot be expressed by one single Lewis formula. Each contributing structure can be represented by a Lewis structure, with normal single, double or triple covalent bonds between every pair of adjacent atoms within the structure.

MESOMORPHIC PHASE: States of matter in which an isometric molecules (or particles) are regularly arranged in one (nematic state) or two (smectic state) directions, but randomly arranged in the remaining direction(s). It is often formed by concentrated systems of surfactants.

MESOMORPHIC STATE: A state of matter in which the degree of molecular order is intermediate between the perfect three-dimensional, long-range positional and orientational order. It is found in solid crystals and the absence of long-range order found in isotropic liquids, gases and amorphous solids.

MESOMORPHIC TRANSITION: A transition that occurs between a fully ordered crystalline solid and an isotropic liquid. Mesomorphic transitions can occur (i) from a crystal to a liquid crystal, (ii) from a liquid crystal to another liquid crystal and (iii) from a liquid crystal to an isotropic liquid.

MESON: A member of a group of unstable elementary particles and part of a larger group of particles called hadrons, composed of smaller particles called quarks and antiquarks with masses intermediate between those of electrons and nucleons and with positive, negative or zero charge. Mesons are emitted by nuclei that have been bombarded by high energy electrons. Some mesons help hold together the particles called neutrons and protons that make up the nucleus of an atom.

MESOPAUSE: A level in the ionosphere, at an altitude of about 85 km, above which temperatures rise as altitude increases instead of falling as they do in the lower ionosphere.

MESOPHASE: The phase of a liquid crystalline compound between the crystalline and the isotropic liquid phase.

MESOPHASE PITCH: A pitch with a complex mixture of numerous essentially aromatic hydrocarbons containing anisotropic liquid-crystalline particles (carbonaceous mesophase) detectable by optical microscopy and capable of coalescence into the bulk mesophase. The carbonaceous mesophase particles are formed from the aromatics of high molecular mass in mesogenic pitch, which have not yet been aggregated to particles detectable by optical microscopy within the apparently isotropic pitch matrix. The carbonaceous mesophase is insoluble in quinoline and pyridine, but the amount of mesophase measured from microscopical observation appears somewhat higher because parts of the carbonaceous mesophase can be extracted by the solvents.

MESOPHASE PITCH-BASED CARBON FIBRES (MPP-BASED CARBON FIBRES): Carbon fibres obtained from mesogenic pitch after it has been transformed into mesophase pitch (MPP) at least during the process of spinning and after the spun mesophase pitch fibres have been made non-fusible (stabilised) and carbonised.

MESOPHILES: Microbes which reproduce and grow best in the temperature range of 20 °C to 40 °C and which have an optimum growth temperature in the range 30 °C to 39 °C. They represent the majority of microbial species on Earth, including all the pathogens of mammals.

MESOPHYLL: The internal supporting tissue of a leaf blade (lamina), consisting of parenchyma cells containing chlorophyll between the epidermal layers. There are two types of mesophyll cells, namely the spongy and palisade layers. The palisade mesophyll lies just beneath the upper epidermis and consists of cells elongated at the right angles to the leaf surface. They contain a large number of chlorophyll and their principal function is photosynthesis. Spongy mesophyll characterised by the presence of wide intercellular spaces occupies most of the remainder of the lamina. It consists of spherical or irregularly shaped cells containing fewer chloroplasts than the palisade mesophyll cells.

MESOPHYTE: Terrestrial plants such as goldenrod, clover, oxeye and daisy adapted to grow in soil that has moderate water supply and mineral salts, but may also grow as desert annuals that flower during the rainy season. Such plants wilt easily when exposed to drought as they are not adapted to conserve water. The majority of flowering plants are mesophytes.

MESOPHYTIC LEAVES: The leaves of plants that grow under moderately humid conditions with abundant soil and water.

MESOPORE: Pore of intermediate size. Those with widths exceeding about 50 nm are called macropores; those with widths not exceeding 2nm are called micropores.

MESOSCALE: In meteorology, it refers to the size or scale of a phenomena smaller than ordinary cyclones or weather systems but larger than such microscale phenomena as the thickness of the boundary layer, the wakes of objects, etc. Thunderstorms involve mesoscale processes and other meteorological events. The size of cities are usually mesoscale processes.

MESOSCOPIC: Pertaining to a size scale intermediate between those of the microscopic and the macroscopic states. Mesoscopic objects and systems require quantum mechanics to describe them. Many electronic devices are mesoscopic.

MESOSOME: A structure found in bacterial cells formed by intrusions of the plasma membrane into the protoplasm. The respiratory enzymes are associated with the mesosome and it has the same function as the mitochondrion of eukaryotic cells. It is also attached to the DNA and is thought to control the separation of replicated DNA molecules during cell fission.

MESOSPHERE: The part of Earth's atmosphere immediately above the stratosphere, where the temperature drops from about 270 K to 180 K. It lies between the stratopause and the mesopause and overlaps the lower part of the ionosphere.

MESOTHELIUM: A single layer of thin plate-like cells derived from mesoderm, which covers the inner surface of the abdominal cavity and the thorax. It surrounds the heart and forms part of the peritoneum and pleura. Mesothelial tissue also surrounds the male internal reproductive organs, the *tunica vaginalis testis* and covers the internal reproductive organs of women, the tunica serosa uteri.

MESOTHELIOMA: Malignant tumour occurring in the pleura, peritoneum or pericardium. Pleural mesothelioma affects the tissue that surrounds the lungs and may occur as a consequence of asbestosis. Peritoneal mesothelioma affects the tissue in the abdomen; and pericardial mesothelioma affects tissue that surrounds the heart. A virus similar to the monkey virus SV40 may trigger mesothelioma.

MESOTHORAX: The middle segment in the thorax of an insect, which bears the second pair of legs and the first pair of wings. The thorax is divided into three segments, the other two are prothorax (the posterior or front segment), and metathorax (the anterior or rear segment). Each of the segments has a pair of legs and hard plates called sclerites.

MESOTROPHIC: A body of water such as a lake that is intermediate between an eutrophic lake and an oligotrophic lake in the amount of nutrients and oxygen contained within it.

MESOXALYLUREA (ALLOXAN): A compound with molecular formula $C_4H_4O_4N_2$ that is an important decomposition product of uric acid. It is obtained by oxidising uric acid with nitric acid. It can react with alkali to take up two molecules of water to produce urea and mesoxalic acid. It can be used experimentally to induce diabetes mellitus in animals by selectively destroying the insulin-secreting islet cells of the pancreas and therefore used in medical research.

MESOZOIC ERA: The geological era that extended from the end of the Palaeozoic era, about 251 million years ago, to the beginning of the Cenozoic era, about 65 million years ago. The Mesozoic era is often known as the **Age of Reptiles** as these animals which included the dinosaurs, pterosaurs and ichthyosaurs, became the dominant animal life form. Most became extinct before the end of the era.

MESSAGE: A discrete piece of information, whose likelihood of accurate transmission through a channel with a given probability law is the subject of study by information theory.

MESSAGE-ID: In computing, it is a globally unique numeric or alphanumeric string given to a digital message and used in email and for controlling traffic in Usenet newsgroups.

MESSAGING APPLICATION PROGRAMMING INTERFACE (MAPI): An interface in Microsoft Windows that allows software applications to send and receive electronic mails (e-mails), provided the sender and receiver's applications are MAPI-enabled. Documents may be attached to the message. Microsoft applications that support MAPI include word processors, spreadsheets and graphics applications.

MESSENGER RNA (mRNA): One of the three types of RNA; the other two are transfer RNA (tRNA) and ribosomal RNA (rRNA). It is synthesised or transcribed from DNA in the genetic material that attaches to ribosomes in the cytoplasm and specifies the primary structure of a protein by specifying amino acid sequences. In eukaryotes, the mRNA carries the genetic information from the nucleus to the site of protein synthesis (ribosome) in the cytoplasm to be translated. In non-eukaryotes such as bacteria, the mRNA is produced and translated in the cytoplasm of the bacterial cell.

MESTOM SHEATH (MESTOME SHEATH): A thin-walled type of bundle sheath present in some grasses.

META: Relating to the 1-3 carbon atoms in a benzene ring, abbreviated to m-. For example, xylene is 1,3-dimethylbenzene. 2. Compounds formed by dehydration, for example, metaphosphoric acid, made by strongly heating orthophosphoric acid.

META LANGUAGE: A language that is used to describe other languages.

META SEARCH ENGINE: The search engine software that is used to search other search engines. It passes queries on to many other search engines and/or directories and returns a summary of the results.

META TAG (META-TAG): A special HTML tag that is used to store information about a Web page but is not displayed in a Web browser. For example, meta tags provide information such as what program was used to create the page, a description of the page and keywords that are relevant to the page. Many search engines use the information stored in meta tags when they index Web pages.

METABOLIC DISEASE: One of a group of diseases that affect the ability of the cell to perform critical biochemical reactions that involve the processing or transport of proteins, carbohydrates or lipids. It can be congenital due to inherited enzyme abnormality or acquired due to disease of an endocrine organ or failure of a metabolically important organ such as the liver. An example is diabetes mellitus in which the pancreas does not produce enough insulin or the body resists its use.

METABOLIC HALF-LIFE (METABOLIC HALF-TIME): Time required for one-half of the quantity of a substance in the body to be metabolised.

METABOLIC PANEL: A series of chemical tests carried out on blood serum (the part of blood that doesn't contain cells) to provide information about a patient's metabolism. Blood is drawn from a vein, usually from the inside of the elbow or the back of the hand. There are two types of metabolic panel: basic metabolic panel (BMS) and comprehensive metabolic panel (CMP).

METABOLIC PATHWAY: A series of chemical reactions in a living system that combine to perform one or more important functions. The product of one reaction in a pathway serves as the substrate for the following reaction. Each reaction is dependent on a specific precursor: a chemical, an enzyme or the transfer of energy and often require dietary minerals, vitamins and other cofactors in order to function properly.

METABOLIC RATE: A measure of the energy used by an animal in a given time period. The metabolic rate of an animal is affected by several interacting factors. These include temperature and the level of activity. The metabolic rate of a resting animal is known as the **basal metabolic rate**. It decreases when an animal is resting and increases during activity.

METABOLIC SYNDROME: A combination of medical disorders that increase the risk of developing cardiovascular disease and diabetes.

METABOLIC WASTE (EXCRETES): The collective waste products of metabolism which cannot be used by the organism and must therefore be excreted because if maintained it may have lethal effects. Metabolic wastes include nitrogen compounds, water, CO_2 , phosphates, sulfates, insoles, medicals, food additives, etc. In plants some of them, primarily the nitrogen compounds, are chemically transformed into useful substances.

METABOLIC WATER: One of the products of aerobic respiration or water formed when food is broken down to release energy, an important source of water for desert animals in habitats where water sources are not available. For instance, the chief source of water for kangaroo rats living in desert is metabolic water. They never have to drink; the metabolic water produced within their cells is sufficient for their survival.

METABOLISABLE ENERGY (ME): The net energy in food or feed that is available to humans or animals by digestion and absorption, and measured as the difference between gross energy content and the energy lost as being undigested or indigestible. It is the energy that is available for metabolism which does not include energy lost in faeces, urine and gasses.

METABOLISM: The series of biochemical and physical processes that occur in living organisms, influenced by exercise, food and environmental temperature, for example, as occurs in respiration when glucose is broken down with or without oxygen to yield new products including adenosine triphosphate (ATP) and CO_2 . Metabolism has two subcategories: **anabolism**, which involves the synthesis of complex organic molecules from simpler precursors; and **catabolism**, the breakdown of complex substances such as food into simpler molecules, including digestion, respiration and the excretion of waste substances often accompanied by the release of energy. Anabolism

and catabolism occur by a number of reaction steps called metabolic pathway. The various organic compounds that take part in or are formed by the processes of metabolism are called **metabolites**. In animals, many metabolites are obtained by the digestion of food, whereas in plants only the basic starting materials such as carbon dioxide, water and minerals are externally derived. Some pathways, such as glycolysis are linear while others such as Krebs cycle are cyclic. The changes at each step in the pathway are usually small and are promoted by enzymes. Thus, the amount of energy released or required are minimal and helps in maintaining a constant internal environment. **Basal metabolism** is the energy spent on activities such as breathing, batting eyelids and sleeping for an organism at rest; it represents the minimum amount of energy required to maintain life at normal body temperature.

METABOLITE: Any substance or small molecule that takes part or is produced in a metabolic reaction or a biochemical reaction in the body such as when the body breaks down food, drugs, chemicals, or its own tissue. Examples of metabolites are glucose, amino acids and fatty acids glycerol (fat). In plants, metabolites are used for both essential functions called primary metabolites and specific functions called secondary metabolites. Primary metabolites (e.g., carbohydrates, lipids, proteins and nucleic acids) are responsible for growth, development and reproduction; and secondary metabolites (e.g., antibiotics and pigments) are responsible for specific functions, such as pollinator attraction or defence against herbivorous. Metabolites can be measured in blood, urine, and other body fluids. Disease and environmental factors, such as diet, drugs, and chemicals, can affect how metabolites are made and used in the body.

METABOLOME: The complete set of molecules of low molecular weight or small-molecule metabolites such as metabolic intermediates, hormones and other signalling molecules and secondary metabolites found within an organism and produced by cells during metabolism. Metabolome varies with the physiological or developmental stage of the cell.

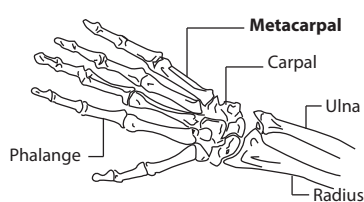
METABOLOMICS: The study of substances called metabolites in cells and tissues and how they change under various physiological or developmental conditions or in response to genetic modification. Metabolomics may help find new ways to diagnose and treat diseases, such as cancer.

METABONOMICS: The evaluation of cells, tissues or biological fluids for changes in metabolite levels that follow exposure to a given substance, in order to determine the metabolic processes involved, to evaluate the disruption in intermediary metabolic processes that results from exposure to that substance, or to determine the part of the genome that is responsible for the changes. Although "metabolomics" and "metabonomics" are frequently used as synonyms, there is a growing consensus that there is a difference in that "metabolomics" places a greater emphasis on comprehensive metabolic profiling, while "metabonomics" is used to describe multiple (but not necessarily comprehensive) metabolic changes caused by a biological perturbation.

METABOTROPIC RECEPTOR: A type of membrane receptor at the surface or in vesicles of eukaryotic cells that, when activated by binding of ligand, triggers changes in cell metabolism via intracellular secondary messenger. Examples are metabotropic glutamate receptors, muscarinic acetylcholine receptors and GABA_B receptors.

METACARPAL: One of the five small, long bones that join wrist bones or carpals to the finger bones or phalanges in the human hand.

METACARPUS: The hand or corresponding part of the



Human hand showing the metacarpal

fore limb in terrestrial vertebrates, consisting of a number of rod-shaped bones that articulate with the bones of the wrist and those of the fingers. The number of metacarpals varies between species: in the basic pentadactyl limb there are five, but this number is reduced in many species.

METACENTRIC: A term used to describe a chromosome in which the chromatid arms are of a similar length either side of the centromere so that the centromere appears to be near the centre of the chromosome.

METACHROMATIC STAIN: A stain that dyes certain tissues a different colour to the dye solution. Examples are methyl violet and thionine. Thionin violet produces colours in sections ranging from blue to reddish violet and is particularly useful for distinguishing chromatin and mucin.

METADATA: Data that describes the characteristics of stored data; descriptive data. It provides information about certain item's content. For example, an image may include metadata that describes how large the picture is, the colour depth, the image resolution, when the image was created and other data. A text document's metadata may contain information about how long the document is, who the author is, when the document was written and a short summary of the document.

METAFEMALE (TRIPLE X SYNDROME; TRIPLO-X; TRISOMY X; XXX SYNDROME; 47,XXX ANEUPLOIDY): A female organism such as fruit fly (*Drosophila* sp.) with an extra X (female) chromosome. For example, the fruit fly has three instead of the normal two X chromosomes. In such females the disproportion of sex chromosomes to that of autosomes (chromosomes not controlling sex) leads to impaired development.

METAFILE: 1. A file that describes the contents of other files. It may contain metadata, which defines a group of other files and gives a summary of what data they contain. 2. A file often used in computer graphics. These files define objects and images using a list of coordinates. They are typically used for vector images, such as Adobe Illustrator, CorelDRAW and EPS files, but can include raster images as well.

METAGENESIS (ALTERNATION OF GENERATIONS; HETEROGENESIS): The occurrence in one life cycle of an organism of two or more different forms, usually an alternation of a sexual with an asexual form. The sexual phase, called the gametophyte, produces gametes or sex cells; the asexual phase or sporophyte, produces spores asexually. The two phases or generations, are often morphologically and sometimes chromosomally, distinct. It occurs more commonly in algae, fungi, mosses, ferns and seed plants; many invertebrates, for example, protozoans, jellyfish and flatworms also have an alternation of sexual and asexual generations, but the alternation of haploid and diploid generations is unknown.

METAL: A substance whose atoms have some of their outermost electrons loosely held. These electrons tend to migrate from one atom to another on application of potential differences or temperature gradient resulting in the formation of positive ions. It can also be defined as any of a class of chemical elements with certain chemical characteristics such as being reactive or stable, has a crystalline structure, is shiny and has high melting and boiling points and physical properties such as good conductors of heat and electricity, malleable, ductile and opaque, but reflect light well. Metals and nonmetals are separated in the periodic table by a diagonal line of elements drawn from boron (B) to polonium (Po), with non-metallic elements on the right-hand-side. Metals can be classified generally as alkali metals, alkaline earth metals, transition elements, noble or precious metals, platinum metals, lanthanoid or rare earth metals and actinoid metals. In chemistry, metals fall into two distinct types. Those of the s- and p-blocks such as sodium and aluminium are generally soft silvery reactive elements. They tend to form positive ions and so are described as electropositive. This is contrasted with typical non-metallic behaviour of forming negative ions. The transition elements such as iron and copper are harder substances; form coordination

complexes and generally less reactive. All metals have oxides that are basic, although some, such as aluminium, have amphoteric properties.

METAL DETECTOR: Electronic device, which uses electromagnetic fields to detect the presence of concealed metallic objects such as in the ground or on the body. Its uses include detecting land mines, in security to detect concealed weapons such as knives and guns, in geophysical prospecting, in archaeology, to detect foreign bodies in food and in the construction industry to detect concealed metals such as reinforced steel in concretes, pipes and wires buried in walls and floors.

METAL FATIGUE: A cumulative effect of repeated applications of stress on metal parts in machines, none of which exceeds the ultimate tensile strength, resulting in fracture under a stress much weaker than that necessary to cause fracture in a single application.

METAL FOIL: Metal rolled or beaten into a very thin sheet. The most common is aluminium foil, which has largely replaced tin foil. It is used extensively in packaging and in the home in the storage or cooking of food. Gold foil is used decoratively and in electronics. Lead foil is used as a radiation shield.

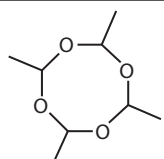
METAL TO LIGAND CHARGE TRANSFER (MLCT) TRANSITION: An electronic transition of a metal complex that corresponds to excitation populating an electronic state in which considerable electron transfer from the metal to a ligand has occurred.

METAL-CARBENE COMPLEXES (CARBENE METAL COMPLEXES): Metal complexes of the type R_2CML_n (M = metal, L = ligand) in which formally a carbene is coordinated to a metal.

METAL-INSULATOR TRANSITION (METAL-NONMETAL TRANSITION): A transition between a metal and an insulator, characterised by a sudden change in electrical transport properties (conductivity) due to a reversible change from localised to itinerant behaviour of the electrons. There are several ways in which a metal-insulator transition or the reverse transition can occur, including increasing or eliminating band gaps by doping, by changing their ambient conditions by applying pressure or gate voltages, etc. Example is the transition at 339 K in VO_2 where it changes from a high-temperature metallic behaviour to a low-temperature semiconductor behaviour. These materials are useful as controlling devices in electric and electronic circuits.

METALANGUAGE: The language in which another language or system of symbols, the object language, is described. For example, when one says that 'P & Q' is true if and only if P and Q are both true; one is defining the connective '&' in the object language (in this case, sentential calculus) in terms of the conjunction of the metalanguage (in this case English).

METALDEHYDE: A solid compound with the molecular formula $C_4O_4H_4(CH_3)_4$. It is formed by polymerisation of ethanal (acetaldehyde) in dilute acid solutions below 0 °C (273 K). The compound, a tetramer of ethanol, is commonly used as a pesticide against slugs, snails and other gastropods and as a fuel for portable stoves.



Chemical structure of Metaldehyde

METALLIC: Constituting, containing or made of metal.

METALLIC BOND (METALLIC BONDING): An electrostatic attraction in a metal between an array or a collection of localised metal nuclei and delocalised valence electrons. In other words the positive atomic nuclei are surrounded by a collection of delocalised electrons which are attracted to the nuclei. The valence electrons move freely and are shared by all the atoms in the lattice. It accounts for many physical properties of metals, such as strength, malleability, ductility, thermal and electrical conductivity, opacity, etc.

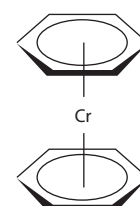
METALLIC CHARACTER: Chemical properties associated with those elements classified as metals in the periodic table. These properties, which arise from the element's ability to lose its outer valence

electrons and form a positive ion in chemical reactions include the displacement of hydrogen from dilute acids, the formation of basic oxide; the formation of ionic chlorides; and reducing reaction, as in the thermite process. The metallic character increases as you go down a group in the periodic table and decreases as you move across the periodic table from left to right.

METALLIC CRYSTAL: A crystalline structure that is held together by metallic bonds, found in some interstitial compounds as well as in metals and alloys. They have unique properties because of the relationship between the positive ions and the electrons of the metal.

METALLIC HYDROGEN: A hypothetical form of hydrogen in which the molecules have been forced by extremely high pressures to assume the lattice structure typical of metals. It is estimated that as much as 40% of Jupiter's mass (but not more than 3% of Saturn's) may be in the form of metallic hydrogen.

METALLOCENE: A compound obtained as a cyclopentadienyl derivative of a transition metal or a metal halide with the general formula $(C_5H_5)_2M$. It consists of two cyclopentadienyl anions (Cp , which is $C_5H_5^-$) bound to a metal centre (M) such as iron (Fe), copper (Cu) and zirconium (Zr) in the oxidation state II. Typical example is ferrocene ($(C_5H_5)_2Fe$ or Cp_2Fe with the systematic name bis(η^5 -cyclopentadienyl) iron(II) or it is a cyclic polyene complexes made up of a metal atom between two polyhepto rings. By IUPAC definition, a metallocene contains a transition metal and two cyclopentadienyl anion ligands coordinated in a sandwich structure, i.e. the two Cp anions are coplanar and with equal bond lengths and bond strength. A notable feature of the metallocenes is their thermal stability which allows them to sublime in vacuum. Metals in the periodic table commonly known to form metallocene complexes are titanium, zirconium, hafnium, vanadium, chromium, molybdenum, tungsten, manganese, iron, ruthenium, osmium, cobalt, rhodium and nickel.



Structure of (Benzene) chromium tricarbonyl, an example of metallocene

METALLOENZYME: An enzyme that, in the active state, contains one or more metal ions which are essential for its biological function.

METALLOGRAPHY: The microscopic study of the structure of metals and their alloys. Both optical microscopes and electron microscopes are used in this work.

METALLOID (SEMI-METAL): An element such as boron, silicon, germanium, arsenic, antimony and tellurium whose properties are of a metal but behave chemically both as a metal and as a nonmetal. They are used to make and their oxides are amphoteric (having both acidic and basic characteristics).

METALLURGICAL COKE: Metallurgical coke is produced by carbonisation of coals or coal blends at temperatures up to 1400 K to produce a macroporous carbon material of high strength and relatively large lump size. Metallurgical cokes must have a high strength to support heavy loads in the blast furnace without disintegration. It is used as filler coke for polygranular carbon products.

METALLURGY: The branch of engineering, usually materials science that studies the physical and chemical behaviour of metallic elements, their intermetallic compounds and their mixtures. It is also concerned with the extraction of metals, purification of metals, the manufacture of alloys and the use and performance of metals. Process metallurgy is concerned with the extraction and production of metals and physical metallurgy concerns the mechanical behaviour of metals.

METAMAGNET: A material that is an antiferromagnet in the absence of an external magnetic field but undergoes a first-order transition to a phase in which there is a non-zero ferromagnetic moment when the

external magnetic field becomes sufficiently large. Iron(II) chloride is an example of a material that can act as a metamagnet

METAMALE (XYY SYNDROME): A defect of the sex chromosomes of males in which there is one extra Y-chromosome totalling 47 chromosomes instead of the usual 46, producing 47 XYY karyotype. This condition is associated with tallness, aggressiveness and acne.

METAMATERIAL: A type of synthetic composite material with a complex nanostructure, constructed to have properties that may not be found in nature. A particular type consists of materials that have a negative refractive index. There has been considerable amount of research into using these as invisibility cloak for microwaves and possibly visible radiation. Its potential uses include remote aerospace applications, sensor detection and infrastructure monitoring, smart solar power management, public safety, radomes, high-frequency battlefield communication and lenses for high-gain antennas, improving ultrasonic sensors and even shielding structures from earthquakes.

METAMATHEMATICS: 1. The abstract science of space, number and quantity. The study of experiences such as measurement, order and patterns as separate attributes of human thought is known as mathematics. 2. In mathematics, the logical study and analysis of the reasoning, principles and rules that govern the use and combination of mathematical symbols, numbers, etc., including questions of both syntax and semantics such as those studied by combinatorial logic and whether axiom systems are independent, consistent or complete.

METAMERIC SEGMENTATION: The repetition of organs and tissues at intervals along the body of an animal, thus dividing the body into a linear series of similar parts or segments (metameres), but not including the head. It is common in animals such as earthworm. In these animals, the segmentation is visible both externally and internally. It also occurs internally in arthropods and in the embryonic development of all vertebrates, in which it is confined to parts of the muscular, skeletal and nervous systems and does not show externally.

METAMICT STATE (METAMICTIZATION; METAMICTIION): The amorphous state of a substance such as zirconium, allanite, ekanite and titanite caused by alpha-particles and the recoil nuclei from radioactive disintegrations resulting in the gradual and ultimately complete destruction of a mineral's crystal structure. The process also lowers a mineral's refractive index, hardness and specific gravity. An example of a metamict mineral is zircon, in which uranium and thorium atoms substitute for zirconium in the crystal structure, resulting in radiation damage.

METAMORPHIC ROCK: Rock which has been altered from its original structure and composition by pressure, heat or chemically active fluids or a type of rock formed when sedimentary rock is heated and pressurised in the earth's crust. For example, limestone is transformed to marble, shale to slate and slate to schists. They are usually hard and crystalline.

METAMORPHISM: A process of change in the physical structure of rocks as a result of pressure and long-term heat, but excluding changes produced by weathering and sedimentation. This applies especially to a change that increases the rock's crystalline structure and hardness.

METAMORPHOSIS: The pronounced change in both the internal and external morphology in the development of some animals such as butterflies and frogs when the juvenile stage is transformed into an adult or a complete change of shape of an animal. For example, metamorphosis steadily changes caterpillar into butterflies or moths and tadpoles into frogs. Metamorphosis often involves considerable destruction of larval tissues by lysosomes and in both insects and amphibians it is controlled by hormones. In amphibians, it is also referred to as **indirect development** (i.e., from egg to larva to adult), compared to higher vertebrates (reptiles, birds and mammals) where development is direct (i.e. from egg to adult).

METANDRY (PROTOGYNY): A condition in which the pistils mature before the stamens. This is usually found in some wind pollinated flowers, such as grasses.

METAPHASE: The second stage of cell division during which the nuclear membrane breaks down. The spindle forms and centromeres attach the chromosomes to the equator of the spindle. In the first metaphase of meiosis pairs of chromosomes are attached, while in mitosis and the second metaphase of meiosis, individual chromosomes are attached.

METAPHLEOM: The part of the primary phloem that develops in a plant after the stem or other part has finished elongating.

METAPHYSICS: The branch of philosophy concerned with the fundamentals of existence or reality.

METAPLASIA: The transformation of a tissue into a different and undesirable one such as tumour, in which tissues return to their normal pattern of differentiation after the causative factor has been removed. It is an abnormal process, which is not dysplasia and not directly considered carcinogenic. For example, metaplasia of the epithelium of the bronchi may be an early sign of cancer.

METAPOPOPULATION: A group of spatially separated populations of the same species, which interact at some level. For example, through human intervention, such as deforestation and road building, individuals may migrate between subpopulations or disperse to unoccupied habitat patches. They may even become extinct, the smaller the population, the more prone it is to extinction. The sizes and composition of the component subpopulations and levels of migration and dispersal are all crucial factors in determining the metapopulation dynamics.

METASTABLE ION: An ion which is formed with sufficient excitation to dissociate spontaneously during its flight from the ion source to the detector.

METASTABLE STATE: A condition of system such as excited state of an atom, nucleus or other system that has a longer lifetime than the ordinary excited states and generally has a shorter lifetime than the ground state. It can also be defined as a condition of system in which it has a precarious stability that can easily be disturbed. It is unlike a state of stable equilibrium in that a minor disturbance will cause a system in a metastable state to fall to lower energy level. For example, a book lying on a table is in a state of stable equilibrium; a thin book standing on edge is in a metastable equilibrium. Supercooled water is also in a metastable state, as it remains liquid below 0 °C (273 °C); and a grain of dust or ice introduced into it will cause it to freeze. An excited state of an atom or nucleus that has an appreciable lifetime is also metastable.

METASTASIS: The spread of cancer cells to distant sites in the body, resulting in the development of secondary tumours.

METASTATIC: Electrons that leave an orbital shell, either entering another shell or being absorbed into the nucleus.

METATARSAL: One of five arching bones that join the ankle bones (tarsals) to the toe bones (phalanges) in the human foot.

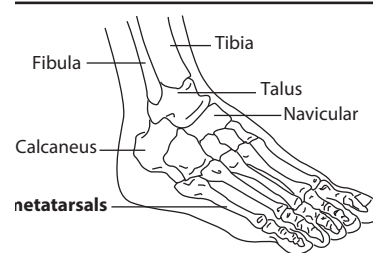
METATARSUS (METATARSAL BONES):

A group of long bones in the foot of terrestrial vertebrates, located between the tarsal bones of the hind-foot and mid-foot and the phalanges

of the toes. It consists of a number of rod-shaped bones that articulates with the bones of ankle and those of the toes. The number of metatarsals varies and is reduced in some species.

METATHERIA: A subclass of mammals containing the marsupials, which includes kangaroos and koala bears, in which the female has a pouch containing the newly born, which obtains nourishment from the mother's mammary glands.

METATHESIS (METATHESIS REACTIONS; DOUBLE DECOMPOSITION): A type of reaction between two compounds,



Human leg showing the metatarsals

which can either be ionic or covalent in which the first and second parts of one reactant are united, respectively, with the second and first parts of the other reactant. It can also be defined as the exchange of bonds between the two reacting chemical species, which results in the creation of products with similar or identical bonding affiliations. An example is $\text{KCl} + \text{AgNO}_3 \rightarrow \text{KNO}_3 + \text{AgCl}$. Metathesis of alkenes is an important type of reaction in synthetic organic chemistry. It involves exchanges of groups. Reactions of this type are catalysed by metal alkylides containing an $\text{M} = \text{CR}_2$ grouping and the intermediate is a four-membered ring containing the metal ion. The catalysts most often used are the Schrock catalysts based on molybdenum and the Grubbs catalysts based on ruthenium. The American chemists Richard Schrock and Robert Grubbs shared the Noble price for Chemistry in 2005 for work in this field.

METATHORAX: The posterior or rear segment in the thorax of an insect, which bears the third pair of legs and the second pair of wings. The thorax is divided into three segments, the other two are prothorax (the anterior or fore segment near the head), and the mesothorax (the middle segment). Each of the segments has a pair of legs and hard plates called sclerites.

METAVARIABLE: In logic, a variable in the metalanguage that can take as values expressions of the object language. So, for example, in the rule of sentential calculus that $A \& B \vdash A$, 'A' is a metavariable that can be replaced by any well-formed formula to yield a valid sequent of the calculus.

METAXYLEM: The part of the primary xylem that develops in plant after the stem or other part has finished elongating. The walls of the metaxylem are more extensively lignified than those of the protoxylem and the lumina are wider in diameter.

METAZOA: The sub-kingdom of living organisms, composed of numerous cells differentiated into tissues and organs and usually a digestive cavity lined with specialised cells. It usually includes the coelenterates and all higher animals but sometimes also includes the sponges and mesozoans.

METEOR: A mass or solid particle from space appearing as a streak of light observable in the night sky, when it enters the Earth's atmosphere and becomes incandescent as a result of friction with atmospheric atoms and molecules. If the apparent brightness of a meteor exceeds that of the planet Venus as seen from Earth, it is called a **fireball** and when a bright meteor is seen to explode, it is called a **bolide**.

METEOR-BURST COMMUNICATIONS (METEOR SCATTER COMMUNICATIONS): Technique for sending messages by bounding radio waves off the trails of meteors that continually enter the Earth's atmosphere at altitudes of between about 85 and 120 km. It is used to broadcast signals on frequencies normally between about 40 and 150 MHz.

METEORITE: A piece of rock or object originating in outer space that survives impact with the Earth, Moon or other body. Most meteorites are derived from meteoroids but they are also sometimes produced by impacts of asteroids.

METEOROID: A small rocky or metallic object in orbit around the Sun or another star. A meteoroid which strikes the Earth or any other large body is called a meteorite.

METEOROID STREAM: A trail of solid particles released from a parent body such as a comet or asteroid, moving on similar orbits. Various ejection directions and velocities for individual meteoroids cause the width of a stream and the gradual distribution of meteoroids over the entire average orbit.

METEOROID SWARM: A clump of particles within a meteoroid stream. Such clumps have been released quite recently from the parent body and have not had time to spread out along the orbit. When Earth passes through a swarm, the visible effect is a meteor storm.

METEOROLOGICAL SATELLITES (WEATHER SATELLITES): Spacecrafts that are used to monitor Earth's atmospheric conditions and provide data to help understand and predict global weather patterns. The first was TIROS-1 in 1960. United States, Russia, Europe

and other nations have launched weather satellites into both polar and geostationary orbits.

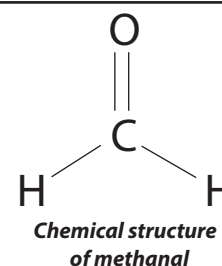
METEOROLOGY: The study of current climate and weather patterns with the aim of making short-term forecasts, or the scientific observation and study of the atmosphere so that weather can be accurately forecast. In contrast, climatology is the study of long-term trends in climate and weather patterns. A **meteorologist** is one who studies processes in the earth's atmosphere that cause current climate and weather patterns in order to make accurate short-term forecasts.

METHACRYLATE: A salt or ester of metacrylic acid (2-methylpropenoic acid). Methacrylates are common monomers in polymerplastics, forming the acrylate polymers. They easily form polymers because the double bonds are very reactive. They are used for windscreen repairs and as bone cement for fixing prosthetic devices in orthopaedic surgery.

METHACRYLATE RESINS: Acrylic resins obtained by polymerising 2-methylpropenoic acid or its esters.

METHADONE (6-DIMETHYLAMINO-4,4-DIPHENYL-3-HEPTANONE, AMIDONE): A synthetic opioid, first developed during World War II by the German chemists Bockmühl and Erhart working for the Hoechst Laboratories of IG Farben. It is similar in effect to morphine with the chemical formula $\text{C}_{21}\text{H}_{27}\text{NO}$. However, it has pharmacological properties that are qualitatively identical to those of natural alkaloid. It is used medically as an analgesic for chronic pains and also as a substitute for heroin in the treatment of addiction.

METHANAL (FORMALDEHYDE): A colourless, flammable, poisonous gas with a suffocating odour with the chemical formula HCHO , melting point -92°C (181 K) and boiling point -21°C (252 K). It is the simplest aldehyde, soluble in water, alcohol and ether. It is prepared commercially by passing methanol vapour mixed with air over a catalyst such as hot copper, to cause oxidation of the methanol. It is also prepared by the oxidation of natural gas. It forms two polymers: methanol trimer and polymethanal. Its uses include in the preparation of dyes, in the production of bakelite and other plastics and synthetic resins.



METHANAL TRIMER: A cyclic trimer of methanol with the molecular formula $\text{C}_3\text{O}_3\text{H}_6$, obtained by distillation of an acidic solution of methanol. It has a six-membered ring of alternating $-\text{O}-$ and $-\text{CH}_2-$ groups.

METHANATION: A method of manufacturing methane from a mixture of various gases out of biomass fermentation or thermochemical gasification, the main components being carbon monoxide or dioxide by high pressure catalytic hydrogenation. It is often used to improve the calorific value of town gas.

METHANE: A lighter than air, colourless, odourless gaseous, saturated hydrocarbon with the chemical formula CH_4 , melting point -184°C (89 K) and boiling point at -161.4°C (111.75). It is the simplest hydrocarbon and the first member of the alkane series. It is released from petroleum while it is being processed and the main constituent of natural gas, forming 99%. It is formed by the decomposition of plant and animal matter. It can be prepared in the laboratory by heating sodium acetate with sodium hydroxide, by the reaction of aluminium carbide with water, by the direct combination of carbon and hydrogen or by the destructive distillation of coal or wood. It is combustible and can form explosive mixtures with air. It is an important raw material for producing other organic compounds. It can be converted into methanol by catalytic oxidation. Methane is a very powerful greenhouse gas. Two kilogrammes (kg) of methane traps 25 times more heat in the atmosphere than a pound of carbon dioxide.

METHANE CARBOXYLIC ACID (ACETIC ACID; ETHANOIC ACID): A liquid carboxylic acid with the chemical formula $\text{CH}_3\text{CO}_2\text{H}$, density 1.049 g cm^{-3} , melting point $16-17^\circ\text{C}$ (289-290 K) and boiling

point 118-119 °C (391-392 K), whose salts and esters are acetates. In its pure or glacial state, it is a clear, syrupy, corrosive liquid that mixes readily with water. It occurs naturally as a metabolic intermediate in body fluids and plant juices. Its dilute solution, from fermentation and oxidation of natural products is vinegar. Industrial production is either synthetic, from acetylene; or biological, from ethanol. Industrial chemicals made from it are used in printing and as plastics, photographic films, textiles and solvents.

METHANE RECOVERY: Any method by which methane emissions, such as from coal mines or waste sites are captured and then reused either as a fuel or for some other economic applications, for example, reinjection in oil or gas reserves, to produce electricity, heat buildings, etc.

METHANE TRICHLORIDE (CHLOROFORM; FORMYL TRICHLORIDE; METHYL TRICHLORIDE; METHENYL TRICHLORIDE; TRICHLOROMETHANE): A colourless volatile sweet-smelling liquid haloform with the chemical formula CHCl_3 , relative density 1.48, melting point -63.5 °C (209.5 K) and boiling point 61.7 °C (334.7 K). It can be made by chlorination of methane followed by separation of the mixture of products or by the haloform reaction. It is an effective anaesthetic but can cause liver damage and it has now been replaced by other halogenated hydrocarbons. It is used as a solvent and raw material for making other compounds.

METHANOGENESIS: The production of methane and carbon dioxide by anaerobic archaea during the process of anaerobic fermentation.

METHANOGEN (METHANOGENIC BACTERIA): Any anaerobic archaea of the family Methanobacteriaceae that require completely anaerobic conditions for growth and produce methane during the process of anaerobic fermentation. Methanogens are different from bacteria in certain aspects of their chemical structure, such as the composition of their cell walls. Methanogens are found in oxygen-deficient environments, such as marshes and swamps, where they produce marsh gas; and in sludge; and in the digestive systems of ruminants, where they are responsible for the methane content of belching in ruminants and flatulence in some humans. They obtain their energy mostly by reducing carbon dioxide and oxidising hydrogen, with the production of methane.

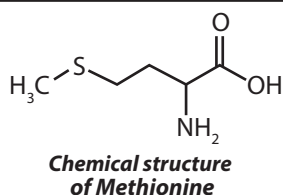
METHANOIC ACID (FORMIC ACID): A colourless, pungent, corrosive liquid with the chemical formula HCOOH , boiling point 100.7 °C (373.85 K) and solidifies at 8.4°C (281.55 K). It can be made by the action of concentrated sulfuric acid on the sodium salt (sodium methanoate) and occurs naturally in ants and stinging nettles. It has the lowest molecular weight and is the simplest of the carboxylic acids. It is used for insecticides, as a refrigerant and in textile and paper industry.

METHANOL (METHYL ALCOHOL; WOOD ALCOHOL): A simple primary alcohol with the chemical formula CH_3OH , melting point -93.9 °C (179 K), boiling point 64.96 °C (338 K) and relative density 0.79. It is a colourless, flammable liquid that is miscible with water in all proportions. It is made by the catalytic oxidation of methane from natural gas, using air. It is used as a solvent and added to ethanol to make methylated spirit. It is also used as a raw material for making methanol, as a solvent for varnishes and lacquers, as antifreeze and as a gasoline extender in the production of gasohol, a type of fuel used in motor vehicles.

METHANOTROPH: An organism capable of oxidising methane as a source of carbon and energy for its metabolism.

METHIONINE (2-AMINO-4-METHYLTHIOBUTANOIC ACID):

A sulfur-containing essential amino acid with the chemical formula $\text{CH}_3\text{S}(\text{CH}_2)_2\text{CH}(\text{NH}_2)\text{COOH}$. It is one of the 20 amino acids commonly found in animal proteins and an important source of dietary sulfur, which the human body cannot synthesise from simpler metabolites. It is found in blue copper protein site. It is hydrophobic and acts as metal donor ligand.



METHOD: In object technology, it is the processing that an object performs. When a message is sent to an object, the method is implemented.

METHOD OF INITIAL RATES: A method of determining the rate-law expression by carrying out a reaction with different initial concentrations and analysing the resultant changes in initial rates.

METHOD OF LINEAR INTERPOLATION (SECANT METHOD): A variation of Newton's method for finding a root of the nonlinear

equation $f(x) = 0$. It is given by $x_{n+1} = x_n - f(x_n) \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right]$, $n \geq 1$.

METHOD OF MIXTURES: A method of determining the specific heat capacities of a substance or a liquid and a solid whose specific heat is known by mixing known masses of the substance at different temperatures and measuring the final temperature of the mixture or finding the equilibrium temperature.

METHOD OF SUCCESSIVE DISPLACEMENTS (GAUSS-SEIDEL ITERATION; LIEBMANN METHOD): An iterative method similar to the Jacobi iteration, which is used to solve a linear matrix system of equations, $\mathbf{Ax} = \mathbf{b}$ by repeatedly solving for $\mathbf{x}_{\text{NEW}}$ from

$$\mathbf{x}_{\text{NEW}(i)} = \left(\frac{1}{a_{ii}} \right) \left[\mathbf{b}_i - \sum_{j < i} a_{ij} \mathbf{x}_{\text{OLD}(j)} - \sum_{j > i} a_{ij} \mathbf{x}_{\text{NEW}(j)} \right]. \text{ This assumes that } a_{ii} \neq$$

0 and will not always converge.

METHODOLOGY: Methods or principles used in doing something or the study of research methods.

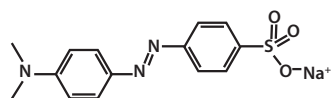
METHYL ALCOHOL (METHANOL; WOOD ALCOHOL): A simple primary alcohol with the chemical formula CH_3OH , melting point -93.9 °C (179 K), boiling point 64.96 °C (338 K) and relative density 0.79. It is a colourless, flammable liquid that is miscible with water in all proportions. It is made by the catalytic oxidation of methane from natural gas, using air. It is used as a solvent and added to ethanol to make methylated spirit. It is also used as a raw material for making methanol, as a solvent for varnishes and lacquers, as antifreeze and as a gasoline extender in the production of gasohol, a type of fuel used in motor vehicles.

METHYL ETHANOATE (METHYL ACETATE; ACETIC ACID METHYL ESTER): A colourless, volatile fragrant liquid with the chemical formula $\text{CH}_3\text{COOCH}_3$, density 0.932 g cm^{-3} , melting point -98 °C (175 K) and boiling point 56.9 °C (330 K). It is a typical ester which can be made from methanol and ethanoic acid and is used mainly as solvent in glues, paints and nail polish removers.

METHYL CHLOROFORM (1,1,1-TRICHLOROETHANE; TRICHLOROETHANE): A volatile colourless non-flammable liquid with a sweet odour and the chemical formula $\text{C}_2\text{H}_3\text{Cl}_3$, density 1.32 g/cm^3 , melting point -33 °C (240 K) and boiling point 74 °C (347 K). It is used as solvents for adhesives, pesticides and lubricants and in industrial cleaning solutions.

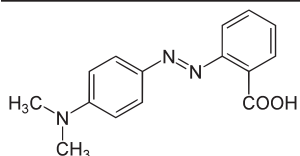
METHYL METHACRYLATE: A methyl ester of methacrylic acid, (2-methylpropenoic acid) with the chemical formula $(\text{C}_5\text{H}_8\text{O}_2)$, density 0.94 g/cm^3 , melting point -48 °C (225 K) and a boiling point of 101 °C (374 K). The compound is manufactured by several methods, the principal one being the acetone cyanohydrin (ACH) route, using acetone and hydrogen cyanide as raw materials. It is a thick, sticky, resin-like material, used in the preparation of the plastic polymethylmethacrylate and other acrylic resins and one of the main ingredients in the paints used to delineate the lanes of roads and highways and also used to make lenses of spectacle frames, false teeth, etc.

METHYL ORANGE: An acid-base or pH indicator used in chemical tests and as a stain in the preparation of slides of biological materials. It is pink when acidic, orange when neutral and yellow when basic. It changes from red below pH 3.1 to yellow above pH 4.4 at 25 °C. Therefore, it has its working range in acidic medium. It is frequently used in titrations involving weak bases.



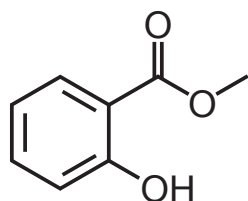
Chemical structure of Methyl Orange

METHYL RED (C.I. ACID RED 2; BENZENEDIAZONIUM 2-CARBOXYLATE): A red organic dye that turns red in pH under 4.4 orange between pH 4.4 and 6.00 and yellow in pH over 6.00, used as pH acid-base indicator. It changes colour in response to the nature of its chemical environment. Methyl Red may be prepared by diazotisation of anthranilic acid, followed by reaction with dimethylaniline.



Chemical structure of Methyl Red

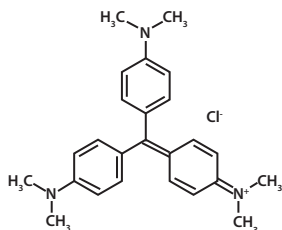
METHYL SALICYLATE (OIL OF WINTERGREEN; METHYL 2-HYDROXYBENZOATE; SALICYLIC ACID METHYL ESTER; BETULA OIL;): A methyl ester of salicylic acid with the chemical formula $C_6H_4OHCOOCH_3$, density 1.174 g/cm³, melting point -9 °C (264 K) and a boiling point of 220-224 °C (493-497 K). It is produced by plants as an anti-herbivore defence. Methyl salicylate can be produced by esterifying salicylic acid with methanol. It is used in medicine as a painkiller.



Chemical structure of Methyl Salicylate

METHYL TRICHLORIDE (CHLOROFORM; FORMYL TRICHLORIDE; METHANE TRICHLORIDE; METHENYL TRICHLORIDE; TRICHLOROMETHANE): A colourless volatile sweet-smelling liquid haloform with the chemical formula $CHCl_3$, relative density 1.48, melting point -63.5 °C (209.5 K) and boiling point 61.7 °C (334.7 K). It can be made by chlorination of methane followed by separation of the mixture of products or by the haloform reaction. It is an effective anaesthetic but can cause liver damage and it has now been replaced by other halogenated hydrocarbons. It is used as a solvent and raw material for making other compounds.

METHYL VIOLET (CRYSTAL VIOLET; GENTIAN VIOLET): A violet dye with the molecular formula $C_{24}H_{28}N_3Cl$. It is a mixture of compounds of rosaniline, made by oxidising dimethyl phenylamine with copper(II) chloride. It is used as a chemical indicator and as a biological stain, as a colouring agent in methylated spirits and textile dye.



Chemical structure of Methyl Violet

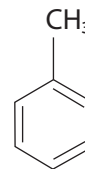
METHYLAMINE: A colourless flammable gas with the chemical formula CH_3NH_2 , melting point -94 °C (179.15 K) and a boiling point of -6 °C (267.2 K). It is produced by the decomposition of organic matter. It can be made by catalytic reaction between methanol and ammonia and is used in the manufacture of other organic chemicals such as dyes and insecticides. The simplest primary amine is a gas, which smells like ammonia and used in making herbicides.

METHYLATED SPIRITS: An alkanol or alcohol consisting mainly of ethanol with added methanol (9.5%), pyridine (0.5%) and blue

or violet dye. These additives are included to make the ethanol undrinkable so that it can be sold without excise duty for use as a solvent and a fuel for spirit burners.

METHYLATION: A chemical reaction through which a methyl group (CH_3) is introduced in a molecule. A particular example is the replacement of a hydrogen atom by a methyl group, as in a Friedel-Crafts reaction. In biological systems, methylation is catalysed by enzymes, which can be involved in modification of heavy metals, regulation of gene expression, regulation of protein function and RNA metabolism.

METHYLBENZENE (TOLUENE; PHENYLMETHANE): A colourless liquid aromatic hydrocarbon, derived from benzene by replacement of hydrogen atom by a methyl group. Its chemical formula is $CH_3C_6H_5$, melting point -95 °C (178 K) and boiling point 110.8 °C (394 K). It is insoluble in water but highly soluble in most organic solvents. It can be obtained from coal tar or made from methylcyclohexane (extracted from crude oil) by catalytic dehydrogenation. Its main uses are as a solvent and as a raw material for producing trinitrotoluene (TNT). It is also used in aviation fuel and other high-octane fuels, in dyestuffs, explosives and as a solvent for gums and lacquers.



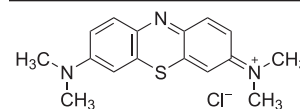
Chemical structure of Methylbenzene (Toluene)

METHYLCARBINOL (ALCOHOL; ETHANOL; GRAIN ALCOHOL; FERMENTATION ALCOHOL; ETHYL ALCOHOL; ETHYL HYDROXIDE): A colourless, limpid, volatile, flammable, toxic liquid with a pleasant odour and a pungent taste and the molecular formula C_2H_5OH , relative density 0.7851 at 20 °C, melting point at -112.3 °C (160.85 K) and boiling point at 78.4 °C (351.55 K). It is soluble in water and most organic liquids. It is produced by chemical synthesis and by fermentation or biosynthetic processes. It is the alcohol found in alcoholic drinks such as beer, wine and spirits. Apart from being a key input into alcoholic beverages, it is used as a solvent, extractant, antifreeze and intermediate in the synthesis of innumerable organic chemicals.

METHYLENE (CARBENE): The bivalent group of atoms derived from methane with the chemical formula CH_2 . It generally refers to a carbon atom bonded to two hydrogen atoms. The highly reactive carbene having a divalent carbon atom with six valence electrons has the formula CH_2 . The $-CH_2$ group includes dichloromethane also known as methylene chloride. The $=CH_2$ group includes methylenecyclopropene.

METHYLENE BLUE (TETRAMETHYLTHIONINE CHLORIDE; 3,7-BIS(DIMETHYLAMINO) PHENOTHIAZINIUM CHLORIDE):

A blue dye with the molecular formula $C_{16}H_{18}N_3S^+Cl^-$ used in optical microscopy to stain nuclei of animal tissues and also suitable as a vital stain and a bacterial stain. Pharmacologically, it is a weak germicide and used as urinary antiseptic.



Chemical structure of Methylene Blue

METHYLOTROPHIC MICROORGANISMS: Organisms that use, as substrates for growth, compounds containing carbon in a lower oxidation state than in carbon dioxide.

METHYLPHENOL (CRESOL): Organic compounds having a methyl group and a hydroxyl group bound directly to a benzene ring with the chemical formula $CH_3C_6H_4OH$. There are three isomeric methylphenols differing in the relative positions of the methyl and hydroxyl groups. A mixture of the three can be obtained by distilling coal tar. It is used as a germicide and antiseptic (a mild vaginal douche).

METHYLXANTHINES: Derivatives of xanthine in which one or more hydrogen atoms have been substituted by methyl groups. The common ones are the trimethylxanthine caffeine and the dimethylxanthines theophylline and theobromine.

METONIC CYCLE: A period of 19 years (235 lunar months), after which the new and full moons return to the same day of the year. It was the basis of the ancient Greek calendar, and is still used for calculating movable feasts such as Easter.

METRE: The SI or metric system unit of length defined as the path travelled by light in vacuum during a time interval of $1/2.99\,792\,458 \times 10^8$ second. This definition was adopted by the General Conference on Weights and Measures in October, 1983 to replace the 1967 definition based on the krypton lamp, which is 1650763.73 wavelengths in a vacuum of the radiation corresponding to the transition between the levels $2p_{10}5d_5$ of the krypton isotope having an atomic weight of 86. This definition in 1958 also replaced the older definition of a metre based on a platinum-iridium alloy bar of standard length kept at Sevres, France, near Paris. When the metre system was introduced in 1791 in France, the metre was intended to be 1/10,000,000 of the Earth's meridian quadrant passing through Paris or 1/10,000,000 of the distance between the equator and either pole, south or north. However, the original geodetic surveys proved the impartibility of such a standard and the original platinum metre bar, the metre des archives, was constructed in 1793.

METRE BRIDGE: A uniform resistance wire, 1 metre in length, mounted above a scale marked in millimetres, with terminals added to make the device usable as either part of a Wheatstone bridge or of a potentiometer. A practical version of the Wheatstone bridge is shown in the illustration. It consists of a wire AC of uniform cross-section area, 1m long, with a resistance of several Ohms and often made of constantan. Heavy brass strips A, B, C are used for the connections whose resistance are generally negligible. To function as a Wheatstone bridge, no current is allowed to flow through the galvanometer (G) by changing the position of the wire AC with the jockey D until no deflection is detected on G. The unknown resistance to be determined is R_1 and the known one is R_2 . The ratio of P to Q is altered by changing the position on the wire of the movable contact or jockey D. The position of D is adjusted until there is no deflection on the galvanometer (G). Then,

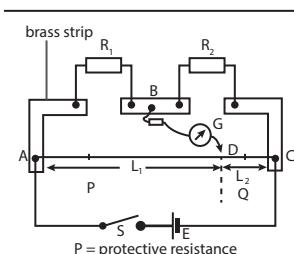
$\frac{R_1}{R_2} = \frac{P}{Q} = \frac{\text{resistance of AD}}{\text{resistance of DC}}$. Since the wire is uniform, its resistance will be proportional to its length and therefore,

$$\frac{R_1}{R_2} = \frac{AD}{DC} = \frac{L_1}{L_2}, R_1 = R_2 \times \frac{L_1}{L_2}.$$

Two practical points are noted. First, the value of R_2 is chosen to give a balance point near the centre of the wire, say between 40 and 60cm. Second, after obtaining a balance, R_1 and R_2 are interchanged and a second pair of values of L_1 and L_2 are obtained. The average values are then used to calculate the result.

METRIC: 1. Of or based on the metre or relating to or using the metric system of measurement. 2. A mathematical function defined for a coordinates system that assigns a value to each pair of elements. It is equal to the distance between them or to a property analogous to distance between points on a line. 3. A standard or a statistic for measuring or quantifying something.

METRIC DENSITY THEOREM: A fundamental theorem of integration, used to determine whether a set of points is measurable and therefore provides condition necessary and sufficient for the existence of integrals in the sense of Lebesgue.



Using a metre bridge to measure an unknown resistance

METRIC OUTER DENSITY: For a given set, A in Euclidean space with respect to Lebesgue outer measure, μ^* , at a point x , the limit, if it exists, as $\mu^*(I_x)$ tends to zero, of the ratio $\mu^*(A \cap I_x)/\mu^*(I_x)$ over all closed intervals I_x containing x . A point with unit outer density is called a **point of density** for A and a point with zero outer density is called a **point of dispersion** for A. Almost all points of any set are points of density of the set. The set is measurable if and only if almost all points in its complement are points of dispersion of the set. If the set is measurable, it is simply called a **metric density**.

METRIC PREFIXES: Symbols used to specify orders of magnitude in the metric system of measurement, using factors of 10^3 between steps. The mostly used prefixes are tera (10^{12}), giga (10^9), mega (10^6), kilo (10^3), hecto (10^2), deca (10^1), deci (10^{-1}), centi (10^{-2}), milli (10^{-3}), micro (10^{-6}), nano (10^{-9}) and picro (10^{-12}).

METRIC SPACE: A set that has a metric; in other words, a kind of space in which the concept of distance has meaning. It induces a topology on the set in which Ω is open if and only if for all X in Ω , there is a positive ϵ such that the open ball $B_\epsilon(x)$ is contained in Ω . The 3-dimensional Euclidean space is an example of metric space.

METRIC SYSTEM: Any physical measuring system such as centimetre-gram-second system (cgs) whose units and subunits are related by multiples of ten. The centimetre, gram and second (the cgs system) is also a metric system. The centimetre-gram-second system (cgs) is superseded by the metre-kilogram-second (MKS) system on which SI units are based for precise scientific work to be carried out.

METRIC TENSOR: A type of function that is defined on a manifold (such as a surface in space) which takes as input a pair of tangent vectors v and w and produces a real number (scalar) $g(v, w)$ in a way that generalises many of the familiar properties of the dot product of vectors in Euclidean space. Metric tensors are used to define the length of and angle between, tangent vectors in the same way as a dot product. A positive-definite metric tensor assigns a positive value to every non-zero vector. A manifold equipped with a positive-definite metric tensor is known as a **Riemannian manifold**. By integration, the metric tensor allows one to define and compute the length of curves on the manifold.

METRIC TON (TONNE): A unit of mass, equal to 1000 kg or approximately the mass of one cubic metre of water at 4 °C. Its symbol is t .

METRICAL: Of or relating to measurement.

METRICATION: The conversion of, for example, an instrument or a measuring system to the metric system.

METRITIS: An infection of the lining of the womb in cattle. The symptoms include a white discharge and/or high temperature.

METRIZABLE: In mathematics, a metrizable topology is one that is compatible with a metric, so that one can define a suitable metric that induces it.

METROLOGY: The scientific study of measurement especially the definition and standardisation of the units of measurement used in science. It is an important part of semiconductor manufacturing, where high-resolution tools are required to measure the tiny elements in a chip that continues to decrease with time.

METRONE: A device used to indicate a given tempo in which work is to be performed by means of a regularly recurring audio or visual signal. The mechanical metronome consists of an up and down pendulum that swings back and forth and ticks with each stroke. Electronic metronomes are more accurate and more durable than mechanical metronomes. They indicate the set tempo with an audible click, a blinking light or both.

METROPOLITAN-AREA NETWORK (MAN): A type of internet network that is larger than Local-Area Networks (LANs), but smaller than Wide Area Networks (WANs), used across institutions or companies with multiple buildings, a town or city.

MEVALONIC ACID: A precursor in the formation of terpenoids and steroids. It has the formula $\text{CH}_2\text{OHCH}_2\text{C}(\text{OH})(\text{CH}_3)\text{CH}_2\text{COOH}$. Biosynthesis of gibberellins, cytokinins and abscisic acid is from mevalonic acid. Certain growth retardants, for example, Amo-1618 and CCC (chlorcholine chloride), are believed to act by blocking the step from geranyl pyrophosphate to kaurene in the mevalonic acid synthesis pathway, so preventing gibberellin synthesis.

MEYER, VIKTOR (1848-97): German chemist, best known for inventing an apparatus for determining vapour densities, the Viktor Meyer apparatus. He also discovered the sulfur compound thiophene, a heterocyclic compound. Meyer also worked on stereochemistry and was the first to identify steric hindrance in chemical reactions.

MEZZANINE BOARD (DAUGHTERBOARD; PIGGYBACK BOARD; DAUGHTER CARD; RISER CARD): In computing, it is a small printed circuit board that connects directly into a motherboard to give it new capabilities.

MHC CLASS I PROTEIN: Any of a class of histocompatibility proteins encoded by the major histocompatibility complex that are expressed by all nucleated cells. Proteins such as normal proteins (self) and microbial invaders (nonself) are continually synthesised in the cell. When an antigen enters a body cell, these proteins within the cell attach themselves to the antigen and transport it to the surface of the cell, where they are presented to T lymphocytes. Some MHC molecules, called class I MHC proteins, present antigens to killer T cells. Other MHC proteins, called class II MHC proteins, present antigens to helper T cells for destruction. Cytotoxic tissue grafts from a mismatched donor are marked for rejection by the recipient's immune system by virtue of the nonself MHC class I proteins carried by all tissue cells.

MHC CLASS II PROTEIN: Any of a class of histocompatibility proteins encoded by the major histocompatibility complex whose expression is restricted to certain immune cells, in particular B cells, macrophages, other antigen-presenting cells, epithelial cells of the thymus and in humans activated T cells. These proteins bind antigenic peptide fragments. Class II proteins consist of an alpha chain and a beta chain and have a peptide-binding site that is open at both ends and can accommodate longer peptides than MHC class I molecules.

MHO: A MKS unit of electrical conductance which is equivalent to 1 siemens or one ampere per volt, used primarily in electronic applications. It is equivalent to the reciprocal of the ohm unit. Mho is derived from spelling ohm anagrammatically backwards and its symbol is inverted ohm sign, $\oslash$. It is now replaced by the siemens in the International System of Units (SI), with the symbol S.

MICA: Any of a group of hydrous aluminium silicate minerals found in flat transparent sheets or with a layered structure. They are composed of linked SiO_4 tetrahedra with cations and hydroxyl groupings between the layers. Mica is widely distributed and occurs in igneous, metamorphic and sedimentary regimes. Large crystals of mica used for various applications are typically mined from granitic pegmatites. The most common micas are muscovite $[\text{KAl}_2(\text{AlSi}_3\text{O}_{10})(\text{OH})_2]$, paragonite $[\text{NaAl}_2(\text{AlSi}_3\text{O}_{10})(\text{OH})_2]$, phlogopite $[\text{K}(\text{Mg},\text{Fe})_3(\text{AlSi}_3\text{O}_{10})(\text{OH})_2]$, biotite $[\text{K}(\text{Fe},\text{Mg})_3(\text{AlSi}_3\text{O}_{10})(\text{OH})_2]$ and lepidolite $[\text{K}(\text{Li},\text{Al})_{2.5-3.0}(\text{Al}_{1.0-0.5}\text{Si}_{3.0-3.5}\text{O}_{10})(\text{OH})_2]$. Micas have perfect basal cleavage and the thin cleavage flakes are flexible and elastic. They are used as electrical insulators. They are good conductors of heat and used in some electrical heaters and capacitors.

MICELLAR CATALYSIS: The acceleration of a chemical reaction in solution by the addition of a surfactant at a concentration higher than its critical micelle concentration so that the reaction can proceed in the environment of surfactant aggregates (micelles).

MICELLAR MASS: The term usually refers to a neutral micelle and therefore includes an equivalent amount of counterions with the surfactant ions.

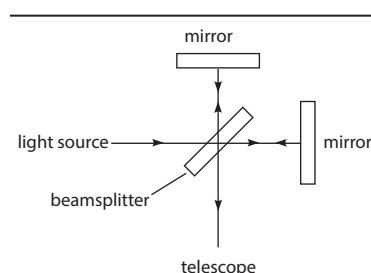
MICELLAR SOLUBILISATION (SOLUBILIZATION): In a system formed by a solvent, an association colloid and at least one other component (the solubilise), the incorporation of this other component into or on the micelles is called micellar solubilisation or, briefly, solubilisation.

MICELLE: An aggregate of electrically charged ions or molecules in a colloid or suspension of soap, detergent and other suspensions or fat digestion dispersed into a colloid by the action of bile salt. A micelle is either polar or non-polar. Micelle is attracted to a similar-type structure in the continuous phase whether it is polar or non-polar. The charges built up on the surface through electrostatic molecular attractions are like charges and therefore repel, separating the particles. The non-polar hydrocarbon groups are in the centre and the hydrophilic polar groups are on the outside solvated by the water molecules. A micelle may take several forms such as distorted spheres, disks or rods, depending on the conditions and composition of the system.

MICHAELIS-MENTEN CURVE: A graph showing the relationship between the concentration of a substrate and the rate of the corresponding enzyme-controlled reaction. The graph can be used to calculate the Michaelis constant, which is the concentration of a substrate required in order for an enzyme to act at half of its maximum velocity. Its symbol is K_m .

MICHAELIS-MENTEN EQUATION: An equation which shows the hyperbolic relationship between the initial velocity, V_o , and the substrate concentration, $[S]$ in a number of enzyme-catalysed reactions such that $V_o = V_{\max}[S]/K_m + [S]$, where $V_{\max}$ is the maximum velocity and K_m is the Michaelis constant.

MICHELSON-MORLEY EXPERIMENT: An experiment, conducted in 1887 by Albert Michelson (1852-1931) and Edward Morley (1838-1923) that attempted to measure the velocity of the Earth through the luminiferous aether. Using a modified Michelson interferometer they expected to observe a shift in the interference fringes formed, when the instrument was rotated through 90° , showing



Michelson-Morley Interferometer

that the speed of light measured in the direction of the Earth's rotation or orbital motion, is not identical to its speed at right angles to this direction. No shift was observed. An explanation was subsequently provided by the Lorentz-Fitzgerald contraction, which provided an important step in the formulation of Einstein's special theory of relativity and the abandonment of the aether concept.

MICRO-: A prefix used in the metric system to denote one millionth (10^{-6}), symbol μ . For example, 10^{-6} metre = 1 micrometre (μm). 2. A prefix denoting very small size, for example, microgamete, micronucleus. 3. In botany, it refers to an axial embryo in minute seeds, less than 0.2 mm long; minute and undifferentiated to almost total size of seed.

MICRO CAVITY: A cavity with reflecting faces on the two sides of a spacer layer or optical medium that is so small that specifically quantum mechanical aspects of quantum electrodynamics, such as a spontaneous emission, can be investigated. They range in sizes from nanometres to micrometres.

MICRO CHANNEL ARCHITECTURE (MCA): The rules that define how subsystems and adapters use the Micro Channel bus in a computer. The architecture defines the services that each subsystem can or must provide.

MICRO LIGHT AIRCRAFT: Very light aircraft with a small engine rather like a powered hand-glider.

MICRO RNA (miRNA): A small RNA molecule that is 21-25 nucleotides long and may be single- stranded or double-stranded encoded by a cell found in animals, including *Drosophila* and *Caenorhabditis elegans* and in plants. They are transcribed as parts of longer RNA molecules and processed in the nucleus by the dsRNA ribonuclease Droscha to hairpin structures 70-100 nucleotides long. These are transported to the cytoplasm where they are digested to 21-23-mers by the dsRNA ribonuclease Dicer. Single-stranded miRNAs bind to complementary sequences in mRNA thereby inhibiting translation.

MICROAEROPHILE: A microorganism that lives in environment that only has a low level of oxygen. Some microaerophiles need some oxygen because they cannot ferment or respire anaerobically and need oxygen when carrying out aerobic respiration. Examples of microaerophiles are *Borrelia burgdorferi*, a species of spirochaete bacteria that causes lyme disease in humans, and *Helicobacter pylori*, a species of proteobacteria that has been linked to peptic ulcers and some types of gastritis.

MICROARRAY: A glass slide or silicon thin-film cell on which are deposited biomolecules or other materials in a regular micro-scale pattern to enable automated simultaneous multiple assays of target substances or activities. They are powerful analytical tools with various types, which include DNA microarrays, which include cDNA microarrays, oligonucleotide microarrays and SNP microarrays; MMChips, for surveillance of microRNA populations; protein microarrays; tissue microarrays; cellular microarrays or transfection microarrays; chemical compound microarrays; antibody microarrays; and carbohydrate arrays or glycoarrays. The microarray can be read by a scanner, which automatically assesses the amount of label at each address and hence the amount of target substance.

MICROAGGREGATE: 1. Clusters of soil particles composed of diverse mineral, organic and biotic materials that are bound together by various physical, chemical and biological processes. Microaggregates are between 2-20 μm in diameter. 2. Microscopic collection of particles, such as platelets, leukocytes or fibrin that occurs in blood stored in a refrigerator.

MICROBALANCE: A sensitive balance instrument capable of weighing masses of the order 10^{-6} to 10^{-9} kg or a balance for precisely weighing extremely small quantities up to 0.0001 g.

MICROBIAL BIOMASS: Total mass of microorganisms living in a given mass or volume of soil organic matter. It is generally expressed in the milligrams of carbon per kilogram of soil or micrograms of carbon per gram of dry weight of soil.

MICROBIAL POPULATION: Total number of microorganisms living in a given mass or volume of soil organic matter.

MICROBIAL INSECTICIDE: Insecticide against fungi, bacteria or other microorganisms that are pathogens or agents of diseases of insects. An example is *Verticillium lecani*, used to control whitefly in glasshouse.

MICROBIAL LEACHING: A process for the solubilisation of metals, mostly from poor ores, by lithotrophic bacteria.

MICROBIAL PROTEIN: A protein source in ruminant from dead rumen microbe, usually forming 70% to 100% of the ruminant's supply of protein.

MICROBIOLOGY: The scientific study of microorganisms such as fungi, protists and prokaryotes, which can only be seen under the microscope. Microbiology includes virology, mycology, parasitology, bacteriology and other branches. Viruses, though not strictly classed as living organisms, are also studied. Microbes are now recognised as important vehicles for the study of biochemical and genetic processes common to all living organisms and their rapid growth enables their laboratory cultures in large numbers for studies in genetics.

MICROBODY: Any of a class of organelles, 0.2-1.5 μm , including peroxisomes, glyoxysomes and microperoxisomes that typically are spherical vesicles, bounded by a single membrane and contain a variety of enzymes responsible for the oxidation of various materials. They originate from the endoplasmic reticulum.

MICROCARRIER: A small, beaded material, derived from silica, glass, dextran or similar materials, used for the immobilisation of biocatalysts or as a support for the culture of anchorage-dependent animal cell lines.

MICROCHIP: A series of electrical circuits built into a tiny wafer of silicon or another semiconductor and typically having millimetre dimensions, fabricated in such a way that it can perform a large number of independent electronic functions. The circuits may be made by exposing the chip to a high temperature vapour of controlled composition. The vapour deposits a thin layer on the silicon. In this way complex layers of materials such as those found in transistors can be built up in a very small area. Some of the uses include in the manufacture of mobile phones and other telecommunication devices, data processing and in other electronic devices.

MICROCIRCULATION: The part of the blood circulatory system comprising of the networks of capillaries and vessels which are embedded within organs and are responsible for the distribution of blood within tissues. They also regulate blood flow and tissue perfusion, blood pressure, tissue fluid, delivery of oxygen and other nutrients and removal of CO_2 and other metabolic waste products and body temperature. At its end, each artery gives rise to a microcirculatory bed of increasingly narrow vessels, arterioles, which in turn subdivide to form metaarterioles and then capillaries. Capillaries are typically 3-10 μm in diameter, just wide enough for erythrocytes to pass through. Blood flow into the capillary bed is controlled by a muscular precapillary sphincter at the end of each arteriole and blood can bypass the capillaries via a direct arteriovenous connection or anastomosis.

MICROCLIMATE: The local atmospheric zone, of a particular habitat such as a woodland or lake, where the climate differs from the larger surrounding geographical area. Microclimate also exists in urban areas where brick, concrete and asphalt absorb the Sun's energy, heat up and radiate the heat to the ambient air.

MICROCLIMATOLOGY: The science that deals with the climate of restricted areas and investigates their phenomena and causes.

MICROCLINE: A common mineral in the feldspar group, occurring in igneous rocks as vitreous crystals of various colours. It has the same chemical composition as orthoclase and differs from it only in the arrangement of the silicon and aluminium atoms.

MICROCODE: Permanent memory that holds the elementary circuit operations a computer must perform for each instruction in its instruction set.

MICROCOMPUTER: A computer in which the central unit is implemented by means of a single semiconductor integrated circuit chip and designed for individual use. Its basic components include central processing unit (CPU), program memory, data memory, output

ports, input ports, monitor, keyboard and mouse. The power of a semiconductor is determined by the speed and power of the processor and the features of the other components in the computer such as the storage capacity of the main memory, the disks used as backing store, the operating system and other software used.

MICROCOSM: A community or any other unit that is representative of a larger community.

MICRODISSECTION: A technique used for the dissection of living cells under the high power of an optical microscope without opening up the body. Tiny mechanically manipulated instruments, such as needles, scalpels, micropipettes and lasers are used. Some examples are **chromosome microdissection**, which uses fine glass needle under a microscope to remove a portion from a complete chromosome; **laser microdissection**, which uses a laser through a microscope to dissect selected cells; and **laser capture microdissection**, which uses a laser through a microscope to cause selected cells to adhere to a film.

MICROELECTRODE: A tiny device filled with a physiologically appropriate electrolyte solution used to electrically or chemically stimulate a living cell and then record the electrical activity that it caused within the cell. The tip is inserted through the plasma membrane into cells without major disturbance to cellular function.

MICROELECTRONICS: A branch of electronic technology concerned with designing and making electronic circuits which are extremely small and consume very little electric power or the use of tiny electric current in integrated circuits or chips to perform tasks in information technology and control.

MICROELECTROPHORESIS: The electrophoresis technique involving the movement of a mass of particles on a small scale, for example, paper electrophoresis.

MICROENVIRONMENT (MICROHABITAT): A small, specific area in a habitat which a microorganism inhabits that differs from the larger environment by certain factors such as vegetation, types of nutrients and availability, temperature, degree of sunlight, soil physical characteristics, etc.

MICROEVOLUTION: Small scale changes in allele frequencies in a population, over a few generations which involve the emergence of new species level, such as race and subspecies. The processes involved are mutation, natural and artificial selection, gene flow and genetic drift.

MICROFAUNA: Animals such as protozoa and nematodes that are too small to be seen with the naked eye and can only be seen by microscope.

MICROFIBRIL: A fibre-like strand or thin thread, about 5nm in diameter consisting of glycoproteins and cellulose found in muscles and the cytoplasm of all cells or a microscopic fibre of cellulose found in plant cell walls. Each microfibril consists of 50-60 parallel cellulose chains that are associated together to form a rod or flat ribbon. Cellulose microfibrils are arranged in layers at right angles to each other.

MICROFILAMENT (ACTIN FILAMENTS): Any of numerous microscopic protein fibres found in the cytoplasm of all eukaryotic cells. Each microfilament is about 7-9 nm in diameter consisting of two helically twisted strands each comprising a chain of globular subunits of the protein actin. They can extend or shorten by the removal or addition of subunits and are linked by cross linking proteins to form three dimensional networks. They are also involved in the transport of materials within the cell and the peripheral flow of cytoplasm and cell organelles known as **cytoplasmic streaming**.

MICROFILTRATION: Pressure-driven membrane-based separation process in which particles and dissolved macromolecules larger than 0.1µm are rejected.

MICROFLORA: 1. Plants and algae that cannot be seen with the naked eye and are observed only under a microscope. 2. Plants and algae that live in a particular microhabitat

MICROFOSSIL: A fossil that ranges in size from less than one millimetre to four millimetres and can only be studied under a microscope. They include bacteria, diatoms and protozoa and parts of organism, such as plant pollen and skeletal fragments. Microfossils are important in the correlation of rocks where only small samples are available. The study of spores and pollen, including fossilised spores and pollen is called **palynology**.

MICROGAMETOGENESIS: In botany, it refers to the process in reproduction where a microgametophyte develops in a pollen grain to the three-celled stage of its development. In flowering plants, it occurs with a microspore mother cell inside the anther of the plant.

MICROGAMETOPHYTE: A stage in the plant life cycle that develops from or within a microspore. It produces sperm in specialised structures known as **antheridia**.

MICROGEL: Particle of gel of any shape with an equivalent diameter of approximately to 0.1µm to 100µm.

MICROGRAPH: A photograph of a specimen taken through a microscope. Photomicrographs and electron micrographs are produced using light microscopes and electron microscopes respectively.

MICROGRAVITY (MICRO-G ENVIRONMENT; µg): 1. The very weak gravitational force that occurs on board a spacecraft in orbit round the Earth. 2. A minute shift in gravity that can occur through geologic factors in a region, such as the movement of the Earth's crust along fault lines.

MICROHABITAT: The local environment that has a unique set of ecological conditions within a larger habitat that supports distinct flora and fauna. A stream microhabitat will have different microhabitats, depending on oxygen content, pH, speed of water flow and other factors.

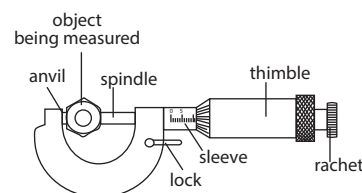
MICROINJECTION: A procedure for injecting substrate cells, cell organelles or other microscopic structures in which a needle about 0.5 to 5 micrometres in diameter is inserted through the cell membrane or the nuclear envelope to introduce a substance such as DNA or dye to help visualise cell components.

MICROMELITOPHILY: The process of pollination by small bees.

MICROMETEORITE: A small extraterrestrial particle that has survived entry into Earth's atmosphere, usually weighing less than a gram.

MICROMETEOROLOGY: The study of the meteorological processes on scales from a millimetre or less up to tens or hundreds of metres, for example, meteorology of a local site that is usually small and often is confined to a shallow layer of air next to the ground.

MICROMETER SCREW GAUGE: A gauge for measuring small diameters or thickness accurately. It consists of a g-shaped device in which a gap between the measuring faces is adjusted by means of an accurately calibrated screw whose end forms one of the measuring faces. This instrument can measure



A micrometer screw gauge

accurately to the nearest 0.01mm. Its operation is described as follows: the object to be measured is placed between the anvil and the spindle and then the ratchet is turned on until the spindle screws up tight against the object to be measured. The spindle moves through a fixed distance during each revolution or one complete turn because of its uniform pitch. The sleeve is graduated in millimetres along the upper scale and half millimetres along the lower scale. The thimble is calibrated with divisions of 0.01mm. The reading for the object is obtained on the sleeve.

MICROMETRE: A unit of length (formerly called a micron, symbol μ), which is equal to one thousandth of a millimetre (10^{-6} metre). It is used in the measurement of cells, cell inclusions, bacteria, etc. Its symbol is μm .

MICROMINIATURISATION (MICROMINIATURIZATION): Reduction in size and weight to an extremely small scale, used especially for electronic components.

MICRON: A unit of linear measurement equivalent to one-millionth (10^{-6}) of a metre.

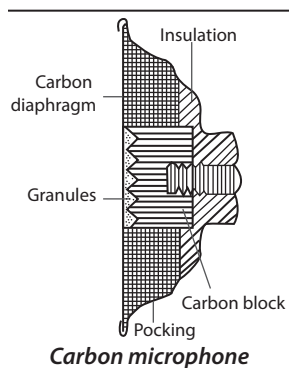
MICRONUCLEUS: 1. The small nucleus in ciliate protozoans, such as the paramecium containing a single copy of the genome that is used for sexual reproduction. 2. The small nucleus that is formed when a chromosome or a fragment of a chromosome is not incorporated into one of the daughter nuclei during cell division.

MICRONUTRIENTS (TRACE NUTRIENTS; MICROMINERALS; TRACE ELEMENTS): Dietary chemical elements required by animals and plants in relatively small quantities throughout life, found in cofactors and coenzymes. They include iron, cobalt, chromium, copper, iodine, manganese, selenium, zinc and molybdenum, etc.

MICROORGANISM (MICROBE): Any non-cellular (no cell) or unicellular (single celled) organism such as bacteria including cyanobacteria, lichens, microfungi, protozoa, rickettsiae, viroids, viroids and viruses, some algae and all prokaryotes. They can be observed only under a microscope. Microorganisms play important roles for humans and other organisms and the environment, such as the carbon and nitrogen cycles and in decomposition. Some are used in fermentation processes such as brewing, winemaking, baking and also in the production of dairy products such as yogurt and cheese, ethanol, biogas and methane. Other uses include in sewage treatment, cultivated or grown by scientists for use in researches in biotechnology, biochemistry, genetics and molecular biology. It is also used in biological warfare. Some microorganisms are also the cause of many infectious diseases such as plague, tuberculosis, anthrax, malaria, sleeping sickness, ringworm, candidiasis, influenza, yellow fever and AIDS.

MICROPHAGOUS: Describing the method of feeding such as ciliary feeding and filter feeding in which heterotrophic organisms take their food in the form of tiny particles or microorganisms.

MICROPHONE: A device which converts sound waves to a changing electric current for amplification and transmission. It basically consists of a diaphragm that responds to the pressure or particle velocity of sound waves. It is used in radio broadcasting, recording and sound amplifying systems. The carbon microphone is a type of microphone in which a diaphragm, vibrated by sound waves, applies a varying pressure to a container packed with carbon granules, altering the resistance of the carbon. A current flowing through the carbon is thus modulated at the frequency of the sound waves.



MICROPHONICS: The transmittance of mechanical vibrations through an electrical component. Basically, it is the noise caused by mechanical vibration of the elements of an electron tube, component or system.

MICROPHOTOMETER: The optical instrument used to measure transmittance in atomic emission spectroscopy.

MICROPHYLL: A type of foliage leaf that has a single unbranched midrib. Such leaves are generally no more than a few millimetres long. Examples are found in clubmosses and horsetail.

MICROPIPETTE: An instrument with an ultrafine tip, typically less than 1 mm in diameter and usually held by a micromanipulator. It is

used to accurately measure and dispense small volumes of liquid. It can be inserted into single cells or other microscopic structures and used, for example, to inject materials.

MICROPORE: Pore with diameter not exceeding 2mm, normally found within structural aggregates. Materials with micropores are often used in contaminated environments to facilitate contaminant-free exchange of gases; for example, mould spores, bacteria, and other airborne contaminants will become trapped, while allowing clean gases to pass through, creating a sterile environment within the contained area.

MICROPORE FILLING: The process in which molecules are adsorbed in the adsorption space within micropores.

MICROPOROUS CARBON: A porous carbon material, usually a char or carbon fibres, which may or may not have been subjected to an activation process to increase its adsorptive properties. A microporous carbon is considered to have a major part of its porosity in pores of less than width and to exhibit apparent surface areas usually higher than 200 to 300 $\text{m}^2 \text{g}^{-1}$.

MICROPROCESSOR: An electronic circuit containing the arithmetic, logic and control circuitry required to interpret and execute instructions from a computer program. It is used as the central processing unit (CPU) of a computer. They can be used to control appliances such as hand held calculators and washing machines in a single integrated circuit.

MICROPROPAGATION: The propagation of plants by cloning a small piece of plant tissue cultured in a growth medium. This involves culturing excised meristematic tissue on a special medium that enhances adventitious shoot development. The new shoots are then separated and cultured and the cycle is repeated until finally the shoots are transferred to a medium that promotes root development to produce plantlets. Micropropagation is used in agriculture, horticulture and forestry as special genotypes can be bred and maintained. The process is rapid and plants can be kept disease free at the initial stage. It is also used to provide a sufficient number of plantlets for planting from a stock plant which does not produce seeds or does not respond well to vegetative reproduction.

MICROPYLE: 1. A small opening or a pore in the ovule of a flower through which the pollen tube enters to release the male gametes for fusion with the female gamete. 2. A pore in the ovum of insects through which the spermatozoa enters.

MICRORELIEF: Small-scale, local differences in the elevation of land, including mounds, swales, or hollows.

MICROSATELLITE: A satellite with an on-orbit mass of 10 to 100 kg.

MICROSATELLITE DNA: A form of highly repetitive DNA consisting of large numbers of very short base sequence scattered throughout the inherited eukaryote genome. Each consists of a short base sequence, generally 2-6 base pairs (bp) long that is repeated 5-30 times in a tandem array at any given locus. For example, one individual might have 17 consecutive repeats of the sequence CA (Cytosine-Adenine) at a given locus, while another individual might have 15 repeats. Overall, the same sequence might occur at different loci and be repeated tens of thousands of times throughout the genome.

MICROSCOPE: A scientific instrument with high resolution for detail, used to magnify image of minute objects so that they can be seen properly and examined scientifically. The simple microscope consists of a biconvex magnifying glass or an equivalent system of lenses, either hand held or in a simple frame. The compound microscope uses two lenses or systems of lenses, the second magnifying the real image formed by the first. The lenses are usually mounted at the opposite ends of a tube that has mechanical controls to move it in relation to the object. An optical condenser and mirror, often with a separate light source, provide illumination of the object. The widely used binocular microscope consists of two separate instruments fastened together so that one eye looks through one while the other eye looks through the other.

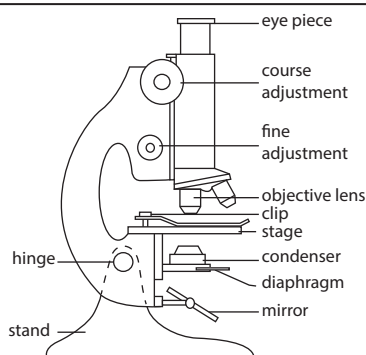


Fig. I: A simple light microscope

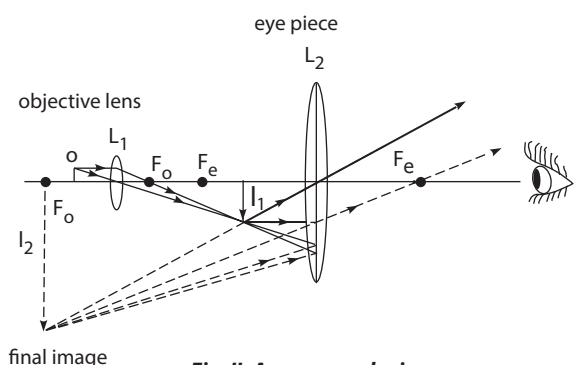


Fig. II: A compound microscope

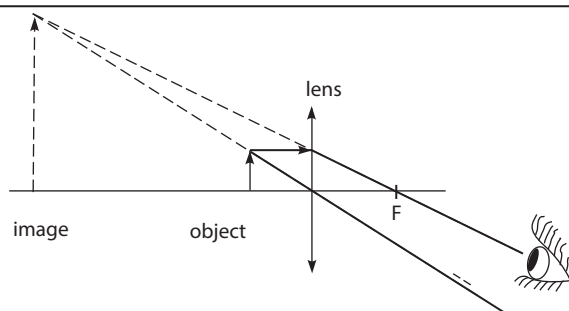


Fig. III: A simple microscope

MICROSCOPIC: Too small to be seen by the unaided eye but large enough to be studied under a microscope or designating a size scale comparable to the subatomic particles, atoms and molecules.

MICROSCOPIC ELECTROPHORESIS: A technique in which the electrophoresis of individual particles is observed with the aid of a microscope or ultra-microscope.

MICROSCOPIC FILM: A film which has lateral dimensions in the range 0.1 – 100 μm .

MICROSCOPIC REVERSIBILITY: A law formulated by R. C. Tolman in 1924 that in a reversible reaction the mechanism in one direction is the exact reverse of the mechanism in the other direction.

MICROSITE: 1. In ecology, it is a small area or habitat where the biological or chemical processes are different from the immediate areas. It is caused by factors such as vegetation, types of nutrients and availability, temperature, degree of sunlight, soil physical characteristics, etc. 2. In information technology, it is a small auxiliary website designed to function as a supplement to a primary website.

MICROSOFT ACCESS (ACCESS): A computer database application software, used for the creation of custom databases to store information in an organised structure based on the Access Jet Database Engine. It also provides a graphical user interface and software development tools.

MICROSOFT DISK OPERATING SYSTEM (MS-DOS): The first operating system developed by Microsoft for IBM-compatible personal computers. It is a command-line-based system, where all commands are navigated by using MS-DOS commands in text form or allows the user to type commands. It is not much used today, as it has been largely replaced by graphical user interfaces, especially Microsoft Windows.

MICROSOFT NETWORK (MSN): A collection of Internet sites and services provided by Microsoft that includes news, sports and entertainment as well as the Bing search engine (www.msn.com). Integration with Windows Live, Facebook and Twitter is also provided. It was launched in concert with Windows 95 as a dial-up service similar to AOL and the brand name for Microsoft's online search, Hotmail e-mail and Messenger instant messaging services.

MICROSOFT OFFICE: Microsoft's suite of desktop business applications for Windows and Macintosh (Mac). The suite comprises a Word-processor (Word), a Spreadsheet (Excel), a Presentation Program (PowerPoint), an Email package (Outlook), a Database program (Access) and a Desktop Publishing package (Publisher).

MICROSOME: A small particle obtained after isolating a cell or ribosomes associated with a fragment of endoplasmic reticulum formed when cells or tissues are disrupted. Microsomes can be isolated by centrifugation and are commonly used to investigate the functional properties of endoplasmic reticulum, such as enzymatic activity and protein synthesis.

MICROSPORANGIA: Sporophyte structures in which microspores are produced by meiosis. In flowering plants the microsporangia are known as **anther sacs**. In gymnosperms and angiosperms or flowering plants, the microsporangium produces the microsporocyte or microspore mother cell. The term is not used for bryophytes.

MICROSPORANGIOPHORE: A stalk-like structure bearing microsporangia.

MICROSPORE: The smaller of the two types of spore formed after meiosis in heterosporous species and usually designated male. In seed plants it is the pollen grain. Microspores are the four haploid cells produced by the meiotic division in the pollen sacs of flowers or microsporangia of gymnosperms. Microspores undergo mitotic division and form pollen grains in seed plants (angiosperms), which are usually less than 200 micrometres. They develop into male gametophytes and megaspores develop into female gametophytes.

MICROSPORE MOTHER CELL (MICROSPOROCYTE): A diploid cell in plants that undergo meiosis to produce four microspores. In flowering plants microspore mother cells are formed within the pollen sac of anthers by meiosis and the microspores they produce develop into pollen grains.

MICROSPOROPHYLL: A modified leaf that bears the microsporangia. In angiosperms and gymnosperms it is represented by the stamens and male scales respectively.

MICROTOME: An instrument used to cut extremely thin slices of plant or animal tissue for microscopical observation. There are various designs of microtome, each basically consisting of a steel knife, a block for supporting the specimen and a device for moving the specimen towards the knife. The specimen is usually supported by being embedded in wax. If a freezing microtome is used, the specimen is frozen. An ultramicrotome is used to cut much thinner sections (20-100nm thick) of materials for electron microscopy.

MICROTUBULE: A microscopic hollow rod of tubulin protein structure enclosed by a membrane, with an external diameter of 24 nm and of variable length found within cytoplasm, cilia, flagella of animal and plant cells. Microtubules are composed of numerous subunits of the globular protein tubulin and occur singly or in pairs or bundles. They help to maintain the shape and movement of a living cell and the transportation of material within it and participate in the formation of the spindle during mitosis or cell division.

MICROTUBULE-ORGANISING CENTRE (MTOC): A structure found in eukaryotic cells from which microtubules emerge. Most animal cells have a single MTOC, the centrosome, which is located near the nucleus and is also involved in organising the spindle during nuclear division. A plant cell has hundreds of MTOCs, producing the girdle-like arrays of microtubules running around the cell's inner periphery.

MICROVILLUS (MICROVILLI, pl.): One of a number of microcell projections on the free surface of epithelial cells that increase the surface area of cells and are involved in functions such as absorption, secretion, cellular adhesion and mechanotransduction. Microvilli are covered with plasma membrane and their cytoplasm is continuous with the main cell cytoplasm. They are found in abundance on the villi of the intestine, where they form a brush border; on the cell surface of white blood cells, aiding in the migration of white blood cells and also in sensory cells of the inner ear in the cells of taste buds and in olfactory receptor cells.

MICROWAVE (MICROWAVE RADIATION): High frequency electromagnetic waves with a frequency range from 1,000 MHz (megahertz) to 300,000 MHz with a wavelength of 1mm to 30cm. They are used in radar, radio transmissions and microwave heating and cooking.

MICROWAVE BACKGROUND RADIATION, MBR (CMB RADIATION, CMBR; RELIC RADIATION): A cosmic background radiation in the frequency range of 3×10^{11} hertz to 3×10^8 hertz, discovered in 1965 by American radio astronomers Arno Penzias and Robert Wilson. It is believed to have been a radiation created during the Big Bang. The radiation has an energy density in intergalactic space of some $4 \times 10^{-4} \text{ J m}^{-3}$. Detailed study of this radiation has been very important in modern cosmology.

MICROWAVE OPTICS: The study of the behaviour of microwaves by comparison with the behaviour of liquid waves.

MICROWAVE SPECTROSCOPY (ROTATIONAL SPECTROSCOPY): The study in the electromagnetic radiation spectrum, especially in the microwave region, of absorption and emission of electromagnetic radiation by molecules associated with a corresponding change in the rotational quantum number of the molecule. Rotational spectroscopy gives information about inter-atomic distances. Electromagnetic radiation interactions with matter involve absorption or emission of electromagnetic radiation energy by means of transitions between quantised or discrete, levels of energy for electrons, bond vibrations, molecular rotations and electron and nuclear spins in atoms and molecules. The transitions between different rotational energy levels in molecules correspond to the microwave and far-infrared regions of the electromagnetic spectrum. It is only possible for there to be transitions between rotational energy levels in pure rotational spectra if the molecule has a permanent dipole moment.

MICROWAVE WINDOW: The frequency band between 1,000 and 10,000 MHz. This is the favoured portion of the electromagnetic spectrum to search for signals that might originate from extraterrestrial civilisations because the background noise (static) due to natural celestial sources is minimal at such frequencies.

MICTURITION: The discharge of urine from the bladder. It is brought about by the reflex contraction of the detrusor muscle after the voluntary relaxation of the sphincter muscle at the junction of a bladder with the urethra.

MID-OCEAN RIDGE (OCEANIC SPREADING RIDGE): A fracture zone along the ocean bottom where molten mantle material comes to the surface, thus creating new crust. This fracture can be seen beneath the ocean as a submarine mountain belt or line of ridges that form as molten rock reaches the ocean bottom and solidifies. The first mid-ocean ridge to be discovered was that running roughly north to south in the Atlantic. Mid-ocean ridges are important sites of earthquakes and volcanic activity.

MID-ORDINATE RULE (RECTANGLE METHOD): Rule for finding the area under a curve, bounded by a horizontal below the curve.

MIDBRAIN (MESENCEPHALON): The portion of the brain located superior to the pons and medulla and containing the motor nuclei of the ocular motor and trochlear nerves and also contains the major pathways and decussations of fibres from the cerebrum and cerebellum. Unlike the forebrain and the hindbrain, the midbrain does not undergo further subdivision to form additional zones. In mammals it becomes part of the brainstem, but in amphibians, reptiles and birds the roof of the midbrain becomes enlarged as the tectum, a dominant centre for integration and may include a pair of optic lobes.

MIDDLE ATMOSPHERE: The combined stratosphere and mesosphere in the atmosphere.

MIDDLE EAR: The part of the ear that consists of the eardrum (tympanic membrane) and an air-filled cavity (tympanum) within the skull of vertebrates. It lies between the outer ear and the inner ear. It is linked to the parynx via the Eustachian tube and in mammals contains three tiny bones called malleus (hammer), incus (anvil), and stapes (stirrup) known collectively as the ossicles, which connect the eardrum to the internal ear and transmit auditory vibrations from the outer ear.

MIDDLE LAMELLA: A layer of intercellular material, consisting mainly of pectins forming a cementing material between walls of contiguous plant cells. The cell plate that is formed during cell division itself develops into middle lamella, made up of calcium and magnesium pectates.

MIDDLE TERM: In logic, the term of a syllogism that appears in both premises but not the conclusion.

MIDDLE WHITE: A breed of pig native to the United Kingdom, now extinct which was a cross between Large White and Small White. It was bred for producing pork rather than bacon or lard.

MIDDLEWARE: Software that connects two otherwise separate applications. For example, software on a Web server that enables the HTTP server to interact with scripting engines like PHP or ASP when processing webpage data. This allows users to request data from the database using forms displayed on a Web browser and it enables the Web server to return dynamic Web pages based on the user's requests and profile.

MIDGUT: 1. The central section of the alimentary canal of a vertebrate, in which the processes of absorption and digestion take place. 2. The middle section of the alimentary canal of an invertebrate such as arthropods, lined with an enzyme-secreting tissue and serving as the main site of digestion and absorption. 3. The middle portion of the gut of an embryo from which the ileum, jejunum and portions of the duodenum and colon develop; also known as **mesenteron**.

MIDI (MUSICAL INSTRUMENT DIGITAL INTERFACE): A standard for the interchange of information between musical instruments such as keyboards, microphones and synthesisers and computers for recording inputs from a variety of audio instruments that also allows files to be converted into a format that can be read by the computer to synchronise all components to create a more harmonious sound.

MIDLINE: 1. An imaginary line running along the surface of the brain, which separates the right and left hemispheres; or a medial line, especially the medial line or plane of the body. 2. In geometry, the median of a trapezoid.

MIDLOBE: The central lobe of a leaf.

MIDNERVE: The central nerve of a leaf.

MIDNIGHT SUN: A phenomenon observed north of the Arctic Circle and south of the Antarctic Circle. Each summer the Sun remains above the horizon for at least one 24-hour period (a corresponding period of darkness occurs in winter), owing to the tilt of the Earth's equator to the ecliptic.

MIDPOINT: The point M on a line segment AB is the midpoint of the segment if $AM = MB$: M is halfway between A and B . The formula for coordinates of the midpoint (M) of a line segment joining the points

(x_1, y_1) and (x_2, y_2) is given by: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$; $|AM| = |BM|$.



Midpoint (M) of a line segment

MIDPOINT THEOREM: The theorem stating that the line joining the midpoints of two sides of a triangle is parallel to the third side and half its length.

MIDRIB (RACHIS): The central conducting and supporting structure of the blade of a simple leaf. This gives support to the leaf tissue and helps keep its shape.

MiFi: A brand name used to describe a portable broadband device, first introduced in May 2009 by Novatel Wireless in the United States. It provides a local ad hoc network and can be connected to up to ten 3G or 4G mobile devices.

MIGRAINE: A severe type of headache, which may last from 2 hours to 2 days. People who get migraines often describe the pain as pulsing or throbbing in one area of the head. During migraines, people are very sensitive to light and sound. They may also become nauseated and vomit. A sufferer may experience only a single attack; more commonly, he or she has recurrent attacks at varying intervals.

MIGRATION: 1. The relocation, either seasonal or as part of a single life cycle of certain animals, mainly birds and fish to distant breeding or feeding grounds or the movement of an individual or a group of people from one region or country to another. In animals, it is usually a response to lower temperatures resulting in a reduced food supply and is often triggered by change in day length. Migrating animals possess considerable powers of orientation. Birds seem to possess a compass sense using the Sun, pole stars and the Earth's magnetic lines of force as reference points. 2. The transportation of a group, atom or double bond from a part of molecule to another. 3. The movement of ions when influenced by the effects of an electric field.

MIGRATION CURRENT: The difference between the current that is actually obtained, at any particular value of the potential of the indicator or working electrode, for the reduction or oxidation of an ionic electroactive substance and the current that would be obtained, at the same potential, if there were no transport of that substance due to the electric field between the electrodes. The sign convention regarding current is such that the migration current is negative for the reduction of a cation or for the oxidation of an anion and positive for the oxidation of a cation or the reduction of an anion. Hence the migration current may tend to either increase or decrease the total current observed. In any event the migration current approaches zero as the transport number of the electroactive substance is decreased by increasing the concentration of the supporting electrolyte and hence the conductivity.

MIGRATORY APTITUDE: The relative tendency of a group to participate in a rearrangement. In nucleophilic rearrangements (migration to an electron-deficient centre), the migratory aptitude of a group is loosely related to its capacity to stabilise a partial positive charge, but exceptions are known and the position of hydrogen in the series is often unpredictable.

MIGRATORY OVERSHOOTING: Migration of wildlife in which they exceed their destination, while being on the normal travel route.

MILDEW: A fungal disease of plants with whitish powdery appearance owing to the superficial fungal mycelium and conidia growing on the surface. Common powdery mildew fungi include *Leveillula taurica* on pepper plants and *Phyllactinia gutatta* on pawpaw.

MILE: A unit of linear measurement in the Imperial and United States (US) Customary systems of measurement, which is equal to 5,280

feet (1,760 yards), or in the International System of Units (SI) is equal to 1.60934 km. Its symbol is m or mi. Although most countries have adopted the metric system and use kilometres instead of the mile, it is still commonly used in the United Kingdom (UK) and the United States (US) and other English speaking countries.

MILK: An opaque white nutritious secretion from the mammary glands of female mammals with which they suckle their young during lactation to provide a balanced diet. It is a major source of calcium and a good source of phosphorus. Milk consists of proteins, mainly casein, fat, salts and milk sugar or lactose, as well as vitamins A, C, D, certain B vitamins and lesser amounts of others. Cow's milk comprises about 87% water, 3.6% lipids, 3.3% protein, 4.7% lactose and in much smaller amounts, vitamins (A and B) and minerals. The milk of certain animals widely used by humans includes cow, mare, goat, ewe, buffalo, camel, ass, zebra, reindeer, llama and yak.

MILK FEVER: A disease in livestock, especially cows, goats and sheep, also occurring in women that have recently given birth. It is caused by reduced blood calcium levels during milk production, when demand for calcium for milk production exceeds the body's ability to mobilise calcium reserves. Symptoms include temporary loss of consciousness or inability to move.

MILK OF MAGNESIA: A common name for suspension of magnesium hydroxide $(\text{Mg}(\text{OH})_2)$ in water, commonly used as an antacid.

MILK SINUS: The space in each teat into which milk is secreted.

MILK TEETH (DECIDUOUS TEETH): The first set of teeth occurring in most mammals, developing during the embryonic stage of development. They become visible in the mouth during infancy. Humans have 20 milk teeth which are replaced during childhood by the larger permanent teeth after five years.

MILK YIELD: The quantity of milk produced each year by a cow.

MILKING MACHINE: An almost automatic machine which initiates the sucking action of a calf, used for extracting milk from the udder of cow, goats, sheep, etc. It is capable of handling as many as 200 cows an hour. The system consists of the cluster which is manually attached to the cow, a milk tube, a pulse tube and pulsator, a vacuum pump or blower and a recorder jar that measures yield.

MILKING PARLOUR: A building for milking cows and also often used for feeding, washing and cleaning.

MILKY: Resembling or looking like milk; having the appearance or consistency of milk.

MILKY BUSH (PENCIL CACTUS; MILK BUSH; PENCIL EUPHORBIA; RUBBER HEDGE; *Euphorbia tirucalli*): A widely distributed species in Africa being prominently present in Northeastern, Central and Southern parts of Africa including the Arabian Peninsula. It is a tropical perennial of the family Euphorbiaceae which is similar in appearance to a cactus but with spines. Its branches grow in a pseudo (false) whorl pattern producing small, deciduous leaves. It produces yellow flowers at the ends of the branches. It is a common source of cattle feed. Milky bush is also used as hedging or as natural fencing (as part of landscaping). It is used in treating several ailments like asthma, warts, ear pain, neuralgia and rheumatism. The sap of the milky bush is poisonous if consumed and can irritate the human skin. The milk-like latex produced by the milky bush can be changed into a fuel similar to gasoline.

MILKY WAY: The galaxy of which the Sun and solar system are a part, seen as a faint band of light crossing the night sky consisting of stars in the plane of our galaxy. It also contains about 400 billion other stars. It has an overall spiral structure, about 100,000 light-years across, along with star clusters, interstellar gas and dust, and vast amounts of dark matter. The Milky Way has four main spiral arms: the Norma and Cygnus arm, Sagittarius, Scutum-Crux, and Perseus. The Sun is located in a minor arm, or spur, named the Orion Spur. The Milky Way is constantly rotating, with the arms moving through space together with the Sun and the solar system.

MILL: A building with machinery for grinding grain into flour. 2. A building or farm equipped with machinery that presses fruits into

juice. 3. A building or factory equipped with machinery to manufacture paper, steel, or textiles. 4. A rolling mill is any machine that modifies the shape or size of a work piece by rotating tools.

MILLENNIUM: A period of thousand years.

MILLER: A moth (*Acronicta leporina*) of the family Noctuidae, whose wings have a powdery appearance. Its wingspan is 35-45 mm.

MILLER INDICES (CRYSTAL INDICES): A set of three numbers that characterise a face of a crystal. The intercepts of a plane on the three crystallographic axes are expressed as fractions of the crystal parameters. The reciprocals of these fractions, reduced to integral proportions, are the Miller indices of the crystal face, named after the British mineralogist William Hallowes Miller (1810-80), who pointed out that crystal faces could be characterised by these indices. If a plane is parallel to one of the crystal axes then its intercept is at infinity and hence its reciprocal is 0. If a face cuts a crystal axis on the negative side of the origin then the intercept, its reciprocal or the Miller index for that axis is negative.

MILLET (PEARL MILLET; *Pennisetum glaucum*): A tall robust annual bunchgrass that can grow up to 4.5 metres high, depending upon the variety and environment. The plant possesses a terminal inflorescence with a spike-like panicle which is dense. Mature panicles are brownish in colour. Millets are usually cross-pollinated. It is well adapted to growing in areas characterised by drought and low soil fertility especially in the Sahel zone of West Africa and the Indian subcontinent. It also performs well in soils with high salinity or low pH. The millet is a staple in the Sahelian regions. It is a high nutritive-value crop, popular among livestock producers for grazing, silage, hay and green chop. It is also used to make drinks in some parts of Africa.

MILLETTIA: A genus of trees in the family, Leguminosae (Fabaceae), with about 150 species found in tropical and subtropical regions such as Congo, Zaire, Tanzania, Mozambique and Cameroon. Some species of *Millettia* are *M. laurentii* (in Zaire) and *M. stuhlmannii* (in Tanzania). These plants tower about 27 metres with straight bole with an unbuttressed trunk. The heartwood of these plants is resistant to termite attack; hence, it is used for strip flooring, general construction and decorative veneer. Its pharmacological properties include antitumour, anti-inflammatory, antiviral, bactericidal, insecticidal and also used as a pesticide.

MILLI-: A prefix used in the metric system to denote one thousandth (10^{-3}). Its symbol is m. For example, 0.001 metre = 1 millimetre (mm).

MILLIBAR: A unit of atmospheric pressure, which is equal to one thousandth (10^{-3}) of a bar or 750.8 mmHg. Standard atmospheric pressure at sea level is about 1,013 millibars. In the standard atmosphere, air exerts a pressure of one bar at sea level.

MILLIGRAM: A unit of mass, which is equal to one thousandth of a gram. Its symbol is mg.

MILLIKAN: The quantity with the dimensions of energy that determines the extent to which thermionic or photoelectric emission of a solid will occur at a given temperature in accordance with Einstein's photoelectric equation or the Richardson equation. It is either expressed as the minimum energy needed to remove an electron in electron volts or joules (symbol W) and called the **work function energy**; or as a potential difference (symbol θ) in volts called the **work function potential**.

MILLIKAN, ROBERT ANDREWS (1868-1953): US physicist, who taught at the University of Chicago and California Institute of Technology. His best-known work, begun in 1909, was to determine the charge on the electron in his oil-drop experiment, which led to the award of the 1923 Nobel Prize for Physics. He then went on to determine the value of the Planck constant and work on cosmic radiation.

MILLIKAN'S OIL-DROP EXPERIMENT: An experiment begun in 1909 by Robert Millikan (1868-1953), an American physicist to measure the charge on an electron and showed that the charge exists only as a whole number of units of that charge. Between

two horizontal oppositely charged plates, Millikan introduced a fine spray of oil droplets. He first measured the mass of the oil drops by measuring their rate of fall under the influence of gravity against the air resistance. Then using x-rays to ionise the air, he noted the changed rate of fall of droplets that were attracted to the lower plate after these droplets had acquired a charge from the ionised air. These figures enabled Millikan to conclude that the acquired charge was always a simple multiple of a basic unit. He found this basic unit to be 4.774×10^{-10} esu, which was the value for the charge on an electron used until it was improved on in 1928.

MILLILITRE (mL): The metric (SI) unit of volume or capacity, which is equal to one thousandth (10^{-3}) of a litre, or 1 cubic centimetre.

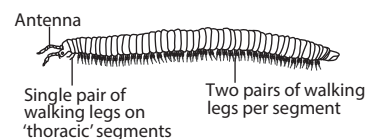
MILLIMETRE (mm): The metric (SI) unit of length, which is equal to one thousandth (10^{-3}) of a metre.

MILLISECOND (ms; msec): A unit of time which is equal to one thousandth (10^{-3}) of a second.

MILLIMETRES OF MERCURY (mmHg): A unit of pressure equal to the pressure exerted under standard gravity by height of one millimetre of mercury or 133.322 pascals. Another standard measure of pressure is called an atmosphere (atm), which equals 760 mm mercury (mm Hg).

MILLING (GRINDING): The mechanical reduction of the particle size of a sample by attribution (friction), impact or cutting. The required particle size of a sample is related to the size of the test portion and the number of particles required ensuring homogeneity among test portions. The reduction in particle size may sometimes result in particles of different hardness and density, which produces inhomogeneity during the preparation of the test sample or during the withdrawal of the test portion.

MILLIPEDE: An invertebrate animal of the class Diplopoda of the phylum Arthropoda with a cylindrical body consisting of many tubular segmented bodies with two pairs of legs on each of its body segment. It belongs to the group of terrestrial



Lateral view of a millipede

arthropods generally found in damp conditions. One species, *Illacme plenipes*, has more than 650 legs.

MILLON'S REAGENT: An analytical reagent used to detect the presence of soluble proteins, consisting of a solution of mercuric nitrate and nitrous acid. It was developed by the French chemist, Auguste Millon (1812-67). A sample is added to the reagent and heated gently for two minutes at 95 °C (368 K). The formation of a reddish-brown precipitate indicates the presence of protein in the sample.

MIME (MULTIPURPOSE INTERNET MAIL EXTENSIONS): An Internet standard for identifying the type of object being transferred across the Internet. MIME types include several variants of audio, graphics and video.

MIMICRY: An advantageous resemblance of one species or group of species by another or to a feature of its natural surroundings, which has evolved as protection from predators. It is most prevalent among insects, particularly butterflies and moths; and also in both plants and animals. In Batesian mimicry, the markings of certain harmless animals (mimics) closely resemble the warning colours of another animal which is harmful (model). Predators that have learnt to avoid the model will also avoid good mimics. **Mullerian mimicry** involves the mutual resemblance of a group of animals, all harmful, such as the wasp, bee and hornet, so that a predator, having experienced one will subsequently avoid them all.

MIMOSA: A type of tropical tree, shrub or herb of the genus *Mimosa* of about 400 species with clusters of small, ball shaped, sweet smelling, yellow flowers making up the subfamily Mimosoideae of

the family Fabaceae. The two notable examples are *Mimosa pudica* (sensitive plant) which folds its leaves when touched or exposed to heat and *Mimosa hostilis* which contains large levels of the powerful hallucinogen, dimethyltryptamine (DMT).

MIND: The mental organ responsible for thinking, reasoning, remembering and will. The existence of mind, as separate from the workings of the brain, is denied in materialism, but affirmed in dualism such as that taught by Descartes.

MINERAL: 1. A naturally occurring inorganic substance with a particular chemical composition and a regularly repeating internal structure. There are over 4,900 known mineral species. Over 4,660 of these have been approved by the International Mineralogical Association (IMA). The silicate minerals compose over 90% of the Earth's crust. 2. The inorganic chemicals such as iron, calcium and phosphorus that must be ingested by animals and plants to remain healthy.

MINERAL ACID (INORGANIC ACID): An inorganic acid such as sulfuric acid (H_2SO_4), hydrochloric or nitric acid (HNO_3). They tend to be very soluble in water and insoluble in organic solvents. Mineral acids are used for the synthesis of other chemicals, both organic and inorganic. They are also used directly for their corrosive properties such as for descaling or removing deposits from the inside of boilers.

MINERAL DEFICIENCY: Nutritional deficiency of any vital mineral such as nitrogen, phosphorus or potassium in living organisms, which can result in mineral deficiency diseases. For example, the lack of calcium causes poor bone development. In plants, calcium deficiency results in stunted growth and chlorosis.

MINERAL DRESSING: Treatment of raw material such as pulverised ore to improve physical or chemical properties in preparation for further processing.

MINERAL EXTRACTION: Excavation of or recovering valuable minerals or ores from the Earth's crust or from the ground. Most minerals are extracted by surface mining, which includes open-pit mining, strip mining and quarrying.

MINERAL MATTER: The solid part of the soil, which is composed of stones, sand, silt and clay. It does not include the vegetable matter, formed from dead or decaying plants

MINERAL NUTRIENT: Nutrients other than carbon, hydrogen and oxygen which are inorganic and are absorbed by plants from the soil.

MINERAL OIL: A liquid by-product of the distillation of petroleum to produce gasoline and other petroleum based products from crude oil. It is also obtained from coal.

MINERAL PROCESSING: The techniques of changing minerals and metallic ores into more useful forms, for example, processing bauxite into aluminium.

MINERAL SALT: A simple inorganic chemical or dissolved salts that is needed to be absorbed by living organisms for healthy growth. Plants usually obtain their mineral salts from the soil while animals get theirs from their food.

MINERAL SOIL: A soil consisting mainly of mineral matter. Mineral soils include sandy, loamy, and clayey types. The organic content of mineral soil is less than 12 - 20%, while organic soil has greater than 12 - 20%.

MINERALISATION (MINERALIZATION): The conversion of organic substance into inorganic substance. It may be a normal biological process which takes place during the life of an organism such as the formation of bone tissue or eggshells, largely with calcium or in the process of petrification. It is also the digestion process, in which bacteria utilise the organic part of the matter, leaving behind the inorganic minerals.

MINERALOCORTICOID: A class of steroid hormones concerned with water retention by control of inorganic ions distribution. These hormones include aldosterone, which increases the reabsorption of

sodium and chloride by the kidney tubules and corticosterone, the precursor of aldosterone.

MINERALOGICAL ANALYSIS: The study of materials to determine mineral composition and mineral structure. This analysis can be used to identify mineral species, and understand their characteristics and properties.

MINERALOGY: The scientific study of minerals. It includes physical properties, chemical composition, internal crystal structure, occurrence and distribution in nature and origins or conditions of formation.

MINICELL: 1. A small anucleate bacterial cell produced by the abnormal division of rod-shaped bacterium that contains no chromosomal DNA and is unable to grow or divide. It is used experimentally to express nucleic acid and protein encoded by DNA that is introduced to the cell. 2. A naturally occurring substance with a characteristic chemical composition and a generally crystalline structure. It may be identified by its physical and chemical properties such as crystal system, hardness, relative density, lustre, colour, cleavage and fracture.

MINICOMPUTER: A type of computer with size, capacity and speed between that of a mainframe computer and a workstation and generally capable of supporting up to about 200 users simultaneously. It was first developed in the 1960s by IBM Corporation and used for process control, data management and communications. Minicomputers have been largely replaced by microcomputers or personal computers.

MINIMAL: In mathematics, the least element in an ordering; it is not greater than any other element in the ordering.

MINIMAL IDEAL: In mathematics, a proper non-zero ideal that is minimal with respect to inclusion.

MINIMAL SUPERSYMMETRIC STANDARD MODEL (MSSM): The smallest possible model that combines the standard model of elementary particles theory with supersymmetry. It predicts that all the fermions and gauge bosons of the standard model should have supersymmetric partners and that should be five Higgs Bosons.

MINIMAL SURFACE: A surface that, bounded by a given closed curve or curves, has the smallest possible area. A minimal surface has a mean curvature of zero at all points.

MINIMAX: In mathematics, being or relating to a strategy or value that minimises the maximum value of a function; for example, the minimax of the family $\{f_i\}$ is a function f_i such that $\max f_i < \max f_j$ for all $j \neq i$. In other words, minimax is the procedure of choosing the strategy that least benefits the most advantaged member of a group.

MINIMAX THEOREM: A theorem justifying the exchange of order in taking the minimum and maximum of a saddle function, $\min_x \max_y F(x, y) = \max_y \min_x F(x, y)$.

MINIMISE: To find or yield, the minimum value of a given function.

MINIMUM: 1. The smallest of a set of values or elements, usually denoted minS. For example, the positive numbers have no minimum, but the nonnegative numbers have minimum 0, although both have 0 as infimum. 2. The smallest value of a function, usually denoted minf. It is called a **global minimum** if it is the smallest value with respect to all other values of the function. A **local minimum** is a value less than any other in a neighbourhood of its argument and is identified in the real differentiable setting by having zero first derivative and positive second derivative since the tangent to the curve changes from falling to rising at this point.

MINIMUM CONDITION: A condition on a module that every non-empty set of submodules has a minimal member.

MINIMUM CONSUMPTION TIME: The time during which nebulisation must be carried out in order to perform one analysis

with a given precision. The minimum consumption time is equal to the minimum volume consumed of the prepared sample divided by the rate of fluid consumption in order to obtain a given precision. It can be reduced by means of integrating the signal or measuring the signal increments (kinetic methods).

MINIMUM DYNAMIC AREA (MDA): 1. The smallest area with a natural disturbance regime, which maintains internal recolonisation sources and hence minimises extinction. 2. The amount of suitable habitat needed to sustain a viable population.

MINIMUM LETHAL CONCENTRATION (LC_{min}; LOWEST LETHAL CONCENTRATION FOUND): Lowest concentration of a toxic substance in an environmental medium that kills individual organisms or test species under a defined set of conditions.

MINIMUM LETHAL DOSE (LD_{min}): Lowest amount of a substance that, when introduced into the body, may cause death to individual species of test animals under a defined set of conditions.

MINIMUM POLYNOMIAL: The polynomial of least degree, unique up to multiplication by a scalar, that is associated with a given matrix, transformation or algebraic element over a field, such that a polynomial function of the given matrix is zero; any other such polynomial is a multiple of the minimum polynomial.

MINIMUM VIABLE POPULATION (MVP): 1. The smallest biological population size in the wild that can avoid extinction due to breeding or environmental problems. 2. The smallest number of individuals needed to perpetuate a population.

MINIMUM-ENERGY REACTION PATH (MERP): Path orthogonal to the equipotential contours of a potential energy surface that connects the energy minima through a saddle point (transition state) from which it slopes downwards along the steepest descending lines in $3N - 6$ configurational space (N is the number of nuclei in the reacting system).

MINING: The means by which economically important minerals and ores are extracted from the Earth. Where the desired minerals lie near the surface, the most economic form of mine is the open pit. Strip mining refers to stripping off a layer of overburden to reach a usually thin mineral seam, often coal.

MINKOWSKI DISTANCE: A distance d_{xy} for two observation $X' = \{x_1, x_2, \dots, x_n\}$ and $Y' = \{y_1, y_2, \dots, y_n\}$ from a set of multivariate data given by $d_{xy} = [\sum_{i=1}^n (x_i - y_i)^r]^{\frac{1}{r}}$. When $r = 2$ this reduces to Euclidean distance and when $r = 1$ to city block distance.

MINKOWSKI FUNCTION (GAUGE FUNCTION): A non-negative, positively homogeneous, convex function with value zero at zero. Given a convex set C an associated gauge is constructed by setting $g_C(x) = \inf\{t > 0 : x \in tC\}$. An everywhere finite (symmetric) gauge is a norm and there is a duality between gauge and support functions.

MINKOWSKI METRIC: The metric tensor of the Minkowski space-time used in special relativity. It is a 4×4 matrix whose nonzero entries lie on the diagonal, with one entry (corresponding to the time coordinate) equal to 1 and three entries (corresponding to space coordinates) equal to -1. Sometimes, the negative of this matrix is used.

MINKOWSKI SPACE (MINKOWSKI SPACE-TIME): Named after, Hermann Minkowski (1864 - 1909) Lithuanian-German mathematician, it is the mathematical setting in which Einstein's theory of special relativity is most conveniently formulated. In this setting the three ordinary dimensions of space are combined with a single dimension of time to form a four-dimensional manifold for representing a spacetime.

MINKOWSKI, HERMANN (1864 - 1909): A Lithuanian-German mathematician, who developed the idea of four-dimensional spacetime, a concept used in relativity theory.

MINKOWSKI'S INEQUALITY: The theorem that the Euclidean norm satisfies the triangle inequality and so is actually a norm.

MINOR: The minor of an element in a matrix is the determinant of matrix formed by crossing out the row and column containing that

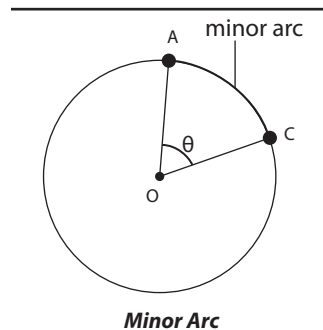
element. For example, the minor element of a_{11} in $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

is the determinant $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{23}a_{32}$.

MINOR ARC: Arc on a circle that is less than half the circumference. It bounds the minor sector or segment of a circle.

MINOR AXIS: The shorter of the two axes of symmetrical geometrical figure.

MINOR PLANETS: An astronomical object in direct orbit around the Sun that is neither a dominant planet nor a comet. The first minor planet discovered was Ceres in 1801. The orbits of more than 540,000 objects have been archived at the Minor Planet Centre.



MINOR PREMISE: In logic, the premise of a syllogism that contains the subject of the conclusion.

MINOR SEGMENT: The smaller section of the circle between a chord and an arc.

MINOR TERM: In logic, the subject of the conclusion of a syllogism.

MINORCA: A black and white domestic chicken, bred around the Mediterranean.

MINORITY CARRIER: The type of carrier, electron or hole that constitutes less than half the total number of carriers in a semiconductor.

MINUEND: The number from which another number, called the subtrahend, is to be subtracted. It is the first number in a subtraction; for example, in $10 - 2 = 8$, 10 is the minuend, 2 the subtrahend and 8 the difference.

MINUS: 1. Reduced by the subtraction of, usually using the symbol '-'; for example, 10 minus 2 is written as $10 - 2$. 2. Of a number, less than zero, having a negative value; usually written with the minus sign as a prefix; for example, minus 2 is written as -2 and read as 'minus two'. 3. Operated upon by any analogous operation, such as the relative complement $A \setminus B$.

MINUTE: 1. One sixtieth of an hour or a unit of time consisting of 60 seconds. 2. Also known as **minute of an arc**, it is a unit angle which is equal to one sixtieth of a degree (angle): $1 \text{ minute} = 1/60^\circ = 2.90332 \times 10^{-4} \text{ rad}$.

MIOCENE: The first epoch of geologic time stretching from the end of the Oligocene, roughly 5 million years ago, during which the modern ocean currents were established and Antarctica was frozen.

MIRABILITE: A yellow or white monoclinic mineral form of sodium sulfate with the chemical formula Na_2SO_4 . It consists of hydrous sodium sulfate, occurring as a deposit from saline lakes, playas and springs and as efflorescence. The pure crystals are known as **Glauber's salt**.

MIRACIDIUM: The small free-living larval stage of trematode worms (flukes), which hatches from eggs excreted in the faeces of the primary host. Miracidium is covered with cilia, enabling it to swim actively in water.

MIRAGE: An optical illusion that occurs as a result of the bending of light rays in the atmosphere during abnormal vertical distribution of

air density or an optical phenomenon caused by refraction and total internal reflection in the air. This is observed when the air close to a surface is warmer than that further out. As light nears the surface it bends away, thus, an image appears on the scale of the surface. For example, an image of the sky gives the effect of a pool of water.

MIRROR: 1. Any polished surface that reflects most of the light falling on it without diffusing it and gives a clear image of anything in front of it or an optical element which bends light by refraction. Different types of mirrors include plane, convex and concave mirrors and give different kinds of images, some virtual, some real. It is often made from a mercury-alloy coating of glass. 2. In computer science, it is a Web or FTP server that has the same files on it as another server. Its purpose is to provide an alternate way to access files when the main server is so swamped with people connecting and downloading files that other people cannot get through.

MIRROR TEST: A test of whether animals possess the ability to recognise themselves in a mirror, used as a measure of self-awareness. It has been proven that over time, dogs become aware of their own image and can in fact see themselves in a mirror.

MIRROR-IMAGE ORIENTATION: The reversal of orientation of migratory route in opposite directions by migratory birds.

MIRRORED VOLUME: A hard drive or other form of storage media that stores an exact copy of the data from another volume. It is used for fault tolerance, which means mirrored volume serves as a backup device in case the primary device fails.

MISCANTHUS (ELEPHANT GRASS): A genus of about 15 species of perennial grasses native to subtropical and tropical regions of Africa and southern Asia related to sugar cane that is grown for use as fuel.

MISCARRIAGE: Spontaneous or involuntary expulsion of a foetus from the uterus before it is capable of independent survival.

MISCELLANEOUS LAND TYPE: A mapping unit for areas of land that have little or no natural soil, or that are inaccessible for orderly examination, or where, for some reason, it is not feasible to classify the soil, for example, rough mountainous land, eroded slopes, and marshes.

MISCH METAL (CERIUM MISCHMETAL; RARE EARTH MISCHMETAL): An alloy of cerium and rare earth metals used with other alloys to achieve certain desirable characteristics. A typical composition includes cerium (50%), lanthanum (25%), neodymium (18%), praseodymium (5%) and other rare earths. It is used as flint stone in lighter and torches and in small quantities to improve the malleability of iron. It is also added to copper alloys to make them harder, to aluminium alloys to make them stronger, to magnesium alloys to reduce creep and to nickel alloys to reduce oxidation.

MISCIBLE: Pertaining to liquids such as water and ethanol that are capable of being mixed in all proportions, forming a homogeneous solution over certain ranges of temperature, pressure and composition.

MISMATCH REPAIR: A form of DNA repair that replaces mismatched bases in newly replicated DNA in living cells. Example is guanine (G) on the template strand pairs with thymine (T) instead of the usual cytosine (C), the resulting abnormality in the DNA duplex is detected by the mismatch repair system. A length of strand containing the mismatched T is cut out and the gap is repaired with new lengths containing the correct base C, using the parental strand as template. This accounts for the vast majority of DNA repairs.

MISSENSE CODON: A particular codon that has been altered so that it codes for an amino-acid residue different from the one for which it normally codes. In some cases the mutant protein may be unstable or less active.

MISSENSE MUTATION (NONSYNONYMOUS MUTATION): A type of point mutation that converts one codon to another by

specifying a different amino acid. This results in incorporation of the wrong amino acid in the polypeptide chain during translation and consequently a potentially faulty protein.

MISSILE: 1. An object or weapon that is fired, thrown, dropped or otherwise projected at a target; a projectile. 2. An unmanned aerial weapon of which there are two basic types, the ballistic missile and the guided missile. The term most often describes the self-propelled, guided weapons developed during or since World War II.

MISSING MASS: The mass of matter in the universe that cannot be observed directly by emitted or absorbed electromagnetic radiation. There are a number of astrophysical observations that suggest that the actual mass of the universe is much greater than that estimated by observations using optical telescopes, radio telescopes, etc. It is thought that there is a considerable amount of dark matter (or hidden matter) causing this discrepancy. Various explanations have been put forward for the missing mass, including black holes, brown dwarfs, cosmic strings, axions, neutrinos, monopoles and various exotic particles, such as weakly interacting massive particles (WIMPs). The universe contains far more missing matter than directly observable matter.

MISSION CONTROL CENTRE: A room or building equipped with the means to monitor and control the progress of a spacecraft during all phases of its flight after launch.

MIST: A qualitative term applied to a suspension of droplets in a gas. In the atmosphere, a mist produces generally thin, greyish veil over the landscape. It reduces visibility to a lesser extent than fog but somewhat more than haze (visibility of less than 2 km but greater than 1 km).

MISTLETOE: Any of many species of semi-parasitic evergreen plants of the families Loranthaceae and Viscaceae common on branches of citrus, cola and cocoa trees, especially those of the genera *Viscum*, *Phoradendron* and *Arceuthobium*. They cause death of parts of the branch beyond point of attachment. The African mistletoe is *Tapinanthus bangwensis*.

MISTRAL: A powerful cold, dry, northerly wind that occasionally blows during the winter on the Mediterranean coast of France.

MITE: A tiny eight-legged invertebrate animal of the spider family which may be free-living in the soil or on stored products or parasitic on animals or plants. They can carry disease, attack plants or cause allergies in human.

MITICIDE: A mite-killing substance.

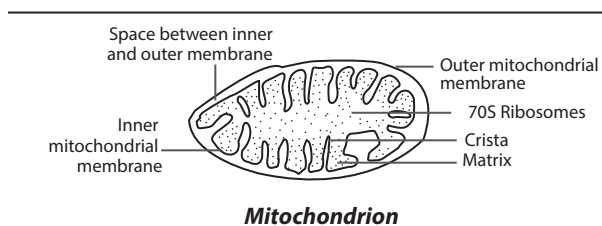
MITOCHONDRIAL DNA (mtDNA): A small circular ring of DNA molecule found in the mitochondria of a cell inherited only from the mother. In mammals mtDNA makes up less than 1% of the total cellular DNA, but in plants the amount is variable. Mitochondrial DNA disease is a rare disease caused by mutations in mitochondrial DNA or by mutations in the nuclear DNA that affect the mitochondria. Such diseases include several muscle disorders.

MITOCHONDRIAL EVE: The hypothetical female claimed by some biologists to be the ancestor of all humankind. Analysis of mitochondrial DNA from groups of people throughout the world suggest that mitochondrial Eve lived around 140,000 years ago, probably in Africa. These assertions are particularly useful for investigating recent genetic history as it mutates quickly (ten times more rapidly than nuclear DNA and in humans, is inherited solely through the female line).

MITOCHONDRIAL MATRIX: The compartment of the mitochondrion enclosed by the inner membrane and containing enzymes and substrates for the Krebs cycle.

MITOCHONDRION: A spherical or rod-shaped mobile, self-replicating microscopic membrane-enclosed organelle located in the cytoplasm of cells of almost all eukaryotic cells. It contains

enzymes responsible for energy production in the form of adenosine triphosphate (ATP) during aerobic respirations. It stores calcium for cell signaling activities, generating heat and mediating cell growth and death. It is the site for Krebs cycle and electron transport chain. They vary in shape, size and number but are commonly oval or sausage shaped and bounded by two membranes, an outer and an inner membrane. The inner is folded into finger-like projections called **cristae**.



MITOSIS: The process by which an eukaryotic cell separates the chromosomes in its cell nucleus into two identical sets in two nuclei. The nucleus of a cell divides in such a way that the resultant daughter cells receive precisely the same numbers and types of chromosomes as the original mother cell. The changes during divisions are clearly visible with a light microscope. Each chromosome divides lengthwise into two chromatids, which separate and form the chromosomes of the resulting daughter nuclei. Mitosis occurs in sequential stages from prophase to metaphase to anaphase and to telophase. Mitosis results in growth and repair or replacement of damaged or worn-out tissues. In protozoans, it is the process that occurs during asexual reproduction usually termed binary fission which occurs during favourable conditions or sporulation during unfavourable conditions.

MITOSIS-PROMOTING FACTOR (MATURATION-PROMOTING FACTOR, MPF; M-PHASE-PROMOTING FACTOR): A protein complex composed of cyclin B or cyclin A and cyclin-dependent kinase responsible for triggering mitosis in somatic cells and for maturation of oocytes into egg cells. MPF triggers prophase and enzymes condense DNA into chromosomes, breakdown the nuclear membrane and form the spindle. Several other activities occur to conclude mitosis.

MITOTIC CROSSING OVER: A process that results in genetic recombination occurring in somatic cells. It was first discovered as a rare event in *Drosophila* (fruit flies) but has since been found to occur in other organisms, especially fungi. In the latter, it sometimes occurs at a frequency that suggests it is a major source of genetic variation. The mechanisms involved are uncertain, since chromosomal pairing (a supposed requirement of crossing over in meiosis) is normally believed not to occur during mitosis.

MITOTIC SPINDLE (NUCLEAR SPINDLE): A system of microtubules first observed at prophase in a dividing cell. It is the fusiform figure characteristic of a dividing cell, consisting of microtubules, some of which become attached to each chromosome at its centromere and provide the mechanism for chromosomal movement.

MITRAL VALVE (BICUSPID VALVE; LEFT ATRIO-VENTRICULAR VALVE): A valve in the heart situated between the left atrium and the left ventricle. When the left ventricle contracts, forcing blood into the aorta, the mitral valve closes the opening to the left atrium thereby preventing blood from flowing back and so reducing the efficiency of the pumping action. The valve reopens to allow blood to flow from the atrium to the ventricle. The mitral valve is the only valve to have two flaps or leaflets; the other three valves in the heart all have three leaflets to close off the inlet and outlet apertures.

MITSCHERLICH'S LAW (LAW OF ISOMORPHISM): It states that compounds crystallising together probably have similar structures and compositions. The law can be used to determine the formula of an unknown compound if it is isomorphous with a compound of known formula. It is named after Eilhard Mitscherlich (1794 -1863), a German chemist.

MIXED BUD: A bud which produces both rudimentary flowers and foliage leaves.

MIXED CONGRUENTIAL METHOD: The specific congruential method used by a random number generator to produce the sequence $n_{i+1} \equiv [an_i + c] \pmod{m}$, where a and c are non-negative integers, m is a large number relative to the size of the specific computer and n_0 is the seed.

MIXED CROPPING (MULTIPLE CROPPING): A type of farming system in which two or more crops are grown simultaneously on the same piece of land at the same time. An example is growing crops like pepper, tomatoes, cassava and garden eggs on the same piece of land at the same time. This type of farming system is done predominantly, for example, in the rural areas in developing countries. Some of the advantages include better soil cover, better weed protection, improved soil fertility and increased crop yield, insurance against failure of one crop (other crop(s) may produce yield), different crops may exploit different parts of the soil profile, etc. The disadvantages include the following: crops may compete for nutrients, difficult to apply correct amounts of fertiliser and pesticides, makes mechanisation difficult, increased problems with pests and diseases, crop production is limited in capacity, etc.

MIXED CRYSTAL (SOLID SOLUTION): A crystal containing a second constituent which fits into and is distributed in the lattice of the host crystal.

MIXED CULTURE: The process of growing several species of trees on the same piece of land.

MIXED FARMING: A type of farming system in which the farmer grows crops and rears animals or keep poultry on the same piece of land. Some of the crops that are cultivated include the following: coconut, plantain, maize, tomatoes, cassava, okra, etc. The animals include cattle, pigs, goats, sheep, giant snails, cane rat (grass cutters), rabbits, etc. Some of the advantages include the following: livestock dung and urine excreted by animals, for example, contain several nutrients such as nitrogen, phosphorus, potassium and organic matter that maintain soil structure and fertility; by-products of the farm are utilised for other purposes on the farm (e.g., crop by-products such as straw and fodder are used for feeding of livestock and in return they provide milk and meat); it supplies most of the food the family needs; intensive cultivation; efficient utilisation of resources, etc. The disadvantages include the following: production levels can be lower than in specialised systems; overgrazing may result in erosion and degradation of land; etc.

MIXED FUNCTION OXIDASE: A family of oxidase enzymes found in animals that oxidise toxic compounds to render them susceptible to metabolism and excretion. Such complexes are induced in a wide range of species following exposure to toxic organic substances such as alkaloids, phenolics, terpenoids and quinines. The enzymes are localised in the liver and in similar tissues, such as the hepatopancreas in invertebrates.

MIXED FUNCTION OXYGENASE (MONO-OXYGENASE): An enzyme that introduces an oxygen atom into a substrate. Mixed function oxidases are vital components of many important metabolic pathways. Cytochrome oxidase is one of such enzymes.

MIXED GRAZING: A grazing system where two or more types of animals graze the same pasture at the same time.

MIXED INDICATOR: An indicator containing a supplementary dye selected to heighten the overall colour change.

MIXED INFLORESCENCE: An inflorescence, which has a number of cymose clusters that are arranged in a racemose manner on an elongated peduncle, for example, Lilac (*Syringa vulgaris*).

MIXED INTERCROPPING: One type of intercropping in which two or more crops are cultivated simultaneously on the same field not in rows but are randomly distributed.

MIXED LABELLED: A specifically labelled compound is referred to as mixed labelled when the isotopically substituted compound has more than one kind of modified atom, for example, $\text{CH}_3\text{-CH}_2\text{-}^{18}\text{O}[^{2}\text{H}]$.

MIXED LACTIC FERMENTATION (HETEROLACTIC FERMENTATION): A form of anaerobic respiration found in certain microorganisms, such as *Lactobacillus brevis*, in which the end products are a molecule each of lactic acid, ethanol and carbon dioxide.

MIXED LAYER: The upper region of the ocean where active turbulence generated by active air-sea exchanges has mixed and homogenised the water to a certain level, which becomes vertically uniform in temperature and salinity, as well as density.

MIXED NUMBER: A figure that consists of an integral and rational part or a whole number and a fraction, for example, $5\frac{1}{4}$.

MIXED PARTIAL DERIVATIVE: A partial derivative consisting of derivatives with respect to more than one variable; for example,

$$\frac{\partial^3 f}{\partial x \partial y \partial z}.$$

MIXED POLYNOMIAL: A polynomial expression or equation having an integral and rational part; for example, $3x^3 + 2x + 2/5x$.

MIXED POTENTIAL: The value of the potential of a given electrode with respect to a suitable reference electrode when appreciable contributions to the total anodic and/or cathodic partial currents are made by species belonging to two or more different couples, but the total current is zero.

MIXED TYPE BOUNDARY CONDITIONS: Boundary conditions for a partial differential equation of the form $g \frac{\partial u}{\partial \mathbf{n}} + u = f$ on the

boundary, where g is some function, possibly constant and $\frac{\partial u}{\partial \mathbf{n}} = \nabla u \cdot \mathbf{n}$ is the normal derivative, for u the dependent variable.

MIXED-VALENCE COMPOUNDS (INTERMEDIATE-VALENCE COMPOUNDS): A chemical compound in which energy levels of f electrons coexist with energy bands of s and/or d electrons. It contains an element, usually a metal in more than one oxidation state. Such compounds have usual thermal and magnetic properties.

MIXER: 1. In electronics, it refers to a device having two or more inputs, usually adjustable and a common output. It is used to combine separate audio or video signals linearly in desired proportions to produce an output signal. 2. Also known as **first detector**; **heterodyne modulator**; **mixer-first detector**, it refers to the stage in a superheterodyne receiver in which the incoming modulated radio-frequency signal is combined with the signal of a local r - f oscillator to produce a modulated intermediate-frequency signal. 3. In optics, it refers to a nonlinear device in which two light beams are combined to form new beams having frequencies equal to the sum or the difference of the input wavelengths.

MIXING: The combining of components, particles or layers into a more homogeneous state. The mixing may be achieved manually or mechanically by shifting the material with stirrers or pumps or by revolving or shaking the container. The process must not permit segregation of particles of different size or properties. Homogeneity may be considered to have been achieved in a practical sense when the sampling error of the processed portion is negligible compared to the total error of the measurement system.

MIXING CONTROL: The experimental limitation of the rate of reaction in solution by the rate of mixing of solutions of the two reactants. It can occur even when the reaction rate constant is several powers of 10 less than that for an encounter-controlled rate. Analogous (and even more important) effects of the limitation of reaction rates by the speed of mixing are encountered in heterogeneous (solid/liquid, solid/gas, liquid/gas) systems.

MIXING HEIGHT: The height to which significant mixing of added

pollutants occurs within the atmosphere. In reference to stack gases, it is considered the height at which stack effluent begins mixing with the atmosphere as it leaves the stack.

MIXING RATIO: In meteorology, it refers to the dimensionless ratio of the mass of a substance such as water vapour in an air parcel to the mass of the remaining substances in the air parcel. For trace substances, this is approximated by the ratio of the mass of the substance to the mass of air. However, in the case of water vapour the mass of dry air is used. In atmospheric chemistry, mixing ratios (molecular, molar, by volume, as well as by weight) are used to describe relative concentrations of atmospheric trace gases and impurities.

MIXTURE: A physical combination of two or more pure substances, where these substances are not bonded to each other and no chemical reaction occurs between the substances. In a mixture, the substances that make up the mixture are not in a fixed ratio; keep their physical properties; and can be separated by physical means such as distillation and crystallisation. Mixtures are grouped into two categories: heterogeneous and homogeneous. A **homogenous mixture** has a definite composition, and specific properties and the different parts cannot be seen. All the atoms or molecules are interspersed, as in a mixture of gases, solutions and colloids. The components of a mixture retain their own chemical properties and may be present in any proportion. Examples of such mixtures are when water dissolves in water; and oxygen made up of a mixture of different gases which are in a constant ratio. **Heterogeneous mixtures** do not have a definite composition but have distinguishable phases, for example, if a substance is mixed with another, they will stay separate in the mixture. for example, soil is composed of a mixture of pebbles, plant matter and sand, which are distinguishable. A heterogeneous mixture can be grouped based on the phases of matter making up the mixture as follows: (i) **liquid-liquid** is a mixture of two phases of liquids, for example, oil and water, called **emulsion**; (ii) **liquid-solid** is a mixture of two different phases of matter, for example, water and mud, called **suspension**; (iii) **gas-liquid** is a mixture of two different phases of matter, for example, fizzy drinks such as coca-cola, called **aerosol**; and (iv) for example, smoke is a mixture of two different phases of matter, **gas-solid**, called **smog**.

MIXOTROPH: An organism that is capable of existing as a heterotroph or autotroph. As an autotroph, it assimilates organic compounds as carbon sources; as an autotroph, inorganic substances are taken in and used to synthesise organic molecules.

MKS UNITS (GIORGI UNITS): A system of units devised by A. Giorgi in 1901. It is based on the metre, kilogram and second as the units of length, mass and time and grew from the earlier centimetre-gram-second (cgs) units. The units of electricity and magnetism (ohm, farad, coulomb, etc.) are MKS (**metre**, **kilogram** and **second**) units. The ampere and the permeability of space (magnetic constant) was taken as 10^{-7} Hm^{-1} . To simplify electromagnetic calculation the magnetic constant was later changed to $4\pi \times 10^{-7} \text{ H m}^{-1}$ to give the rationalised MKSA system. This system, with some modifications, formed the basis of SI units, now used in most scientific work.

MOBBING: An aggressive stand taken by some animals to ward off a potentially dangerous predator, usually to protect their offspring by attacking or harassing it in their numbers.

MOBILE ADSORPTION: A situation in which molecules of adsorbate are free to move about the surface. Mobility of the adsorbate will increase with temperature and mobile adsorption may be either localised or nonlocalised. In localised mobile adsorption the adsorbate spends most of the time on the adsorption sites but can migrate or be desorbed and readsorbed elsewhere. In non-localised adsorption the mobility is so great that a small fraction of the adsorbed species is on the desorption sites and a large fraction at other positions on the surface.

MOBILE COMMERCE (M-COMMERCE): Conducting commercial activities such as buying and selling of goods and services, online banking and bill payment through wireless handheld devices such as laptops, cellular telephone and personal digital assistants (PDAs).

MOBILE ION: An ion that is free to move. They are found only in aqueous solutions or melted (molten masses) in electrolyte.

MOBILE NODE: An Internet connected device such as a portable computer or cellular telephone whose location and point of attachment to the Internet may be frequently changed. Every node has a unique network address, sometimes called a Data Link Control (DLC) address or Media Access Control (MAC) address.

MOBILE NUMBER PORTABILITY: A service that allows cell phone or smartphone users to retain same mobile telephone numbers when changing from one mobile network carrier to another.

MOBILE PHASE: A fluid which percolates through or along the stationary bed, in a definite direction. It may be a liquid (liquid chromatography) or a gas (gas chromatography) or a supercritical fluid (supercritical-fluid chromatography). In gas chromatography the expression carrier gas may be used for the mobile phase. In elution chromatography the expression 'eluent' is also used for the mobile phase.

MOBILE VIRTUALISATION: A system of mobile device management in which mobile hardware such as a mobile phone or connected wireless device is partitioned into various logical components and devices. It enables multiple operating systems or virtual machines to run simultaneously.

MOBILITY: A quantity that gives a measure of how readily a particle moves. The term is frequently used to characterise the motion of a charged entity, such as an ion, electron or hole, in an electric field or the terminal speed of an ion in an electric field divided by the field strength. Its symbol is u .

MOBILITY EDGE: The energies in an energy band in a disordered solid, at which there is a transition between localised states which do not contribute to the electrical conductivity of the system at absolute zero and extended states, which can contribute to the electrical conductivity.

MÖBIUS AROMATICITY: A monocyclic array of orbitals in which there is a single out-of-phase overlap (or, more generally, an odd number of out-of-phase overlaps) reveals the opposite pattern of aromatic character to Hückel systems. With $4n$ electrons it is stabilised (aromatic), whereas with $4n + 2$ it is destabilised (antiaromatic). In the excited state $4n + 2$ Möbius π -electron systems are stabilised and $4n$ systems are destabilised. No examples of ground-state Möbius π -electron systems are known, but the concept has been applied to transition states of pericyclic reactions. The name is derived from the topological analogy of such an arrangement of orbitals to a Möbius strip.

MÖBIUS INVERSION FORMULA: The formula that, given any arithmetic function f , one considers the related arithmetic function

$$F(n) = \sum_{d|n} f(d); \text{ where the summation is over the divisors } d \text{ of } n; \text{ then}$$

one recovers f as $f(n) = \sum_{d|n} F(d) \mu\left(\frac{n}{d}\right)$, where μ is the Möbius function and f is multiplicative exactly F is.

MÖBIUS FUNCTION: The multiplicative arithmetic function $\mu(r)$ defined by $\mu(r) = 1$ for $r = 1$; $\mu(r) = -1$ for prime r , $\mu(r) = (-1)^{\lambda}$, where r is the number of prime factors of n , provided none of them is repeated; and $\mu(r) = 0$ for numbers whose prime decomposition includes repeated primes. For example, $\mu(30) = \mu(3) \cdot \mu(2) \cdot \mu(5) = -1$ and $\mu(32) = \mu(2^5) = 0$.

MÖBIUS STRIP (MOBIUS BAND): A flat strip which is twisted halfway and the ends joined together. It was named after the German mathematician August Ferdinand Möbius, who was a pioneer in topology in the 1800s. It can be formed by taking a long, rectangular strip of paper, rotating the ends 180° with respect to one another and joining the ends together to form a loop. It has only one side and edge.

MÖBIUS TRANSFORMATION (FRACTIONAL LINEAR TRANSFORMATION): A rational invertible transformation of the

complex plane of the form, $w = \frac{az + b}{cz + d}$ with $ad \neq bc$. These are conformal mappings.

MÖBIUS, AUGUST FERDINAND (1790- 1868): German mathematician and astronomer, who was Professor of Astronomy in Leipzig and best known for his discovery of the Möbius strip. Some mathematical concepts such as Möbius function, Möbius inversion formula, Möbius plane, Möbius strip and Möbius transformations have been named after him.

MOCKINGBIRD: 1. A group of passerine birds from the Mimidae family, found mainly in tropical America. Some species are noted for their mimicry of the calls and songs of other birds and animals, much like a parrot. 2. Software that intercepts communications, especially login transactions between users and hosts and provides system-like responses to the users while saving their responses.

MOD: Abbreviation and symbol for modulus or modulo.

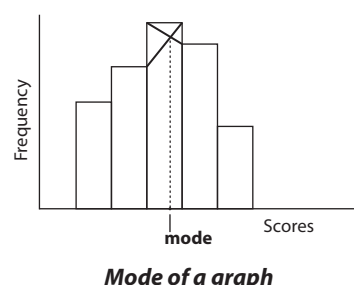
MODAL: 1. In statistics, of, relating to or comprising of a mode. 2. In logic, of or relating to, a modality.

MODAL CLASS (MODAL GROUP): In a grouped frequency distribution, it is the class with the highest frequency.

MODALITY: In logic, the classification of logical propositions according to their asserting or denying the possibility, impossibility, contingency or necessity of their content.

MODAL LOGIC: 1. The logical study of such concepts as possibility, necessity, etc. and of formal systems of which the intended interpretation includes such concepts. 2. The study of analogous families of concepts, such as moral, epistemological and psychological concepts and of systems intended to represent them. 3. Any formal system capable of being interpreted as a model for the behaviour of analogous families of concepts.

MODE: The element or component that occurs most often in a given set of data or the most common number or item in a set of numbers and one of the ways of finding an average. For example, the mode for the data 1, 1, 7, 7, 7, 9, 9, 14, 28, 38 is 7. For a grouped frequency distribution, it can be estimated using a histogram and also by the formula: Mode = 3 Median - 2 Mean, when the mean and median are known. When a distribution has two modes, it is known as **bimodal**.



MODEL: 1. A basic and simplified version or representation of a complicated system designed to help with understanding it in the real world. For example, an atomic model represents what the structure of an atom could look like, based on what we know about how atoms behave. 2. The mathematical or formal theory of something that reflects some aspect of a particular physical, social, technological or natural phenomenon or process and enables predictions to be made about its behaviour. 3. A copy of an object, especially one made on a smaller scale.

MODEL OF COMPUTATION: An idealised version of a computing device that usually has some simplifications such as infinite memory. A Turing machine and the lambda calculus are models of computation.

MODEL THEORY (SEMANTICS): The study of mathematical structures that satisfy a particular set of axioms, especially in the field of logic. Its fields of study include the concepts of truth, satisfaction and validity, which are defined extrinsically for a formal system, as contrasted with proof theory, which is concerned with the study of only the intrinsic property of syntactic deducibility.

MODEM: 1. Acronym for **modulator/demodulator**, an electronic device that connects computers via a telephone line, allowing the

exchange of information. It converts digital data into analogue (modulated-wave) signals suitable for transmission over analogue telecommunications circuits. It demodulates received analogue signals to recover the digital data transmitted. Modems can be either installed externally or internally. 2. An **external modem** is a self-contained unit located outside a personal computer (PC) having lights indicating various modem functions and connected to the PC by a cable via the serial or USB port. An **internal modem** is contained on a printed circuit board within the computer.

MODERATELY COARSE TEXTURE: Soil profile consisting mainly of coarse particles. In soil textural classification, it includes all the sandy loams except the very fine sandy loam. **Moderately fine texture** consists predominantly of intermediate-sized soil particles with or without small amounts of fine or coarse particles. It includes soil types of clay loam, sandy clay loam, and silty clay loam.

MODERATION: Reduction of the neutron energy by scattering without appreciable capture.

MODERATOR: A substance such as light water, heavy water, graphite or beryllium that reduces the energy of free neutrons down to thermal levels, hence the speed, in a nuclear reactor making them more likely to cause fissions of atoms of uranium -235 and less likely to be absorbed by atoms of uranium -238.

MODERN EVOLUTIONARY SYNTHESIS (NEO-DARWINISM; NEW SYNTHESIS; MODERN SYNTHESIS; NEO-DARWINIAN SYNTHESIS): The current theory of the process of evolution, formulated between 1920 and 1950 that combines evidence from classical genetics with the Darwinian theory of evolution by natural selection. It makes use of modern knowledge of genes and chromosomes to explain the source of genetic variation upon which selection works.

MODIFIED NEWTONIAN (MOND): A hypothesis that proposes a modification of the Newtonian theory of gravity that has been used as an alternative to the idea of dark matter to explain why the motion of stars in galaxies is not in accord with the expectations of standard Newtonian theory. It was proposed by Mordehai Milgrom, an Israeli physicist in 1983.

MODIFIED REPEATED ADDITION: A multiplication algorithm based on adding a to itself b times to find $a \times b$. One of the factors is separated into parts and the partial products of the other factor and those parts are then added. For example, to compute 67×53 , think of 10 [67s] as 670 and add five of them to get 50×67 . Then add the remaining 3 [67s] to the result.

MODIFIED SAMPLE: A sample or a known fraction of the parent population in which the analyte has been isolated or (usually) concentrated before being submitted to the laboratory. If the isolation or concentration occurs in the laboratory, the procedure is usually considered part of the preparation of the test sample from the laboratory sample (in-laboratory processing).

MODIFIER: A substance added to a solvent to improve its properties, for example, by increasing the solubility of an extractant, changing interfacial parameters or reducing adsorption losses. Additives used to enhance extraction rates should be called **accelerators** or **catalysts**.

MODIFIER GENE: A gene that influences the expression of another gene. For example, one gene controls whether eye colour is blue or brown but other genes can also influence the colour by affecting the amount or distribution of pigment in the iris.

MODIFIER KEY: In computer science, it is a key on a keyboard that is pressed at the same time as another key to perform some function. For example, pressing the shift key at the same time as an alphabet key produces the upper case letter of the key being pressed. Examples of such keys include shift, function, control, alt, command and option.

MODULAR ARITHMETIC: A branch of arithmetic that deals with the

remainders of integers or whole numbers after the numbers have been divided by a modulus, introduced by Carl Friedrich Gauss (1777-1855), German mathematician. An example is the 12-hour clock system, in which the day is divided into two 12 hour periods. If the time is 8:00 now, then 8 hours later it will be 4:00. Addition of $8 + 8 = 16$, which should give the time as 16:00, but because of modular arithmetic, the time is based on 12 hours and does not continue beyond 12; the present time, therefore, would not be 16:00 hours, but 4:00 pm.

MODULAR EQUATION: An identity of the form $f(x) = f(x^n)$ for integral n , where n is referred to as the order of the modular

equation. For example, $f(x) = \frac{\sqrt[2]{f(x^2)}}{1 + f(x^2)}$ is a second-order modular equation.

MODULAR FIELD: In mathematics, a field of finite non-zero characteristic n , the smallest positive integer such that the n -fold sum of the multiplicative unit of the given field is zero. It can be shown that n must be prime. Any finite field is modular; for example, in $\mathbb{Z}_p$, the integers modulo some prime p , the multiplicative unit is 1 and the sum of p such units is zero (mod p), so that the integers mod p are a field with characteristic p . If no such integer exists the field is said interchangeably to have characteristic 0 or ∞ .

MODULAR FUNCTION: 1. A function that is meromorphic in the upper half of the complex plane and that is automorphic for the modular group or one of its subgroups. There are more general modular forms that are merely required to be automorphic forms of dimension $-2m$ with respect to the modular group: $f\left(\frac{az+b}{cz+d}\right) = (cz+d)^{-2m} f(z)$ for

some positive half-integer m . 2. A function that satisfies the conditions of a modular equation.

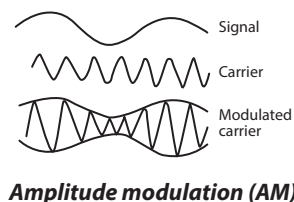
MODULAR GROUP: In mathematics, the group comprising of all Möbius transformations with integral coefficients and determinant equal to unity.

MODULAR OBJECT-ORIENTED DYNAMIC LEARNING ENVIRONMENT (MOODLE): An open source course management

system, used by thousands of educational institutions around the world to provide an organised interface for learning over the Internet. Moodle allows educators to create online courses, which students can access as a virtual classroom. A typically Moodle home page will include a list of participants including the teacher and students and a calendar with a course schedule and list of assignments. Other Moodle features include online quizzes, forums, where students can post comments and ask questions, glossaries of terms and links to other Web resources.

MODULATION: The process of superimposing the characteristics, of a periodic signal onto a carrier wave so that the information contained in the signal can be transmitted by the carrier wave. For example, Amplitude Modulation (AM) uses the information signal to alter the amplitude of the carrier wave. **Demodulation** is the process by which the original signal is recovered from the wave produced by modulation.

MODULE: 1. One of a set of standardised parts or independent parts that can be connected or combined to build or complete a more complex structure. 2. Any self-contained part of a spacecraft, designed to perform a particular task or function, which may be detachable. 3. A selected unit of measure, ranging in size from a few centimetres to several metres, used as a basis for the planning and standardisation of building materials. 4. In mathematics, an Abelian group with a set of



left or right operators forming a ring such that for any two operators and any group element the result of having the first operator act on the element, giving a second element and the second operator act on the second element is equal to the result of having a single operator, formed by adding or multiplying the two operators, act on the first element. 5. A self-contained unit of a computer or computer program that does a particular job. A module may be independent, self-contained and detachable. Large complex computer programs are formed by one or more modules.

MODULO: The integers a and b are said to be congruent modulo m if $a - b$ is divisible by m . For example, 18 is congruent to 42 modulo 12 because both 18 and 42 leave 6 as a remainder when divided by 12.

MODULUS: 1. Values of a real number irrespective of its sign whether negative or positive or the absolute value of a complex number or the square root of the sum of the squares of the real numbers in a complex number. For example, the absolute value of the complex number $z = x + iy$ is $|z| = \sqrt{x^2 + y^2}$. 2. A number that can be divided exactly into the difference of two other numbers, so that they are congruent modular that divisor. For example, 7 is a modulus of 11 and 25. 3. The factor by which a logarithm of one base must be multiplied to become the logarithm of another base. 4. A coefficient expressing the degree to which a substance exhibits a particular property. 5. The variable, denoted k , of the complete elliptic integrals and elliptic functions. The quantity is $\sqrt{1-k^2}$ the complementary modulus.

MODULUS OF ELASTICITY (ELASTIC MODULUS; COEFFICIENT OF ELASTICITY; ELASTICITY MODULUS): The mathematical description of an object or a substance's tendency to be deformed elastically when a force is applied to it or the ratio of the increment of some specified form of stress to the increment of some specified form of strain, such as Young's modulus, the bulk modulus or the shear modulus.

MODUS PONENS (MODUS PONENDO PONENS; RULE OF DETACHMENT): In logic, the principle or rule of inference that whenever a conditional statement and its antecedent are both given to be true, then its consequent may be validly inferred. In other words, a single conditional statement is made and a hypothesis (P) is stated. The conclusion (Q) is then deduced from the statement and the hypothesis. The most basic form is listed below:

$P \rightarrow Q$ (conditional statement)

P (hypothesis stated)

Q (conclusion deduced)

For example, "If an angle satisfies $90^\circ < A < 180^\circ$, then A is an obtuse angle." $A = 120^\circ$. A is an obtuse angle.

MODUS TOLLENS (MODUS TOLLENDO TOLLENS): In logic, the rule used to make inference about that whenever a conditional statement and the negation of its consequent are both given to be true, the negation of its antecedent may be validly inferred. It may be expressed as if p is true, q is true; but q is false; therefore A is false.

MOHAIR: Fine wool from Angora goat or a large group of fabrics made from it, either wholly or in combination with wool, silk or cotton, used in the manufacture of sweaters, upholstery plushes, heavy cloaking and cloths such as astrakhan.

MOHOROVICIC DISCONTINUITY (MOHO): A discontinuity within the Earth that marks the boundary between the Earth's crust and the mantle. It lies at a depth of about 10-12km below the ocean floors and about 33-35km below the continents. It was first identified in 1909 by Andrija Mohorovicic, a Croatian seismologist. Below the discontinuity earthquake seismic wave undergo a sudden increase in velocity.

MOHS SCALE: A scale used to measure a material's ability to resist abrasion or scratching, determined by the ability of harder materials to scratch softer ones. Gemstones and other minerals are ranked according to their Mohs hardness. It was devised by German mineralogist, Friedrich Moh in 1812. A mineral with a given hardness rating will scratch other minerals of the same hardness and all samples

with lower hardness ratings. On the Mohs scale talc has a hardness of 1, gypsum 2, calcite 3, fluorite 4, apatite 5 orthoclase 6, quartz 7, topaz 8, corundum 9 and diamond 10.

MOIETY: A characteristic part of a molecule. For example, in the ester $C_2H_5COOCH_3$ the OCH_3 can be regarded as the alcohol (methanol) moiety.

MOIRÉ PATTERN: A pattern produced when two sets of line grids are slightly inclined to one another at an angle of less than about 45° .

MOISSANITE: A clear form of silicon carbide used as an inexpensive substitute for diamond.

MOIST ADIABATIC LAPSE RATE: The rate at which humid air cools as it rises upward, usually at less than $6^\circ C/km$.

MOISTURE: 1. Water or other liquids in small quantities or in the form of steam or mist. **Moist** is being slightly wet or damp. 2. Condensed or absorbed liquid.

MOISTURE EQUIVALENT: The water content expressed as a percentage of the dry weight that a soil can retain against a centrifugal force 1000 times the force of gravity. It is used as a convenient laboratory measure of soil moisture conditions.

MOISTURE-HOLDING CAPACITY: The amount of water held by soil between field capacity and the permanent wilting point. The amount of moisture will vary according to the texture and structure of the soil.

MOISTURE-RETENTION CURVE (WATER RETENTION CURVE): A graph showing the water content of soil and the applied tension or pressure. It gives the relation between soil moisture suction and soil moisture content. Points on the graph are usually obtained by increasing or decreasing the applied tension or pressure over a specified range.

MOISTURE-VOLUME PERCENTAGE: In soils, it is the ratio of the volume of water in a soil to the total bulk volume of the soil.

MOISTURE-WEIGHT PERCENTAGE: In soils, it is the moisture content expressed as a percentage of the oven dry weight of soil.

MOL FILE FORMAT: The MOL file format is used to encode chemical structures, substructures and conformations as text based connection tables. It is used by MDL Information Systems Inc., for example, in their MACCS or ISIS programs.

MOLALITY: A unit of concentration of a solution defined as the number of moles of solute dissolved in 1000 grams of solvent or the amount of entities of a solute divided by the mass of the solvent. Its symbol is m or b .

MOLAR: 1. Denoting that an extensive physical property is being expressed per amount of substance usually per mole. For example, the molar heat capacity of a compound is the heat capacity of that compound per unit amount of substance. In SI units it would be expressed in $JK^{-1} mol^{-1}$. 2. Having a concentration of one mole per dm^3 . 3. In anatomy, it is the rearmost and most complicated kind of tooth in most mammals, with ridges or cusps for grinding food. In adult humans, it consists of four groups of three at the back of the mouth, totalling twelve.

MOLAR CONDUCTIVITY: Conductivity of a volume of an electrolyte that contains one mole of solution between electrodes which are placed one metre apart. It is also defined as the conductivity of an electrolyte solution divided by the molar concentration of the electrolyte and so measures the efficiency with which a given electrolyte conducts electricity in solution. Its symbol is Λ or Λ_m and the units are siemens per metre per molarity or siemens metre-squared per mole.

MOLAR GAS VOLUME: The volume occupied by one mole of gas at standard temperature and pressure (s.t.p), which is equal to $22.4 dm^3$, with the symbol V_m . It is equal to the molar mass (M) divided



Molar

by the mass density (ρ): $V_m = M/\rho$. Its unit is cubic metres per mole (m^3/mol) or cubic decimetres per mole (dm^3/mol) for gases and cubic centimetres per mole (cm^3/mol) for liquids and solids.

MOLAR HEAT CAPACITY (MOLAL HEAT CAPACITY; MOLECULAR HEAT CAPACITY): The heat required to increase the temperature of one mole of a substance by 1 kelvin, expressed in joule(s) (J) per kelvin (K). In the metric system, heat capacity is often expressed in units of calories per gram per degree Celsius ($\text{cal/g}\cdot^\circ\text{C}$).

MOLAR HEAT OF FUSION: The amount of energy required to melt 1 mole of a solid substance.

MOLAR HEAT OF SUBLIMATION (ENTHALPY OF SUBLIMATION; HEAT OF SUBLIMATION): The energy absorbed when one mole of a solid is transformed to vapour at constant temperature and pressure, expressed in kJ/mol.

MOLAR HEAT OF VAPORISATION/VAPORIZATION: The energy absorbed in converting one mole of a liquid to vapour at its boiling point.

MOLAR LATENT HEAT: The amount of heat released or absorbed by a chemical substance or a thermodynamic system during a change of state such as the melting of ice or the boiling of water that occurs at constant temperature. Its symbol is L . For example, the latent heat of vaporisation is the energy a substance absorbs from its surroundings in order to overcome the attractive forces between its molecules as it changes from a liquid to a gas and in order to do work against the external atmosphere as it expands.

MOLAR MASS (GRAM-MOLECULAR WEIGHT): The mass of one mole of any chemical substance/entity such as atoms, molecules or particles in units of grams per mole or g/mol ($\text{g}\cdot\text{mol}^{-1}$). Its symbol is M .

MOLAR SOLUTION: A solution that contains one mole of a solute substance per litre of a solution. Its unit is mol/litre.

MOLAR VOLUME (MOLECULAR VOLUME): The volume occupied by a substance per unit amount of the substance at standard temperature and pressure (STP). According to Avogadro's law, at a given temperature and pressure a given volume of any gas contains the same number of molecules. At standard temperature and pressure (STP), 1 mole of gas occupies 22.414 litres.

MOLAR-MASS EXCLUSION LIMIT: In polymers, it is the maximum value of the molar mass or molecular weight of molecules or particles, in a specific polymer solvent system that can enter into the pores of the porous non-adsorbing material used in size-exclusion chromatography. For particles with molar mass or molecular weight larger than the exclusion limit the separation effect of the size-exclusion chromatography vanishes.

MOLASSES: Thick dark syrup drained from raw sugar after heat crystallisation of sucrose in the process of refining. Sugarcane is the major source of molasses.

MOLE: 1. A small, often slightly raised blemish on the skin made dark by a high concentration of melanin. Change in a mole, such as increase in size, change of colour or bleeding should lead to suspicion of melanoma. 2. The SI unit of amount of substance, equivalent to the amount of a substance that contains as many elementary particles (atoms, molecules, ions, electrons or other particles) as there are in 0.012 kg (12 g) of the isotope of carbon-12 or the number of atoms there are in 12 g of carbon-12 isotopes, which is approximately 6.02×10^{23} , called **Avogadro's number**. For example, one mole of water (H_2O) is $(1 + 1 + 16) = 18 \text{ g}\cdot\text{mol}^{-1}$. Its unit is mol. 3. An insectivorous mammal of the order Insectivora.

MOLE FRACTION (MIXING RATIO): The ratio of the number of moles of a substance in a mixture or solution to the total number of moles of all the components in the mixture or solution, with the symbol X . The mole fraction of component A is given by $X_A = n_A/N$, where n_A is the amount of substance of A (for a given entity) and N is the total amount of substance of the mixture for the same entity.

MOLE METHOD: The relative amount of any two substances involved in a chemical reaction.

MOLECULAR: 1. In chemistry, relating to or consisting of molecules. 2. In logic, of a sentence, formula, etc., capable of being analysed into atomic formulae of the appropriate kind. A conjunction or implication is thus a molecular statement.

MOLECULAR BEAM: A beam of atoms, ions or molecules at low pressure, in which all the particles are travelling in the same direction and there are few collisions between them. They are formed by allowing a gas or vapour to pass through an aperture into an enclosure, which acts as a collimator by containing several additional apertures and vacuum pumps to remove any particles that do not pass through the apertures. Molecular beams are used in studies of surfaces and chemical reactions and in spectroscopy.

MOLECULAR BIOLOGY: The branch of biology concerned with the study of the molecular basis of life, including the biochemistry of molecules such as DNA, RNA and proteins and the molecular structure and function of the various parts of living cells.

MOLECULAR CHAPERONE: Any of a group of proteins in living cells that ensures that newly synthesised or denatured proteins are folded into their functional three-dimensional structures. The chaperones bind to the protein and prevent improper interactions within the polypeptide chain, so that it assumes the correct folded orientation. This process may require energy in the form of adenosine triphosphate (ATP). Molecular chaperones assist the translocation of proteins across the membranes of cell organelles and binding denatured proteins under stress conditions or in degenerative disease.

MOLECULAR CLOCK (GENE CLOCK; EVOLUTIONARY CLOCK): The use of rates of mutation in genetic material to calculate the length of time elapsing since two related species diverged from each other during evolution. The method can be based on the DNA or of widely occurring proteins, such as haemoglobin. Comparative studies of different proteins in various groups of organisms tend to show that the average number of amino-acid substitutions per site per year is typically around 10^{-9} .

MOLECULAR DESIGN: The application of all techniques leading to the discovery of new chemical entities with specific properties required for the intended application.

MOLECULAR DISTILLATION: Distillation in high vacuum with the condensing surface so close to the surface of the evaporating liquid that the molecules of the liquid travel to the condensing surface without collisions.

MOLECULAR DYNAMICS: A branch of physical chemistry concerned with molecular mechanisms of the elementary physical and chemical processes that control rates of reaction.

MOLECULAR ENTITY: Any constitutionally or isotopically distinct atom, molecule, ion, ion pair, radical, radical ion, complex, conformer, etc, identifiable as a separately distinguishable entity.

MOLECULAR EQUATION: A method of writing the essential features of a chemical reaction using chemical symbols or conventions. By convention, reactants (present at the start) are on the left, products (present at the end) on the right. A single arrow between them denotes an irreversible reaction, a double arrow a reversible reaction. The law of conservation of matter requires that every atom on the left appear on the right for the equation to balance; only their arrangements and combinations change. For example, one oxygen molecule combining with two hydrogen molecules to form two water molecules is written as $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. The dissociation of salt into sodium and chloride ions is written as $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$.

MOLECULAR FLOW: The flow of a gas through a pipe in which the mean free path of gas molecules is large compared to the dimensions of the pipe. A gas molecule thus migrates along the path independent of other gas molecules present.

MOLECULAR FORMULA: A chemical formula specifying the actual number of atoms of each element present in a single molecule of a chemical compound. For example, the molecular formula of water is H_2O , which means that one molecule of water contains two atoms of

hydrogen and one atom of oxygen. This also represents the molar mass, which is the sum of the gram atomic mass of all the atoms. Discrete molecules can be described by this formula. As it represents one molecule of the substance giving the names and number of atoms of the various elements present, it denotes the molecular mass of the substance. The chemical formula identifies each constituent element by its chemical symbol and indicates the number of atoms of each element found in each subscript after the chemical symbol. For example, glucose has six carbon atoms, twelve hydrogen atoms and six oxygen atoms, so its molecular formula is $C_6H_{12}O_6$. Molecular formula of a compound can be derived from its empirical formula if the molecular mass or weight is known. The product of the mass of a compound from its empirical formula and a factor equals the molecular mass of the compound.

MOLECULAR FOSSIL (CHEMICAL FOSSILS): Complex organic compounds found in oil, bitumen, rocks and sediments that is indicative of a particular species or group of organisms that lived when the deposits were formed. Examples include the isoprene residues indicative of ancient archaeans and oleanane, a defence chemical found only in flowering plants.

MOLECULAR GENETICS: The study of the mechanisms by which genetic information is expressed and the ways in which the expression is regulated.

MOLECULAR GEOMETRY (MOLECULAR STRUCTURE): The three-dimensional arrangement of atoms but not lone pairs of electrons around a central atom of a molecule or polyatomic ion, which determines several properties of a substance including its reactivity, polarity, phase of matter, colour, magnetism and biological activity.

MOLECULAR GRAPH: The graph with differently labelled or coloured vertices (chromatic graph) which represent different kinds of atoms and differently labelled or coloured edges related to different types of bonds. Within the topological electron distribution theory, a complete network of the bond paths for a given nuclear configuration.

MOLECULAR GRAPH THEORY: The theory which deals with analyses of all consequences of connectivities inherent in molecular structure and chemical transformations. The theory does not produce numerical data but uses available data and searches for regularities that can be attributed to combinatorial and topological origins.

MOLECULAR GRAPHICS: A technique for the visualisation and manipulation of molecules on a graphical display device.

MOLECULAR IMPRINTING: The phenomenon in which there is differential expression of the gene depending on which parent it is inherited from. Genes are now thought to carry either a female or male imprint, which influences their expression in offspring. For example, the development of Huntington's disease in human is delayed until midlife in the gene if inherited from the father, whereas the inheritance from a mother causes the onset of symptoms early in childhood.

MOLECULAR ION: An ion formed by the removal from (positive ions) or addition to (negative ions) a molecule of one or more electrons without fragmentation of the molecular structure. The mass of this ion corresponds to the sum of the masses of the most abundant naturally occurring isotopes of the various atoms that make up the molecule (with a correction for the masses of the electron(s) lost or gained). For example, the mass of the molecular ion of ethyl bromide $C_2H_5^{79}Br$ will be 2×12 plus 5×1.0078246 plus 78.91839 minus the mass of the electron (m_e). This is equal to $107.95751 \text{ u} - m_e$, u being the unified atomic mass unit based on the standard that the mass of the isotope $^{12}C = 12 \text{ u}$ exactly.

MOLECULAR KINETICS: This term is applied to a kinetic study in which there is emphasis on the behaviour of the system at the molecular level.

MOLECULAR KNOT (KNOTANE): A type of compound in which one or more chains of atoms in the molecule are looped in the configuration of a knot. The molecule may have only one closed chain forming knot. If the knot is a trefoil knot the compound is chiral. Alternatively, molecular knots may have two or more separate loops tied together in a knot. In such cases there is no formal chemical bonding between

the rings and they are held by mechanical bonding between the rings. Molecular knots can be produced synthetically and also occur naturally in certain proteins.

MOLECULAR LASER: A gas laser which makes use of electronic, vibrational or rotational energy levels in molecules. The molecules can be excited by an electric discharge, as in the case of the N_2 or CO_2 laser, by chemical reactions, as in the HF or DF lasers or by another laser, as in the CH_3F far IR laser.

MOLECULAR MARKER: 1. Any site in the genome of an organism at which the DNA base sequence varies among the different individuals of a population. Such markers generally have no apparent effect on the phenotype of the individual, but they can be determined by biochemical analysis of the DNA. They are used for a variety of purposes, including chromosome mapping, DNA fingerprinting and genetic screening. Genetic tools such as restriction enzymes and the polymerase chain reaction plus the ever growing abundance of DNA sequence data coupled with automated high-throughput assays, have revealed several classes of molecular markers, including restriction fragment length polymorphisms (RFLPs), variable number tandem repeats (VNTRs), short tandem repeats, microsatellite DNA and single nucleotide polymorphisms (SNPs). 2. A substance that is introduced in an organism as a means to examine something. For example, rubidium chloride is used as a radioactive isotope to evaluate perfusion of heart muscle. 3. A substance native to the organism whose detection indicates a particular disease state, for example, the presence of an antibody may indicate an infection.

MOLECULAR MASS (RELATIVE MOLECULAR MASS; MOLECULAR WEIGHT): The mass or weight of a molecule calculated relative to one-twelfth the mass of an atom of carbon -12; or the sum of the atomic weights of the atoms making up the molecule expressed in atomic mass units (amu). It is found by adding all the relative atomic masses of the atoms that make up the molecule. For example, water has the molecular formula H_2O , indicating that there are two atoms of hydrogen and one atom of oxygen in a molecule of water. Rounded to three decimal places, the atomic weight of hydrogen is 1.008 amu and that of oxygen is 15.999 amu. The molecular weight of water is therefore,
 $(2 \times 1.008) + (1 \times 15.999) = 2.016 + 15.999 = 18.015 \text{ amu}$.

MOLECULAR MECHANICS CALCULATION (FORCE-FIELD CALCULATION): An empirical method for calculation intended to give estimates of structures and energies for conformations of molecules. The method is based on the assumption of 'natural' bond lengths and angles, deviation from which leads to strain and the existence of torsional interactions and attractive and/or repulsive van der Waals and dipolar forces between non-bonded atoms.

MOLECULAR METAL: A non-metallic material whose properties resemble those of metals, usually following oxidative doping, for example, polyacetylene following oxidative doping with iodine.

MOLECULAR MIMICRY: The close resemblance of an antigenic molecule of an invading pathogen to a host tissue molecule, which may result in the production of antipathogen antibodies by the host's immune system, cross-reacting against self tissue and leading to autoimmune diseases, such as in some forms of rheumatic fever.

MOLECULAR MODELLING: The use of computer software to produce simulations of molecular structure. Various chemical drawing programs exist to allow a graphic representation of chemical formulas in two dimensions. More sophisticated three-dimensional modelling programs also exist. The information about the molecule is stored in a data file giving the numbers and types of atoms present and the coordinates of these atoms. The software converts this into an on-screen three-dimensional view of the molecular structure in some specified format such as ball-and-stick and wireframe. The structure can be rotated on the screen and depending on the software; calculations may be done on the molecules.

MOLECULAR NUCLEATION: The initial crystallisation of a small portion of a macromolecule, after which further crystallisation is

thermodynamically favoured. Molecular nucleation may give rise to a new crystal or increase the size of a pre-existing one.

MOLECULAR ORBITAL: A region in space in a molecule where there is a high probability of finding bonding electrons. A molecular orbital is an orbital or wavefunction of a molecule's electron. Electrons around a molecule can be associated with more than one atom and are often expressed as a combination of atomic orbitals.

MOLECULAR ORBITAL THEORY: A method of computational chemistry in which the electrons are not assigned to individual bonds between atoms, but treated as moving under the influence of the nuclei in the whole molecule. The molecule has a set of molecular orbitals. The usual technique is to obtain molecular orbitals by a linear combination of atomic orbitals.

MOLECULAR OXYGEN: Normal diatomic molecular form of oxygen.

MOLECULAR REARRANGEMENT: The term is traditionally applied to any reaction that involves a change of connectivity (sometimes including hydrogen) and violates the so-called 'principle of minimum structural change'. According to this oversimplified principle, chemical species do not isomerise in the course of a transformation, for example, substitution or the change of a functional group of a chemical species into a different functional group is not expected to involve the making or breaking of more than the minimum number of bonds required to effect that transformation. For example, any new substituents are expected to enter the precise positions previously occupied by displaced groups. The simplest type of rearrangement is an intramolecular reaction in which the product is isomeric with the reactant (one type of 'intramolecular isomerisation'). An example is the first step of the Claisen rearrangement.

MOLECULAR RECOGNITION: Exhibition of complementarity between two or more molecules resulting in highly specific response between them as a result of specific and unique interaction between them through noncovalent bonding such as hydrogen bonding, metal coordination, hydrophobic forces, van der Waals forces, pi-pi interactions, electrostatic and/or electromagnetic effects.

MOLECULAR RYDBERG STATE: An excited electronic state which is composed primarily of atomic orbitals with principal quantum numbers greater than that of the ground state and the valence excited states. Such electronic states typically have a large polarisability.

MOLECULAR SHAPE: An attribute of a molecule dealing with spatial extension, form, framework or geometry. It is often described by, for example, principal axes, ovality or connectivity indices.

MOLECULAR SIEVE: Porous crystalline substances consisting of aluminosilicate minerals, clays, porous glasses, microporous charcoals, zeolites, active carbons or synthetic compounds that have open structures through which small molecules, such as nitrogen and water can diffuse. The general formula of these substances is $MnO \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O$, where M is metal ion and n is twice the reciprocal of its valency. As they form regularly spaced cavities, they provide a high surface area for the adsorption of smaller molecules, while larger molecules are not adsorbed. Molecular sieves are often utilised in the petroleum industry, especially for the purification of gas streams and in the chemistry laboratory for separating compounds and drying reaction starting materials. They can also be loaded with chemical substances, which remain separated from any reaction that is taking place around them, until they are released by heating or strongly adsorbed substance. They can thus be used as cation exchange mediums and as catalysts and catalyst supports. They are also used as the stationary phase in certain types of chromatograph (molecular-sieve chromatograph).

MOLECULAR SOLID: A solid such as hydrocarbons, ice, sugar, fullerenes, sulfur and solid carbon dioxide composed of molecules that are held together by van der Waals forces or relatively weak inter-molecular forces. Pure molecular solids are electrical insulators but they can be made conductive by doping. Molecular solids are soft and have relatively low melting temperature.

MOLECULAR SYMMETRY: The set of symmetry operations such as inversion, rotations and reflections that can be applied to a molecule or the symmetry present in molecules and the classification of molecules according to their symmetry. This set forms the point group of the molecule. For an isolated molecule the Schoenflies system rather than the Hermann-Mauguin system is used to denote its symmetry. Molecular symmetry is analysed systematically using group theory. It is possible to make definite statements about certain properties of molecules, such as whether they can have an electric dipole moment or exhibit optical activity on the basis of symmetry.

MOLECULAR SYSTEMATICS (BIOCHEMICAL TAXONOMY; MOLECULAR PHYLOGENETICS): The use of amino acid or nucleotide sequence data in determining the evolutionary relationship of different organisms. Basically, it involves comparing the sequence of functionally homologous molecules from each organism being studied and determining the number of differences between them. The greater the number of differences, the more distantly related the organisms are likely to be. In microbiology, molecular systematic, has transformed bacterial phylogeny, in particular, establishing the view that there are two quite distinct lineages of bacteria, the archaeobacteria and eubacteria.

MOLECULAR WEIGHT (RELATIVE MOLECULAR MASS; MOLECULAR MASS): The mass or weight of a molecule calculated relative to 1/12 the mass of an atom of carbon-12 or the sum of the atomic weights of the atoms making up the molecule expressed in atomic mass units (amu). It is found by adding all the relative atomic masses of the atoms that make up the molecule. For example, water has the molecular formula H_2O , indicating that there are two atoms of hydrogen and one atom of oxygen in a molecule of water. Rounded to three decimal places, the atomic weight of hydrogen is 1.008 amu and that of oxygen is 15.999 amu. The molecular weight of water is therefore,

$$(2 \times 1.008) + (1 \times 15.999) = 2.016 + 15.999 = 18.015 \text{ amu.}$$

MOLECULARITY: The number of molecules colliding to form product(s) in an elementary process. Reactions may be unimolecular (involving one molecular entity), bimolecular (involving two molecular entities) or termolecular (involving three molecular entities).

MOLECULE: The smallest particle of an element or a compound that has all the chemical properties of that element or compound and can take part in a chemical reaction. It is sufficiently stable, electrically neutral, consisting of at least two atoms in a definite arrangement held together by very strong chemical bonds. A molecule may consist of atoms of the same chemical element, for example, O_2 (oxygen) or of different elements, for example, H_2O (water).

MOLENDINACEOUS: Having large, wing-like developments; resembling the wind mill.

MOLFILE FORMAT: A format using a connection table and atom coordinates to hold information about chemical structures. It was produced by Elsevier MDL and is widely used in chemical drawing programs.

MOLISCH'S TEST (ALPHA-NAPHTHOL TEST): A standard procedure for detecting the presence of carbohydrates in solution. A small amount of alcoholic alpha-naphthol is added to the test solution in a test tube. Concentrated sulfuric acid is then poured slowly down the side of the tube. A violet ring forming at the junction of the liquids indicates a positive reaction.

MOLLUSCA: One of the largest phyla of invertebrate animals comprising more than 50,000 living species and about 35,000 fossil species dating back to the Cambrian period. There are six classes including the Gastropoda (snails, slugs, limpets), Bivalvia (bivalves, oysters and mussels) and Cephalopoda (squids and octopuses).

MOLLUSCICIDE (SNAIL BAIT; SNAIL PELLETS): A substance usually used in agriculture or gardening to control molluscs such as snail.

MOLLUSCS: Invertebrate animals with soft bodies wholly or partly enclosed in calcium carbonate shells secreted by the mantles. They occupy habitats ranging from the deep ocean to shallow waters to moist terrestrial niches. They are characterised by an unsegmented body differentiation into a head, a ventral muscular foot used in locomotion and a dorsal visceral hump covered by a fold of skin, the mantle which secretes a protective shell in many species. Respiration is by means of gills and feeding organ is a radula. Examples are snails, shellfish, octopuses, etc.

MOLTEN: State of a solid that has been heated until it melts. Hot molten solids are sometimes called melts.

MOLYBDENITE: A soft, grey sulfide mineral (MoS_2), the chief ore of molybdenum. Purified molybdenite has properties very similar to those of graphite and is used as a lubricant.

MOLYBDENUM (Mo): A silvery hard metallic transition element found in molybdenite (MoS_2) with the chemical symbol Mo, atomic number 42 and melting point 2610°C (2883 K). Chemically, it is unreactive, being unaffected by most acids. It oxidises at high temperature and can be dissolved in molten alkali to give a range of molybdates and polymolybdates. It does not occur uncombined in nature. The metal is extracted by roasting to give the oxide followed by reduction with hydrogen. It is used in alloys, often to harden steels and a necessary element in human nutrition and also used in lamps, vacuum tubes. Molybdenum(IV)sulfide (MoS_2) is used as lubricant.

MOMENT: 1. In mathematics, also called **moment about the origin**, it is the expectation of a specified power of the deviations from zero of all the values of a random variable. The power of the deviations is the order of the moment; thus, the second moment is the expectation of the squares of the deviations. 2. The product of a quantity and a distance or some power of the distance to some point associated with that quantity. The most theoretically useful moments are moments of masses, areas, lines and forces, including magnetic force. If a force tends to rotate a body about some point, then the moment or turning effect, is the product of the force and the distance from the point to the direction of the force. The moment of inertia of a body about a point is the sum, for each particle in the body, of the mass of the particle and the square of its distance from the point. The angular momentum of a body about a fixed axis is equal to the product of the momentum and the length of the moment arm (distance from the body to the axis).

MOMENT GENERATING FUNCTION: In statistics, the expected value of $\exp(t'X)$, usually written $M_X(t)$, where X is a given random variable. The moment-generating function does not always exist even for real-valued arguments, unlike the characteristic function. When it exists, it is related to the characteristic function by: $M_X(t) = C_X(-it)$.

MOMENT OF A FORCE (TORQUE): A measure of the turning effect produced by a force about an axis or fulcrum. It is equal to the product of the force and the perpendicular distance from the axis to the line of action of the force. It can be clockwise or anticlockwise. Its unit is the Newton metre. Mathematically, it is expressed as: Moment = force $\times$ the perpendicular distance from the turning point. If the magnitude of the force is F newtons and the perpendicular distance is d metres then: moment = Fd .

MOMENT OF INERTIA (MASS MOMENT OF INERTIA; ROTATIONAL INERTIA; ANGULAR MASS): A measure of an object's resistance to changes to its rotation, which is the sum of all the products formed by multiplying the magnitude of each element of mass by the square of its distance from the axes. Its symbol is I . The moment of inertia of an object about a given axis describes how difficult it is to change its angular motion about that axis. If a body consists of particles of mass m , a perpendicular distance r from an axis of rotation, the moment of inertia I of the body about that axis of rotation is given by $I = \sum mr^2$. The basic equation is $T = I\alpha$, where T is the torque causing angular acceleration α about the specified axis. The moment of inertia of a body is an important quantity because it is the analogue for rotational motion of mass for linear motion.

MOMENT OF RELATIVE MOMENTUM (RELATIVE ANGULAR MOMENTUM): Of a particle about a point with position vector $\vec{p}$, the quantity $m(\vec{x} - \vec{p}) \times (\dot{\vec{x}} - \dot{\vec{p}})$, where m is the mass and $\vec{x}$ is the position vector of the particle.

MOMENT MAGNITUDE SCALE: A scale that estimates the total energy released by an earthquake. It is used to estimate the magnitudes for all moderate to large earthquakes.

MOMENT PROBLEM: The problem of whether a given sequence of real or complex numbers is a moment sequence for some measure or metric density. The Stieltjes moment problem seeks a measure on $[0, \infty]$; the Hamburger moment problem seeks a measure on $[-\infty, \infty]$; and the Hausdorff moment problem seeks a measure on a finite interval.

MOMENT SEQUENCE: The sequence of integrals, with respect to a given measure, of either x^n (the n^{th} moment) or of $\exp(in\theta)$, which is a trigonometric moment sequence.

MOMENTUM: The product of the mass of a body and its linear velocity. For a moving object, its linear momentum (p) is calculated by the product of the object's velocity (v) and its mass (m): $p = mv$. Its SI unit is kilogram-metre per second (kgm/s) or Newton $\times$ second (Ns). Momentum is a vector quantity, it has both a magnitude and a direction, the direction being the same as that of the velocity vector. When an external force acts upon a body or a system of bodies in motion, it causes a change in the momentum of the body.

MOMENTUM SPACE: A vector space in which every point $\{k_x, k_y, k_z\}$ corresponds to a possible value of the momentum vector $\vec{k}$. Representing a problem in terms of the momenta of the particles involved, rather than in terms of their positions, can greatly simplify some problems in physics.

MOMENTUM SPECTRUM: A spectrum which is obtained when a beam of ions is separated according to the momentum-to charge ratios of the ionic species contained within it. A sector magnetic field achieves separation of the various ionic species in this way. If the ion beam is homogeneous in translational energy, as is the case with sector instruments, separation according to the ratios is also achieved and a mass spectrum is obtained.

MOMORDICA: A genus of about 60 species of herbaceous annual or perennial climbers or rarely small shrubs belonging to the family Cucurbitaceae. It is native to tropical and subtropical Africa, Asia and Australia. *Momordica grosvenori* swingle is a perennial vine, cultivated in northern Guangxi Province in China. It produces a versatile natural sweetener, whose flavour or sweetness is not affected by cooking or baking and used as a flavouring and low calorie sweetening agent in juices or drinks, can be made into a beverage and also used as a remedy for colds and congestion of the lung; *Momordica dioica*, commonly known as spiny gourd, is a climbing herb with tuberous roots, cultivated in Indo-Malaysia region, used as a vegetable in India and used medicinally in the prevention and cure of various diseases such as biliousness, asthma, leprosy, bronchitis, fever, tumours, urinary discharges, excessive salivation, heart problems, inflammation, etc. *Momordica charantia*, commonly known as bitter melon, bitter gourd or balsam pear is a prostrate hairy perennial herb or climber, which reaches up to 5 metres long with very thin angled stems and spiral tendrils opposite to the leaves. It is commonly seen growing on walls and shrubs in tropical and subtropical regions. The leaves are simple, glabrous and palmate, separated into three to seven lobes. The flowers are yellow and tubular and have five simple petals. The fruit is a long drupe and soft, with green or yellow colour, but occasionally orange and red when ripe with black seeds with a red covering. *Momordica charantia* is believed to have medicinal properties for the treatment of diabetes, obesity, cancer, malaria, dysentery, colic, fevers, burns, painful menstruation, etc.

MONAD: 1. In biology, it is a single-celled microorganism, especially a flagellate or protozoan, especially of the genus *Monas*. 2. A single individual that is free from other individuals of the same kind, rather than being fused into a group. 3. In chemistry, an element, atom or group having a valence of one. 4. In mathematics, the set of hyper-real

numbers whose difference from a given real number is infinitesimal. It is expressed as $\mu(a) = \{b \in \mathbb{R}^* : a - b \text{ is infinitesimal}\}$.

MONADELPHOUS: Having stamens that are united by their filaments to form a tube around the gynoecium, as seen, for example, in hibiscus. The stamens of flowers of leguminous plants are usually monadelphous.

MONADIC (SINGULARY; UNARY): Of an operator, predicate, etc., having only a single argument place. Negation, inversion and set complementation are monadic operators. A monadic relation is a (one place) predicate.

MONANDROUS: A flower bearing a single stamen.

MONANTHOUS: Producing only a single flower.

MONAZITE: A brown crystalline mineral consisting of a phosphate of cerium, lanthanum, other rare earth elements and thorium. Monazite occurs widely in granitic and other rocks and usually forms small brown crystals in the monoclinic system.

MOND PROCESS: A method of obtaining pure nickel by heating the impure metal in a stream of carbon monoxide at 50-60 °C (323-333 K). Volatile nickel carbonyl ($\text{Ni}(\text{CO})_4$) is formed and this can be decomposed at higher temperature, 220 °C (493 K) to give pure nickel.

MONEL METAL: An alloy of nickel (60-70%), copper (25-35%) and small quantities of iron, manganese, silicon and carbon. It is used to make acid-resisting equipment in the chemical industry.

MONERA: A taxonomic kingdom that comprises the prokaryotes including bacteria and cyanobacteria or blue-green algae, which are a group of unicellular organisms without a membrane bound nucleus in their cells.

MONIC POLYNOMIAL: A polynomial in one variable in which the coefficient of the highest power is 1 (unity). The polynomial $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is monic if $a_n = 1$.

MONILIFORM: Jointed or constricted at the regular intervals, thereby giving the shape of a string of bead. This is seen in the stems and roots of certain plants and in the antennae of some insects.

MONITOR (COMPUTER SCREEN; DISPLAY): The screen, which displays the computer's user interface and open programs, allowing the user to interact with the computer, typically using the keyboard and mouse.

MONITORING: Continuous or repeated observation, measurement and evaluation of health and/or environmental or technical data for defined purposes, according to prearranged schedules in space and time, using comparable methods for sensing and data collection.

MONKEY: Any member of two tropical anthropoid primate groups, Old World monkeys and New World monkeys, of the class Mammalia. It is characterised by grasping hands (opposable digits), forward-facing eyes (binocular vision), prehensile tail and a highly-developed brain. The tail is used as a fifth limb while foraging for food or climbing. Monkeys are highly skilled climbers and spend most of their lives in trees. They are highly social omnivores organised in clans as large as several hundred individuals headed by an old male. Sexually mature males of all species are always potent and like all human females, all matured females, which are not pregnant have a monthly menstrual cycle. Most species bear a single young, which is reared by the mother for years.

MONO: A prefix used to denote one.

MONO-ENERGETIC RADIATION: Radiation consisting of particles of a single kinetic energy or photons of a single energy.

MONOAMINE OXIDASE (MAO): An enzyme responsible for the destruction of transmitter substances at neuronal synapses by oxidation. Monoamine oxidase inhibitors are drugs that block degradation of amine transmitters within the cell, used for controlling hypertension, to treat forms of depression, Alzheimer's disease and Parkinson's disease.

MONOAMINE-OXIDASE INHIBITOR (MAOI): Any of a family of antidepressant medications that inhibit the activity of monoamine oxidase. It is used to treat diseases such as depression, anxiety, migraine and some other conditions.

MONOAMINE: An amine compound containing one amino group, especially a compound that functions as a neurotransmitter.

MONOATOMIC: A substance made up of single separate atoms of an element or having one atom in the molecule or chemical group that can be replaced during a chemical reaction. Gases which are monatomic are noble gases including helium, neon, xenon, krypton or radon.

MONOATOMIC ION: An ion that contains only one atom, for example, K^+ , H^+ , Zn^{2+} , O^{2-} and H^- .

MONOATOMIC MOLECULE: A molecule such as argon or helium consisting of only one atom.

MONOBASIC ACID: An acid that has only one hydrogen atom per molecule which can be replaced by a metal. Examples are HCl (hydrochloric acid) and HNO_3 (nitric acid).

MONOBIONTIC: A life cycle in which there is only a single independent generation, there is no alternation of generations.

MONOCARBOXYLIC ACID: A carboxylic acid such as acetic acid with the chemical formula CH_3COOH and only one carboxylic group.

MONOCARPIC (HAPAXANTH; SEMELPAROUS): Referring to a plant that flowers only once during its lifetime. After fruiting, the leaves senesce and the plant dies completely. Examples include some species of bamboo, annual plants including all grain crops and most domestic vegetables, long lived plants such as century plant (agave) and *Corypha utan*, commonly known as Gebang Palm or Cabbage Palm, a giant palm tree reaching up to 20 metres high, found in some parts of Asia such as Malaysia, Indonesia and the Philippines. The term is also used for an animal or fish that reproduces only once during its lifetime and dies, for example, Pacific salmon, squid, mayflies, Atlantic eel, some species of butterflies, cicadas, some molluscs such as squid and octopus and many arachnids.

MONOCARPOUS: Having only one carpel.

MONOCHASium (MONOCHASIAL CYME): A cymose inflorescence in which only one axillary bud develops into a lateral branch at each node. Laterals may arise on alternate sides of the stem (a scorpioid monochasium) or may arise on one side only (a helicoid monochasium), which gives an asymmetrical appearance to the inflorescence.

MONOCHLAMYDEOUS: A flower with only a single floral envelope. Such a flower has a calyx without a corolla or sometimes, a corolla without a calyx.

MONOCHORD (SONOMETER): A device used to verify the relationship between the vibration frequency of the sound produced

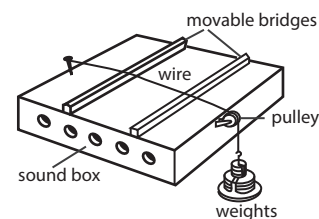
by a plucked string, the tension, length and mass per unit length of the string. It consists essentially of a hollow sounding box with two bridges attached to its top. The string, fixed to the box at one end is stretched between the two bridges so that the free end can run over a pulley and support a measured load. The relationships

are usually called Mersenne's laws, named after Marin Mersenne (1588-1648), who investigated and codified them. The Mersenne's laws state that for small amplitude vibration, the frequency is proportional to the following: (i) the square root of the tension of the string; (ii) the reciprocal of the square root of the linear density of the string; and (iii) the reciprocal of the length of the string.

MONOCHROMATIC: Having only one colour.



Monochasium



A monochord (sonometer)

MONOCHROMATIC RADIATION: Denoting radiation, especially light, of one wavelength or colour. Pure monochromatic radiation is not possible, although the light from a laser occupies a very narrow band of wavelengths and is virtually monochromatic, as are some spectral lines.

MONOCHROMATIC TRIANGLE: A triangle with the same colour of vertices.

MONOCHROMATOR: An optical device that transmits a mechanically selectable narrow band of wavelength of light or other radiation chosen from a wider range of wavelengths available at the input. It is used to produce monochromatic light in science experiments and in optical equipment.

MONOCHROME: In computing, it refers to a two-tone image, rather than one with several shades of a single colour. For example, a monochrome monitor uses one colour for the background and another to display text or images on the screen. Before colour monitors became standard, most computers had monochrome displays. These displays often had a black background with green text, though some displayed text in other colours, such as red or orange.

MONOCISTRONIC: A type of messenger RNA (mRNA) that can encode only one polypeptide per RNA molecule. In eukaryotic cells, virtually all messenger RNAs are monocistronic.

MONOCLINIC: A crystal form in which all three axes are unequal, with one of them perpendicular to the other two, which intersect at an angle inclined at other than a right angle.

MONOCLINOUS (HERMAPHRODITE): Having both male and female reproductive parts in the same flower. This condition is thought to be derived from a primitive unisexual floral arrangement involving wind pollination and to have developed simultaneously with animal pollination. The hermaphrodite arrangement lends itself to self pollination and inbreeding but many species have intricate self incompatibility systems.

MONOCLONAL ANTIBODY (mAB; moAb): A specific antibody, usually a protein, produced artificially by a genetic engineering technique. It is one of numerous identical cells derived from a single parent cell. The parent cell is derived from the fusion of a normal antibody-producing cell with a cell derived from a malignant tumour of lymphoid tissue of a mouse. Monoclonal antibodies are used to identify a particular antigen within a mixture and can, therefore, be used to identify a blood group.

MONOCOLPATE: A term used to describe a pollen grain having one colpus, as is commonly found in most petaloid monocotyledon species.

MONOCOTYLEDON (MONOCOTYLEDONEAE; MONOCOTS): A class of flowering plants distinguished by having one seed leaf (cotyledon) within the seed. The monocotyledons generally have parallel leaf veins, scattered vascular bundles within the stems, lack vascular cambium and flower parts in threes or multiples of three. Monocotyledon species include cereals such as maize and millet, bulbs such as onions and garlic ornamentals such as tulips, lilies and a very limited number of trees such as palms.

MONOCROPPING: A system of farming in which a single crop is grown on the same piece of land for several years, without crop rotation through other crops. Monocropping is economically very efficient, allowing for specialisation in equipment and crop production. However, it often leads to depletion of soil nutrients and problems with weeds and pesticides.

MONOCULAR VISION: Vision in which the eyes are situated on either side of the head giving a visual field of almost 250°, enabling an animal to see two objects at once. Examples are found in rabbits and other herbivores.

MONOCULTURE: The growth of only one species such as rice or other agricultural crops in a given area.

MONOCYCLIC: 1. In chemistry, it refers to having a molecular structure with only one ring. Examples are benzene and toluene. 2.

In biology, having a life cycle that is completed in a year. 3. Arranged in a single whorl, as seen in certain flowers and the shells of some vertebrates.

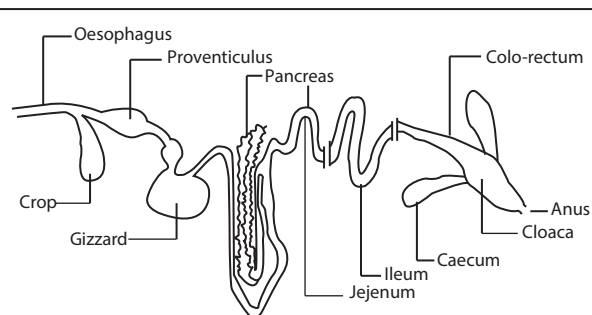
MONOCYTE: A type of leucocyte or white blood cell, the largest in the human body's immune system. It has a kidney-shaped nucleus and is actively phagocytic, ingesting bacterial and cell debris. Monocytes' function in the immune system include replenishing resident macrophages and dendritic cells under normal states and responding to inflammation signals and moving quickly to sites of infection in the tissues and differentiating into macrophages and dendritic cells to elicit an immune response.

MONODROMY THEOREM: The theorem asserting the uniqueness of analytic continuation that if a function of a complex variable is analytic in a disc inside a simply connected domain and if the function can be analytically continued along all polygonal arcs in the domain, then the function extends to a single-valued analytic function on the entire domain.

MONODROMY THEOREM OF DARBOUX: The theorem that if a function of a complex variable, analytic in a domain bounded by a simple closed curve, is continuous in the closed region and is one-to-one on the boundary curve, then the function is one-to-one throughout the region.

MONOECIOUS: Flowering plants such (e.g., maize plant) or animals with separate male and female reproductive organs or flowers on the same individual or plant. On the other hand, **dioecious** means a plant or animal having separate reproductive organs in different individuals of the same species. An example is the pawpaw (*Carica papaya*).

MONOGASTRIC: A term used to describe animals such as humans, cats, pigs and dogs with a simple single-chambered stomach, which cannot digest the sugar molecules raffinose and stachyose. The digestive tract of chicken, which is monogastric, is shown in the following diagram:



The digestive tract of chicken (monogastric)

MONOGYNOUS: Having only one carpel.

MONOHYBRID CROSS: A genetic cross between two individuals of the same species that are homozygous for a single trait or heterozygous for a single trait or parents that differ in the alleles they possess for one particular gene, one parent having two dominant alleles and the other two recessive alleles. All the offspring or the first filial generation (F1) have one dominant and recessive allele. For example, the allele for green pod colour (G) is dominant and the allele for yellow pod colour (g) is recessive. The cross-pollination between a P generation green pod plant and a P generation yellow pod plant will result in all green offspring. All genotypes are (Gg).

MONOHYBRID INHERITANCE: The inheritance of one pair of contrasting characteristics. The different forms of the characteristic are usually controlled by different alleles of the same gene. For example, a monohybrid cross between two pure-breeding plants (homozygous for their respective traits), one with yellow seeds (the dominant trait) and one with green seeds (the recessive trait), would be expected to

produce an F1 (first) generation with only yellow seeds because the allele for yellow seeds is dominant to that of green.

MONOHYDRATE: A chemical compound such as calcium chloride monohydrate, $\text{CaCl}_2 \cdot \text{H}_2\text{O}$ that contains one mole of water of crystallisation in each of its formula unit.

MONOHYDRIC: A term used especially of alcohols describing a chemical compound that has one replaceable hydroxyl (-OH) group, for example, ethanol ($\text{CH}_3\text{CH}_2\text{OH}$). Monohydric alcohol is an alcohol having only one hydroxyl group per molecule.

MONOID: 1. A semigroup or a set with a binary operation that is closed and associative with an identity element. 2. A groupoid, which is usually commutative.

MONOKARYON: Fungal mycelium or hypha in which the cells contain only a nucleus each.

MONOLAYER: A single, closely packed layer of atoms or molecules. The term floating monolayer is used for certain spread monolayers or films which are in the condensed, often solid, state.

MONOLAYER CAPACITY: For chemisorptions, it is the amount of adsorbate which is needed to occupy all adsorption sites as determined by the structure of the adsorbent and by the chemical nature of the absorptive. For physisorption, it is the amount needed to cover the surface with a complete monolayer of atoms or molecules in close-packed array, the kind of close packing having to be stated explicitly when necessary.

MONOLAYER CULTURE: A culture in which all the cells are grown in a single layer on a flask or petri-dish containing the culture medium.

MONOLETE: Describing a spore that has a simple linear scar marking the point at which it was joined in the tetrad. The microspores of members of the Isoetales are monolete, which sets them apart from other lycopsids. However their megaspores are trilete.

MONOLITH: 1. Shaped, fabricated, intractable article with a homogeneous microstructure which does not exhibit any structural components distinguishable by optical microscopy. The article is usually fabricated by cold pressing or hot pressing of a polymeric material or by using a reactive processing technique such as reaction injection moulding, crosslinking, sol-gel processing, sintering, etc. 2. **Soil monolith** is a vertical section of the soil profile that is not disturbed with the natural appearance of the soil, including its horizonation, colour, and structure preserved and mounted in a cabinet for display and teaching purposes.

MONOLOCULAR (UNILOCULAR): Having a single chamber or cavity.

MONOMER: A chemical compound composed of simple molecules that can react with other molecules of the same or other compounds to form very large molecules called polymers. Examples of monomers and their polymers are ethylene, whose polymer is polythene, styrene whose polymer is polystyrene and amino acids whose polymer is protein.

MONOMER MOLECULE: A molecule which can undergo polymerisation thereby contributing constitutional units to the essential structure of a macromolecule.

MONOMEROUS: Having one member, as in a floral whorl with only one carpel.

MONOMIAL: 1. An expression in algebra consisting of a single term such as $3y$ or xz that consists of the multiplication or division of one or more variables by a constant. If several variables are considered, for example, x, y, z , then each can be given by an exponent, so that any monomial is of the form xyz with a, b, c nonnegative integers. Note that any exponent 0 makes the corresponding factor equal to 1. 2. A

scientific name that consists of one element only such as the names of most families of plants and animals.

MONOMODE FIBRE (SINGLE-MODE OPTICAL FIBRE, SMF; MONOMODE OPTICAL FIBRE; SINGLE-MODE OPTICAL FIBRE; SINGLE-MODE OPTICAL WAVEGUIDE; UNIMODE FIBRE):

An optical fibre designed to carry only a single ray of light (mode). The modes define the way the wave travels through space. It has a narrow glass core of about $5\mu\text{m}$ in diameter with a large amount of cladding of a material with a smaller refractive index than the core. The 2009 Nobel Prize in Physics was awarded to Charles K. Kao for his theoretical work on the single-mode optical fibre. Waves can have the same mode but have different frequencies.

MONOMORPHIC: 1. Existing in a single form. 2. In biology, it refers to possessing the same genotype, form or structure throughout the entire life.

MONOMORPHISM: In mathematics, a one-to-one or injective homomorphism. A monomorphism from A to B is often denoted $A \hookrightarrow B$.

MONONUCLEAR PHAGOCYTE SYSTEM (RECTICULOENDOTHELIAL SYSTEM): The network of both free and fixed macrophages derived from bone marrow precursor cells by way of monocytes. Their substantial phagocytic activity is mediated by immunoglobulin and by the serum complement system. They are concentrated in the bone marrow, liver, spleen and lymph nodes.

MONONUCLEOSIS (INFECTIOUS MONONUCLEOSIS; GLANDULAR FEVER): A common virus infection of adolescence causing a variety of symptoms including severe sore throat, headache, fever, malaise and enlargement of lymph nodes and spleen. Skin rashes, hepatitis with jaundice, pericarditis and involvement of the nervous system may also be prominent. A typical lymphocytes in the blood and specific agglutination reactions are diagnostic. Severe cases may require steroids and convalescence may be lengthy. It can be transmitted in saliva and has thus been nicknamed the "kissing disease."

MONOOXYGENASE (MIXED FUNCTION OXIDASE; MIXED FUNCTION OXYGENASE): Any of a large group of enzymes that perform oxidation-reduction reactions in which one atom of the oxygen molecule is incorporated into the reaction.

MONOPETALOUS (GAMOPETALOUS): Having the petals united so as to form a tube.

MONOPHAGOUS: An organism that feeds on only one kind of food. An example is the koala, which exists solely on eucalyptus.

MONOPHYLETIC: Describing a group of organisms that are descended from a single stock or a particular single common ancestor. Examples are birds and crocodiles, which are descended from the monophyletic group, Archosauria and are more closely related to each other than other living reptilian descendants.

MONOPHYLLOUS: Consisting of only one leaf.

MONOPHYDONT: A type of dentition that consists of a single set of teeth that lasts for the entire life span of the animal. These animals have open-rooted teeth, which have wide openings at the root that permit dentin-forming cells to grow and multiply. Most rodents, such as beavers and herbivores have open-rooted teeth. The gnawing habits of these animals wear down their teeth; otherwise the teeth would grow too long.

MONOPLANETIC: A term used to describe fungi that produce only one type of zoospore (planospore).

MONOPODIAL BRANCHING: A type of growth exhibited by many plants in which secondary shoots or branches arise behind the growing

point but remain subsidiary to the main stem, which continues to grow indefinitely. The largest secondary shoots are furthest from the apex of the main stem and the size of the shoots decreases regularly towards the top of the plant. This results in the pyramidal form of growth typical of many conifers, for example, the spruce (*Picea*). The secondary shoots also tend to show the same pattern of branching along their length.

MONOPODIUM: Vascular plants such as orchids and pine trees which grow upward from a single point. It consists of a single main stem that continues to grow from the tip and give rise to lateral branches.

MONOPROPELLANT: A single liquid, used in a liquid-propellant rocket engine that serves as both fuel and oxidiser. A monopropellant decomposes into a hot gas when an appropriate catalyst is introduced. The commonest example is hydrazine, often used in spacecraft attitude control jets.

MONOPROTIC ACID: An acid that has only one ionisable hydrogen atom in each molecule.

MONOPTEROUS: Having a single wing.

MONOSACCHARIDE (SIMPLE SUGAR): The most basic units of carbohydrates that cannot be split into smaller units, having the chemical formula $C_x(H_2O)_y$. They are crystalline solids, usually colourless and water-soluble and the building blocks of disaccharides such as sucrose and polysaccharides (cellulose and starch). They are named on the basis of the number of carbon atoms present: trioses (3 carbons), tetroses (4 carbons), pentoses (5 carbons), hexoses (6 carbons), etc. Each of these is further divided into aldoses and ketoses, depending on whether the molecule contains an aldehyde group (-CHO) or a ketone group (-CO-). Examples of monosaccharides are glucose (dextrose), fructose (levulose) and galactose. Monosaccharides can exist as either straight chain or ring shaped molecules. They also exhibit optical activity giving rise to dextrorotatory and laevorotatory forms.

MONOSEPALOUS (GAMOSEPALOUS): Having sepals fused or united to form one unit.

MONOSODIUM GLUTAMATE, MSG (SODIUM GLUTAMATE; SODIUM 2-AMINOPENTANEDIOATE): White crystalline powder having the chemical formula $C_5H_8NNaO_4 \cdot H_2O$ and melting point 232 °C (505 K). It is the single sodium salt of glutamic acid (an amino acid). It was originally produced from seaweed in East Asia, but now made principally from cereal glumens, such as those of wheat, corn and soybeans, from solutions evolved in the manufacture of beet sugar and by microbiological fermentation of carbohydrates. It is used to enhance the flavour of many packaged and fast foods. It can cause an allergic reaction in some susceptible people.

MONOSOMY: Possessing only one copy of a particular chromosome instead of the normal two copies. Turner syndrome is an example of a condition caused by monosomy. It typically arises by fertilisation occurring between a normal gamete (n) and one deficient in the said chromosome (n-1). Monosomics are more common in polyploids where the deleterious consequences of missing a chromosome can be masked by chromosomes in other genomes. It is caused by non-disjunction.

MONOSPERMOUS: Having only a single seed.

MONOSPORE: A spore formed by certain red algae of the Bangioideae as a means of asexual reproduction.

MONOSTACHOUS: Having flowers that are arranged in a single spike.

MONOSTICHOUS: Arranged in a single vertical rank or row. This is common in the flowers of grasses of the chloridae.

MONOSTYLOUS: Having only one style.

MONOSYMMETRICAL (ZYGOMORPHIC): Capable of being bisected into many similar halves along only one plane. This is seen in the flowers of the sweet pea.

MONOSYNAPTIC REFLEX: A simple reflex consisting of only two neurons (one sensory neuron and one motor neuron) in an animal, which transmits information from sensory neuron to the appropriate motor neuron across a single synapse in the spinal cord. The knee-jerk reflex action is an example of a monosynaptic reflex.

MONOTECTIC REACTION: The reversible transition, on cooling, of a liquid to a mixture of a second liquid and a solid: $\text{liquid}_1 \rightleftharpoons \text{liquid}_2 + \text{solid}$.

MONOTONE (MONOTONIC): A sequence or function is monotone if its terms are always strictly increasing called **isotone** or strictly decreasing called **antitone**, but not both.

MONOTONE CONVERGENCE THEOREM: In mathematics, the theorem about the convergence of monotonic sequences that are also bounded that the integral of the limit of a monotone increasing sequence of nonnegative measurable functions is equal to the limit of the integrals of the functions in the sequence. If $\{f_n\}$ is monotone increasing sequence of nonnegative measurable functions defined on a measurable set E, then $\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu$, where f is the limit of a finite or infinite sequence.

MONOTONE RELATION: A subset of the product of a Banach space and its dual with the property that if (x_1, y_1) and (x_2, y_2) lie in the subset, then $\langle (y_1 - y_2), (x_1 - x_2) \rangle \geq 0$. A **monotone multifunction** is the T if $y \in T$.

MONOTREMES: Egg-laying mammals such as spiny anteater or echidna (*Tachyglossus aculeatus*) and the duck-billed platypus (*Ornithorhynchus anatinus*) classified in the phylum Chordata, subphylum Vertebrata, class Mammalia order Monotremata. They possess mammalian features of hair and mammary glands, but lay eggs rather than giving birth to live young. The eggs resemble those of reptiles, with large yolks and leathery shells. Like birds and reptiles, monotremes have a single opening, the cloaca, for the passage of liquid and solid wastes, the transfer of sperm and, in the female, the laying of eggs.

MONOTRICHIOUS FLAGELLUM: A long single thread-like appendage or flagellum located on one end of the bacterial cell, enabling the bacteria to rotate around counter-clockwise, propelling it forward.

MONOTYPIC: 1. Species which are the only member of a taxonomic category and show only minor variations throughout the entire geographical range of the species so that there are no recognised races or subspecies. 2. An artwork created by pressing on paper laid on an inked metal plate or sheet of glass. Although similar prints can be made, each one will be unique.

MONOVALENT: 1. A chemical element or isotope that has a valency of one. 2. Containing antigens from a single strain of a microorganism, for example, a monovalent vaccine. 3. Describing an antibody or antigen, which has only one site of attachment.

MONOZYGOTIC TWINS (IDENTICAL TWINS): Twins formed at some stage during early development when the embryo from a single fertilised egg divides into two genetically identical twins that are always of the same sex.

MONSOON: A wind pattern that changes direction with change of season bringing seasonally heavy rain to South Asia. It blows towards the sea in winter and towards the land in summer. The change of wind direction is caused by the differences in temperatures of landmasses in contrast to that of oceans.

MONSTER GROUP: The largest irregular finite simple group, of order $246 \cdot 320 \cdot 59 \cdot 76 \cdot 112 \cdot 133 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$, whose value is approximately 8.0817×10^{53} .

MONTANE ZONE: The biogeographic zone made up of relatively moist, cool upland slopes below timberline and characterised by the presence of large evergreen trees such as pine, deodar, cedar trees and large coniferous trees. The higher the altitude, the lower the canopy height, and the plant community transitions to grasslands or tundra, with vegetation that includes moss-draped bushes and epiphytes such as orchids, ferns, moss, lichen, and liverworts. Montane ecosystems are found in higher latitude of the northern hemisphere and high mountains in Europe, Asia, north Canada and USA.

MONTE CARLO SIMULATION/METHOD: A technique for estimating the solution, x , of a numerical mathematical problem that involves random sampling for the mathematical simulation of physical systems. Monte Carlo calculations are applied to problems that can be formulated in terms of probability and are usually carried out by computer. Applications of Monte Carlo methods include calculations in the theory of liquids, phase transitions and quantum mechanical systems. The name comes from the casino at Monte Carlo and alludes to the use of random numbers in this technique.

MONTH: A period of time connected with the motion of the Moon around Earth. The familiar calendar (or civil) month is an artificial unit consisting of a whole number of days. For astronomical purposes there are several other types of months: **Synodic month** or **lunar month** is the average interval from one new Moon to the next (29.53059 days). **Anomalistic month** is the interval in which the Moon passes from perigee to perigee (27.55464 days). **Sidereal month** is the interval in which the Moon passes from a fixed position with respect to the stars back to the same position (27.32166 days). **Draconic month** or **nodal month** is the interval between successive passages of the Moon through its ascending node. **Tropical month** is the interval between successive passages of the Moon through the vernal equinox, a mean.

MONTMORILLONITE: A clay mineral of variable composition. It is hydrated aluminosilicate cations and is the principal constituent of bentonite and fuller's earth. One type of montmorillonite readily absorbs water and another type swells in water to form a gel. They are used in bonding moulding sands and for oil-well drilling muds, to decolourise oils and as a source of petroleum cracking catalysts.

MONTREAL PROTOCOL ON SUBSTANCES THAT DEplete THE OZONE LAYER: An international treaty on substances that deplete the ozone layer which was adopted in Montreal, Canada in 1987. It controls the consumption and production of numerous substances that are responsible for ozone depletion. Such substances include chlorine- and bromine-containing chemicals, such as chlorofluorocarbons (CFCs), methyl chloroform, carbon tetrachloride, and many others.

MOODLE (MODULAR OBJECT-ORIENTED DYNAMIC LEARNING ENVIRONMENT): An open source course management system, used by thousands of educational institutions around the world to provide an organised interface for learning over the Internet. Moodle allows educators to create online courses, which students can access as a virtual classroom. A typically Moodle home page will include a list of participants including the teacher and students and a calendar with a course schedule and list of assignments. Other Moodle features include online quizzes, forums, where students can post comments and ask questions, glossaries of terms and links to other Web resources.

MOON: 1. The Earth's only natural satellite, which it orbits at a mean distance of about 384,400 km. It is less than one-third the size of Earth with a radius of about 1,738 km at its equator and a surface area of

37,958,621 km sq. Its gravity is 1.624 m/s^2 ($1/6$ th that of the Earth). It has a mass of $7.35 \times 10^{22} \text{ kg}$ (one-eightieth that of the Earth) and a volume of $21.958 \times 10^9 \text{ km}^3$. It has no atmosphere or water and it is illuminated by sunlight. It rotates on its axis in 27.3 days, in exactly the time it takes to orbit Earth and it therefore always presents the same face to Earth. 2. A large round object like the moon that circles around a planet other than the Earth. Mercury and Venus are the only planets in the solar system that do not have moons.

MOON PHASES (LUNAR PHASES; PHASES OF THE MOON):

The appearance of the illuminated portion of the moon by the sun as seen by an observer on earth. This is caused by the angle from which an observer on earth can see the moon illuminated by the sun as it orbits the earth. The phase changes as a result of the relative positions of the Earth, Sun and Moon. The eight phases of the moon are: (i) New Moon when the moon is not visible in the sky except during a solar eclipse that occurs when the moon is between earth and the sun, called conjunction; (ii) Waxing Crescent when the moon is less than one-half illuminated by the sun but less than one quarter illuminated that occurs when the moon's illumination is increasing; (iii) First Quarter when the moon is one-half illuminated by the sun that occurs when the moon's illumination is increasing; (iv) Waxing Gibbous when the moon is more than one-half illuminated by the sun but not completely illuminated that occurs when the moon's illumination is increasing; (v) Full Moon when the moon is totally visible in the sky except during a lunar eclipse; (vi) Waning Gibbous when the moon is more than one-half illuminated by the sun but not completely illuminated that occurs when the moon's illumination is decreasing; (vii) Third Quarter when the moon is one-half illuminated by the sun that occurs when the moon's illumination is decreasing; (viii) Waning Crescent when the moon is less than one-half illuminated by the sun but less than one quarter illuminated that occurs when the moon's illumination is decreasing. The cycle of Moon phases takes approximately 29.53 days and explains the origin of the word month.

MOONSTONE: A translucent variety of alkali or plagioclase feldspar, showing opalescence caused by sodium and potassium unable to mix. It is a gemstone.

MOOR: An area of often high land that is not suited for cultivation. It is formed of acid soil covered with grass and shrub such as heather.

MOORE SPACE: In mathematics, a topological space in which there exists a sequence $\{G_n\}$ satisfying the conditions that each G_i is a family of open sets whose union is the whole space; that for each i , G_{i+1} is contained in G_i ; and that if x, y are distinct members of an open set R , then there is an n for which x is a member of a set U whose closure is in G_n , but which does not contain y .

MOORE-OSGOOD THEOREM: The theorem guaranteeing that the order of taking iterated limits may be changed. For example, suppose $f: X \times Y \rightarrow Z$ is a mapping between pseudo-metric spaces; if

$$\lim_{x \rightarrow a} f(x, y) = f(a, y) \text{ uniformly for } y \neq b$$

$$\lim_{y \rightarrow b} f(x, y) = f(x, b) \text{ uniformly for } x \neq a,$$

then

$$\lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x, y) = \lim_{y \rightarrow b} \lim_{x \rightarrow a} f(x, y).$$

The most common application has Y as the extended natural numbers with b as ∞ . The theorem has been named after Eliakim Hastings Moore (1862-1932), who originally proved it and William Fogg Osgood (1864-1943), who generalised it.

MOORE-SMITH CONVERGENCE (NET CONVERGENCE): In topology, the property of a net of a sets with respect to a directed set

D , that a sequence $\{x_d\}$, where x_d is the element of S associated with d in D , is eventually in every neighbourhood V of some x , in the sense that there exists d such that x_e lies in V for all e in D with $e \geq d$; the net is then said to converge to x .

MOORE'S LAW: The statement that the number of transistors that can be placed on an integrated circuit doubles every year, named after Intel co-founder Gordon E. Moore, who described the trend in his 1965 paper. It describes a long-term trend in the history of computing hardware.

MORACEAE: Also known as **mulberry** or **fig family**, it is a family of flowering plants in the rose order (Rosales) comprising about 38 genera and over 1000 species. The family is composed of species of deciduous or evergreen trees and shrubs, occurring most commonly in tropical and subtropical regions. Examples are fig trees (*Ficus carica*), banyan (*Ficus benghalensis*), mulberry (genus *Morus*), breadfruit (*Artocarpus communis*), jackfruit (*Artocarpus heterophyllus*), paper mulberry (genus *Broussonetia*), and Osage orange (*Maclura pomifera*). Some species have medicinal properties. For example, *Ficus exasperate* with the common names sandpaper (leaf) tree or the white fig tree is used for the treatment of various ailments such as arthritis, asthma, bleeding, breast cancer, gonorrhoea, hypertension, epilepsy, intestinal pain, piles, rheumatism, and wounds. The bark of the paper mulberry (genus *Broussonetia*) has been used in the manufacture of cloth and paper products.

MORAINE: An accumulation of boulders and rock fragments carried or deposited by a glacier. **Ground moraine** is the deposit of clay, sand and varying sizes and shapes of rock that builds up underneath a glacier. **Terminal moraines**, also called **end moraines**, are deposition of rocks, soil and sediment at the furthest point reached by a glacier. **Lateral moraines** are formed along the sides of a glacier composed of soil and varying sizes of rock; and **medial moraine** is a type of moraine formed from two lateral moraines when two glaciers merge

MORBIDITY: A state of being sick or being diseased. Proportional morbidity is the proportion of all of the diseased animals in the population that have a particular disease. Morbidity rate is the ratio of diseased to healthy animals in the population.

MORDANT: An acidic or basic compound used in certain dyeing processes to fix certain dyes in cloth. Mordants are often inorganic oxides or salts, which are absorbed on the fabric. The dyestuff then forms a coloured complex with the mordant, the colour depending on the mordant used as well as the dyestuff. Acid mordants such as tannic acid are employed with basic dyes. Basic mordants such as alum, chrome alum and certain salts of aluminium, chromium, copper, iron, potassium and tin are employed with acid dyes.

MORERA'S THEOREM: In mathematics, the consequence of Cauchy's integral formula that a complex-valued function that is continuous on a region and of which the contour integral over all triangles is zero, must be analytic. It is named after Giacinto Morera (1856-1909), Italian engineer and mathematician.

MORINDA: A genus of about 80 species in the family, Rubiaceae that is distributed in the tropical areas growing between 3 to 12 metres high. The species may be trees, shrubs or climbers. The leaf arrangement is opposite. Inflorescences are terminal and are made up of white flowers. Each flower's corolla is infundibular or hypocrateriform. *Morinda citrifolia*, with the common names noni, great morinda, Indian mulberry, beach mulberry and cheese fruit is a small

to medium sized tree which grows in the tropics and Pacific areas. It has white flowers that occur in the leaf axils in clusters and produces aggregate and multiple fruits about the size of a potato, which can be fleshy or dry. Both fruits and leaves have useful purposes. For example, the fresh leaves are used to cure ulcer, while the fruits are used as an emmenagogue to promote menstrual discharge. The leaves are applied to heal wounds and ulcers, as well as internal administration as a tonic and febrifuge. Another species, *Morinda lucida*, with the common name brimstone tree that grows in topical Africa (from Senegal to Sudan and southward to Angola and Zambia) to about 18 metres high with a grey/brown bark. Its leaves are simple and opposite. It bears scented white/green-yellow flowers in heads. Its fruits are clustered in heads with green colour and become soft and black. It has medicinal properties, for example, the leaves and bark are used to treat venereal diseases. In some traditions, the root bark is used to dye textiles red. The wood is used for making charcoal, in construction; as well as for making furniture, canoes, etc.

MORINGA (*Moringa oleifera*): A slender, deciduous, perennial tree in the family, Moringaceae that grows primarily in tropical, subtropical and semiarid climates to a height of about 10 metres. It has drooping branches; and brittle stems, with corky bark. The leaves are feathery, pale green, compound, tripinnate, 30–60 centimetres long, with many small leaflets. The lateral leaves are somewhat elliptic and the terminal ones are obovate and slightly larger than the lateral ones. The flowers are fragrant, white or creamy-white, 2.5 centimetres in diameter, borne in sprays, with five at the top of the flower and have yellow stamens. The pods are pendulous, brown, triangular and about 30–120 centimetres long and 1.8 centimetres wide and tapering at both ends. It splits lengthwise into 3 parts when dry and contains about 20 dark brown seeds embedded in the pith. The plant is commonly planted as a living fence and on graves to keep predatory animals like hyenas away, while its branches are used as charms against witchcraft. Much of the plant is edible by both humans and farm animals. The leaves contain lots of food nutrients like vitamins A, B and C, proteins and mineral salts. The oil in the seeds is considered a potential for biodiesel. The plant also serves as windbreak and also to reduce soil erosion.

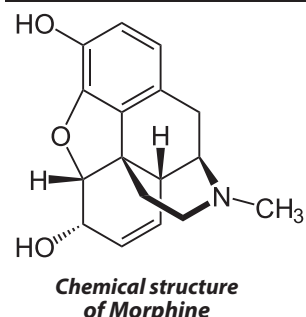
MORNING GLORY (PURPLE MORNING GLORY; TALL MORNING GLORY; *Ipomoea purpurea*): A trailing or twinning plant species in the genus *Ipomoea*, native to Mexico and Central America with tendrils that wrap themselves around supports. It is a perennial plant by cutting and an annual plant by sowing from seeds. The stem is hairy. The leaf blade is ovate, entire or 3-lobed, acuminate or gradually tapering to a sharp point at the apex, cordate at the base and glabrous or pubescent. The flowers are solitary or in few-flowered cymes. The stalk of the inflorescence is up to 12 cm long. Sepals are finely pubescent all over; corolla is white, pink or magenta and white below. The morning glory plant is so named because the flowers blossom early mornings and last for a day. They remain open in overcast weather, but shrivel quickly in hot daytime sun. New flowers are produced daily. Therefore, there is constant supply of startling blue flowers each morning. It grows well in moist and loamy soil in warm temperate and subtropical regions. It is sometimes grown for its beautiful purple and white flowers even though it is considered as a harmful weed.

MORPH: Any of a distinct common form found in a population displaying polymorphism.

MORPHACTIN: Any of a group of synthetic compounds, all derivatives of fluorene-carboxylic acid, that affect many aspects of

plant growth and differentiation. Morphactins have little effect on mature tissues but in seedlings and growing shoots apical dominance is removed and internode elongation inhibited, giving a dwarfed bushy appearance. Expansion of the leaf lamina is also prevented and flowers, if produced, may be deformed.

MORPHINE (MORPHIA): A sedative narcotic alkaloid, which is the principal derivative of opium. Alkaloid was first isolated from opium in 1803 by the German pharmacist, F. W. A. Sertürner. Opium is the juice in the unripe seed pods of the opium poppy, *Papaver somniferum*. It acts directly on the central nervous system and is highly addictive. It is used medically for the relief of severe pains. It also causes a feeling of euphoria or well-being. It also decreases hunger, inhibits the cough reflex, produces constipation and usually reduces sex drive. In women it may interfere with the menstrual cycle.



MORPHISM (ARROW): The elements of an arbitrary category which play the role of mappings of one set into another.

MORPHOGEN: A hypothetical substance that determines the pattern of tissue development and, in particular, the positions of the various specialised cell types within a tissue. During development, different morphogens are produced within the embryo or by cell bodies of the surrounding maternal tissue.

MORPHOGENESIS: The origin and development through growth and differentiation of form and structure in an organism. It is one of the three fundamental aspects of developmental biology along with the control of cell growth and cellular differentiation.

MORPHOLINE: A clear liquid with an ammonia-like smell with the molecular formula C_4H_9ON , density 1.007 g/cm^3 , melting point -5°C (268 K) and boiling point 129°C (402 K). It features both as amine and ether functional groups. It may be produced by the dehydration of diethanolamine with sulfuric acid. Because of the amine, morpholine is a base. Its conjugate acid (with HCl) is morpholium chloride. Morpholine is a common additive in fossil fuel to provide corrosion protection and also used as a solvent.

MORPHOLOGICAL CONVERGENCE (CONVERGENT EVOLUTION): The evolution of basically dissimilar structures such as the wings of birds and insects to serve a common function.

MORPHOLOGY: The study of the physical structure and form of organisms, including their soft tissue external form. **Soil morphology** is the branch of soil science that deals with the characteristics and properties of the soil observable in the field within the various soil horizons. It includes the various types, thickness, and arrangement of the horizons in the profile, and by the texture, structure, consistence, and porosity of each horizon.

MORPHOSPECIES: A group of biological organisms that differs in some morphological respect from all other groups and defined by its anatomical features.

MORPHOTROPIC TRANSITION: An abrupt change in the structure of a solid solution with variation in composition.

MORREY SYSTEM: A paddock system of rotational grazing, used in the management of dairy herds.

MORSE CODE: A signal system devised in 1838 by Samuel Morse for use in the wire telegraph, now used in radiotelegraphy and elsewhere. Letters, numbers and punctuation are represented by combination of dots (brief taps of the transmitting key) and dashes (three times the length of dots).

MORSE POTENTIAL: The potential energy of a diatomic molecule as a function of the difference $(r - r_e)$, where r is the variable

interatomic distance and r_e the equilibrium interatomic distance. The Morse potential $U(r - r_e)$ is given by $De(1 - \exp[-\beta(r - r_e)])^2$, where De is the dissociation energy at the minimum of the curve (i.e. when $r = r_e$) and β is a constant. The Morse potential was used by the US physicist Philip M. Morse in 1929 in solving the Schrodinger equation. The Morse potential is a reasonably good representation of potential-energy function except that as r approaches 0, U does not approach infinity as it should for a true potential energy function. Modifications of the Morse potential have been suggested to improve on this aspect.

MORTALITY: The number of deaths that occur at a specific time, in a specific group/species or from a specific cause.

MORTAR: 1. A hard heavy bowl designed to hold substances to be ground into small pieces or powder by means of a club-shaped tool called pestle. Mortar is usually made of materials such as wood, porcelain or ceramic. The shapes and sizes are varied to meet specific needs. Together with a pestle, they are used in crushing and grinding objects such as foodstuffs; in the chemistry laboratory to prepare reactants for a solid-state synthesis; and by pharmacists to grind substances for medical use. 2. A cast-iron bowl in which ore is crushed. 3. A mixture of cement, sand and water that sets and is used to hold together the bricks and blocks in walls and building. 4. A small canon that fires shells in high arcs.

MORULA: An early stage in the development of the embryo in animals when the cells are in the process of splitting before the blastula stage. The morula consists of a number of blastomeres, the cells formed from the fertilised ovum as a result of cell division.

MOSAIC: 1. An organism made up of cells that have different genotypes but have developed from the same zygote. 2. A disease in plants due to virus infection. The disease is characterised by white mosaic patches on leaves, wrinkled leaves and loss of chlorophyll in leaves, giving a variegated appearance.

MOSAIC DEVELOPMENT: A pattern of development, such as that of a mollusc, in which the early blastomeres each give rise to a specific part of the embryo. In some animals, the fate of the blastomeres is established in the zygote.

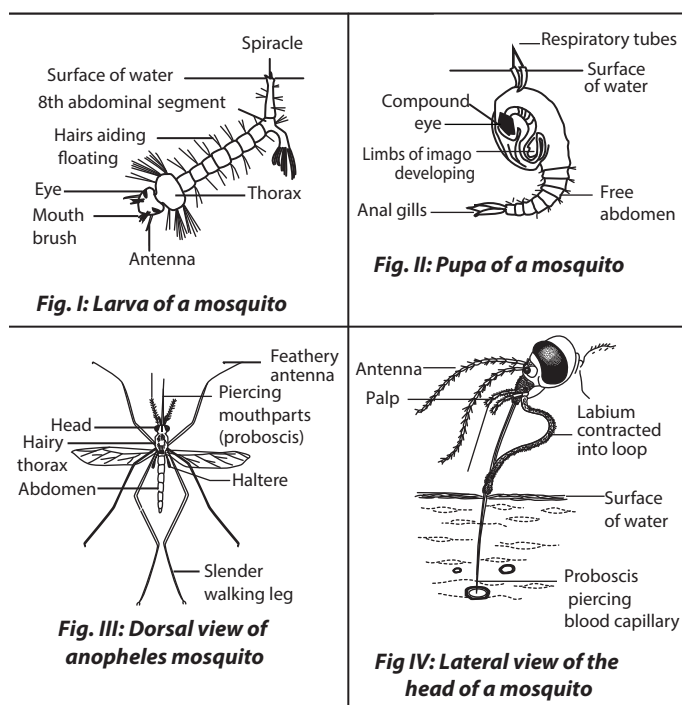
MOSAIC EVOLUTION: The evolution of different parts of an organism at different rates. Many aspects of the human phenotype have evolved relatively slowly or not at all since the hominid diverged from their primate ancestors. One notable exception is the nervous system which has given humans their overwhelming selective advantage. At molecular level, some proteins evolve very rapidly while others remain unchanged over millions of years. This high degree of evolutionary independence among different aspects of the phenotype permits flexibility, for example, when a population is faced with new selection pressures in a changing environment, only the most crucial components need to evolve, not the entire phenotype.

MOSCHATE: Having a musky colour.

MUSCULARIS MUCOSA (LAMINA MUSCULARIS MUCOSAE): A thin layer of smooth muscle that forms the outermost part of the mucosa of the mammalian intestine or stomach.

MOSELEY'S LAW: An empirical law concerning the characteristic x-rays that are emitted by atoms, which states that the frequencies of the lines in the x-ray spectra of the elements are related to the atomic numbers of the elements.

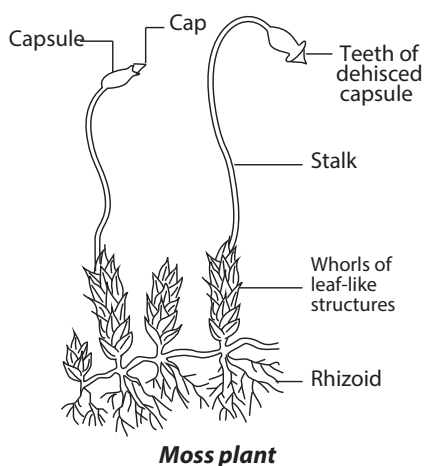
MOSQUITO: A small insect with long legs, a long slender abdomen and narrow wings, belonging to the order Diptera of the family Culicidae. Some females of various species (genera *Aedes*, *Anopheles* and *Culex*) have long proboscis for sucking blood of birds and mammals, including humans and transmit diseases, including dengue fever, encephalitis, filariasis, malaria, yellow fever and elephantiasis. It is found mainly in the tropics.



MOSS: A type of plant belonging to the division Bryophyta that has thin stems, no flowers and no proper roots but reproduces by spores. It usually grows on soil, rocks and the bark of trees and in bogs and shallow streams.

MÖSSBAUER EFFECT: An effect occurring when certain nuclides decay with emission of gamma radiation, discovered in 1957 by the German physicist, Rudolph Mossbauer (1929). For an isolated nucleus, the gamma radiation would usually have a spread of energies because the energy of the process is partitioned between the gamma-ray photon and the recoil energy of the nucleus. In 1957, Rudolph Mossbauer (1929-) found that in certain solids, in which the emitting nucleus is held by strong forces in the lattice, the recoil energy is taken up by the whole lattice. Since this may typically contain $10^{10} - 10^{29}$ atoms, the recoil energy is negligible and the energy of the emitted photon is sharply defined in a very narrow energy spread. The effect is exploited in Mossbauer spectroscopy in which gamma-ray source is mounted on a moving platform and a similar sample is mounted nearby. A detector measures gamma rays scattered by the sample. The source is moved slowly towards the sample at a varying speed, so as to continuously change the frequency of the emitted gamma radiation by the Doppler Effect, a sharp decrease in the signal from the detector levels. In chemistry, Mossbauer spectroscopy can also give information about the bonding and structure of compounds because chemical shifts in the resonance energy are produced by the presence of surrounded atoms.

MOST-SIGNIFICANT BIT (MSB): In computing, it is the extreme left hand digit representing the largest power or value. For example,



in the binary number 11100111 (231 in decimal), the 1 at the extreme left is the Most-Significant Bit.

MOST SIGNIFICANT DIGIT: The leading digit, when correcting to a given number of decimal places or significant figures. For example, 25.307 corrected to 3 significant figures is **25.3**, with the emboldened **2** as the most significant digit. It holds its place value of 20 and maintains the magnitude of the original number.

MOTHER CELL: A cell that divides to form other differentiated cells and hence loses its identity, for example, the cell that gives rise to a sieve element and companion cell.

MOTHER NUCLIDE (PARENT NUCLIDE): Nuclide that undergoes nuclear decay to form the daughter nuclide.

MOTHER-OF-PEARL (NACRE): The iridescent substance of which pearls and the inner coating of bivalve mollusc shells are made. It consists of alternate thin layers of aragonite (calcium carbonate) and conchiolin, a horny substance. Mother-of-pearl is valued for its beauty and is used in thin sheets for ornament, jewelry and buttons.

MOTHERBOARD (MAINBOARD; LOGIC BOARD): The main circuit board of a computer to which the CPU, ROM, memory RAM expansion slots, PCI slots and USB ports are attached. It also includes controllers for devices like the hard drive, DVD drive, keyboard and mouse. Basically, the motherboard is what makes everything in the computer work together. Each motherboard has a collection of chips and controllers known as the chipset.

MOTILE: A term describing organisms or parts of organisms that are capable of movement by independent means. It especially applies to single-celled and simple multicellular organisms, as well as to some mechanisms of fluid flow in multicellular organs, in addition to animal locomotion. Motility is the ability of a cell to move from one place to another.

MOTION: A change in the position or the direction of a body. Only the relative motion of a body can be measured. Absolute motion is meaningless.

MOTIVATION: The internal condition responsible for temporary reversible changes in the responsiveness of an animal to external stimulation. These responses could be due to hunger, thirst, sex, etc. Responsiveness to maturation, injury, learning, are not considered to be due to changes in motivation.

MOTOR: A machine that converts energy into motion and can be used as a power source, such as to drive another machine or to move a vehicle. An example is electric motor, which converts input electric energy to motion. In most cases, a central current carrying coil rotates cutting across a magnetic field. The reaction of the current carrying coil and the magnetic field produces motion. The structure of a simple motor is just like that of a generator. Some other types of motors include the **hydraulic motor** and **pneumatic motor** also known as **compressed air engine**. The **hydraulic motor** is driven by the flow of hydraulic fluid or water or other liquid under pressure and forms part of hydrostatic power transmission systems to convert hydraulic energy into mechanical energy. Hydraulic motors can prove more efficient, suitable and economical over their electrical or pneumatic counterparts. Hydraulic motors are used for many applications such as driving winches and cranes, excavators, conveyor and feeder drives, drilling rigs trench cutters, etc. **Pneumatic motor** or **compressed air engine** is a type of motor which does mechanical work by converting compressed air to mechanical work through either linear or rotary motion. Small pneumatic motors are used to power some hand-held tools such as power ratchet wrenches, drills, sanders, grinders and cutters. Some types are also used to start ship and aeroplane engines, etc.

MOTOR CELL: A type of plant cell that acts like a hinge joint to enable the movement of plant parts such as the opening and closing of leaflets in a carnivorous plant or in response to light intensity. Motor cells adjust their internal concentration of potassium ions (K^+)

to alter their turgidity and hence the cell shape. The movements resulting from the changes in motor-cell turgor are relatively gradual, taking minutes or hours.

MOTOR CORTEX: The region of the cerebral cortex involved in the planning, control, and execution of voluntary movements.

MOTOR EFFECT: The tendency of a wire carrying an electric current in a magnetic field to move or the force experienced by a current-conductor in a magnetic field. This effect is applied in a wide range of devices including the action of moving-coil loudspeaker, meters and electric motors. The force F of length l in a magnetic field B is expressed as $F = BIl \sin \theta$, where θ is the angle between the direction of the current and the field.

MOTOR ENDPLATE: The end of a nerve fibre against a skeletal muscle fibre, to which it transmits nerve signals.

MOTOR NERVE: Any nerve that transmits impulses from the central nervous system to muscles or body organs. They cause voluntary and involuntary muscle contraction and stimulate glands to secrete hormone.

MOTOR NEURONE: A neurone that transmits nerve impulse from the central nervous system to the effector organ and thereby initiates a physiological response.

MOTOR OUTPUT: A response to the stimuli received by the nervous system. A signal is transmitted to organs that can convert the signals into action, such as a movement or a change in heart rate. It is expressed in function as muscle contraction of the smooth and striated muscle and as secretion of the exocrine and endocrine glands and, in effect, represents the total behavioural activity of human life.

MOTOR PATHWAY (EFFERENT NERVE; EFFERENT NERVE FIBRE; MOTOR NEURON; EFFECTOR NEURON): The portion of the peripheral nervous system that carries nerve impulses away from the central nervous system to effectors such as muscles or glands.

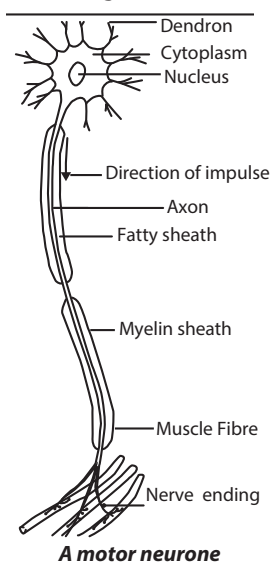
MOTOR UNITS: They consist of lower motor neuron, motor end-plate and muscle fibres supplied by the end-plate forming the units into which skeletal muscles are organised to enable muscles to contract on a graded basis. Each motor unit supplies from four to more than a hundred muscle fibres.

MOTOR-GENERATOR SET (M-G SET): A device for converting electrical power to another form. Motor-generator sets are used to convert frequency, voltage or phase of power. They may also be used to isolate electrical loads from the electrical power supply line.

MOTT INSULATOR: A substance that is expected to conduct electricity but becomes an insulator although the highest occupied energy band is not necessarily full. This effect is due to electron-electron interactions which are not considered in the formulation of conventional band theory. Certain transition metal oxides in which there are narrow bands are Mott insulators. The concept was put forward and developed by the British physicist, Sir Nevill Francis Mott (1905-96).

MOTT TRANSITION: A type of metal-insulator transition in condensed matter, in which the key parameter is the ratio of electron repulsion to bandwidth. Above a certain critical value of this parameter the metal becomes a Mott insulator, even though the uppermost band is not full. It was proposed by British physicist, Sir Nevill Francis Mott (1905-96) in 1949.

MOTTLED: Marked with coloured spots or blotches. It is usually used to describe plants or the skin of animals. In animals, mottling may



either be a sign of a disease or a hereditary trait. In plants, it is usually a sign of a disease or malnutrition. In humans, it is used to describe uneven, discoloured patches on the skin which results from lowered blood flow to the surfaces of the skin or *Herpes zoster* infections. In soils, **mottled zone** is a layer that is marked with spots or blotches of different colour or shades of colour.

MOTZKIN'S THEOREM: In mathematics, the theorem that if two disjoint finite sets, S and T , of points in Euclidean plane do not both lie on the same line, then there exists a line in either S or T containing exactly two of the points. It was named after Theodore Samuel Motzkin (1908-1970), Israeli-American mathematician.

MOULD: 1. A rapidly growing fungus found growing on dead and decomposing organic material such as bread, cheese, kenkey and citrus. The organic material serves as the substrate in which root-like structures called rhizoids penetrate to provide support and carry out nutrition by extracellular digestion. Mould belongs to the phylum zygomycota because of the production of characteristic zygospores contained in sporangium. It can reproduce both sexually and asexually. Sexual reproduction is by conjugation of positive (+) and negative (-) strains of hyphae and asexual reproduction is by means of spores. 2. A pattern, hollow form or matrix used to give a specific shape to something in a plastic or molten state. It is commonly used for shaping metals, plastics, clays, glass and in building construction to cast materials such as bricks, blocks and concrete. 3. Soil that is rich in humus and easily crumbled or worked.

MOULDING: The use of a pattern, hollow form or matrix to give a specific shape to something in a plastic or molten state. It is commonly used for shaping plastics, clays, glass and in building construction to cast materials such as bricks, blocks, concrete, etc.

MOULT MIGRATION: A movement of birds, usually taking place at the end of the breeding season when a flock of birds fly to specific locations to moult (shed their feathers, plumage, skin, etc.).

MOULTING (ECDYSIS; SLOUGHING; SHEDDING): A process in which an animal routinely sheds off a part of its body such as feathers, plumage, skin, etc., for regrowth, either at specific times of the year, or at specific points in its life cycle. For example, arthropods shed off exoskeleton and cuticle to allow regrowth by the secretion of a larger exoskeleton; a reptile sheds off the outer skin as a sign of growth; and birds shed off their feathers as a sign of rejuvenating the reproductive system and causing growth of new feathers.

MOUND: 1. A mass of earth or rock that rises above the surface of the surrounding land. 2. The soft pad of fat over the pubic bone, just above the vulva.

MOUNTAIN: A feature of the Earth's surface, which is a natural elevation, rising abruptly to a summit, attaining an altitude greater than that of a hill (usually more than 610 metres) and sometimes rising to thousands of metres. They generally have steep slopes and a relatively small summit area. They often occur in groups forming linked ranges, such as the Alps or the Himalayas. Kilimanjaro, the highest mountain in Africa, located in north-eastern Tanzania, near the border with Kenya is 5,895 metres above sea level. Mount Everest, a mountain located in the Himalayas of southern Asia is the highest mountain in the world with a height of 8,850 metres above sea level.

MOUNTAIN PASS LEMMA: A theorem giving conditions for a potentially bounded differentiable function to have a critical point that if $f(0) < \inf \{f(x) : \|x\| = 1\} > f(a)$ for some $\|a\| \geq 1$, there will be a critical point b with $f(b) \geq \inf \{f(x) : \|x\| = 1\}$ if f satisfies some 'growth condition' such as that $f(x)$ tends to infinity with x . Geometrically, this critical point is located in the 'mountain pass'.

MOUNTING: The final stage in the preparation of permanent slides for microscopic examination. The mount, which may be a section, wholemount, squash or smear, is immersed in a mounting medium, usually a liquid that later solidifies. The medium permeates entirely through the specimen, leaves no air spaces and does not support the growth of bacteria and fungi. Examples are glycerol, Canada balsam and osmium tetroxide.

MOUSE: 1. A hand held button-activated input device usually moved on a pad, used to control a pointer on a computer screen, enabling the operator to select operations or manipulate text or graphics. A **mouse pad**, **mouse mat** or **mousing surface**, is a surface designed for tracking the motion of a computer mouse. 2. A small furry rodent, belonging to the families Muridae, Cricetidae and Platacanthomyidae of the order Rodentia. The common house mouse is classified as *Mus musculus*, which breeds every 10 to 17 weeks throughout the year, producing five to ten young in a litter. They often have soft grey or brown fur, long hairless tails and large ears and live in houses and in fields. It relates to a rat but smaller. Mice, like rats, are considered pests as they consume and damage large quantities of food and spread diseases.

MOUSE HOVER: It triggers an event such as popping up a graphic or changing the cursor from a pointer to a small hand when a user hovers or places a mouse over a designated area, such as a hyperlink on a Web page.

MOUSE MODEL: A laboratory mouse useful for medical research because it has specific characteristics that resemble a human disease or disorder. Strains of mice having natural mutations similar to human ones may serve as models of such conditions. Scientists can also create mouse models by transferring new genes into mice or by inactivating certain existing genes in them.

MOUTH (ORAL CAVITY): The opening of the alimentary canal to the digestive and respiratory tracts, which in most animals is used for the ingestion of food and for making sounds. It is located at the bottom of the face, defined by the lips, jawbone, teeth and gums. In the rear it merges with the throat. The roof of the mouth is composed of the hard and soft palates and the floor of the mouth is formed by the tongue, a muscular structure that contains the organs of taste called taste buds.

MOUTHPARTS: Modified paired appendages on the head segments of arthropods, used for feeding. A typical insect has a labium, the lower lip, one pair each of mandibles and maxillae and a labrum, which is the projecting upper lip. In some arthropods, the mouthparts are modified to form piercing stylets or a sucking proboscis. Arachnids such as scorpions, spiders and horseshoe crabs have chelicerae (specialised mouth parts resembling fangs or pincers modified for grasping and piercing) and pedipalps, which are a pair of segmented appendages located behind the chelicerae and used for various reproductive, predatory or sensory functions.

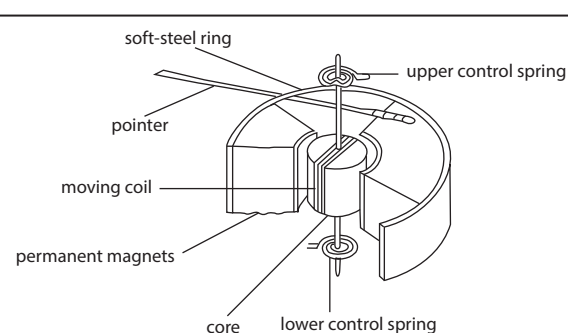
MOVEMENT LICENSE: A license which is needed in order to move animals from areas of infectious disease.

MOVING AVERAGE: In statistics, a sequence of averages derived from a given sequence of values by taking the averages of the subsequences that start with each member of the given sequence in turn and consist of the same number of adjacent members. It is often used in time series to even out short-term fluctuations and make a trend clearer. An example is the 3-term moving average of 4, 6, 8, 7, 9, 8, which is 6, 7, 8, 8. When the set of numbers $y_1, y_2, y_3, \dots, y_n$ are considered, the moving average of the order n is given by the following set means: $\frac{y_1 + y_2 + \dots + y_n}{n}, \frac{y_2 + y_3 + \dots + y_{n+1}}{n}, \frac{y_3 + y_4 + \dots + y_{n+2}}{n}$.

MOVING-COIL INSTRUMENT (MOVING COIL INSTRUMENT):

A measuring instrument in which current or voltage is determined by the couple on a small coil pivoted between the poles of a magnet with curved poles, giving a radial magnetic field. Moving-coil instruments have been largely replaced by solid state and digital ones. Moving coil instruments are of two types: the permanent magnet type (for dc only); and the dynamometer type (for both ac and dc). The permanent magnet type is based on the principle of permanent magnet moving coil mechanism (when a current carrying conductor is placed in a magnetic field, mechanical forces acts on the conductor). The coil placed in magnetic field and carrying current is attached to the moving system. With the movement of the coil the pointer moves over the scale. It consists of a powerful magnet with soft iron piece and light rectangular coil of many turns of fine wire wound on an aluminium

former inside which is an iron core. The purpose of the coil is to make the field uniform. The coil is mounted on spindle and acts as the moving element. The current is fed into and out of the coil by means of springs which also provides controlling torque. The dynamometer type instruments are modified form of permanent magnet type and operate based on the principle that mechanical force exists between the current carrying conductors. The operating field is produced not by a permanent magnet but by another fixed coil. The moving system and the control system are similar to those of permanent magnet type. Such instruments can be used for both ac and dc circuits. It essentially consists of a fixed coil and a moving coil. The fixed coil is split into two equal parts which are placed close together and parallel to each other. The moving coil is pivoted in between two fixed coils. The fixed and moving coil may be excited separately or may be connected in series. The controlling torque is provided by two springs. The deflection of the pointer is directly proportional to the square of the operating current. Hence, the scales of these instruments is non uniform in nature.



The permanent-magnet moving-coil instrument

MOVING-IRON INSTRUMENT (MOVING IRON INSTRUMENTS):

A measuring instrument, generally used to measure alternating voltages and currents in which current or voltage is determined by the force of attraction on a bar of soft iron pivoted within the magnetic field of a fixed coil or by the repulsion between the poles induced in two soft iron rods within the coil. It consists of a moving element of a small piece of soft iron in the form of a vane or rod; a coil to produce the magnetic field due to current flowing through it and also to magnetise the iron pieces (in repulsion type, a fixed vane or rod is also used and magnetised with the same polarity); a control torque is provided by a spring or weight (gravity); a damping torque is normally pneumatic, the damping device consisting of an air chamber and a moving vane attached to the instrument spindle; and a deflecting torque produces a movement on an aluminium pointer over a graduated scale. Moving-iron instruments have scales that are nonlinear and somewhat crowded in the lower range of calibration.

MOVING-IRON MICROPHONE: A transducer in which sound waves cause a light armature to oscillate so that it varies the reluctance of a magnetic circuit. In a coil surrounding this path the varying reluctance is experienced as a variation in the magnetic flux within it, which induces a corresponding emf.

MOWER: Machine for cutting or **mowing** grass and other upright crops.

MOZILLA FIREFOX (FIREFOX): A very popular open source Web browser for Windows, Mac and Linux from the Mozilla project. Firefox is a free web browser that is developed and distributed by Mozilla Foundation.

MP3 (MPEG-1; MPEG-2 AUDIO LAYER III): A popular compressed audio file format, which compresses CD-quality sound by a factor of roughly 10, while maintaining nearly the same CD-quality sound. Because of their small size and good fidelity, MP3 files have become a popular way to store music files on both computers and portable devices like the iPod.

MPEG (MOVING PICTURE EXPERTS GROUP): A group that works with the International Organization for Standardization (ISO) to develop standards for digital video and digital audio compression. Each of the MPEG standards is designed for a different purpose. Some of their compression formats and standards include the following: (i) MPEG-1, developed for storage and retrieval of moving pictures and audio on storage media such as Compact Disc (CD). MPEG-1 and provides VHS quality at 1.544 Mbps and is compatible with single-speed CD-ROM technology; (ii) MPEG-2 that provides broadcast quality video with resolutions up to 1920 x 1080 and supports a variety of audio/video formats, including legacy TV, HDTV and five channel surround sound; (iii) MPEG-4 offers a variety of compression options, including low-bandwidth formats for transmitting to wireless devices as well as high bandwidth for studio processing.

mRNA: Abbreviation for messenger RNA, one of the three types of RNA; the other two are transfer RNA (tRNA) and ribosomal RNA (rRNA). It is synthesised or transcribed from DNA in the genetic material that attaches to ribosomes in the cytoplasm and specifies the primary structure of a protein by specifying amino acid sequences. In eukaryotes, the mRNA carries the genetic information from the nucleus to the site of protein synthesis (ribosome) in the cytoplasm to be translated. In non-eukaryotes such as bacteria, the mRNA is produced and translated in the cytoplasm of the bacterial cell.

MRSA (METHICILLIN-RESISTANT STAPHYLOCOCCUS AUREUS): A type of bacteria that is highly resistant to methicillin, penicillin, and certain other antibiotics. If infected, mostly through contact, it causes infections of the skin, blood and lungs.

MS-DOS (MICROSOFT DISK OPERATING SYSTEM): The first operating system developed by Microsoft for IBM-compatible personal computers. It is a command-line-based system, where all commands are navigated by using MS-DOS commands in text form or allows the user to type commands. It is not much used today, as it has been largely replaced by graphical user interfaces, especially Microsoft Windows.

MSI (MEDIUM-SCALE INTEGRATION): The technology that enables hundreds of electronic components to be etched into a single integrated circuit that is just a few square millimetres.

MU MESON (MUON): An elementary particle in the lepton family (not a meson), having a mass 209 times that of the electron, a negative electric charge and a mean lifetime of 2.2×10^{-6} seconds.

MUCIGEL (SLIME): A mixture of plant secretion, bacterial cells and soil particles that surrounds the tip of plant roots. It contains complex polysaccharide called **mucilage** which is secreted by the root cap and helps to lubricate the passage of growing root tip through the soil. However, the root tip also releases amino acids and sugars, which encourage the growth of bacteria, thought to assist in mobilising soil nutrients for uptake by the root.

MUCILAGE: A sticky substance frequently present in the cell walls of aquatic plants and in the seed coats of certain other species. They harden when dry and shiny when wet.

MUCK SOIL: A soil that contains at least 50% of highly decomposed organic matter; used for growing crops such as onions, carrots, celery, and potatoes.

MUCOPOLYSACCHARIDES: Polysaccharides composed of alternating units from uronic acids and glycosamines and commonly partially esterified with sulfuric acid.

MUCOPROTEIN: A glycoprotein in which the carbohydrate component is a relatively large polysaccharide. Mucoprotein readily forms gel with water and constitutes the mucin in mucus. The mucoprotein chains are joined together end to end by disulfide bonds, forming very long mucin strand. When released into water, the network of strands expands rapidly to produce a large volume of mucus.

MUCORMYCOSIS (MUCOR): A serious fungal infection of the sinuses and the lining of the nose, caused by fungi in the order Mucorales.

MUCOSAL DISEASE: A livestock disease caused by a virus. It is often fatal.

MUCOUS MEMBRANE (MUCOSA): A moist lining in the body passages of a mammal that contains mucus-secreting cells that secrete mucus, a moistening, lubricating and protective fluid. It is opened directly or indirectly to the external environment. They line various body cavities such as the nostrils, the mouth, the lips, the eyelids, the ears, the genital area and the anus that are exposed to the external environment and internal organs. Examples of mucous membranes are glans penis (head of the penis) and glans clitoridis, along with the inside of the prepuce (foreskin) and the clitoral hood.

MUCRO: A short, sharp, pointed projection from certain organs or parts. This is usually seen at the tip of a leaf or shell. **Mucronate** means tip shaped with a short, sharp, point, as occurs in mucronate feathery or a mucronate leaf; or having a length/width ratio of less than 3:1; straight and stiff. **Mucronulate** means tipped with very small mucro; or having a length/width ratio of 1:1; broader than long.

MUCUNA: A genus of about 100 accepted species of climbing vines and shrubs in the family Fabaceae, occurring in the woodlands of tropical areas. The leaves are alternative, occurring in groups of three (trifoliate) and show "sleep-movement" (in the night they fold together) like other legumes. The white, lavender or purple flowers are pea-like but larger, occurring in racemes with distinctive curved petals. The fruit is a pod and in some species are covered with numerous reddish-orange hairs or very fine spicules (trichomes), which cause itching when touched. Some of the numerous species include *Mucuna pruriens*, commonly known as velvet bean, cowhage or cowitch; *Mucuna acuminata*, *Mucuna amblyodon*, *Mucuna monosperma*, *Mucuna paniculata*, etc. The seeds of some species such as *Mucuna pruriens* are high in protein, carbohydrates, lipids, fibre and minerals and used as an effective fertiliser and also as a source of food and medicine for both humans and animals. The seeds also contain a high level of Levodopa (L-Dopa), which is a precursor to dopamine, a neurotransmitter found in the brain and have been used to treat Parkinson's disease. Medicinally, the seeds of some species such as *Mucuna pruriens* have also been used to treat human ailments such as obesity, cellulite, high cholesterol levels, maintain blood sugar levels and increase energy levels. It also contains growth hormones, testosterone and several amino acids used to help enhance muscle growth and toning. It has been also used to help improve the sex drive in men and women due to the increased levels of dopamine.

MUCUS: A slimy fluid rich in glycoprotein secreted by goblet cells in vertebrate mucous membranes and submucosal glands, immunoglobulins and inorganic salts. It protects and lubricates the mucous membranes in many parts of the body, such as intestines, vagina and trachea. Mucus traps dust and bacteria in mammalian air passages, lubricates the surfaces of internal organs and facilitates the movement of food through the gut, while preventing digestive enzymes from reaching and digesting the gut itself. Snails, slugs, bony fish, hagfish and some other invertebrates also produce external mucus, which serves as a protective function against infectious agents and also provides protection against toxins produced by predators. It also facilitates movement and may play a role in communication.

MUDSTONE (MUDROCK): Fine-grained sedimentary rock up to 0.0625mm made up of clay-to-silt sized particles. It is similar to shale but with less developed lamination.

MUFFLE FURNACE: An insulated furnace usually heated by electricity and used for producing controlled high temperatures. In the laboratory, muffle furnaces are used for drying solids, sintering and studying high-temperature reactions. They typically operate in the range 100-1200°C (373-1473 K).

MULBERRY: Any tree making up the family Moraceae of the order Urticales, genus *Morus*, which contains about 48 genera and 1200 species. They are woody flowering dicot plants with dark fruits, similar to blackberries with small, clustered, unisexual flowers. Examples are fig and breadfruit trees. White mulberry, *Morus alba*, has been cultivated for centuries in China and its leaves are the main food of the silkworm.

MULBERRY HEART DISEASE: A pig disease caused by lack of vitamin E and selenium. It is characterised by sudden death or by dyspnoea, cyanosis and recumbency.

MULCH: Deposits of organic materials or compost placed around crop plants to provide nutrients and to protect the soil and the roots of plants from the effects of soil crusting, erosion or freezing and also used to retard the growth of weeds. **Mulch farming** is a system of cultivation in which the organic residues are placed on the surface around the plants, a process called **mulching**.

MULE: 1. A hybrid and usually sterile offspring of a male ass and mare or female horse, bred as a work animal or the offspring of a male horse and female donkey, called hinnie, considered inferior to mule. The mule resembles a horse in size (weighs about 500-640 kg) and strength but has many characteristics as that of a donkey, for example, long ears, a tufted tail, slender legs, small hooves and a loud bray. 2. A cross breed sheep from a blue faced Leicester ram and a Waledale ewe.

MULL: A type of humus found in deciduous forests produced by the decomposition of grass or deciduous-forest litter. Earthworms and other soil fauna mix this humus thoroughly with the mineral content of the soil.

MÜLLER CELL (MULLER GLIA): Any of the cells collectively known as the Müller glia found in the vertebrate retina that supports the visual sensory cells, notably the rods and cones.

MÜLLERIAN MIMICRY: The sharing of a common characteristic in organisms such as a biological resemblance in which two or more unrelated noxious or dangerous organisms exhibit closely similar warning systems, such as the same pattern of bright colours.

MULTI: A prefix used to denote many or several.

MULTILINEAR FUNCTION (MULTILINEAR FORM): A function of several vector space variables that is linear in each variable separately, especially in finite dimensions. The degree is then the number of such variables. Such a form is alternating if all underlying vector spaces are identical and if interchanging two variables merely changes the sign of the expression.

MULTI-CORE PROCESSOR: A single computing component with two or more independent actual central processing units, called cores, which are the units that read and execute program instructions. The instructions are ordinary CPU instructions such as add, move data and branch, but the multiple cores can run multiple instructions at the same time, increasing overall speed for programs amenable to parallel computing.

MULTI-PROCESSING: A mode of operation in which two or more processes (programs) are executed concurrently by separate CPUs that have access to a common main memory. Multiprocessing can be implemented in two different ways: (i) using more than one physical processor or (ii) using a processor with multiple cores.

MULTI-PROGRAMMING (PARALLEL PROCESSING): A mode of operation in which two or more programs are executed in an interleaved manner by a single CPU.

MULTI-STRAND CHAIN: In polymers, a chain that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through more than four atoms, more than two on at least one side of each constitutional unit. A chain that comprises constitutional units joined to each other through atoms on at least one side of each constitutional unit is termed an n-strand chain, for example, three-strand chain. If an uncertainty exists in defining, the highest possible number is selected.

MULTI-STRAND MACROMOLECULE: A macromolecule that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through more than four atoms, more than two on at least one side of each constitutional unit.

MULTI-VALUED FUNCTION (CARRIER; SET-VALUED FUNCTION; MULTIFUNCTION; POINT-TO-POINT MAPPING): A mapping that associates at least one input element with multiple output elements. It is a mapping from a set into the power set of another set. In mathematics, it is a mapping that associates a number of different elements of the output with the same element of the input;

a mapping from a set into the power set of another set. The inverses of trigonometric, hyperbolic, exponential and integer power functions are multivalued. For example, the function z^2 maps each complex number z to a well-defined number z^2 , while its inverse function $\sqrt{z}$ maps, for example, the value $z = 1$ to $\sqrt{1} = \pm 1$

MULTIACCESS SYSTEM: A computer system in which computational and data resources are made available simultaneously to a number of users who access the system through terminal devices, normally on an interactive or conversational basis.

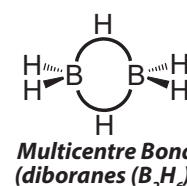
MULTIANT: In mathematics, the generalised form of determinant for arrays of more than two dimensions.

MULTICAST: A type of communication in which data is sent from one or more points to a set of designated recipients.

MULTICAST BACKBONE (MBONE): In computing, it a virtual network on the Internet that supports the IP multicast protocol. It is composed of many networks and delivers packets of multi-media data to multiple users at the same time, enabling video and audio communication as well as video conferencing.

MULTICELLULAR: Describing tissues organs or organisms that consist of many cells specialised to do a certain job for that organism. Most lifeform that can be seen with the naked eye is multicellular. For example, most animals and plants are multicellular.

MULTICENTRE BOND: A bond formed between three and sometimes more atoms that contains only a single pair of electrons. The structure of boranes can be explained by considering them to be electron-deficient compounds containing multicentre bonds. A typical example of multicentre bond occurs in diboranes (B_2H_6) shown in the diagram.



MULTICILIATE: Having many cilia. This is common in the protozoan of the phylum ciliophora. An example is the paramecium.

MULTICIPITAL: Possessing many heads or stems from one crown or root.

MULTICOLLINEARITY: The multicollinearity problem in multiple regression arises when two or more independent variables are highly correlated. In that case, it is difficult to determine the individual effects of the different variables. In the extreme case where two independent variables are perfectly correlated, the multiple regression calculation cannot be performed because it would involve dividing by zero.

MULTICOSTATE: Possessing numerous ribs, as the leaf of a plant or certain shells and corals.

MULTICOSTATE CONVERGENT: Having veins that converge towards the tip, as seen in zizyphus and the leaves of smilax.

MULTICOSTATE DIVERGENT: A type of venation whereby the veins move away from each other and never meet, as they go towards the tip of the leaf, as seen, for example, in cotton.

MULTICRITERIA OPTIMISATION (MULTIOBJECTIVE OPTIMISATION): The study of optimisation problems with more than one objective criterion to satisfy. An example is attempting in an industry to minimise environmental damage while maximising profit. Typically, there is no maximum and one must settle for an efficient point or use additional external criteria.

MULTICROP: To grow more than one crop on the same piece of land in one year.

MULTIFOIL: A symmetric plane figure constructed by placing congruent arcs of a circle around a regular polygon so that the ends of the arcs bisect the sides. A trefoil has three sides, a quatrefoil has four and a hexafoil has six sides.

MULTICS (MULTIPLEXED INFORMATION AND COMPUTING SERVICE): The first time-sharing operating system, developed by MIT and Bell Labs in the mid-1960s and used until 2000 MULTICS. It was one of the first to implement symmetric multiprocessing. It informed the development of Unix operating system.

MULTIDIMENSIONAL SPECTROSCOPY: A type of spectroscopy used to study complex systems and large molecules such as proteins, for which traditional spectroscopy does not give clear result. The basic technique is to excite all possible resonances in the system and analyse the radiation produced using Fourier analysis.

MULTIENZYME SYSTEM: A complex of enzymes within a cell that forms a reaction sequence of a biochemical pathway so that the product of the first enzyme reaction is transferred directly to the next enzyme and instantly undergoes a second reaction and so on. The rate of an enzyme reaction often depends on the concentration of the enzyme and the substrate, both being required in relatively high amounts. Multi enzyme systems such as those involved in RNA and protein synthesis, help maintain a high rate of cellular metabolism since the intermediate products are transferred directly to the next enzyme and are therefore not required in large concentrations.

MULTIFACTORIAL INHERITANCE: Inheritance involving many factors, of which at least one is genetic but none is of overwhelming importance, as in the causation of a disease by multiple genetic and environmental factors.

MULTIFOLIATE: Having many leaves or leaflets.

MULTIGENE FAMILY: A collection of genes with similar or identical sequences, presumably of common origin. Such genes may be clustered together on the same chromosome or dispersed on different chromosomes. Examples of multigene families include those that encode the haemoglobins, immunoglobulins, histocompatibility antigens, actins, tubulins, keratins, collagens, heat shock proteins, salivary glue proteins, chorion proteins, cuticle proteins and yolk proteins.

MULTIGRADE: Sets of equations in which the sums of powers of two different sets of numbers are the same for several different exponents. The simplest example is:

$$1 + 6 + 8 = 2 + 4 + 9; 1^2 + 6^2 + 8^2 = 2^2 + 4^2 + 9^2.$$

MULTIMEDIA: A combination of various media such as text, audio, graphics, animation and full-motion video, now often held on CD-ROM. The user can make use of the different media in an integrated way.

MULTIMEDIA MESSAGING SERVICE (MMS): A mobile phone service that allows users to send multimedia messages such as images, videos and sound files to each other. MMS is an extension of SMS, which is used to send and receive text messages.

MULTIMER: An aggregate molecule that consists of two or more component molecules which may or may not be identical, held together by non-covalent bonds. Multimers can be named variously according to the number of constituent subunits, hence a dimer has two subunits, a trimer has three and a tetra has four.

MULTIMETER: An electrical measuring instrument designed to measure several functions such as alternating and direct current potential differences (voltages), current and resistance, capacitance with several full-scale ranges provided for each quantity. **Analogue multimeters** use a microammeter whose pointer moves over a scale calibrated for all the different measurements that can be made. **Digital multimeters** display the measured value in numerals.

MULTIMODE FIBRE (MULTI-MODE OPTICAL FIBRE): An optical fibre in which there is a relatively large glass core of about 50-100 micrometres in diameter surrounded by material with a smaller refractive index than the core. It is used for short distance communications such as LANs within a building or on a campus. Typical multimode links have data rates of 10 Mbit/s to 10 Gbit/s over link lengths of up to 600 metres. The value of the refractive index in the surrounding material can either suddenly drop or gradually drop to a lower value than that of the core.

MULTINOMIAL (POLYNOMIAL): An algebraic expression consisting of two or more terms. An example is $9x^3 + 2x^2 + 5$. The operations involved in forming a multinomial are addition, subtraction, multiplication and division (+, -, ×, ÷). The sum or product of two or more multinomials is also a multinomial, but subtraction or division may not result in a multinomial.

MULTINOMIAL COEFFICIENT: The number of ways that q distinguishable items can be distributed into p containers so that there are q_i items in the i th container. This is given by, $\frac{q!}{q_1!q_2! \dots q_n!}$.

MULTINOMIAL DISTRIBUTION: A principle of the binomial distribution to situations in which k outcomes can occur on each of n trials, where $k > 2$. Specifically, the distribution is given by

$$P(n_1, \dots, n_k) = \frac{n!}{n_1! \dots n_k!} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}, \text{ where } n_i \text{ is the number of trials with outcomes } i \text{ and } p_i \text{ is the probability of the outcome } i \text{ occurring on a particular trial. Thus, } \sum_{i=1}^k n_i = n \text{ and } \sum_{i=1}^k p_i = 1. \text{ The expected value of } n_i$$

is np_i and its variance is $np_i(1 - p_i)$. The covariance of n_i and n_j is $-np_i p_j$.

MULTINOMIAL THEOREM: The generalisation of the binomial theorem to polynomials of n variables. For any positive integer m and any nonnegative integer n , with m terms, the expansion when raised to an arbitrary power n is as follows:

$$(x_1 + x_2 + \dots + x_m)^n = \sum \frac{n!}{n_1! n_2! \dots n_m!} x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}, \text{ where the sum is over all non-negative integers summing to } n.$$

MULTINUCLEATE (MULTINUCLEATED; COENOCYTIC): A cell having more than one nucleus per cell, which is the result of nuclear division not being followed by cytokinesis.

MULTIPAROUS: 1. Bringing forth more than one offspring at a birth. 2. A cyme with many lateral axes.

MULTIPERENNIAL (PLIESTESIAL): Monocarpic; living several years before flowering and dying promptly thereafter. An example is *Agave*.

MULTIPLE: 1. A number that can be divided exactly by a particular smaller number. In more formal words, any number or polynomial that is the product of a given number or polynomial and an integral multiplier; for example, $x^2 - y^2$ is multiple of $x + y$. 2. A system of wiring so arranged that a group of communication lines are accessible at a number of points.

MULTIPLE ALLELES: Three or more alternative forms of a gene that can occupy the same locus controlling one particular character. Multiple alleles occur and are maintained within a population, but any individual possesses only two such alleles. For example, the ABO system of the blood group is controlled by three alleles, only two of which are present in an individual.

MULTIPLE BIRTHS: The delivery of more than two babies from one pregnancy in humans. Identical or monozygotic twins develop from a single fertilised egg, which splits into two genetically identical embryos. Fraternal or dizygotic twins develop from two eggs fertilised by two sperm. Repeated twinning produces triplets, quadruplets, etc, which may be identical, fraternal or a combination. The use of fertility drugs such as clomiphene (used when the cause of infertility is lack of released ova) sometimes cause several ova to be released and fertilised simultaneously leading to a rise in the incidence of multiple births, including sextuplets, septuplets and octuplets.

MULTIPLE BOND: A bond between two atoms that contains more than one pair of electrons. Such bonds primarily involve sigma bonding with secondary contribution from pi bonding or, sometimes, delta bonding.

MULTIPLE DOCUMENT INTERFACE (MDI): A Microsoft software program interface that allows a program to have multiple windows called child windows being displayed within the main or parent window. This makes a user to view and work with different documents such as a spreadsheet or a text document, by moving the cursor from one space to another without the need to switch between windows.

MULTIPLE FRUIT (COMPOSITE FRUIT): A fleshy fruit that incorporates a complete inflorescence and is thus derived from the ovaries of many flowers. It may also incorporate other floral parts and the receptacles. Examples are fig, pineapple and mulberry.

MULTIPLE GATEWAYS: More than one gateway serving to connect the same two systems network architecture (SNA) networks for cross-network sessions.

MULTIPLE INTEGRAL (REPEATED INTEGRAL): An expression requiring more than one integration. **Multiple integration** or **repeated integration** is the integration of functions of more than one real variable. Integrals of a function of two variables over a region in R^2 are called **double integrals** and integrals of a function of three variables over a region of R^3 are called **triple integrals**.

MULTIPLE MYELOMA (MYELOMA): Malignant disease of the bone marrow in which the bone marrow does not produce normal plasma cells, resulting in weakened immune system. Plasma cells make proteins called antibodies that are important in the immune system for fighting infections. Myeloma usually occurs in older people. The symptoms include fatigue, muscle weakness, severe bone pain and backache, loss of appetite, nausea, restlessness and confusion. Myeloma can cause anaemia, vertebral collapse, clotting problems and damage to the eyes and internal organs. It is treated with chemotherapy and radiotherapy.

MULTIPLE REGRESSION: A term used to represent models in which dependent variable, y , is regressed on a number of independent variables, $x_1, x_2, \dots, x_n$. The fitted model is given by

$E(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$, where E represents the expected value. The model can be represented in vector form by $y' = [y_1 y_2 \dots y_n]$ and an

$n \times (p + 1)$ matrix X given by $X = \begin{bmatrix} 1 & \dots & X_{1p} \\ \vdots & \ddots & \vdots \\ 1 & \dots & X_{np} \end{bmatrix}$. The model can

be written in the form $y = X\beta + \varepsilon$, where $\varepsilon' = [\varepsilon_1 \varepsilon_2 \dots \varepsilon_n]$ contains the residual error terms and $\beta' = [\beta_0 \beta_1 \beta_2 \dots \beta_p]$. Least squares

estimation of the parameters involve the equation, $\hat{\beta} = (X'X)^{-1}X'y$. The regression coefficients $\beta_1, \beta_2, \dots, \beta_p$ give the change in the response variable corresponding to a unit change in the independent variable with the other variables held constant.

MULTIPLE ROOT (REPEATED ROOT): One of a number of equal roots of the same polynomial or equation that occurs when the polynomial has $(x - a)^n$ as a factor, for $n > 1$. The largest such n is the multiplicity or order of a . A root of order 1 is a simple root. Equivalently, it is a root of one or more derivatives of the polynomial.

MULTIPLE SCATTERING: A process in which a particle undergoes a large number of collisions and the total change in its momentum is the sum of the many small changes occurring during individual collisions.

MULTIPLE SCLEROSIS (MS): A central nervous system disease that affects the brain and spinal cord. It damages the myelin sheath, the material that surrounds and protects the neurons (nerve cells). This damage slows down or blocks messages between the brain and the rest of the body, leading to the symptoms of MS. These can include visual disturbances, muscle weakness, trouble with coordination and balance, sensations such as numbness, prickling or "pins and needles" and thinking and memory problems.

MULTIPLE SEQUENCE: A sequence indexed over a number of index sets, such as $\{x_{ijk} : 0 \leq i < n, 1 \leq j \leq m, 0 \leq k \leq p\}$.

MULTIPLE SERIES: A series indexed over a number of index sets, the sum of all the elements of a multiple sequence, which in the finite case is independent of the order.

MULTIPLE SUCKLING: A system in dairy breeding where nurse cows suckle several beef calves at the same time.

MULTIPLE-DOMAIN NETWORK: 1. A network with more than one system services control point. 2. An advanced peer-to-peer network (APPN) with more than one network node.

MULTIPLE-FACTOR INHERITANCE: The determination of an inherited characteristic by polygenes, so that it shows approximately continuous variation from one extreme to another. The size of beans, ear-length in maize and many other characteristics show multiple-factor inheritance and do not, at first sight, behave like Mendelian characters.

MULTIPLY: 1. A group of elementary particles that are identical in all respects but have a different electric charge. 2. A special line in a spectrum formed by more than two component lines, caused by slight variations in atomic or molecular energy levels.

MULTIPLEXED INFORMATION AND COMPUTING SERVICE (MULTICS): The first timesharing operating system, developed by MIT and Bell Labs in the mid-1960s and used until 2000 MULTICS. It was one of the first to implement symmetric multiprocessing. It informed the development of Unix operating system.

MULTIPLEXER: A device which takes information from any of several sources and places it on a single line or sends it to a single destination.

MULTIPLEXING: The transmission at the same time of two or more analogue and digital signals that have been combined into a single channel. A device that performs the multiplexing is called a multiplexer (MUX).

MULTIPLICATION: 1. The binary arithmetic operation of obtaining the product of two quantities, which for positive integers consist of adding one of them to itself as many times as the other states; usually written as $a \times b$, $a \cdot b$ or ab . For example, in $7 \times 5 = 35$ it consists of adding 7 to itself 5 times ($7 + 7 + 7 + 7 + 7 = 35$). In this case, 7 is called the **multiplicand**, 5 the **multiplier** and 35 the **product**. Multiplication can also be done with numbers other than positive integers, including negative numbers, fractions or irrational numbers, etc., observing the appropriate rules. 2. In biology, the act or process of reproduction in animals, plants and people.

MULTIPLICATION CONSTANT (MULTIPLICATION FACTOR): The ratio of the total number of neutrons produced by fission in a given time to the number absorbed or escaping in the same period in a nuclear reactor.

MULTIPLICATION PRINCIPLE (FUNDAMENTAL COUNTING PRINCIPLE; COUNTING RULE): A way of determining the total number of possible outcomes for two or more separate choices. It is determined by multiplying the events together to get the total number of outcomes. For example, if the first event has 3 choices of occurring, the second event can occur 4 ways, and the third event can occur 5 ways, then you can find out the number of unique combinations by multiplying: $3 \times 4 \times 5 = 60$ unique combinations..

MULTIPLICATION LAW OF PROBABILITY: Theory that the probability of two independent events will occur is equal to the probability that one independent event will occur times the probability that a second independent event will occur. For example, on a single toss of two dimes (each dime having a head and a tail), the probability

that both will land on their tails is equal to $\frac{1}{4} = \frac{1}{2} \times \frac{1}{2}$.

MULTIPLICATION PRINCIPLE: If two choices are to be made, one from a list of m possibilities and the second from a list of n possibilities and any choice from the second list, then the fundamental principle of counting says that there are $m \times n$ total ways of making choices.

MULTIPLICATION/MULTIPLICATIVE PROPERTY OF -1: A property of multiplication that says multiplying any number by -1 gives the opposite of a number. For example, $-1 \times 5 = -5$ and $-1 \times -3 = -(-3) = 3$. In symbols, for any number a , $-1 \times a = -a$.

MULTIPLICATION SIGNS/SYMBOLS: The symbol $(\cdot)$, $(\times)$, or $(*)$ between two mathematical expressions, denoting multiplication of the second expression by the first. In certain algebraic notations the sign is suppressed and multiplication is indicated by immediate juxtaposition, as in ab .

MULTIPLICATION TABLE: A table consisting of the result of a multiplication operation of a range of numbers or elements of a group, ring or other algebraic structure. In any place-value notation, only a

finite number (equal to the base) of distinct multiplication tables are needed to enable any product to be calculated.

MULTIPLICATIVE: 1. Of a function, distributing over multiplication, so that $f(xy) = f(x)f(y)$. An arithmetic function multiplicative in this case is called **totally multiplicative**. 2. Of an arithmetic function, multiplicative only for relatively prime arguments, so that $f(xy) = f(x)f(y)$ if x and y have no common divisors.

MULTIPLICATIVE COMPARISON: A numerical comparison that indicates how many times one quantity is larger than another or what fractional part one quantity is of another.

MULTIPLICATIVE IDENTITY: The number 1 is a multiplicative identity, because $(1)(y) = (y)(1) = 1$ for all y .

MULTIPLICATIVE INVERSE: The multiplicative inverse of a number x (written as $1/x$ or x^{-1}) is the number that, when multiplied by x , gives a result of $(x)(1/x) = (1/x)(x) = 1$. The multiplicative inverse is also called the reciprocal. For example, $1/5$ is the reciprocal of 5. There exists multiplicative inverse for every number except zero.

MULTIPLICATIVE NAME: A name that expresses the multiple occurrence of identical parent structures, two or more of which are connected by a symmetrical structure expressible by means of a multivalent simple or composite prefix.

MULTIPLICITY: A quantity used in atomic spectra to describe the energy levels of many-electron atoms characterised by Russell-Saunders coupling given by $2S + 1$, where S is the total electron spin quantum number. The multiplicity of an energy level is indicated by a left superscript to the value of L , where L is the resultant electron orbital angular momentum of the individual electron orbital angular momental.

MULTIPLIER: 1. A fixed resistance used with a voltmeter, usually a moving coil instrument to vary its range. 2. The number by which another number, a multiplicand, is multiplied. For example, the number 4 is the multiplier in the statement $2 \times 4 = 8$.

MULTIPLIER METHOD: A class of optimisation methods that use variants of the Lagrange multipliers in combination with penalty function methods to solve constrained optimisation problems.

MULTIPLY LABELLED: A selectively labelled compound may be multiply labelled when in the unmodified compound there is more than one atom of the same element at the position, where the isotopic modification occurs, for example, H in CH_4 or there are several atoms of the same element at different positions where the isotopic modification occurs, for example, C in $\text{C}_4\text{H}_8\text{O}$.

MULTIPLY OUT: To expand an expression by applying the distributive law to all the multiplications over any other operations, for example, expressing $(x + a)^2$ as $x^2 + 2ax + a^2$.

MULTIPOINT NETWORK: 1. A network in which there are precisely two endpoint nodes, any number of intermediate nodes and only one path between any two nodes. 2. In data communication, a configuration in which more than two terminal installations are connected. The network may include switching facilities.

MULTIPOLAR NEURON (MULTIPOLAR NEURONE): A neuron that has one axon and several dendrons extending from its cell body in different directions, allowing for the integration of a great deal of information from other neurons. Most vertebrate motor neurons and interneurons are multipolar neurons.

MULTIPOLE INTERACTION: Interactions between arrays of point charges (multipoles) associated with potential energies that depend on the distance. Multipole interactions are an important feature of intermolecular forces. An n -pole is a set of n charges having an n -pole moments but not a lower moment. A monopole is a single charge and the monopole moment is the charge. A dipole consists of two opposite charges and thus has no overall charge (and hence no monopole moment). Higher multipoles are the quadrupole and the octupole. The interaction potential energy between multipoles falls off with distance increasingly rapidly as the order of the multipoles increases. If a

$2m$ -pole interacts with a $2n$ -pole, the interaction potential energy, V , varies with distance r as $V = \frac{C}{(rm + n - 1)}$, where C is a constant. The

rapid fall-off with distance, as the order increases, is due to the set of charges tending to move towards neutrality, as seen from outside. The multipole interactions with the largest interactions are: ion-ion, ion-dipole, dipole-dipole and dispersion interactions. The influence of each multipole at the point where the potential is observed is dependent on the distance between. Thus the contribution of a multipole to a potential at a point decreases with increasing distance.

MULTIPOLE RADIATION: Electromagnetic radiation which has characteristics equivalent to those of radiation generated by an oscillating electric or magnetic multipole and is made up of photons of well-defined angular momentum and parity.

MULTIPURPOSE INTERNET MAIL EXTENSIONS (MIME): An Internet standard for identifying the type of object being transferred across the Internet. MIME types include several variants of audio, graphics and video.

MULTIRADIATE: Having many rays.

MULTISEPTATE: Having many septae or chambers. An example is the pith of pokeweed.

MULTITASKING: A mode of operation that provides for concurrent performance or interleaved execution of two or more tasks.

MULTIVALENT: An association of three or more homologous chromosomes observed during meiosis in polyploid or polysomic organisms.

MULTIVARIATE: Having or involving a number of distinct statistical mathematical random variables, though not necessarily independent. A **multivariate normal distribution** is the generalisation of the normal distribution to two or more variables. The joint distribution of any n normal random variables has probability density function

of $P(x, u, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$, where μ is the

vector of means and the variance-covariance matrix, Σ is positive definite.

MULTIVERSE (META-UNIVERSE, METAVERSE): A hypothetical cosmos consisting of all possible universes and space-times, of which our universe is one example.

MULTIVIBRATOR: A monostable circuit that rapidly switches between two states: stable (having a steady voltage) and unstable (having an unsteady or variable voltage) because of positive feedback between multiple states producing a harmonic output. Normally, the circuit is in the stable condition. The output of one is coupled to the input of the other through resistance-capacitance elements or other elements to obtain in-phase feedback voltage. The circuit can be removed from the stable condition temporarily by an input pulse called a triggering signal. After a certain period of time, which can be adjusted or set by choosing specific component values, the circuit returns to the stable state until another triggering signal is applied. There are three types of multivibrators: (i) Astable multivibrator in which the circuit is not considered stable in either of the two possible states and switches from one of the potential states to the other, but does not require a clock pulse or other input in order to operate. (ii) Monostable in which the circuit's two states is considered transient and the other stable. (iii) Bistable in which the circuits are stable in both possible states and external stimuli can change one state to the other. Multivibrators are used as pulse generators, frequency dividers, pulse-shaping circuits, and contactless switches for equipment used in automation, computer technology, and measurement technology, such as time relays, drivers, and shapers in digital computers.

MUNGING: The deliberate alteration of an e-mail address on a web page to hide the address from spam-bot programs that scout the Internet for e-mail addresses. It also includes taking the data from its original state and transforming and mapping into another format.

MUMETAL (MUMETAL; MUSHIELD): A ferromagnetic alloy, containing 78% nickel, 17% iron, plus copper, chromium and molybdenum. It has a high permeability and a low coercive force. It is used in some transformer cores and for shielding various devices from external magnetic fields.

MUNICIPAL SOLID WASTE, MSW (TRASH; GARBAGE; RUBBISH): The total waste generated by households and commercial establishments in a locality. MSW management includes the collection, transfer, resource recovery, recycling, and treatment of the waste.

MUNSELL COLOUR SYSTEM: A method of designating colours that is used to visually identify and match colour. It is based on three colour components as follows: (i) Chroma is the quality of a colour's purity, intensity or saturation, for example, grey is a neutral colour, defined as an extreme low chroma. (ii) **Hue** has a similarity to a colour such as red, yellow, green, and blue which is dependent on its dominant wavelength and independent of intensity or lightness. (iii) **Value** is how light or dark a colour is, ranging from 0 (black) to 1 (white). The system was developed by the American art instructor and painter, Albert H. Munsell (1858 – 1918).

MUNTZ METAL: A form of brass containing 60% copper, 39% zinc and small amount of lead and iron, named after George Fredrick Muntz, a metal-roller in Birmingham, England. It is stronger than alpha-brass and used for hot forgings, brazing rods and large nuts and bolts.

MUON (MU MESON): The second lightest lepton. The muon has an electric charge of -1. It is identical to the electron but is 207 times more massive. It decays with a mean lifetime of 2.2×10^{-6} second. Muons occur commonly in cosmic rays. Despite its name, a muon (short for "mu meson") is no longer classified as a meson.

MUON DECAY EXPERIMENT: An experiment comparing the number of muons in cosmic rays that reach the Earth's surface without decaying as compared to the number predicted to hit the surface calculated without using relativity theory.

MUON NEUTRINO: A type of neutrino associated with the muon, often created in particle interactions involving muons (such as muon or pion decay). The muon neutrino has a mass no greater than 0.49 times that of the electron and has no charge.

MUON-CATALYSED FUSION (MCF): A process allowing nuclear fusion to take place at temperatures much lower than the temperatures required for thermonuclear fusion, even at room temperature or lower. Muons are leptons or are unstable subatomic particles similar to electrons with a mass about 207 times that of an electron with a negative charge and a half-life of two-millionths of a second. It acts very much like an electron to bond atoms together into molecules, but muon is 207 times heavier than an electron and removes the electron and replaces it. If a muon captures two hydrogen atoms it forces them to fuse, resulting in energetic particle release and helium, a process which is repeated about 300 times. This fusion gives energetic neutrons, which are easily converted to heat in a pressurised water reactor resulting in steam which can be harnessed to create electricity.

MUONIC ATOM (MU-MESIC ATOM): An atom in which one of the electrons has been replaced by a muon. Since the mass of muon is 207 times that of an electron, the average radius of the orbit of a muon is much smaller than that of a corresponding electron. Muonic atoms provide test for quantum electrodynamics. They are also used in research into muon catalysed fusion.

MUONIUM: A short-lived entity formed by a muon and its antiparticle attracted together and revolving about a common centre. It consists of a positive muon and an electron.

MURICATE: Having a surface covered by sharp points or prickles or hard short projections.

MUSCARINIC: Describing one of the two main classes of acetylcholine receptors, so called because the effect of acetylchlorine on them can be mimicked by muscarine, a toxic alkaloid produced by *Amanita*

muscaria and certain other gill fungi. Muscarinic receptors occur on target cells innervated by fibres of the vertebrate parasympathetic nervous system, for example, in smooth muscles, cardiac muscles and glands.

MUSCLE: Animal tissue made up of long cells that can contract as a result of nerve impulses to produce movement both in the organism as a whole and maintaining movement of body substances or internal organs. It is made from cells called muscle fibres and classified as striated or skeletal or voluntary muscle, smooth or involuntary muscle or cardiac muscle, according to its structure and function. Striated or skeletal (voluntary) muscle forms the bulk of the body's muscle tissue and gives the body its general shape. It produces voluntary movement and attached to the bones via tendons at joints. Smooth or involuntary muscles are spindle-shaped and are generally arranged in dense sheets. They are regulated by the autonomic nervous system and mainly effects the movements of hollow organs such as intestine and bladder. Cardiac muscle occurs only in the heart, where it forms that organ's thick walls. It is striated like skeletal muscle but, like smooth muscle, is controlled involuntarily. The contractions of cardiac muscle are stimulated by a special clump of muscle tissue located on the heart (the pacemaker), although the rate of contractions is subject to regulation by the autonomic nervous system.

MUSCLE SPINDLE: A sensory receptor in vertebrate muscle, which primarily detects changes in the length of this muscle. They run parallel to normal muscle fibres, each consisting of a capsule containing small striated muscle fibres. Muscle spindles are responsible for the adjustment of muscle tone and play an important role in the subconscious maintenance of posture and movement.

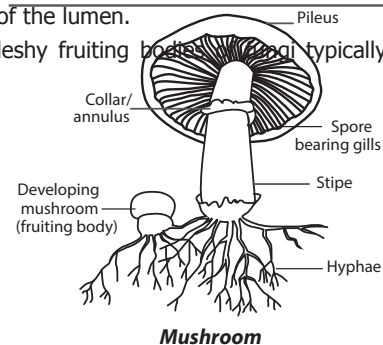
MUSCOVITE (WHITE MICA; POTASH MICA): A mineral form of potassium aluminosilicate, $KAl_2(AlSi_3O_{10})(F,OH)_2$, one of the most important members of the mica group of minerals. It is chemically complex and has a sheetlike crystal structure. It is usually silvery-grey in colour, sometimes tinted with green, brown or pink. Muscovite is a common constituent of certain granites and pegmatites. It is also common in metamorphic and sedimentary rocks. It is widely used in industry, for example in the manufacture of electrical equipment and as filler in roofing materials, wallpapers and paint.

MUSCULAR DYSTROPHY: Any of a group of inherited chronic muscle disorders that can affect people of all ages; characterised by weakening and wasting of the skeletal or voluntary muscles which control movement. The genes mutate and interfere with the production of dystrophin, a protein in muscle that is needed to form healthy muscle. This causes the muscle fibres to degenerate, to be replaced by fatty tissue.

MUSCULAR SYSTEM: One of eleven major body organ systems in animals controlled mainly through the nervous system. It allows movement and locomotion, powers the circulatory, digestive and respiratory systems and plays a role in regulating temperature.

MUSCULARIS MUCOSAE (LAMINA MUSCULARIS MUCOSAE): A thin layer of smooth muscle that forms the outermost part of the mucosa of the mammalian stomach and intestine. It is composed of several thin layers of smooth muscle fibres oriented in different ways to expel contents of glandular crypts and enhance contact between epithelium and the contents of the lumen.

MUSHROOM: Any of the fleshy fruiting bodies of fungi typically produced above ground on soil, on their food sources such as decaying woods and even on standing trees. Some species are cultivated. It often has a white stalk called a stipe which bears a convex or oven flattish expanded portion called the pileus, which is whitish and silky or somewhat scaly



above and bears on the under side radiating gills called lamellae, on the underside of the cap. The lamellae are at first flesh-coloured, but gradually become brown. The term also applies to any large fungus, especially one of the genus *Agaricus*; a toadstool. Several species are edible, but many are very poisonous.

MUSHROOM COMPOST: A special growing medium for the commercial production of mushroom.

MUSHROOTING: Fruiting body of certain fungi, consisting of an upright stem and a spore-producing cap with radiating gills on the under surface.

MUSICAL INSTRUMENT DIGITAL INTERFACE (MIDI): A standard for the interchange of information between musical instruments such as keyboards, microphones and synthesisers and computers for recording inputs from a variety of audio instruments that also allows files to be converted into a format that can be read by the computer to synchronise all components to create a more harmonious sound.

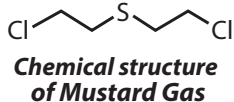
MUSICAL SCALE: A series of musical notes arranged in order of ascending pitch according to a scheme of intervals making them suitable for music.

MUSK: A strongly-scented substance used in the manufacture of perfume. The term is strictly applied to that obtained from the musk glands of the male musk deer, but also covers other similar secretion such as civet musk and badger musk.

MUSLIN: A fabric made with cotton in various degrees of fineness, especially a cotton fabric of an open plain weave that allows for airflow. Dressmakers often use cheap muslin for sewing patterns and test fitting before cutting an expensive fabric to prevent potential mistakes. It is also used for upholstery, curtains, lining, bags, blankets for babies, etc.

MUSTARD: 1. Common name for the family Brassicaceae (formerly Cruciferae) and its representative genus *Brassica*. The mustard family contains a great variety of food plants such as salad and vegetable crops, such as cabbage, cauliflower, Brussels sprouts and broccoli. 2. Any of yellow-flowered Eurasian plants of the cabbage family whose seeds are used to make a paste into a paste or sauce. 3. A pungent-tasting yellow or brown paste made from the crushed seeds of certain plants, typically eaten with meat or used as a cooking ingredient.

MUSTARD GAS (2,2'-DICHLORODIETHYL SULFIDE): A poisonous blistering gas used as a chemical warfare agent with the chemical formula $(\text{ClCH}_2\text{CH}_2)_2\text{S}$, density 1.27 g/ml (liquid), melting point 14.4 °C (287.6 K) and boiling point 217 °C (490 K). It is highly irritating to the eyes and quickly causes conjunctivitis and blindness. If inhaled, it attacks the respiratory tract and lungs, causing pulmonary oedema.



MUTAGEN (MUTAGENIC AGENTS): Any physical or chemical agent capable of altering the structure of the hereditary material or DNA causing mutation in living cells. They change the DNA in cells or they cause damage to chromosomes by interfering with the coding system. Example of chemical mutagens are colchicines or ethylmethylsulfonate (EMS), while physical radiations such as cosmic rays, x rays, gamma rays and ultraviolet light are also mutagens. The effects of some mutagens are increased or suppressed in some organisms by the presence of certain other nonmutagenic substances. Oxygen, for example, makes cells more sensitive to the mutagenic effects of x rays.

MUTAGENESIS: The introduction of permanent heritable changes (mutations into the DNA of an organism).

MUTANT: An organism, animal, gene or cell that has undergone a heritable change, especially one with visible effects. Thus, changes in genotypes occur in the phenotypes.

MUTAROTATION: A change in the specific rotation of a cyclic monosaccharide, such as glucopyranose, as it reaches an equilibrium between its α and β anomeric forms. For example, a solution of β -D-glucopyranose when dissolved in water has a specific rotation of +18.7. Over time, some of the β -D-glucopyranose, will convert to α -D-glucopyranose which has an optical rotation of +112.2. Thus the

rotation of the solution will increase from +18.7 to +52.2 as some of the β -form is converted to α -form.

MUTATION: A random change in the structure of DNA or chromosomes or a sudden random change within the genetic material of a cell that may cause it and all cells derived from it to differ in appearance or behaviour from the normal type. Mutation can be a source of beneficial genetic variation or it can be neutral, it is mostly disadvantageous on the organism inheriting them. An example of a chromosome mutation is that giving rise to Down's syndrome. Mutation is essential for evolution, because it is the ultimate source of genetic variation. They can result in chemical variations within populations which can lead to evolution. Somatic mutations affect the non-reproductive cells and are therefore restricted to the tissues of a single organism. Germline mutations, which occur in the reproductive cells or their precursors, may be transmitted to the organism's descendants and cause abnormal development. Mutations occur naturally at a low rate but this may be increased by radiation and by some chemicals. Most are point mutations which involve invisible changes in the DNA of the chromosomes, but some affect the number or appearance of the chromosomes. To **mutate** is to undergo genetic change that can be inherited.

MUTATION FREQUENCY: The average frequency with which a particular mutation occurs in a population. This frequency can be increased by radiation or by exposure to chemicals such as mustard gas or hydrogen peroxide.

MUTATION THEORY: The theory, first proposed by Hugo de Vries in 1903, that suggests that new forms of organisms (mutations) arise suddenly in a population and that evolution proceeds by natural selection operating on these.

MUTIGERM SEED: A seed which occurs as a cluster of seeds fused together and which produces more than one plant when it germinates. An example is the sugar beet seed. The multiple plants must be reduced to one by a process called singling.

MUTISM (DUMBNESS): Inability to speak or unwilling to speak, which may be caused by a psychological disorder or disease.

MUTUAL INDUCTION: The production of an electromotive force (emf) or voltage in an electric coil of wire caused by a changing magnetic flux placed near to the first coil. The coil in which current is changed is called primary coil and the coil in which emf is induced is called secondary coil. The magnitude of the induced emf depends on the number of turns of wire in each of the coils. Therefore, mutual inductance can be defined as the ratio of the emf induced in the secondary coil to the rate of change of electric current in the primary coil. Some of the applications of mutual inductance are found in transformers, motors and generators.

MUTUAL INFORMATION: In information science, it is the relative entropy between the joint distribution and the product of marginal distributions. Mathematically, it is given as

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)q(y)}, \text{ where } p(x,y) \text{ is the joint distribution}$$

and $p(x)q(y)$ is the product distribution.

MUTUALISM: An association in the form of symbiosis between two organisms of different species whereby both benefit from the relationship. For example, the intestinal bacteria of herbivores digest the cellulose of plant cell walls, the products of which are then used by the herbivore. The protozoans inhabiting the guts of termites have shelter and food, while the termites benefit from the digestion of the cellulose. Another example is *Rhizobium* bacteria found in nodules on the roots of many legume (bean) species fix atmospheric nitrogen into a form (NH_3) that can be taken up by plants. The plant provides the bacteria with carbon in the form of dicarboxylic acids. The carbon is utilised by the bacteria as energy for nitrogen fixation.

MUTUALLY EXCLUSIVE: Describing events that could never occur together. For example, tossing a coin once can result in either heads or tails, but not both. The probability of occurrence of two or more mutually exclusive events is the sum of the probabilities of the individual events.

MUZZLE: 1. The projecting part of an animal's face, made up of its jaws and nose. 2. The front open end of the barrel of a fire arm. 3. A device strapped over the nose and jaws of an animal to prevent it from opening its mouth.

MYALGIC ENCEPHALOMYELITIS, ME (CHRONIC FATIGUE SYNDROME, CFS): A chronic degenerative neuro-immune disease, having serious immune and cardiovascular abnormalities. Its causes are not known but have been linked to viral infectious diseases such as herpes or hepatitis process, exposure to chemicals, bacterial infection, etc. Signs and symptoms are varied and include short-term memory loss, concentration difficulties, muscle and nerve pain, muscle weakness, noise and light sensitivity, sleep and temperature disturbance, inability to sustain upright activity such as standing, sitting or walking and sensitivity to food, alcohol, chemicals and medicines.

MYASTHENIA GRAVIS: A rare condition characterised by weakness and loss of muscle power, especially in the face and neck. Some of the signs and symptoms include drooping of one or both eyelids, double vision, altered speech, difficulty in swallowing and difficulty chewing.

Mycelia sterilia (AGONOMYCETALES): An order of imperfect fungi that do not produce spores, although some develop sclerotia. It contains about 200 species and 28 genera including Rhizoc-tonia, certain species of which cause-off disease and Sclerotium.

MYCELIUM (MYCELIA): The densely branched network of hyphae including rhizoids and stolon in a fungus or a cluster of hyphae that constitutes the vegetative body of the thallus of a fungus and is often submerged in another body such as a soil or organic matter or the tissues of a host of a fungus. It is the conspicuous part in most cases is the fruiting body of fungi.

MYSTICETES (BALEENS): Large whales of the suborder Mysticeti, characterised by baleen plates used for feeding, two blowholes, symmetrical skull, and lower jawbones that are solid and do not join in the middle. They include blue whales, finback whales, grey whales, humpback whales, right whales, bowhead whales, etc. The females are larger than the males. They lack teeth and use the baleen plates to filter great quantities of krill (small fish, crustaceans or plankton) in one gulp.

MYCOBIONT: The fungal partner in lichen, which is usually an ascomycete fungus, especially one of the class Discomycetes.

MYCOLOGY: 1. The study of all kinds of fungi including mushrooms, moulds, truffles, yeasts, lichens, plant pathogens and medically important fungi. It includes their genetic and biochemical properties, their taxonomy and their use to humans as a source of medicinals such as penicillin, food (for example, beer, wine, cheese, edible mushrooms) and entheogens, as well as their dangers such as poisoning or infection. A biologist who studies mycology is called a mycologist. 2. The fungi native to a region. 3. The composition or characteristics of a particular fungus.

MYCOPHAGOUS (FUNGIVOROUS): Organisms that feed on fungi. Many different organisms feed on fungi, for example, birds, mammals, insects, plants, amoebas, gastropods, nematodes, and bacteria. Some derive energy for their metabolism from consuming only fungi; whilst some others eat it only sometimes and feed on other things.

MYCOPLASMAS: A genus of the bacteria that lack a rigid cell wall, are gram-negative, do not require oxygen and are among the smallest of all living cells with a diameter of about 0.1µ micron – 0.8µ micron. Because they lack a cell wall, they are unaffected by many common antibiotics such as penicillin or other beta-lactam antibiotics that target cell wall synthesis. They are either parasitic or saprotrophic. They are parasites of joints and the mucous membranes lining the respiratory, genital or digestive tracts of cud-chewing animals, carnivores, rodents and humans. Several species are pathogenic in humans, including *Mycoplasma pneumoniae*, which is an important cause of atypical pneumonia and other respiratory disorders and *Mycoplasma genitalium*, which is believed to be involved in pelvic inflammatory diseases. They are thought to be responsible for certain yellows diseases of plants and have been implicated in witches broom of alfalfa and maize stunt.

MYCORRHIZA (MYCORRHIZAE; MYCORRHIZAS): A symbiotic or mutually beneficial association of a fungus with the roots of certain green plants in which the fungus supplies the plant with water and minerals and receives sugar and vitamins from the plant in return. Mycorrhizas are generally divided into ectotrophic and endotrophic and there is now a nomenclature identifying seven mycorrhizal types. Ectotrophic mycorrhizas form a network of hyphae around the root and grow intercellularly between the cells epidermis and cortex of the root. The hyphae of endotrophic mycorrhizas, on the other hand enter the cortical cells of the host roots, where they form branching structures called arbuscles, which can fill individual host cells and constitute the organ of interchange of materials between the two partners.

MYCOSIS (MYCOSES): A condition in which fungi pass the resistance barriers of the human or animal body and establish infections. Some fungi reproduce through tiny spores in the air. People inhale those spores or they land on the skin. Mycoses can result when the climate is hot and humid and when people live in crowded conditions.

MYCOTA: An older classification system of a kingdom comprising the fungi.

MYCOTIC DERMATITIS (LUMPY WOOL): A disease affecting sheep, caused by the fungus *Dermatophilus congolensis*. Infection by the bacterium causes the skin to become inflamed and release an exudate, forming scabs and lumps in the wool.

MYCOTOXIN: Any of the mould-produced substances that may be injurious to vertebrates upon ingestion, inhalation or skin contact, causing diseases called mycotoxicoses. Some common mycotoxins are aflatoxin, ergot toxin and the agents of mushroom poisoning.

MYCOTROPHIC: Describing a symbiotic association between a fungus and the whole of a plant. Micotrophy occurs when a mycorrhizal fungus extends into the aerial parts of a plant, as in certain heathers and orchids. Examples are found in striped coral root (*Corallorhiza striata*) and spring coral root (*Corallorhiza wisteriana*).

MYCOVIRUS (MYCOPHAGE): A virus that infects fungi.

MYELIN: A whitish material made up of protein and fats that forms concentric sheath around some nerve cells in the membrane of Schwann cells and certain neuroglial cells. It insulates nearby nerve fibres and enables transmission of nerve impulses.

MYELIN SHEATH (MEDULLARY SHEATH): The layer of fatty acid material that surrounds and electrically insulates the axons of nerve cells of most vertebrates and some invertebrate neurones, formed from the cell membrane of the Schwann cell in the peripheral nervous system and from oligodendroglia cells. It allows for a more rapid transmission of nerve impulses (at speeds of up to 120 ms⁻¹). If damaged, the nerve impulse is impaired and can cause multiple sclerosis.

MYELINATED NERVE FIBRE: Nerve fibre enveloped by a sheath of myelin that mainly forms the somatic nerves that control voluntary motion.

MYELENCEPHALON (MEDULLA OBLONGATA): The lowest part of the vertebrate brain stem. It is continuous with the spinal cord, forms the major pathway for nerve impulses entering and leaving the skull and contains the cardiac, respiratory, vomiting and vasomotor centres and also controls involuntary vital functions such as heart beat and breathing.

MYELITIS: 1. Inflammation of the spinal cord. It occurs in diseases such as meningitis. 2. Inflammation of the bone marrow.

MYELOID TISSUE: Tissues within red bone marrow that produce the blood cells. They are found around the blood vessels and contain various cells that are precursors of the blood cells.

MYELOMA (MULTIPLE MYELOMA): Malignant disease of the bone marrow in which the bone marrow does not produce normal plasma cells, resulting in weakened immune system. Plasma cells make proteins called antibodies that are important in the immune system for fighting infections. Myeloma usually occurs in older people. The symptoms include fatigue, muscle weakness, severe bone pain and backache, loss of appetite, nausea, restlessness and confusion. Myeloma can cause anaemia, vertebral collapse, clotting problems and damage to the eyes and internal organs. It is treated with chemotherapy and radiotherapy.

MYIASIS: The infestation of vertebrates by the larvae or maggots, of numerous species of flies.

MYOCARDIAL INFARCTION (HEART ATTACK): A sudden onset of gripping central chest pain, often accompanied by sweating and vomiting, when the flow of blood to the heart is blocked due to death of a portion of the heart muscle following obstruction of a coronary artery by thrombosis (formation of a blood clot). This is most often caused by a build-up of fat, cholesterol and other substances, which form a plaque in the coronary arteries, obstructing the circulation of blood. Some other symptoms include nausea, indigestion, heartburn or abdominal pain, shortness of breath, fatigue, lightheadedness or sudden dizziness. A heart attack is a major cause of death and disability worldwide and death can occur within the first two hours, if immediate medical help is not sought.

MYOCARDIUM: The middle and thickest layer of the heart wall composed of cardiac muscle. The thickness of the myocardium varies, reflecting the magnitude of the pressure generated by the heart during contraction. It is thickest around the left ventricle which does most of the work in the heart. The inner surface of the myocardium is lined by a layer of endothelium which is continuous with the endothelium lining the blood vessels.

MYOCHROUS: Having the colour of a mouse.

MYOCYTE: A contractile cell especially a muscle cell, arising from myoblasts. Each myocyte contains myofibrils, which are long chains of sarcomeres, the contractile units of the cell.

MYOFIBRIL: A unit of thick and thin filaments in a muscle fibre. They are bundles of actomyosin filaments that run from one end of the cell to the other and are attached to the cell surface membrane at each end.

MYOGENIC: Originating from or produced by muscle cells. For example, the contractions of cardiac muscle fibres are described as myogenic since they are produced spontaneously, without requiring stimulation from nerve cells.

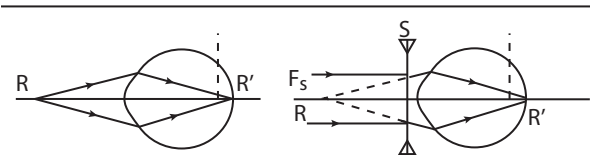
MYOGLOBIN: A globular protein, located in the vertebrate muscle to which oxygen binds. The oxygen is released only when haemoglobin can no longer supply adequate oxygen to muscle cells. It occurs widely in muscle tissue as an oxygen carrier and it comprises a single polypeptide chain and a haem group which reversibly binds a molecule of oxygen.

MYOMETRIUM: The thick middle layer of the uterine wall consisting of smooth muscle cells and supporting stromal and vascular tissue. The myometrium stretches during pregnancy to allow for the harbouring of the pregnancy and contracts in a coordinated fashion, via a positive feedback effect during the process of labour. It contracts to expel the placenta and reduce blood loss.

MYONEME (MYOFIBRIL): A series of microscopic filaments in the covering of the body of some simple ciliate organism that can contract and shorten the body.

MYOPHILY: Pollination of plants by flies, done for example, by flies that feed on nectar and pollen such as bee flies (Bombyliidae) and hoverflies (Syrphidae). Myophilous plants do not usually emit a strong scent, are typically purple, violet, blue and white and have open dishes or tubes.

MYOPIA (NEARSIGHTEDNESS): An eye condition in which a person sees short-distance objects but cannot see long distance objects or close objects are seen clearly, but objects farther away appear blurred. The defect results from the lens of the eye refracting the parallel rays of light entering it to focus in front of the retina, instead of directly on it, generally because of an abnormally long eyeball or the cornea is too steep. The symptoms include headaches, eyestrain, squinting or fatigue. It is corrected by using diverging eyeglasses or contact lenses to move the image back to the retina or by having refractive surgery.



Myopia (Short Sightedness)

MYOPIC ALGORITHM: Any of a class of algorithms that are so called because they look only at very local information at each iteration, as with steepest descent and often pay a price for this short sightedness.

MYOSIN: A contractile protein that interacts with actin to bring about contraction of muscle or cell movement. It is the main protein of the thick filaments in a muscle myofibril. It is composed of two coiled subunits that can aggregate to form a thick filament that is globular at each end.

MYOTOME: One of a series of segmented muscle blocks found in fishes and lancelets. Myotomes are arranged in pairs on either side of the body. It works automatically against the backbone, providing a means of locomotion by causing the tail to sweep from side to side.

MYOTUBE: An embryonic precursor of a skeletal muscle fibre. During development in the early embryo, numerous individual precursor cells called myoblasts, proliferate and migrate to sites where muscle formation will take place. There, they fuse to form myotubes, each containing numerous nuclei and bounded by a single plasma membrane. The process of differentiation continues as the myotube manufactures the characteristic muscle proteins that will form the contractile myofibrils and eventually matures into a muscle fibre.

MYRIAPOD: Any of several arthropods, such as the centipede or millipede, having segmented bodies, one pair of antennae and at least nine pairs of legs. Each segment has one or two pairs of legs. Myriapods live mostly in humid environments and can be found in soils, in leaf litters or under stones and wood. They play an important role in the breakdown of dead vegetable material because they are mainly phytosaprophagous, subsisting on dead plants; although some are carnivorous.

MYXOMATOSIS: Contagious, usually fatal, virus infection of rabbits causing swelling of the mucous membranes and inflammation and discharge around the eyes. It is caused by a poxvirus of the species Myxoma virus and transmitted by infected mosquitoes, biting flies and direct contact of a rabbit with an infected one. It has been deliberately introduced in some countries such as UK and Australia to reduce the rabbit population.

N

N-(4-HYDROXYPHENYL) ACETAMIDE [ACETAMINOPHEN; 4-ACETAMIDOPHENOL; N-(4-HYDROXYPHENYL) ETHANAMIDE; PARACETAMOL]:

A synthetic analgesic drug similar to aspirin having the chemical formula $C_8H_9NO_2$, molecular mass 151.163 g/mol, density 1.263 g/cm³, melting point 169 °C and boiling point of 420 °C. It is effective for relieving pain such as headache, muscle pain and lower back pain; and also used in reducing fever. It has weak anti-inflammatory properties. An overdose can cause severe, often irreversible or even fatal, liver and kidney damage.

n: Abbreviation and symbol for nano- (a factor of 10^{-9} or 0.000000001), often used in the Systeme International (S.I.) for symbols for fractions of units such as length and time.

n-space: In mathematics, a space with n dimensions.

n-tuple: In mathematics, an ordered set of n elements, where n is a non-negative integer.

N: Symbol for newton, the Systeme International (S.I.) unit of force, named after Sir Isaac Newton.

n-: A prefix for an unspecified but specific finite number of variables, dimensions, terms, etc. For example, an **n-gon** is a polygon with n sides, a 6-gon is a polygon with six sides known as a hexagon.

n-gon: A polygon with n sides. For example, an 8-gon is a polygon with 8 sides, known as an **octagon**.

n-type conductivity: Electrical conductivity caused by the flow of electrons in semiconductors.

n-type semiconductors: Semiconductors that contain donor impurities. They are formed when a pure (intrinsic) semiconductor like silicon is doped with a pentavalent element like phosphorous.

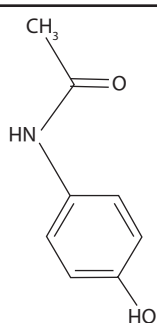
N'DAMA: A breed of hardy cattle of the Bos taurus type, known for its resistance to diseases, found in parts of West Africa. It is medium sized with a large and strong head and lyre-shaped horns.

NABLA (DEL OPERATOR; DIFFERENTIAL OPERATOR): The differential vector operator, with the symbol Δ , used in vector analysis and defined as $i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$, where i, j, k are unit vectors in the

x, y and z directions, respectively and $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ and $\frac{\partial}{\partial z}$ are the respective partial derivatives of the given function of x, y and z .

NACREOUS: Having a pearly lustre; exhibiting rainbow-like colours.

NAD+: Abbreviation for the oxidised form of nicotinamide adenine dinucleotide (NAD), $C_{21}H_{27}N_7O_{14}P_2$. NAD is a coenzyme derived from the B vitamin nicotinic acid and found in all living cells and involved in various metabolic processes by accepting and donating electrons.



Chemical Structure
of Paracetamol

NAD⁺ accepts electrons from other molecules and used in redox reactions in the cell and reduces to NADH. NADH donates electrons to the electron transport chain to generate adenosine triphosphate (ATP) via oxidative phosphorylation.

NADIR: In astronomy, it is the point on the celestial sphere directly below the observer and directly opposite the zenith.

NADP+: Abbreviation for **nicotinamide adenine dinucleotide phosphate**, an enzyme to which electrons are transferred from photosystem I during photosynthesis. The addition of electrons reduces NADP, which acquires a hydrogen ion to form NADPH, which is the reduced form of NADP, a storage form of energy that can be transferred to the Calvin Cycle for the production of carbohydrate.

NAGWARE (ANNOYWARE): In computing, it is any software that is free for a trial period and afterwards periodically reminds (usually through pop-up windows or alerts) the user to register or purchase the product.

NAIL: 1. In biology, it is a hard, flat, flexible outgrowth of the digits of primates (humans, monkeys, apes) or layer of horny substance over the outer tip of a finger or toe. It consists of dead cells containing the fibrous protein keratin, pushed outward by dividing cells in the root, a fold of epidermis at the base of the nail. Each nail consists of a root, concealed within a fold of the skin; a body or exposed part attached to the surface of the skin; a free anterior extremity called the edge; and the matrix, which is the skin underneath the root and body of the nail. 2. A thin piece of metal that is sharp at one end and flat at the other end. It is important in woodwork and construction and used to fasten objects to things such as wood and concrete.

NAIVE SET THEORY (CANTOR SET THEORY): The presentation of the theory of sets in the style of an informal mathematical theory, regarding it as a body of given knowledge rather than the consequences of a set of uninterpreted axioms. In naive set theory, a set is a collection of objects called members or elements that is regarded as being a single object. An object x that is a member of a set A , is written as $x \in A$ and $x \notin A$ indicates that x is not a member of A .

NAKED BUD: A bud that lacks scales.

NAME SERVER: A network server that translates domain names that is presented to it into IP addresses. This makes it possible for a user to access a website by typing in the domain name instead of the website's actual IP address, which are harder to remember. For example, Domain Name Servers (DNS) on the Internet look up the numeric IP address of an alphabetic domain name such as "www.firefox.com," and returns the IP address of Firefox's website. Name servers are a fundamental part of the Domain Name System (DNS).

NAND CIRCUIT: A logic circuit that produces an output that is the inverse of the output of an AND gate. Its output signal is a logical 1 if any of its inputs is a logical 0 and if the output signal is a logical 0 all of its inputs are logical 1.

NANO: A prefix used in SI units of measurement to denote one-billionth part. For example, 10^{-9} metre = 1 nanometre. Its symbol is nm.

NANOCOMPOSITE: Composite in which at least one of the phase domains has at least one dimension of the order of nanometres.

NANOCOMPUTER: A computer smaller than the microcomputer, having microscopic physical dimensions (usually about the size of a credit card) with circuitry so small that it can only be seen through a microscope. Its technology comes from the emerging field of nanotechnology. A common example of a nanocomputer is the Raspberry Pi.

NANOFILTRATION: Pressure-driven membrane-based separation process in which particles and dissolved macromolecules smaller than 2 nm are rejected.

NANO GEL: Particle of gel of any shape with an equivalent diameter of approximately 1 nm to 100 nm.